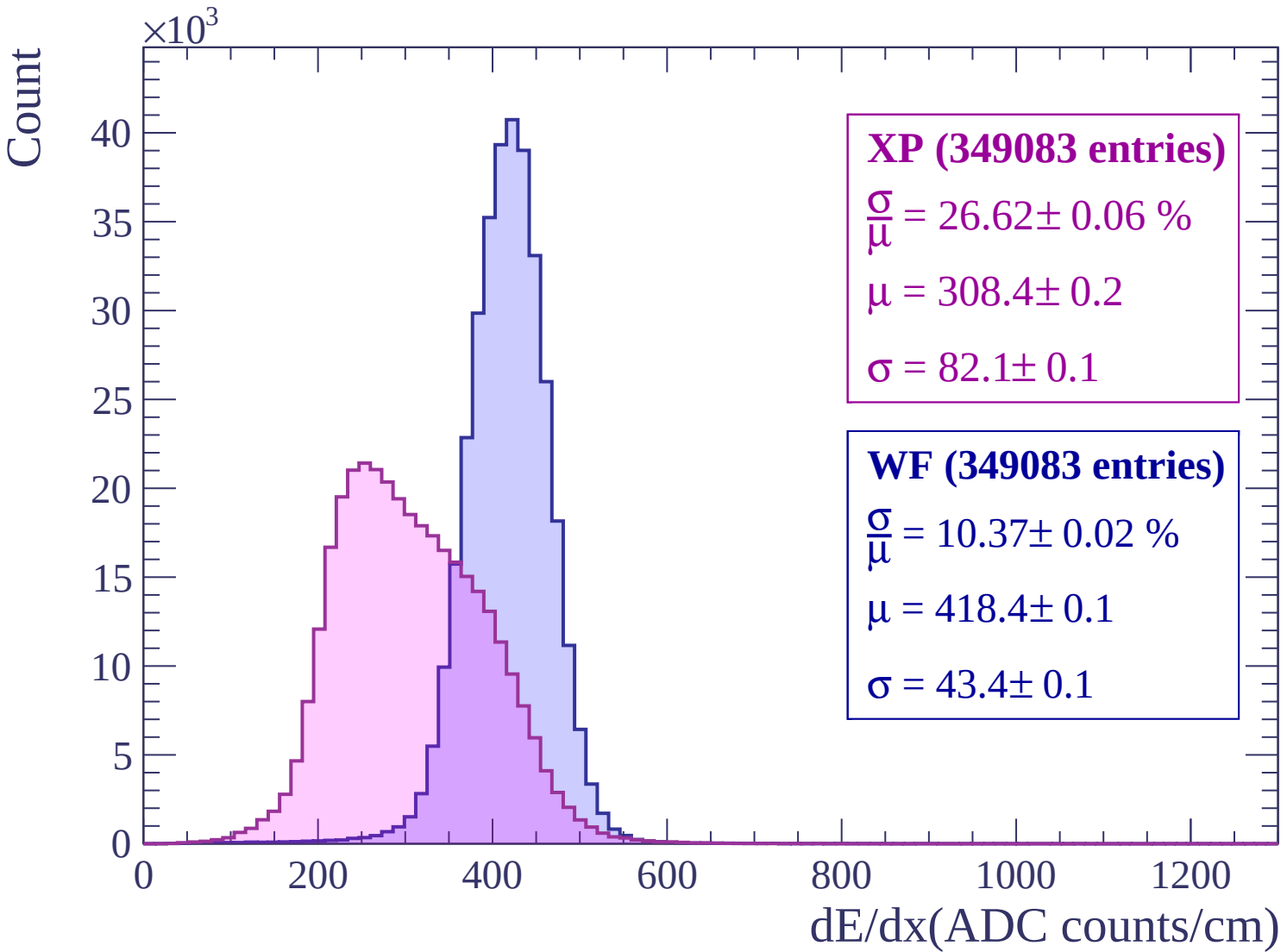
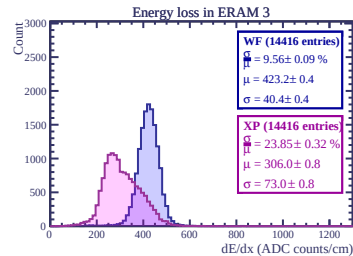
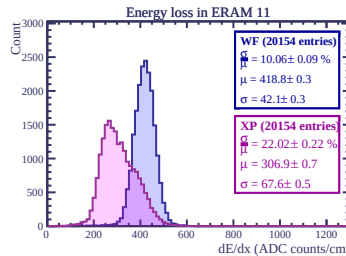
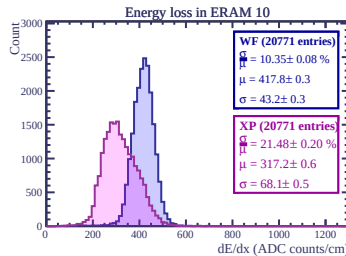
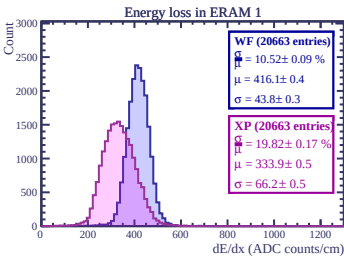
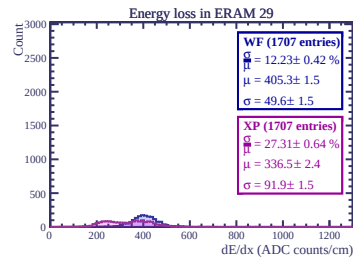
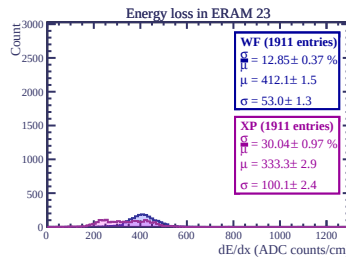
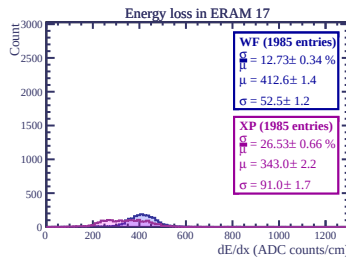
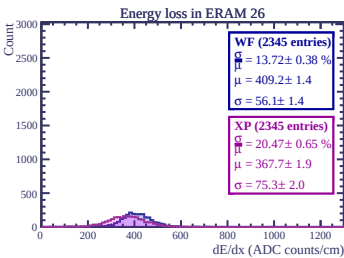
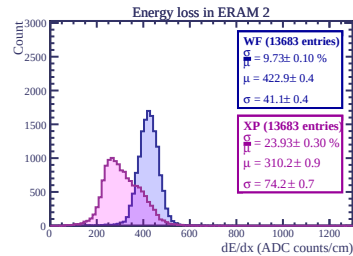
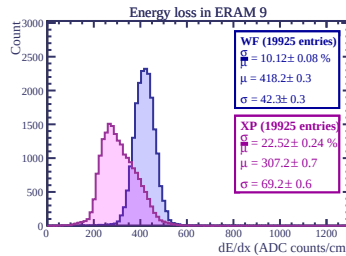
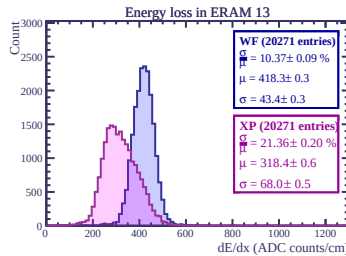
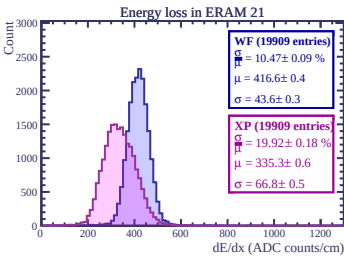
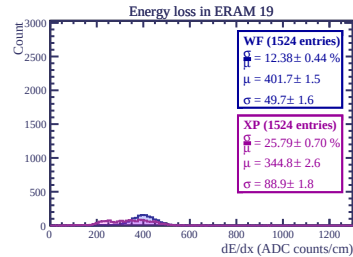
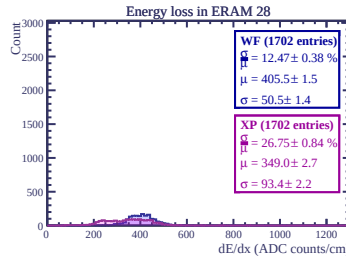
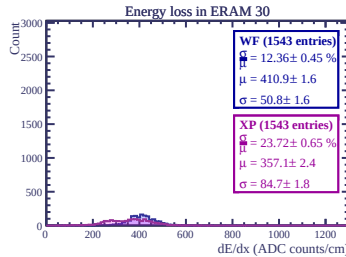
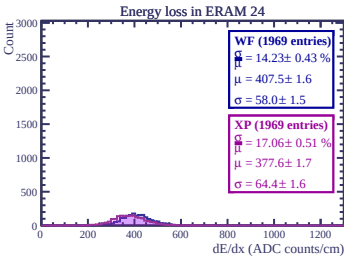
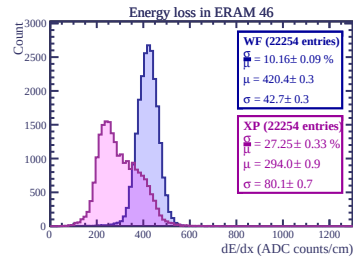
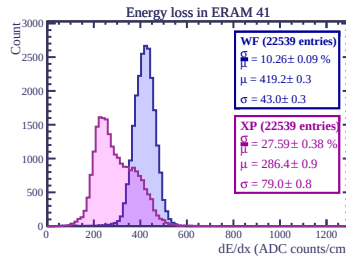
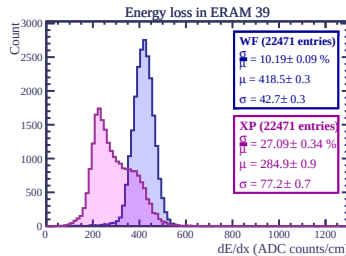
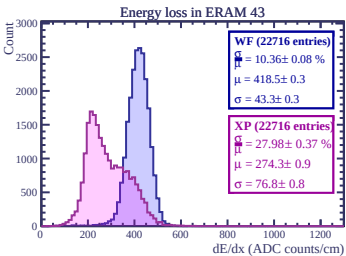
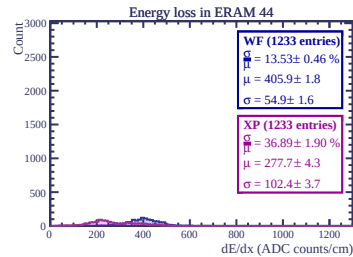
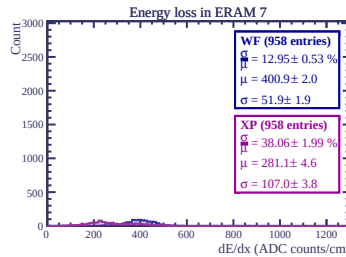
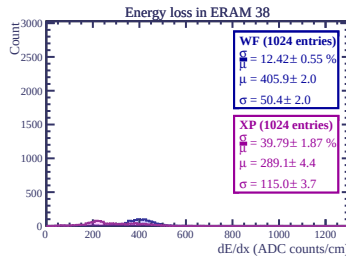
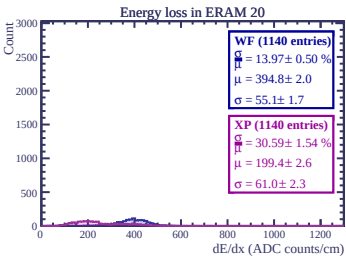
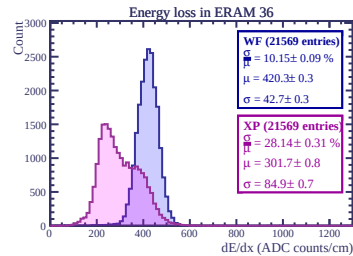
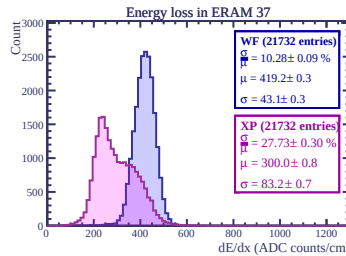
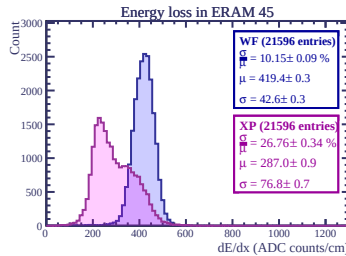
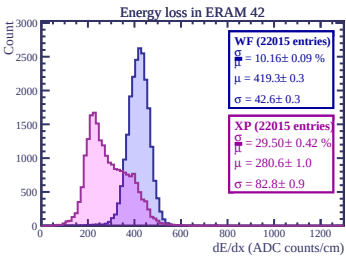
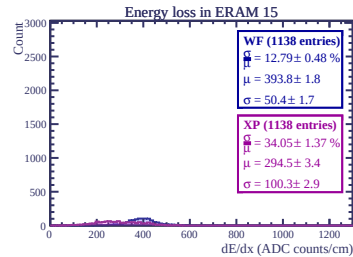
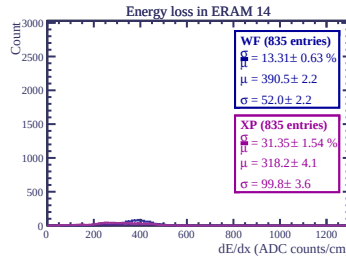
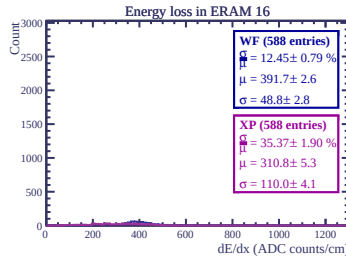
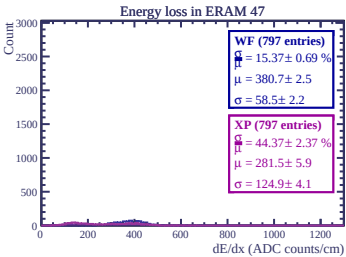
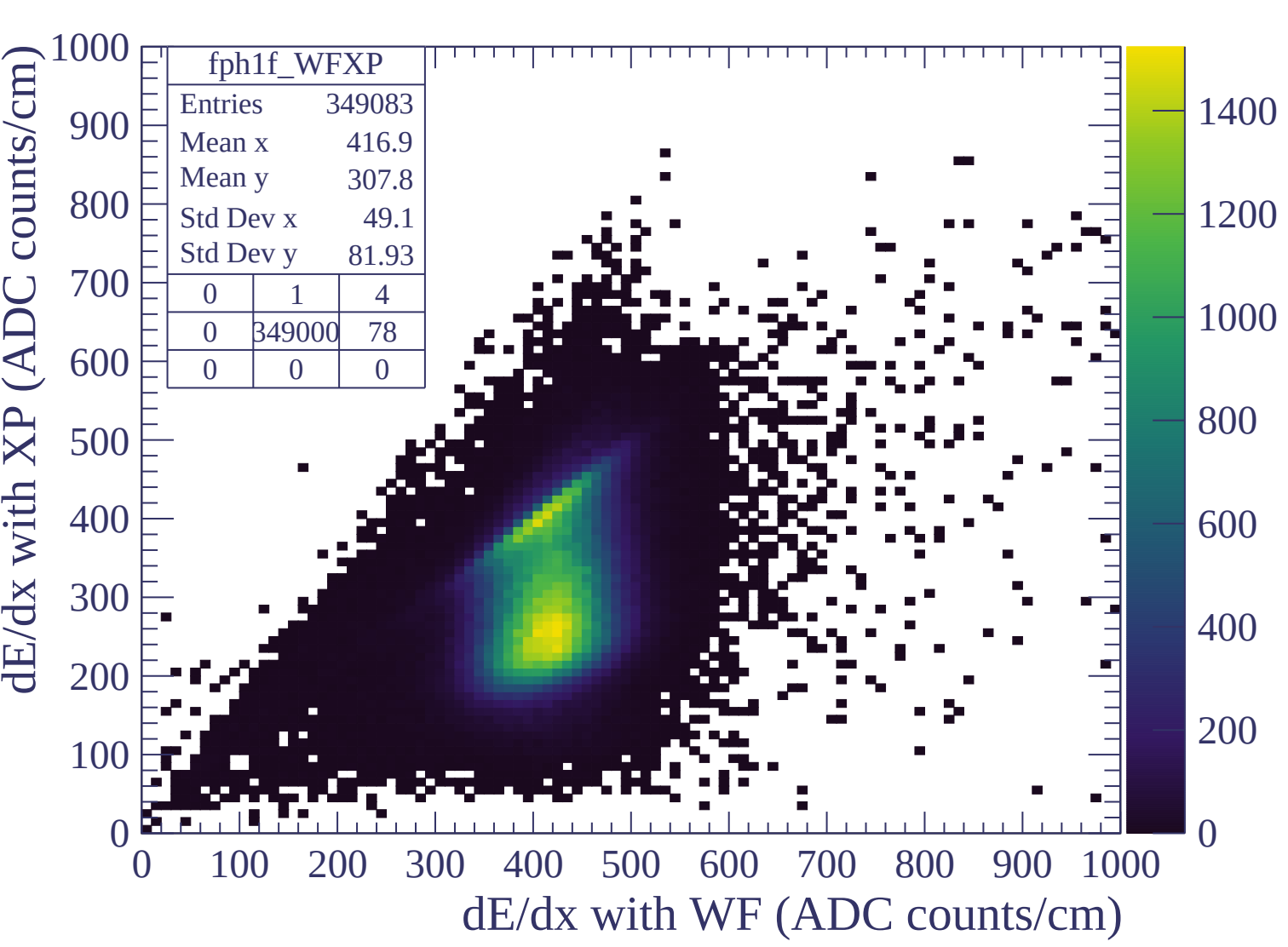
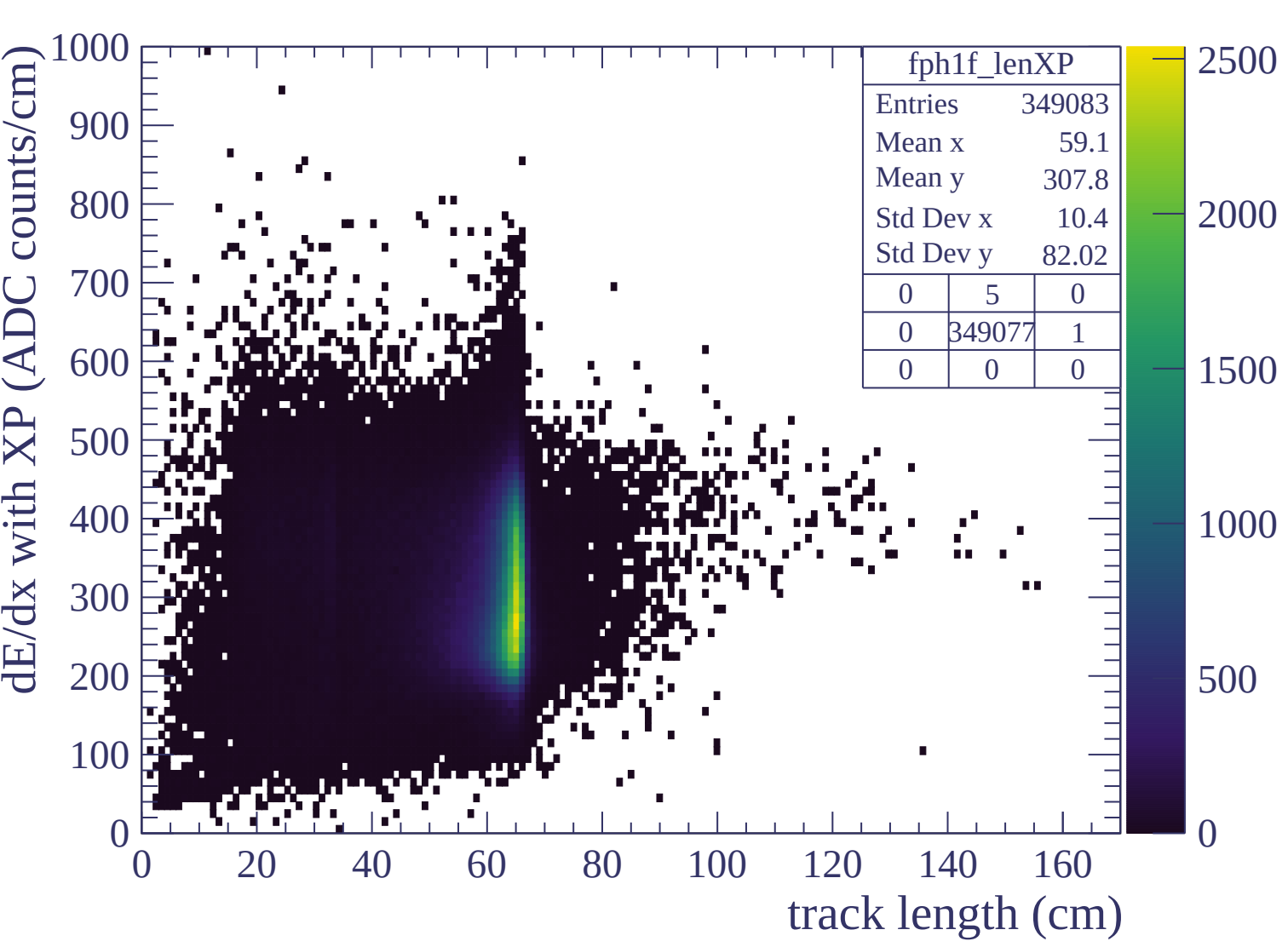


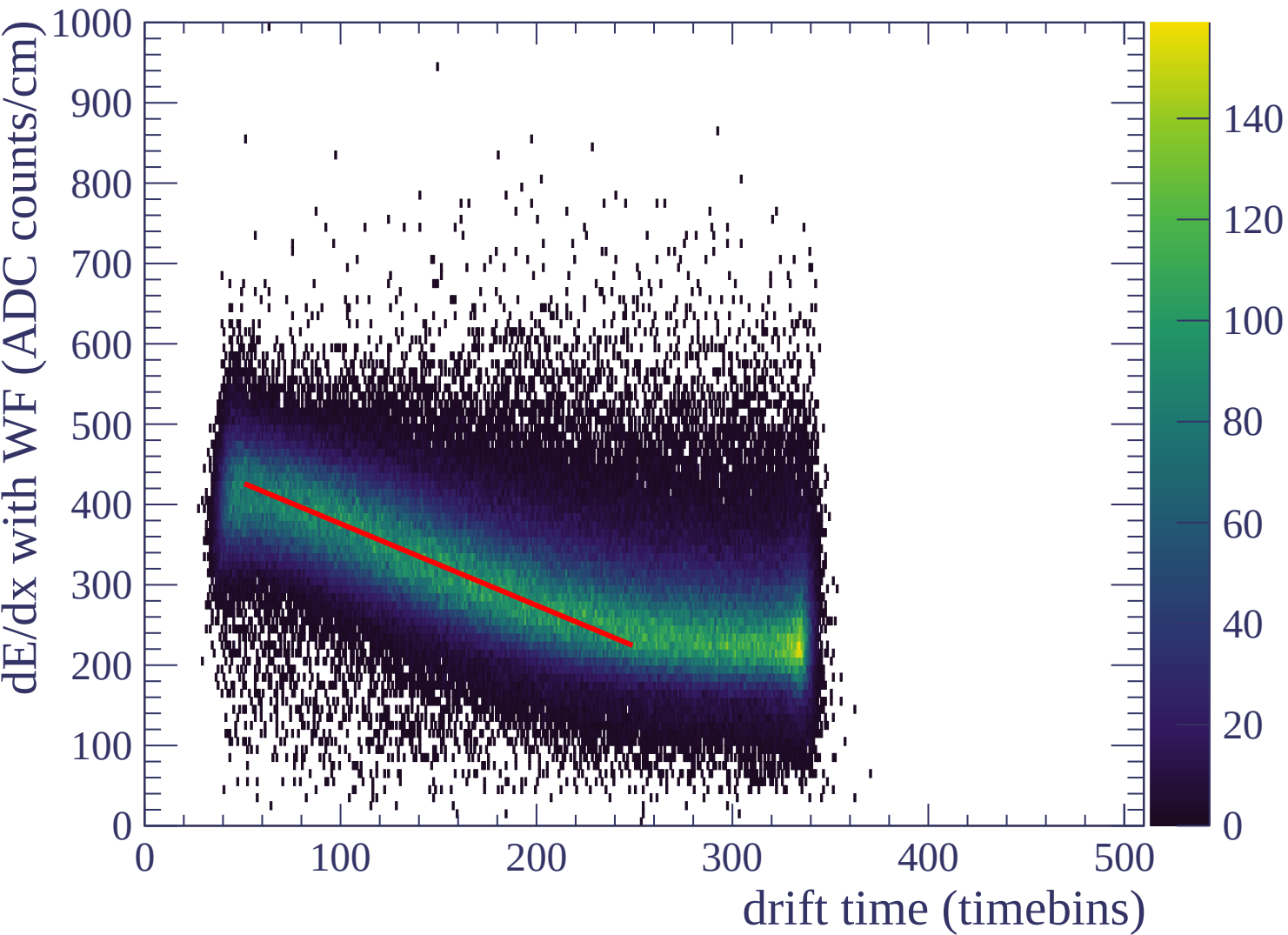


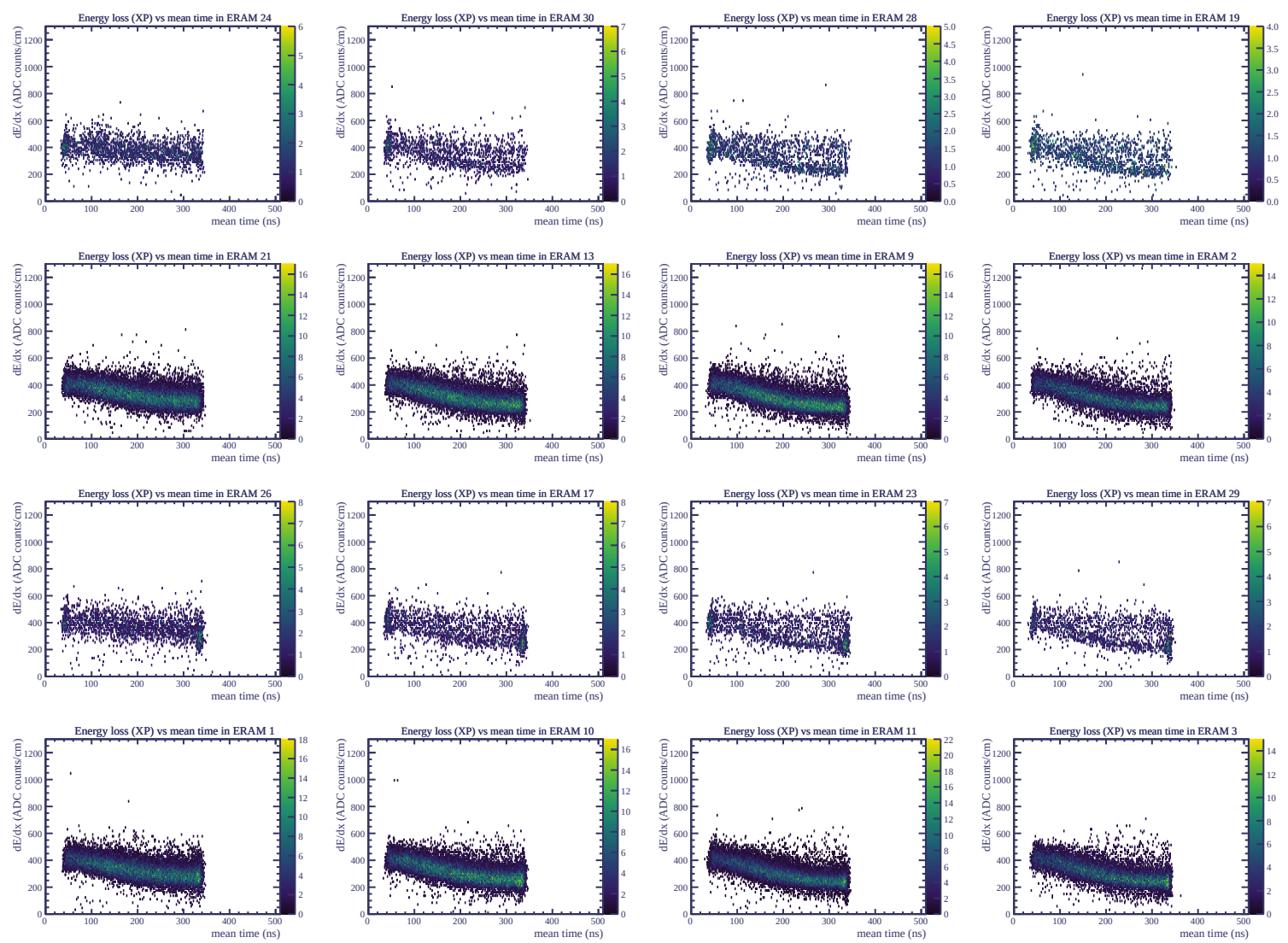


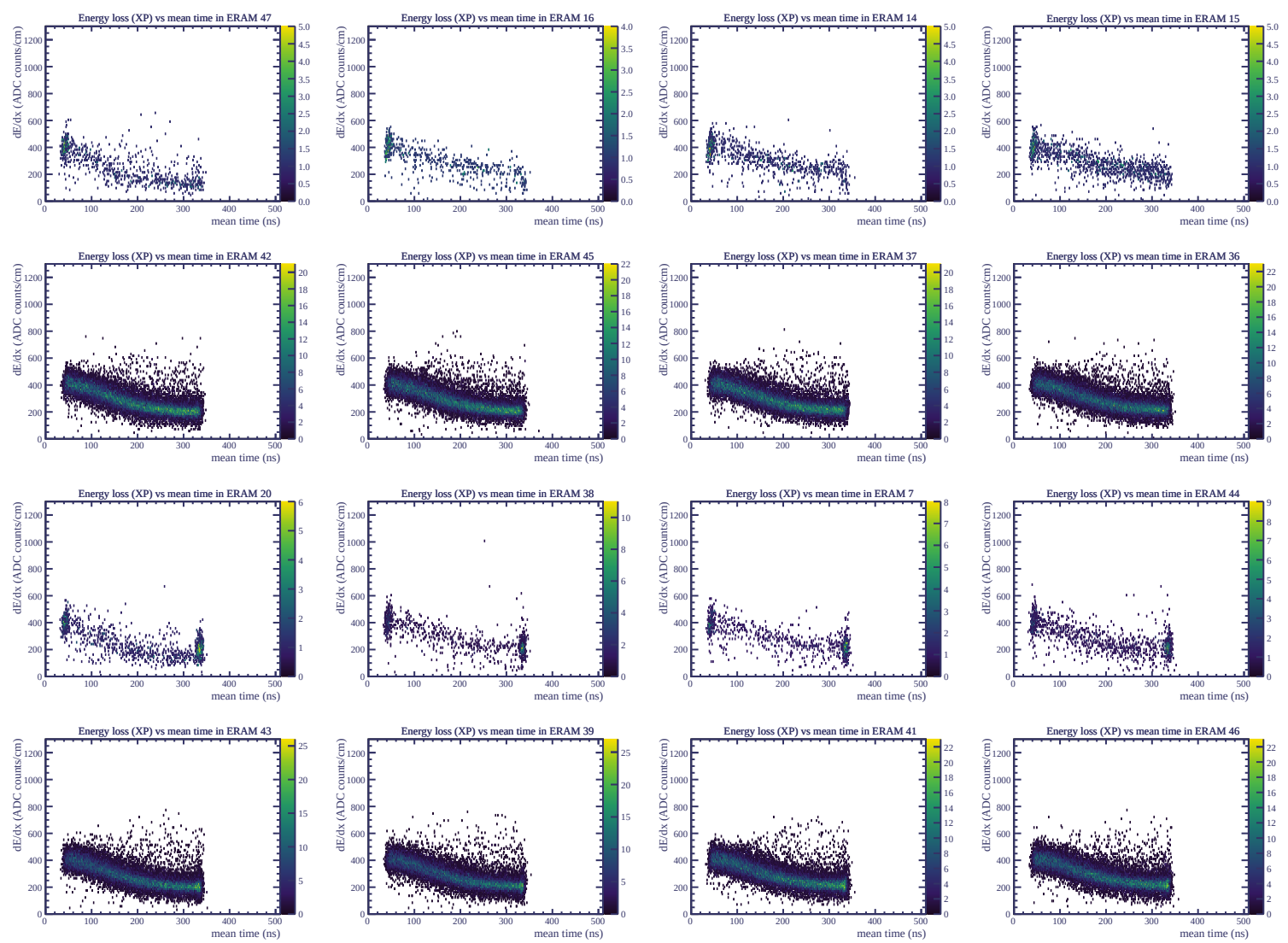




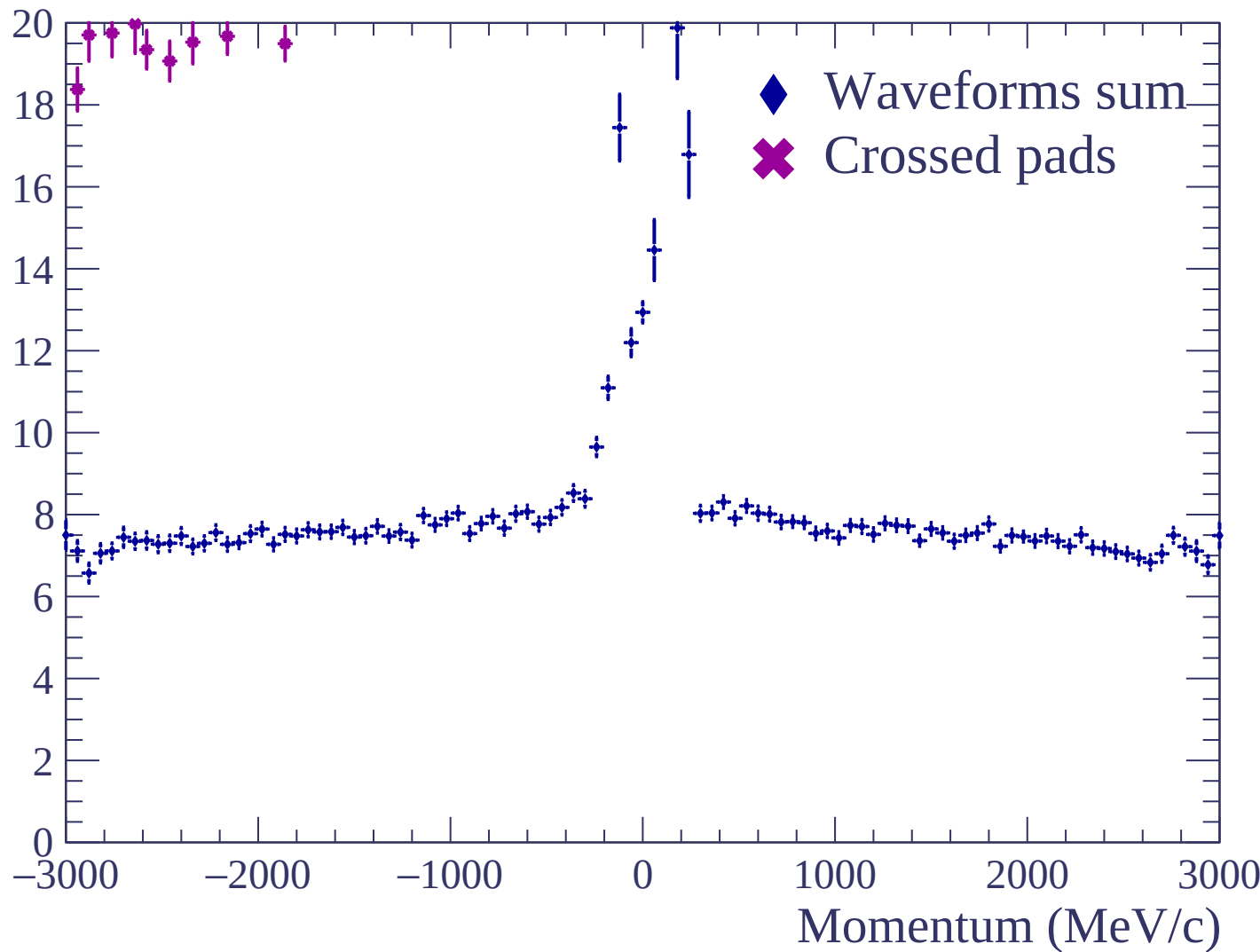


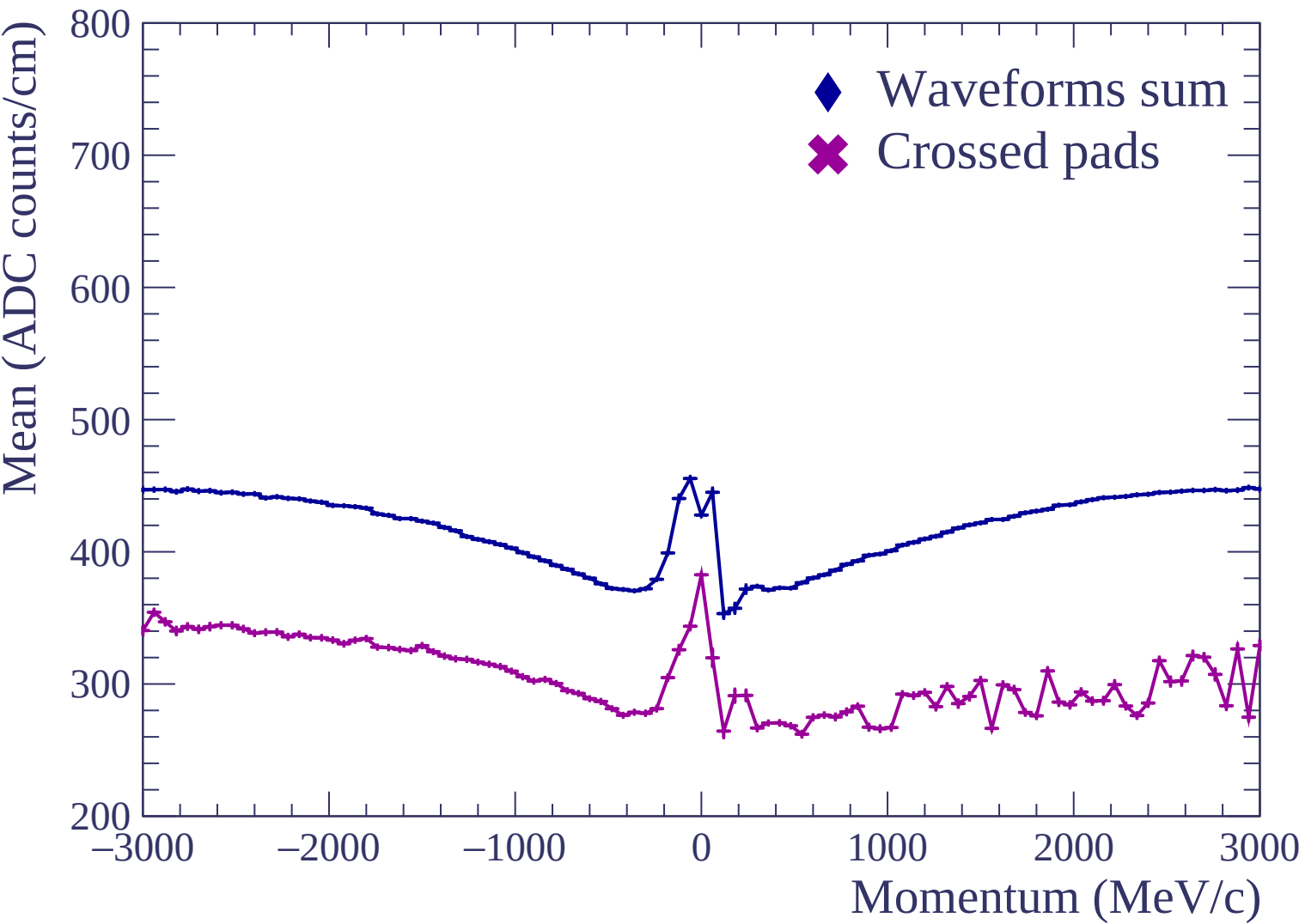




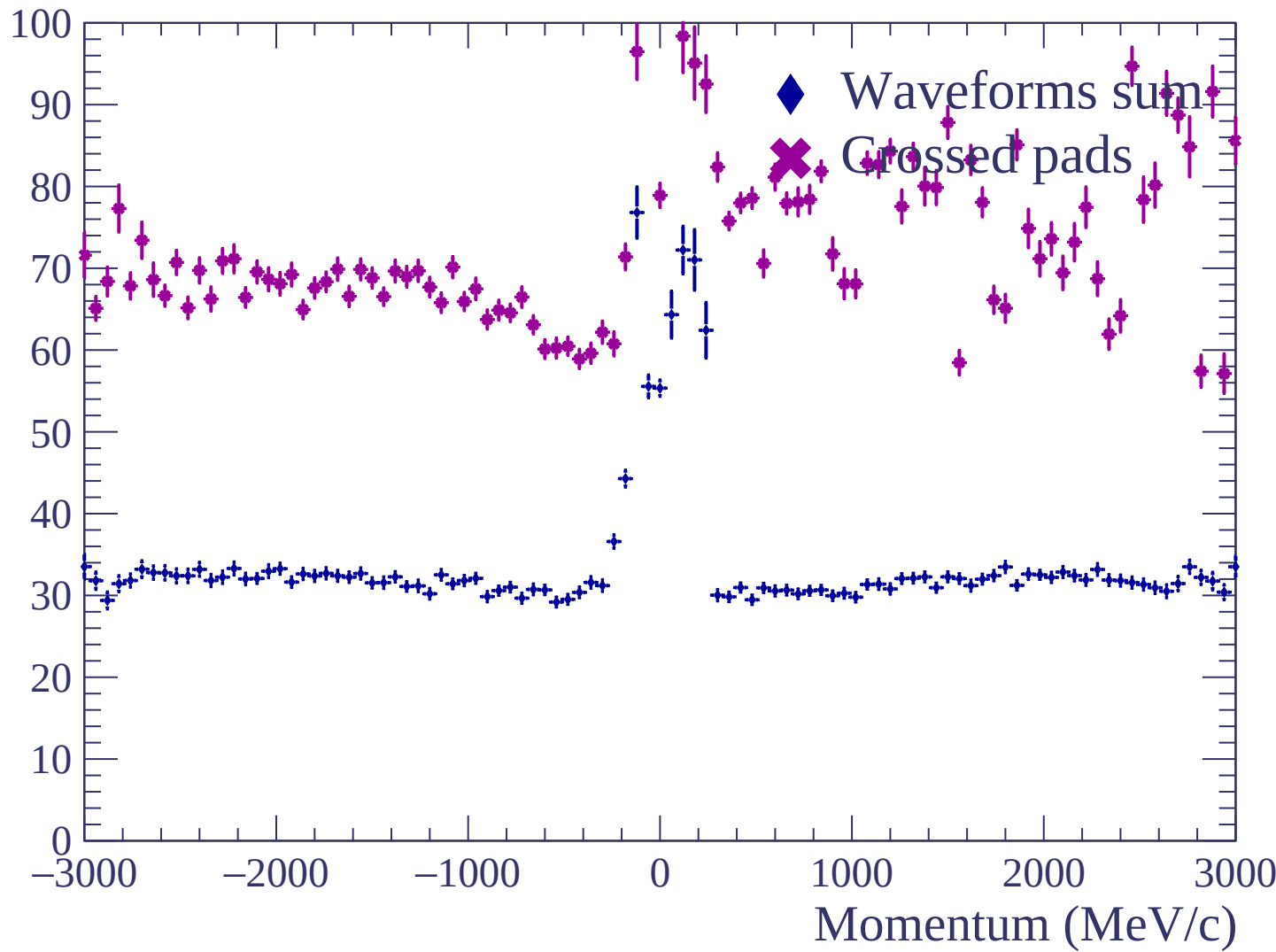


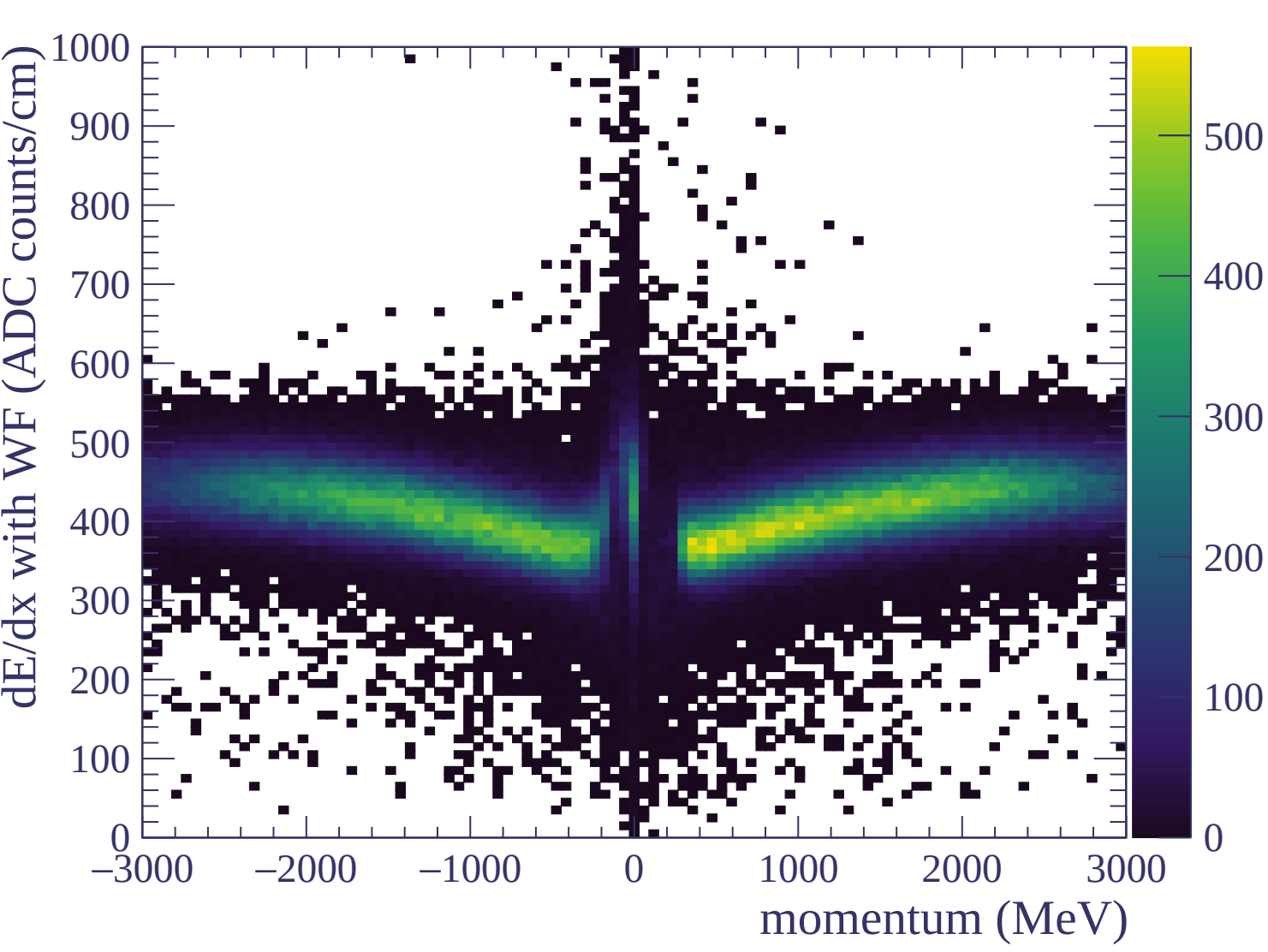
Resolution (%)

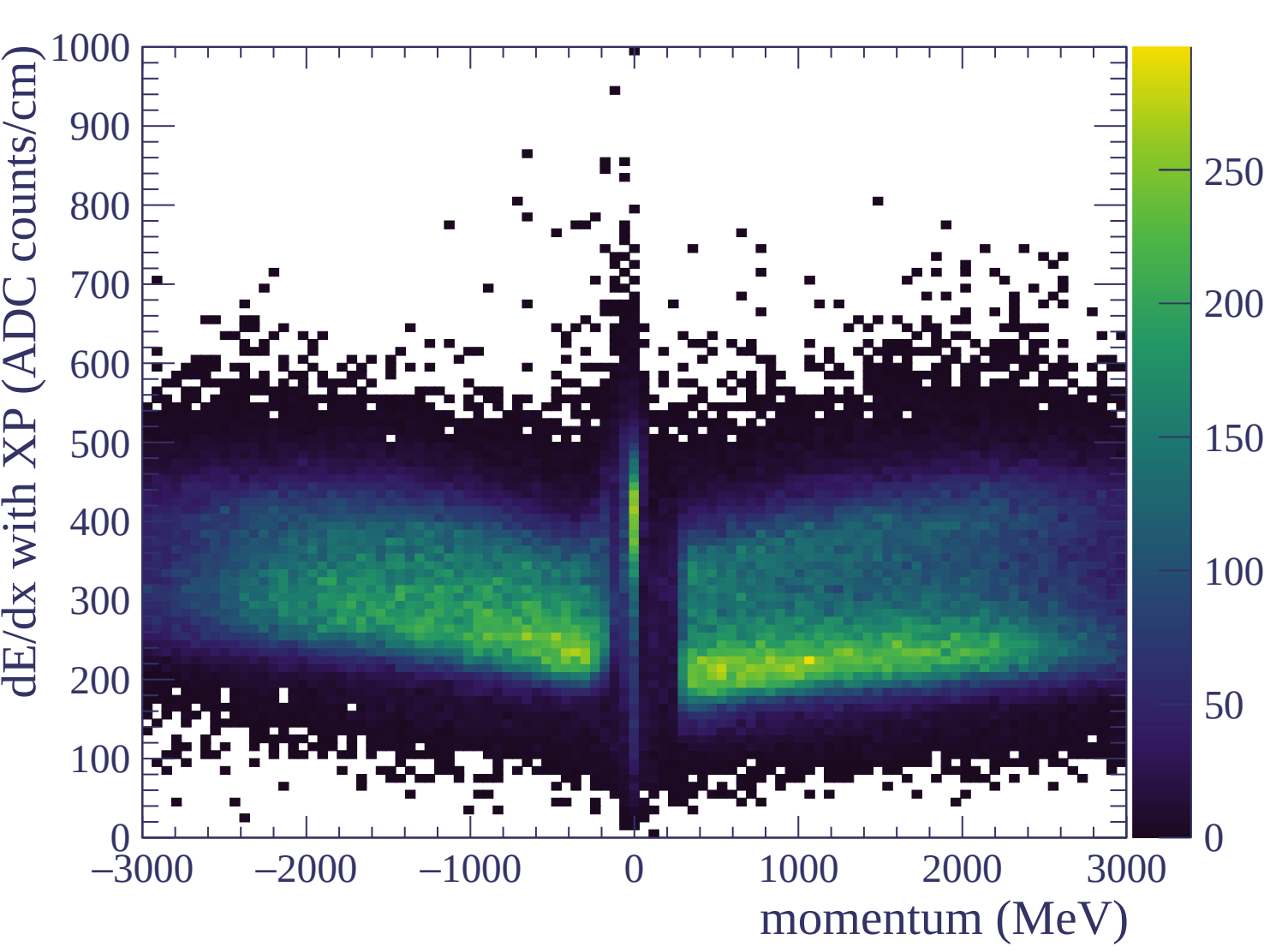


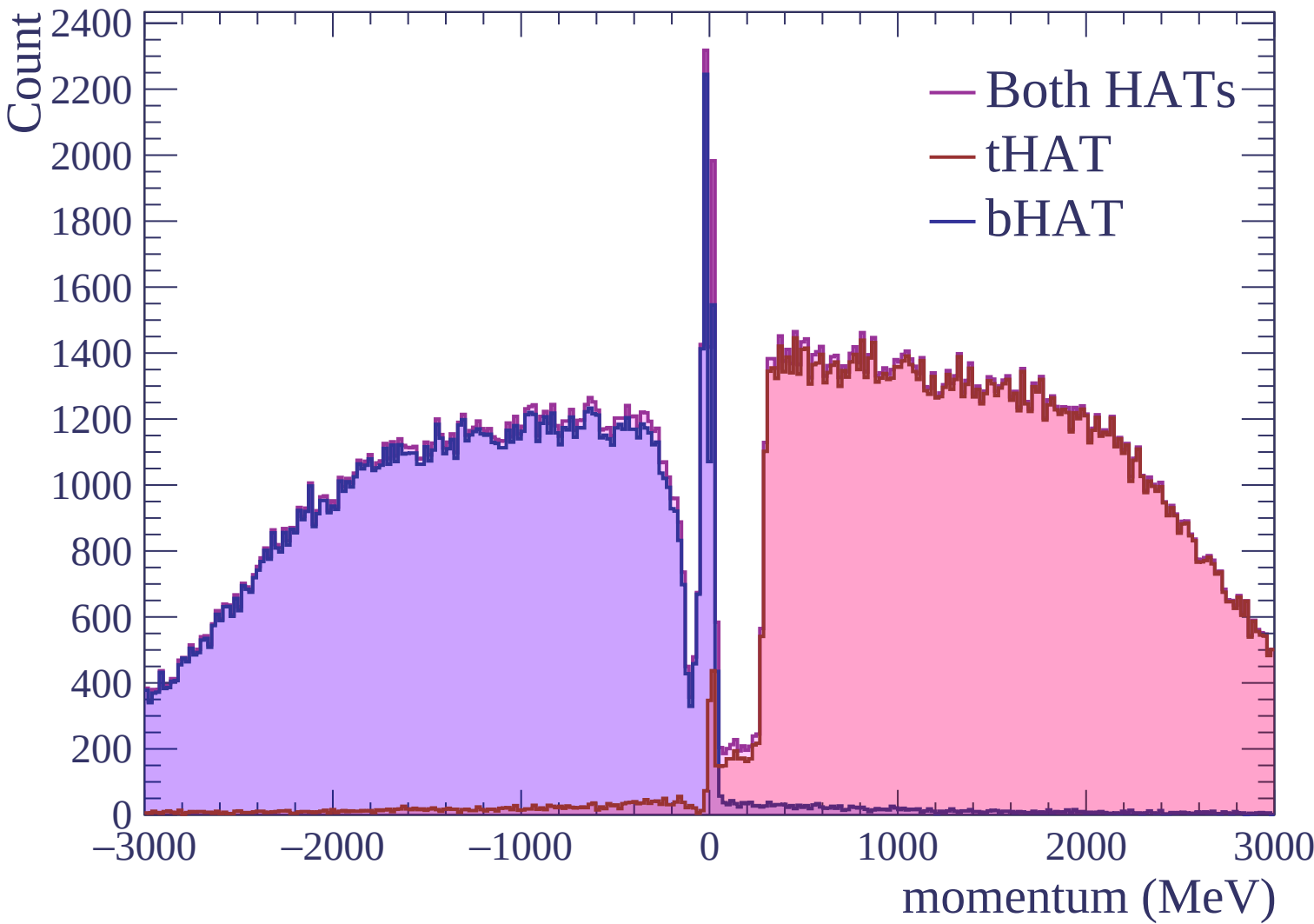


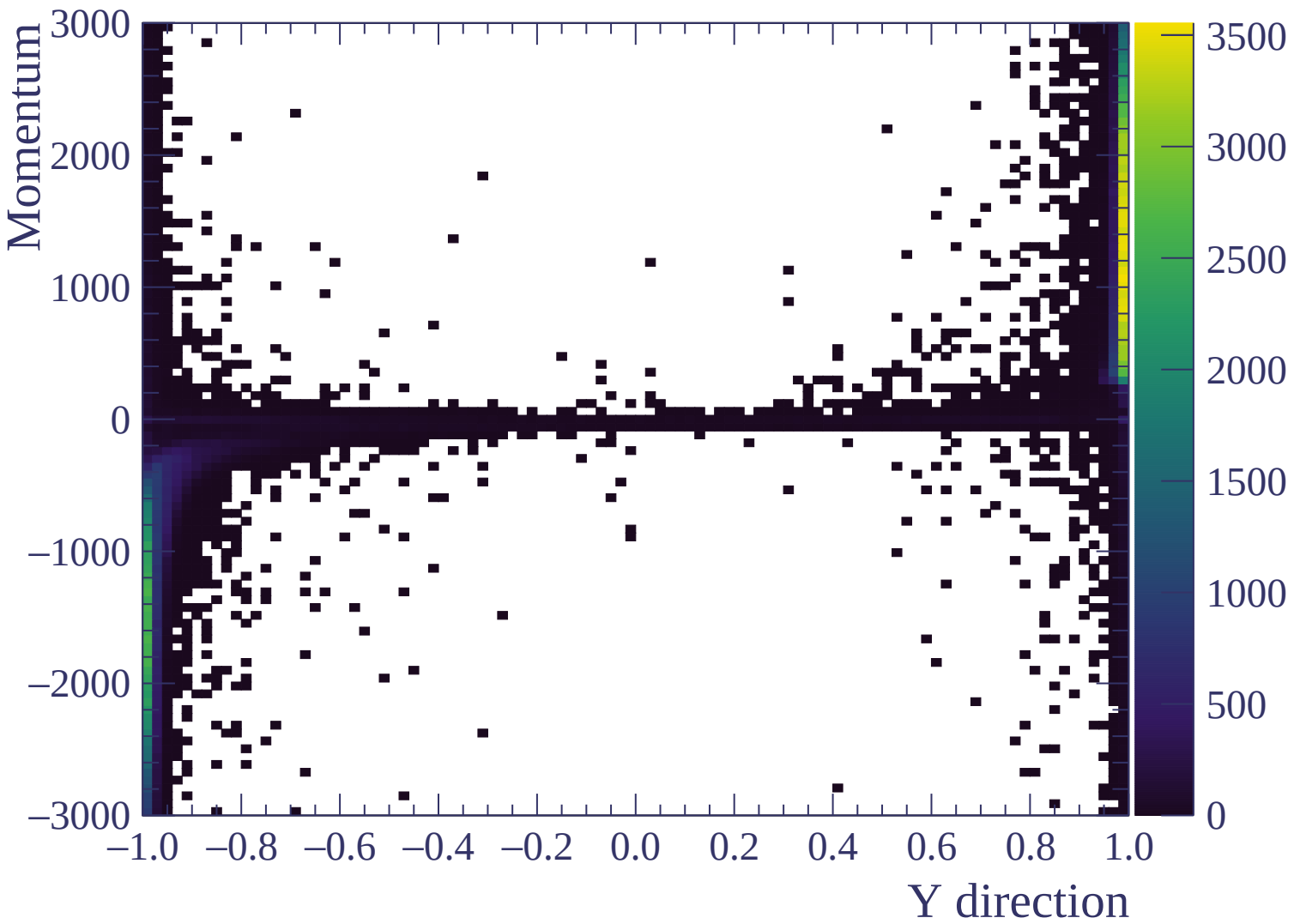
Standard deviation (ADC counts/cm)

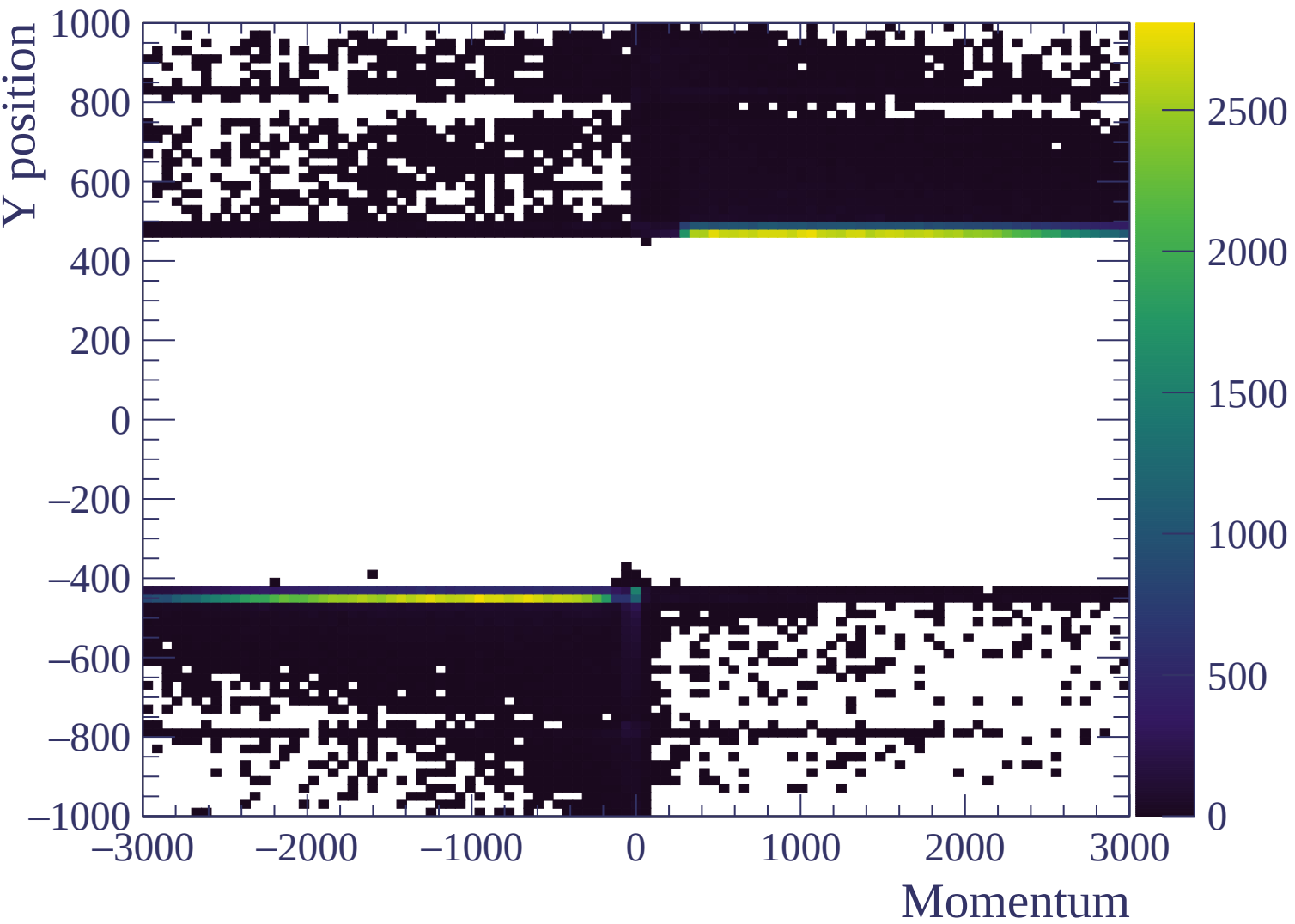


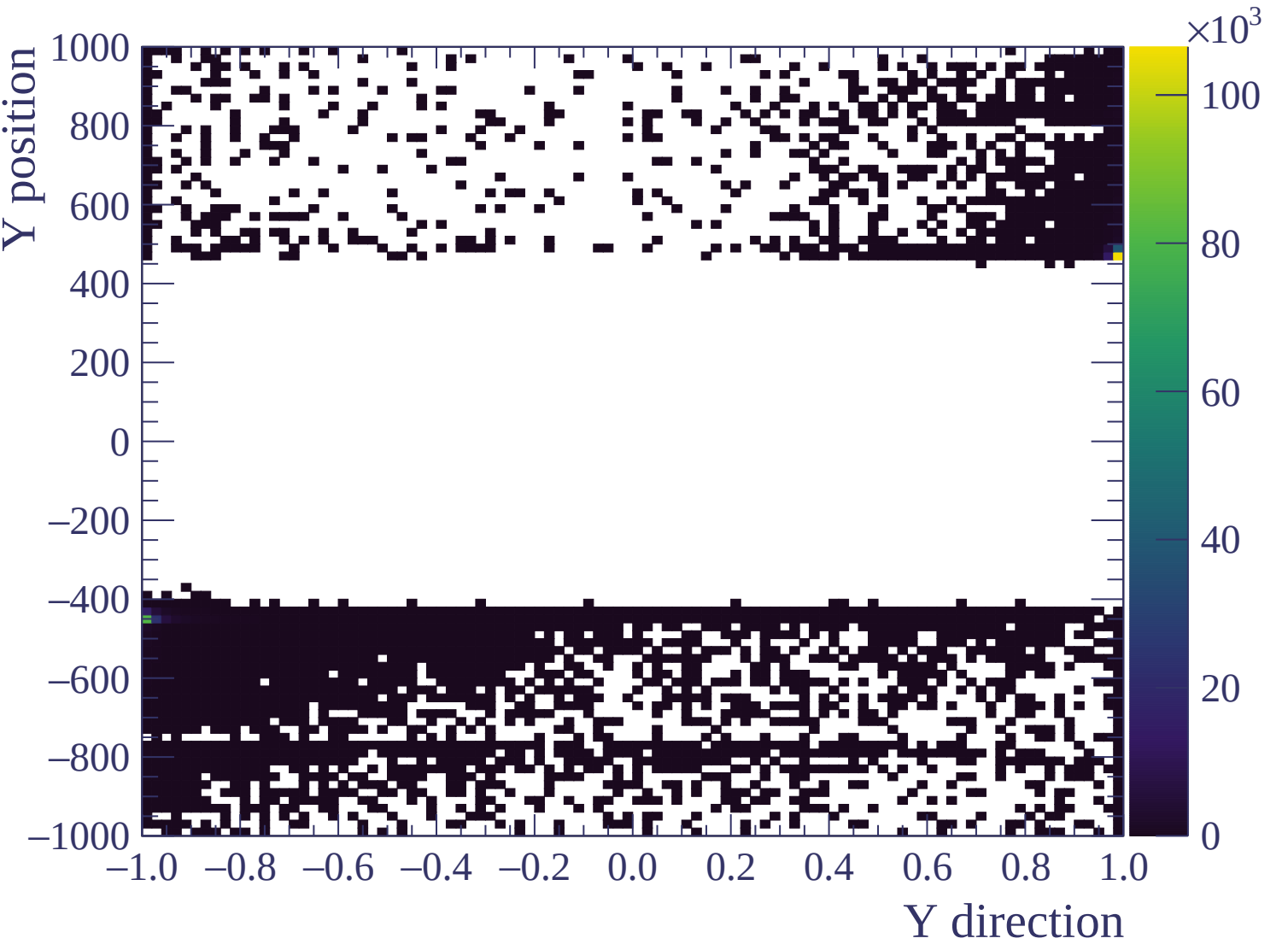






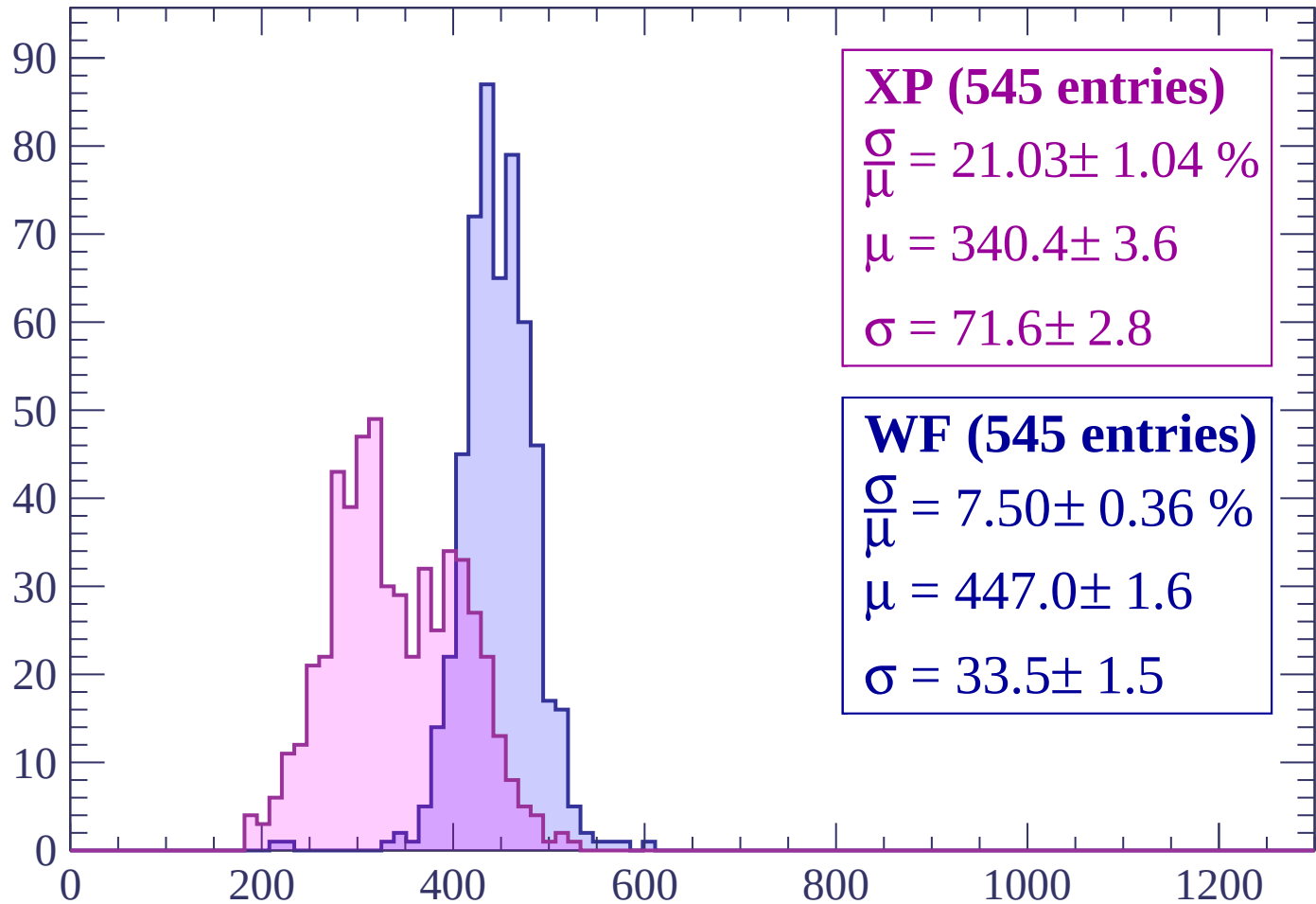






Energy loss | $-3000 < p < -2940$

Count



XP (545 entries)

$$\frac{\sigma}{\mu} = 21.03 \pm 1.04 \%$$

$$\mu = 340.4 \pm 3.6$$

$$\sigma = 71.6 \pm 2.8$$

WF (545 entries)

$$\frac{\sigma}{\mu} = 7.50 \pm 0.36 \%$$

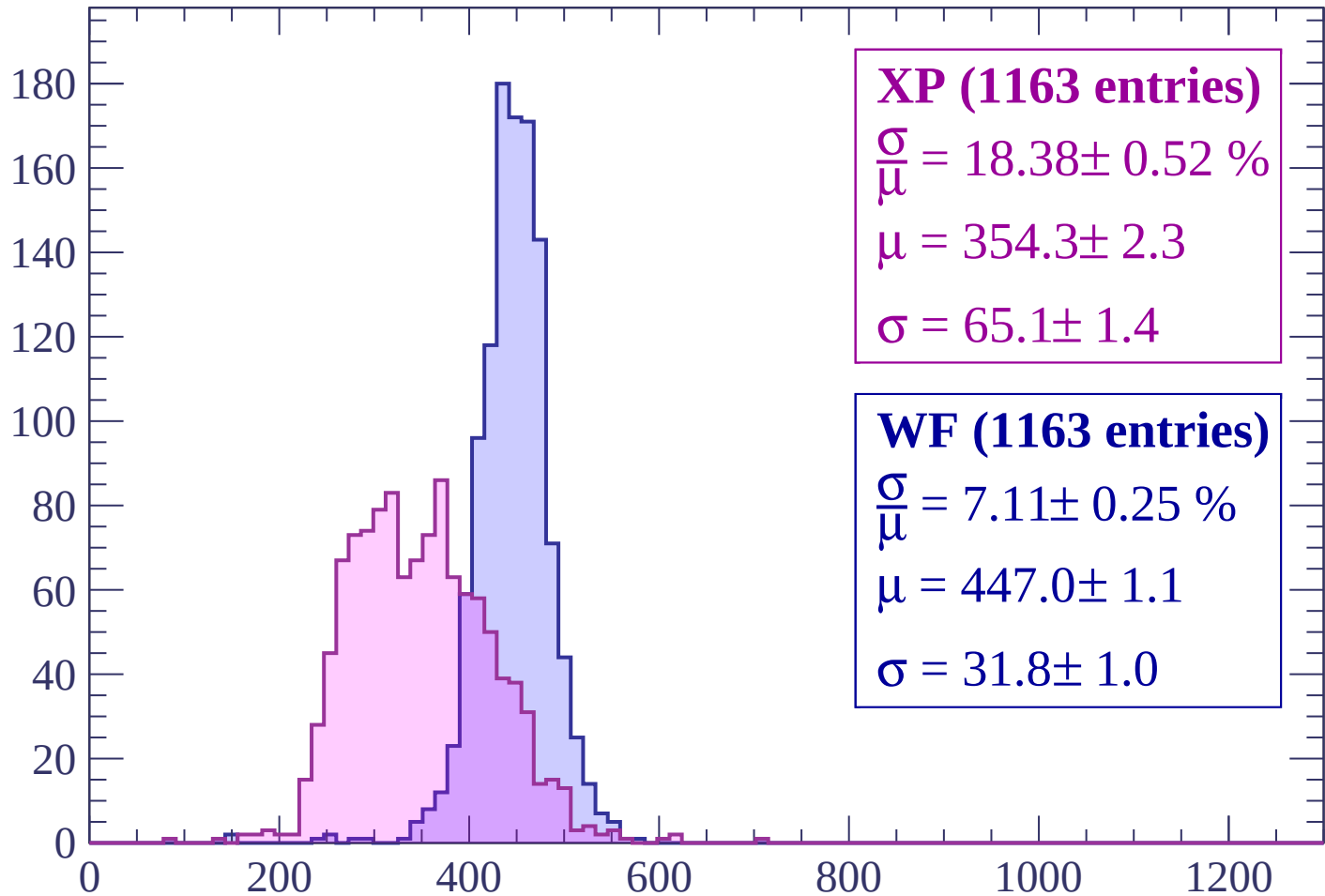
$$\mu = 447.0 \pm 1.6$$

$$\sigma = 33.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-2940 < p < -2880$

Count



XP (1163 entries)

$$\frac{\sigma}{\mu} = 18.38 \pm 0.52 \%$$

$$\mu = 354.3 \pm 2.3$$

$$\sigma = 65.1 \pm 1.4$$

WF (1163 entries)

$$\frac{\sigma}{\mu} = 7.11 \pm 0.25 \%$$

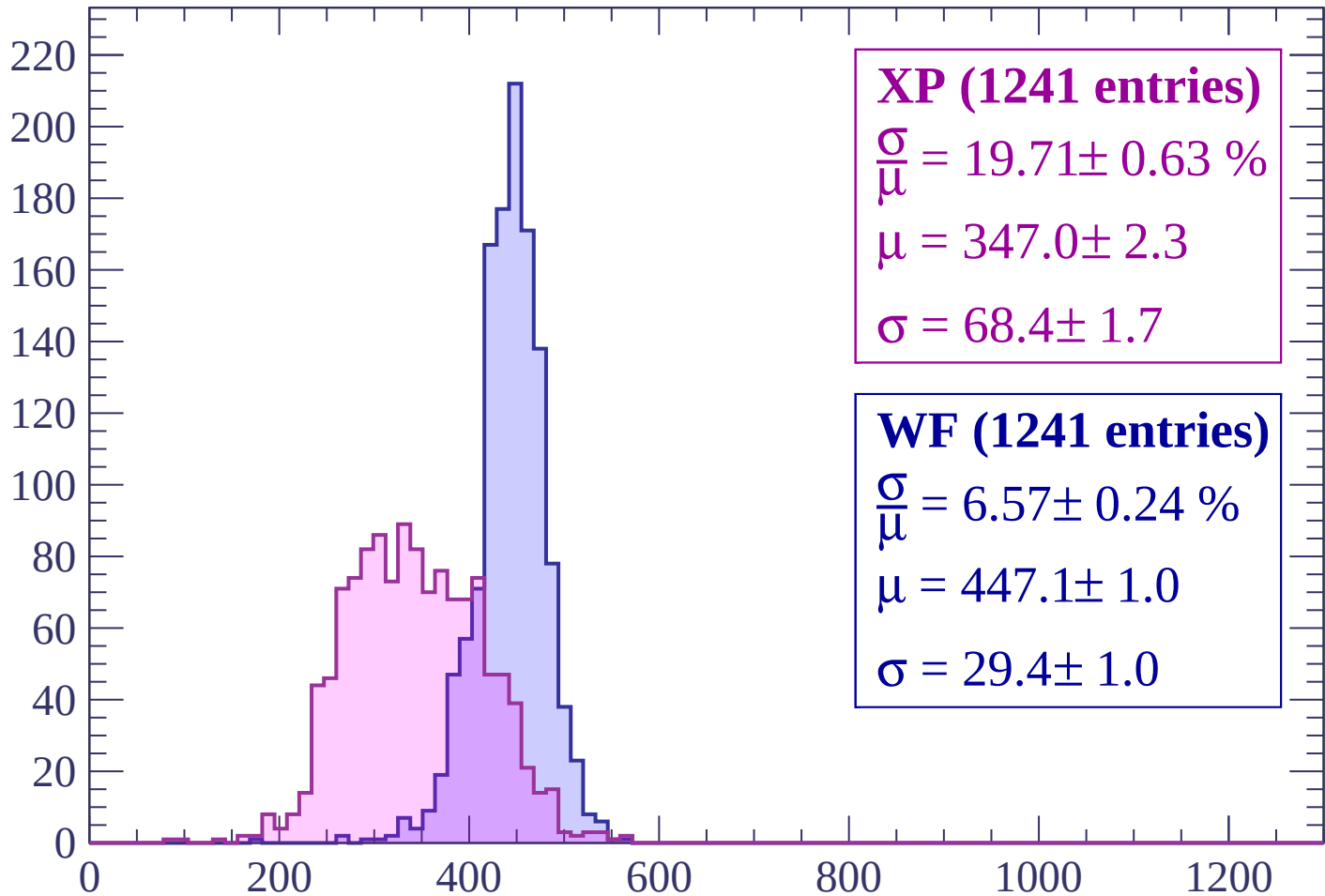
$$\mu = 447.0 \pm 1.1$$

$$\sigma = 31.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-2880 < p < -2820$

Count



XP (1241 entries)

$\frac{\sigma}{\mu} = 19.71 \pm 0.63 \%$

$\mu = 347.0 \pm 2.3$

$\sigma = 68.4 \pm 1.7$

WF (1241 entries)

$\frac{\sigma}{\mu} = 6.57 \pm 0.24 \%$

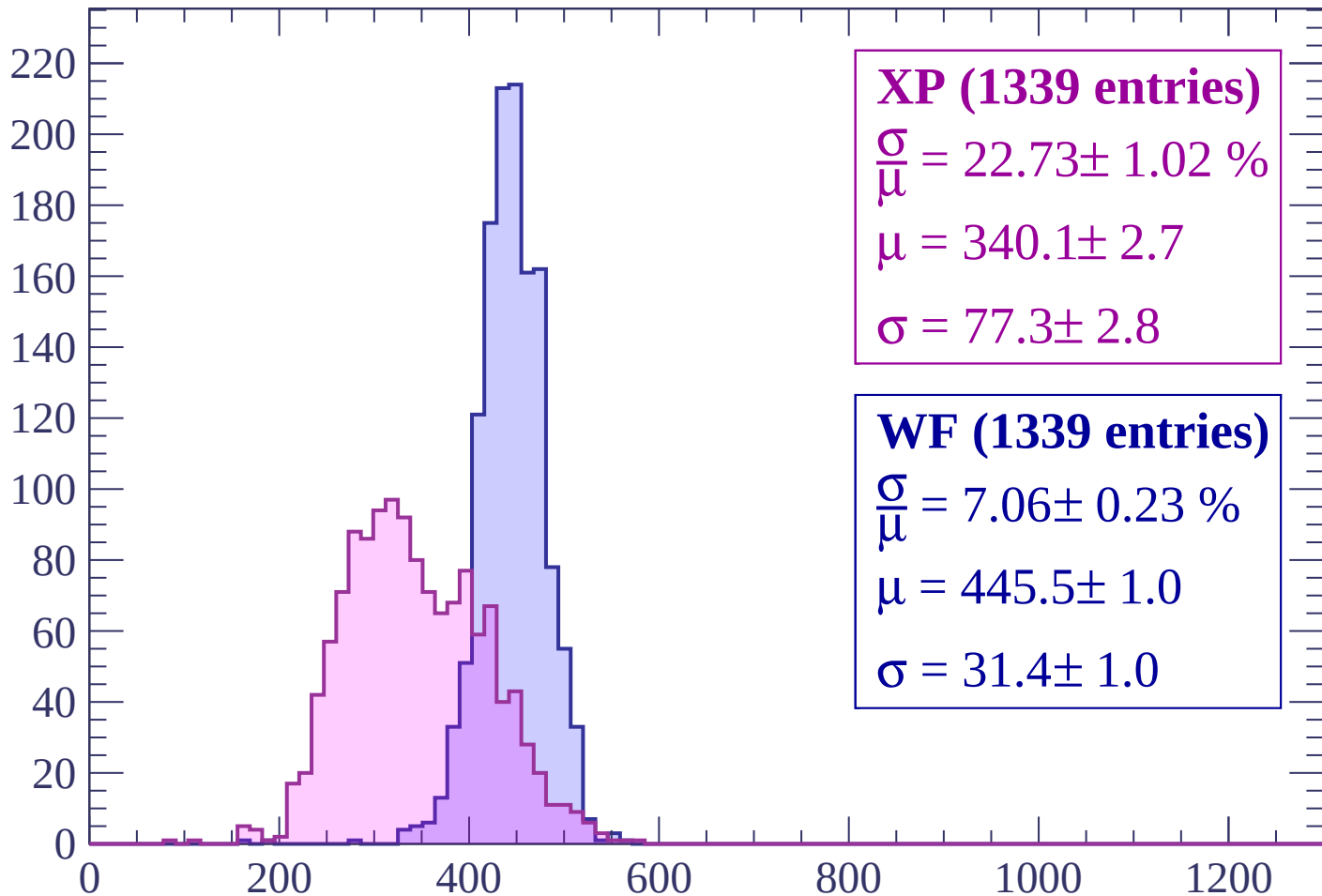
$\mu = 447.1 \pm 1.0$

$\sigma = 29.4 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $-2820 < p < -2760$

Count



XP (1339 entries)

$$\frac{\sigma}{\mu} = 22.73 \pm 1.02 \%$$

$$\mu = 340.1 \pm 2.7$$

$$\sigma = 77.3 \pm 2.8$$

WF (1339 entries)

$$\frac{\sigma}{\mu} = 7.06 \pm 0.23 \%$$

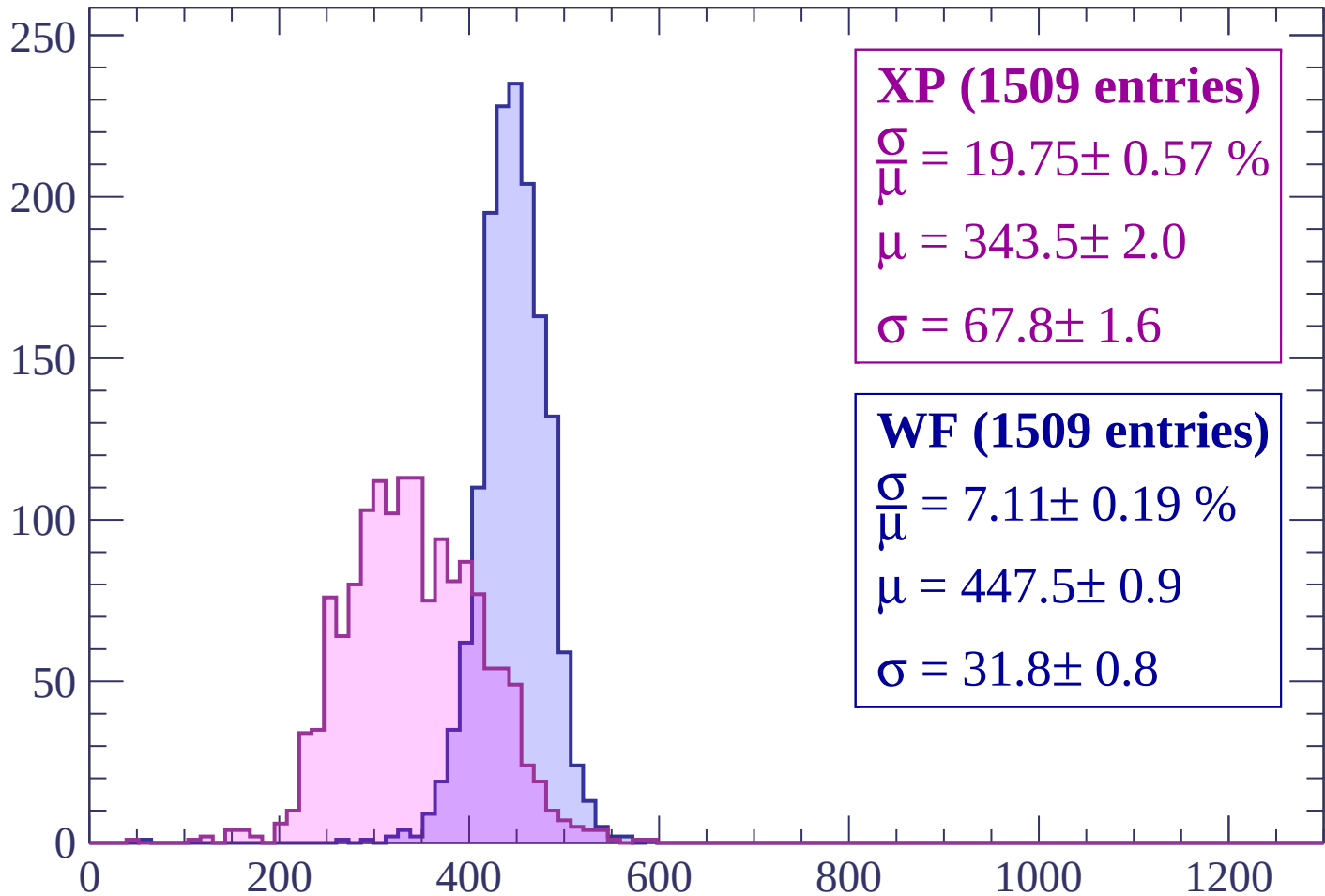
$$\mu = 445.5 \pm 1.0$$

$$\sigma = 31.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -2760 < p < -2700

Count



XP (1509 entries)

$$\frac{\sigma}{\mu} = 19.75 \pm 0.57 \%$$

$$\mu = 343.5 \pm 2.0$$

$$\sigma = 67.8 \pm 1.6$$

WF (1509 entries)

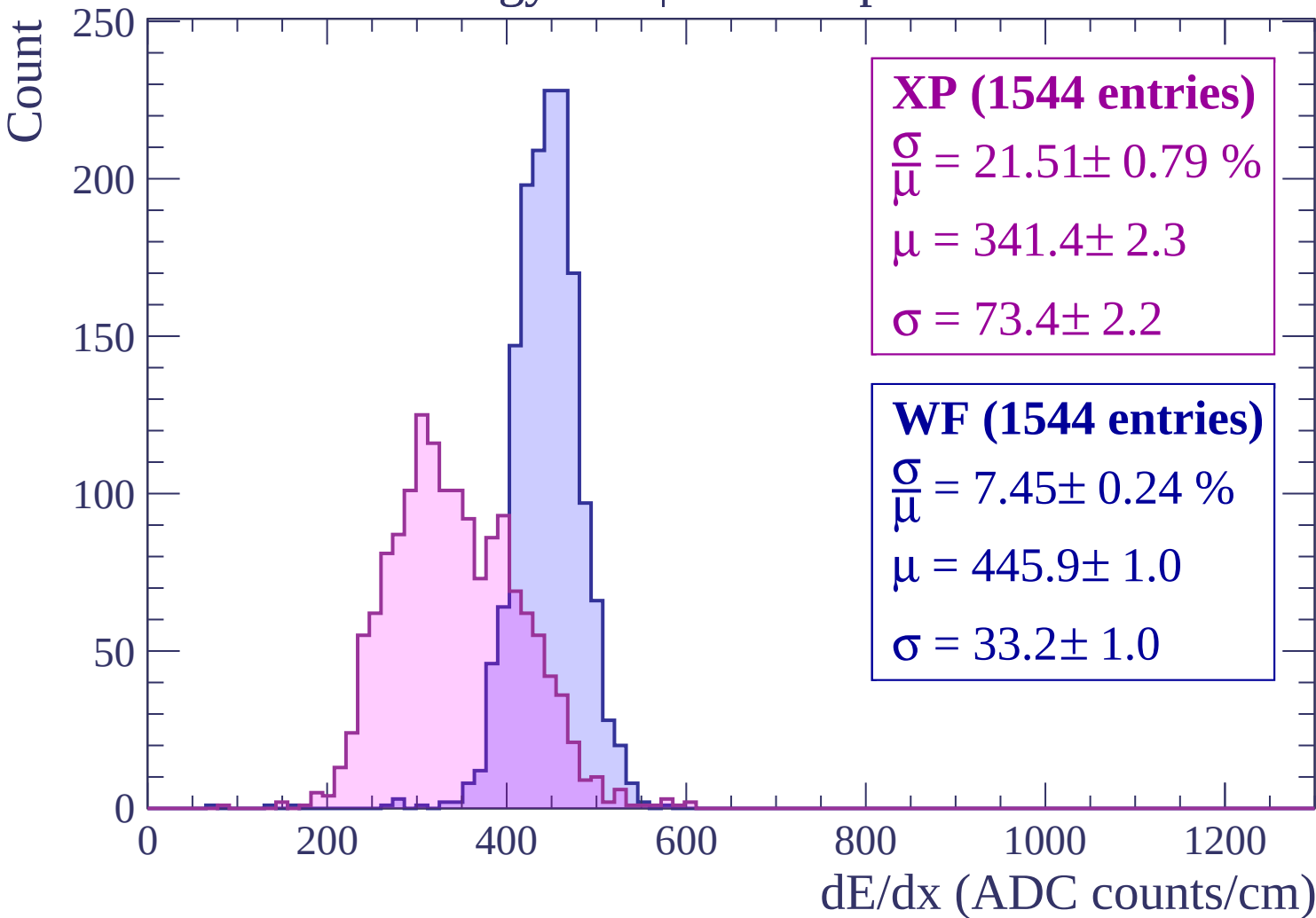
$$\frac{\sigma}{\mu} = 7.11 \pm 0.19 \%$$

$$\mu = 447.5 \pm 0.9$$

$$\sigma = 31.8 \pm 0.8$$

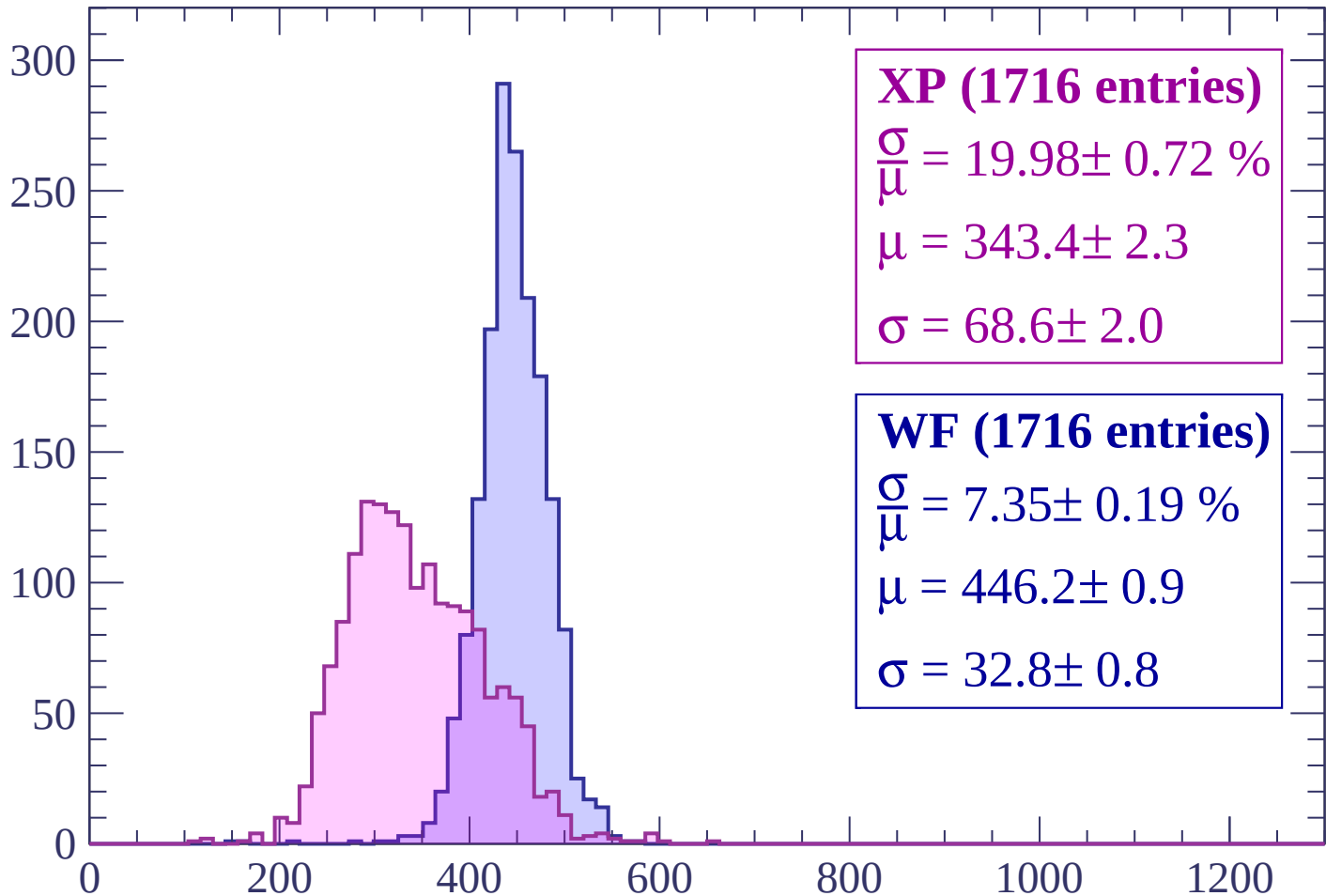
dE/dx (ADC counts/cm)

Energy loss | -2700 < p < -2640



Energy loss | $-2640 < p < -2580$

Count



XP (1716 entries)

$\frac{\sigma}{\mu} = 19.98 \pm 0.72 \%$

$\mu = 343.4 \pm 2.3$

$\sigma = 68.6 \pm 2.0$

WF (1716 entries)

$\frac{\sigma}{\mu} = 7.35 \pm 0.19 \%$

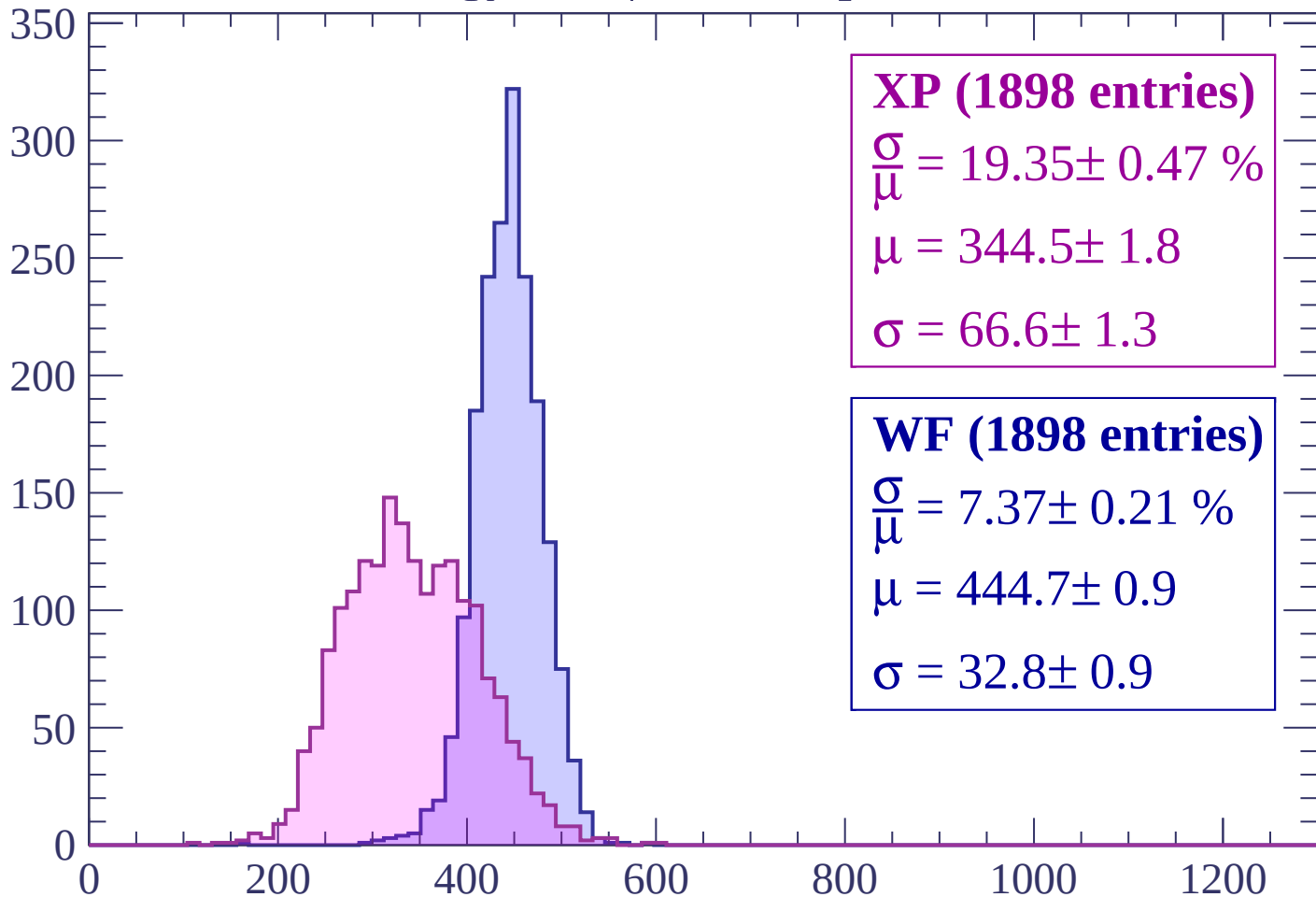
$\mu = 446.2 \pm 0.9$

$\sigma = 32.8 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-2580 < p < -2520$

Count



XP (1898 entries)

$$\frac{\sigma}{\mu} = 19.35 \pm 0.47 \%$$

$$\mu = 344.5 \pm 1.8$$

$$\sigma = 66.6 \pm 1.3$$

WF (1898 entries)

$$\frac{\sigma}{\mu} = 7.37 \pm 0.21 \%$$

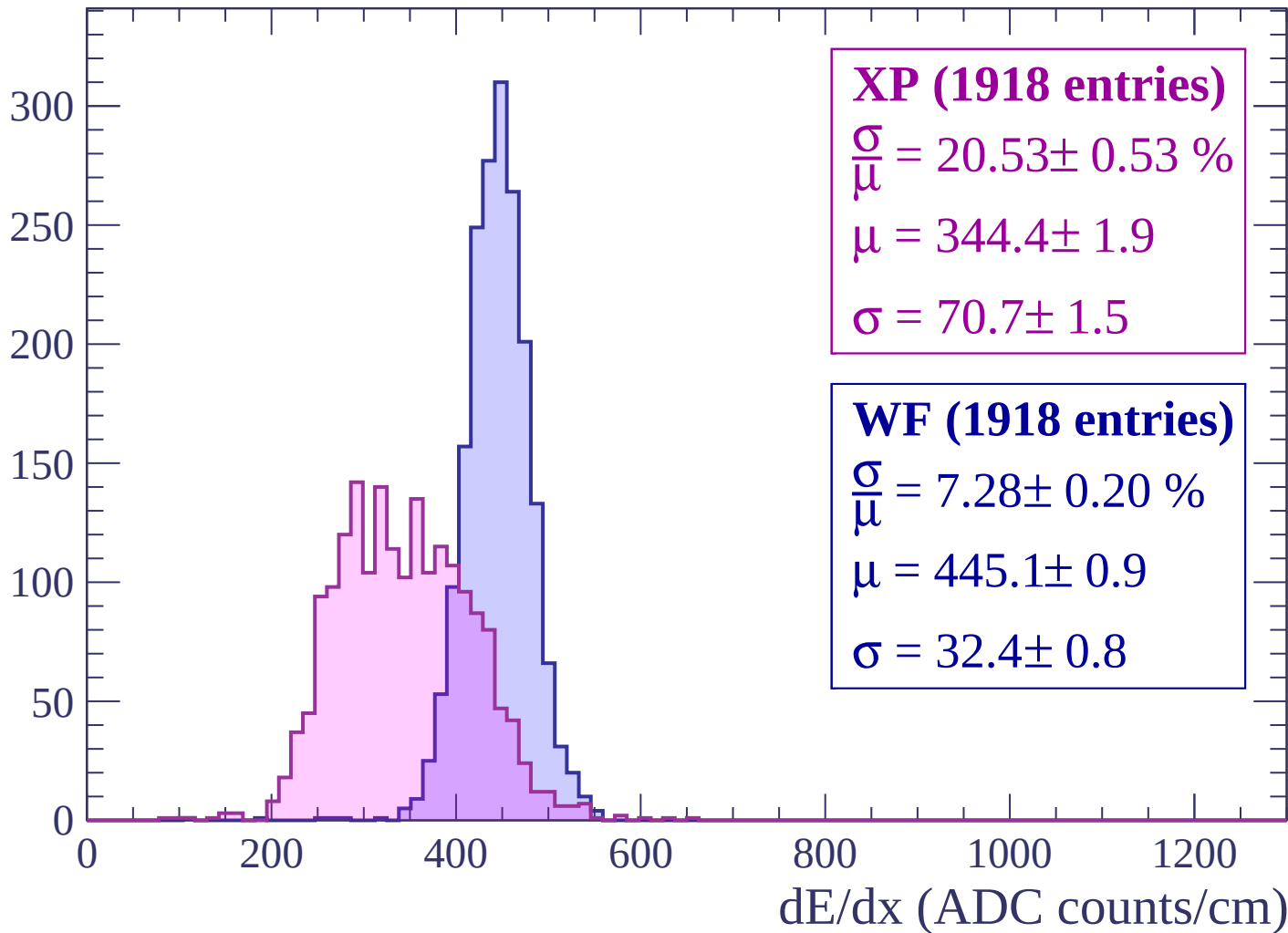
$$\mu = 444.7 \pm 0.9$$

$$\sigma = 32.8 \pm 0.9$$

dE/dx (ADC counts/cm)

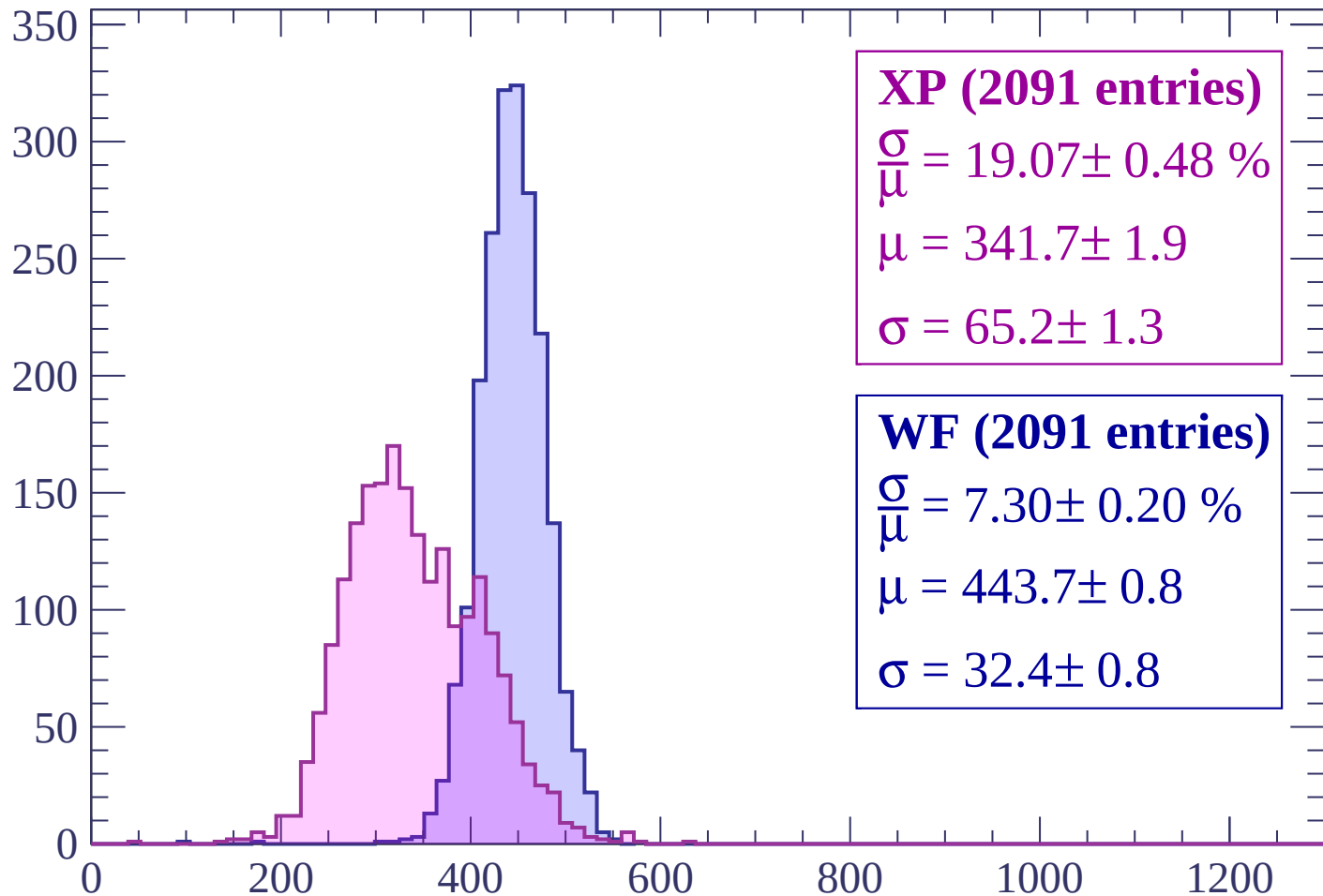
Energy loss | -2520 < p < -2460

Count



Energy loss | -2460 < p < -2400

Count



XP (2091 entries)

$$\frac{\sigma}{\mu} = 19.07 \pm 0.48 \%$$

$$\mu = 341.7 \pm 1.9$$

$$\sigma = 65.2 \pm 1.3$$

WF (2091 entries)

$$\frac{\sigma}{\mu} = 7.30 \pm 0.20 \%$$

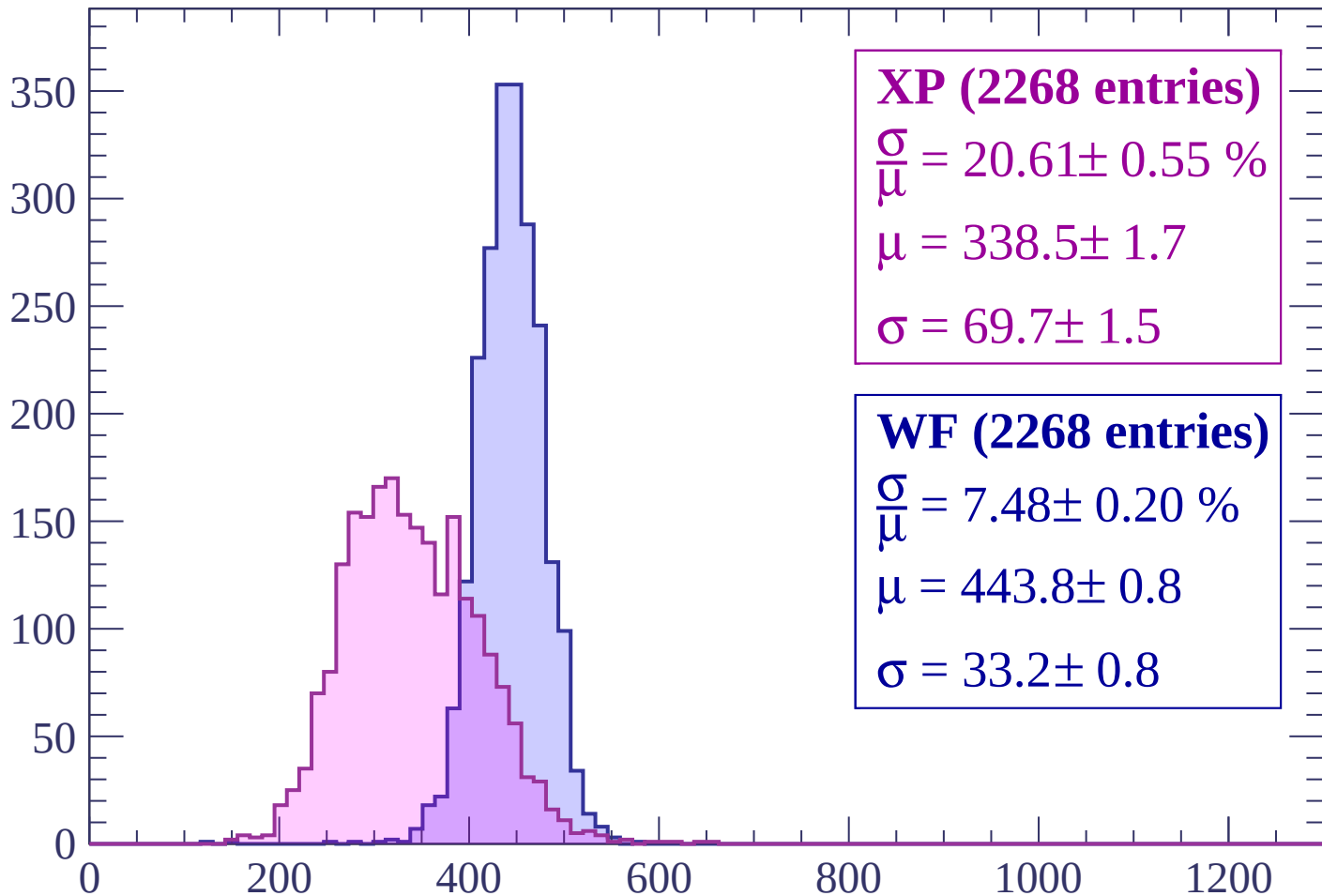
$$\mu = 443.7 \pm 0.8$$

$$\sigma = 32.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -2400 < p < -2340

Count



XP (2268 entries)

$$\frac{\sigma}{\mu} = 20.61 \pm 0.55 \%$$

$$\mu = 338.5 \pm 1.7$$

$$\sigma = 69.7 \pm 1.5$$

WF (2268 entries)

$$\frac{\sigma}{\mu} = 7.48 \pm 0.20 \%$$

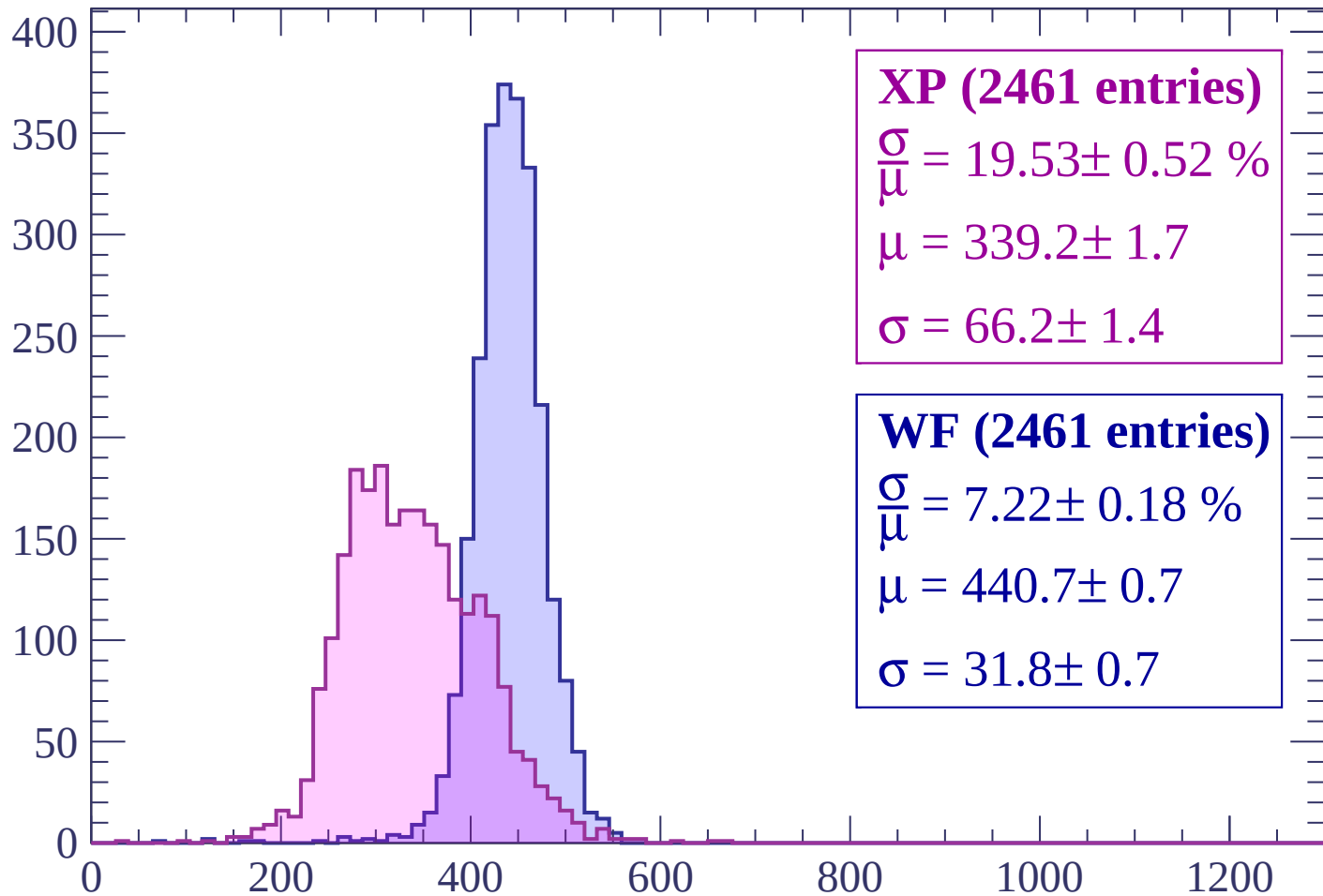
$$\mu = 443.8 \pm 0.8$$

$$\sigma = 33.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-2340 < p < -2280$

Count



XP (2461 entries)

$\frac{\sigma}{\mu} = 19.53 \pm 0.52 \%$

$\mu = 339.2 \pm 1.7$

$\sigma = 66.2 \pm 1.4$

WF (2461 entries)

$\frac{\sigma}{\mu} = 7.22 \pm 0.18 \%$

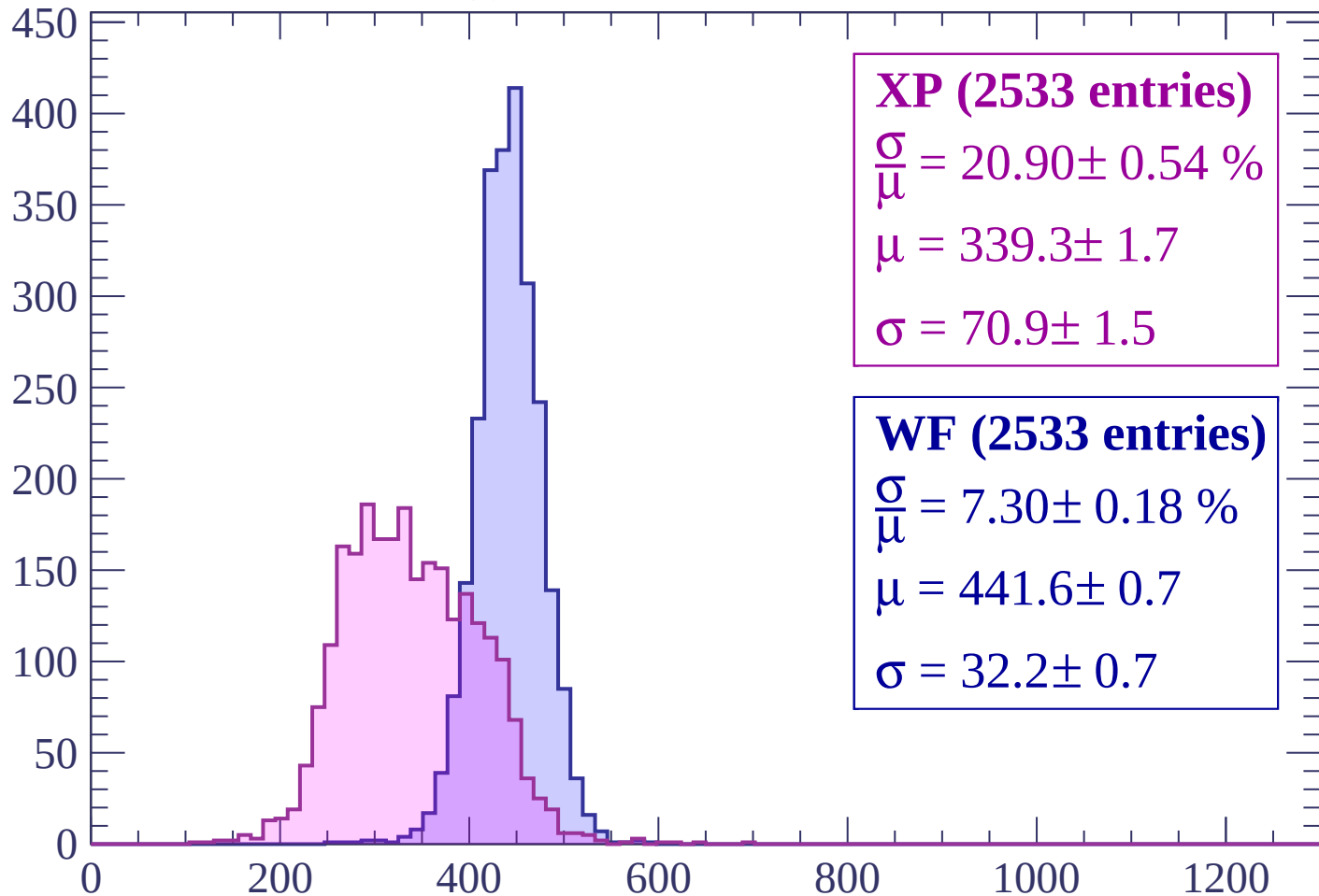
$\mu = 440.7 \pm 0.7$

$\sigma = 31.8 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-2280 < p < -2220$

Count



XP (2533 entries)

$$\frac{\sigma}{\mu} = 20.90 \pm 0.54 \%$$

$$\mu = 339.3 \pm 1.7$$

$$\sigma = 70.9 \pm 1.5$$

WF (2533 entries)

$$\frac{\sigma}{\mu} = 7.30 \pm 0.18 \%$$

$$\mu = 441.6 \pm 0.7$$

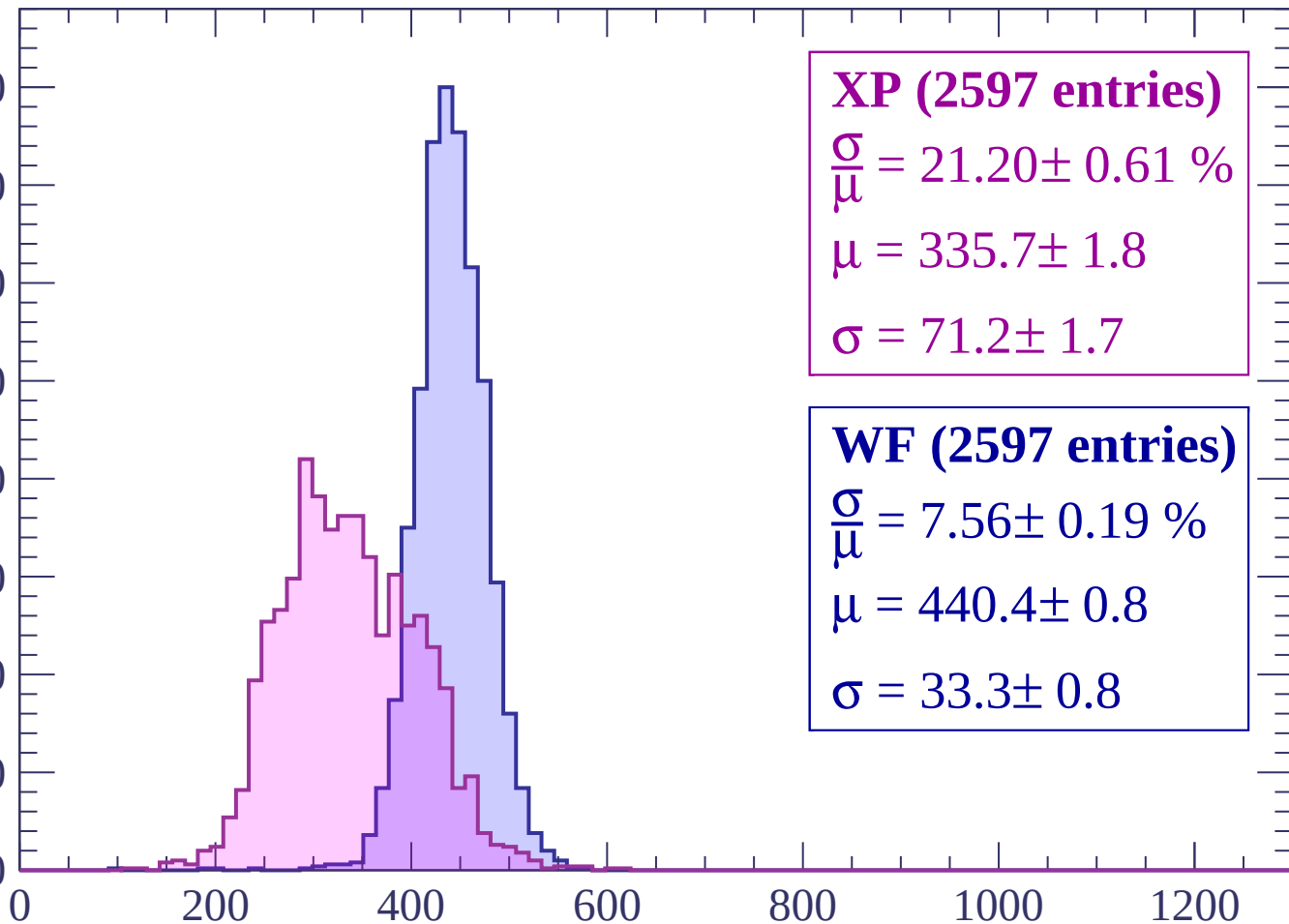
$$\sigma = 32.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160

Count

400
350
300
250
200
150
100
50
0



XP (2597 entries)

$\frac{\sigma}{\mu} = 21.20 \pm 0.61 \%$

$\mu = 335.7 \pm 1.8$

$\sigma = 71.2 \pm 1.7$

WF (2597 entries)

$\frac{\sigma}{\mu} = 7.56 \pm 0.19 \%$

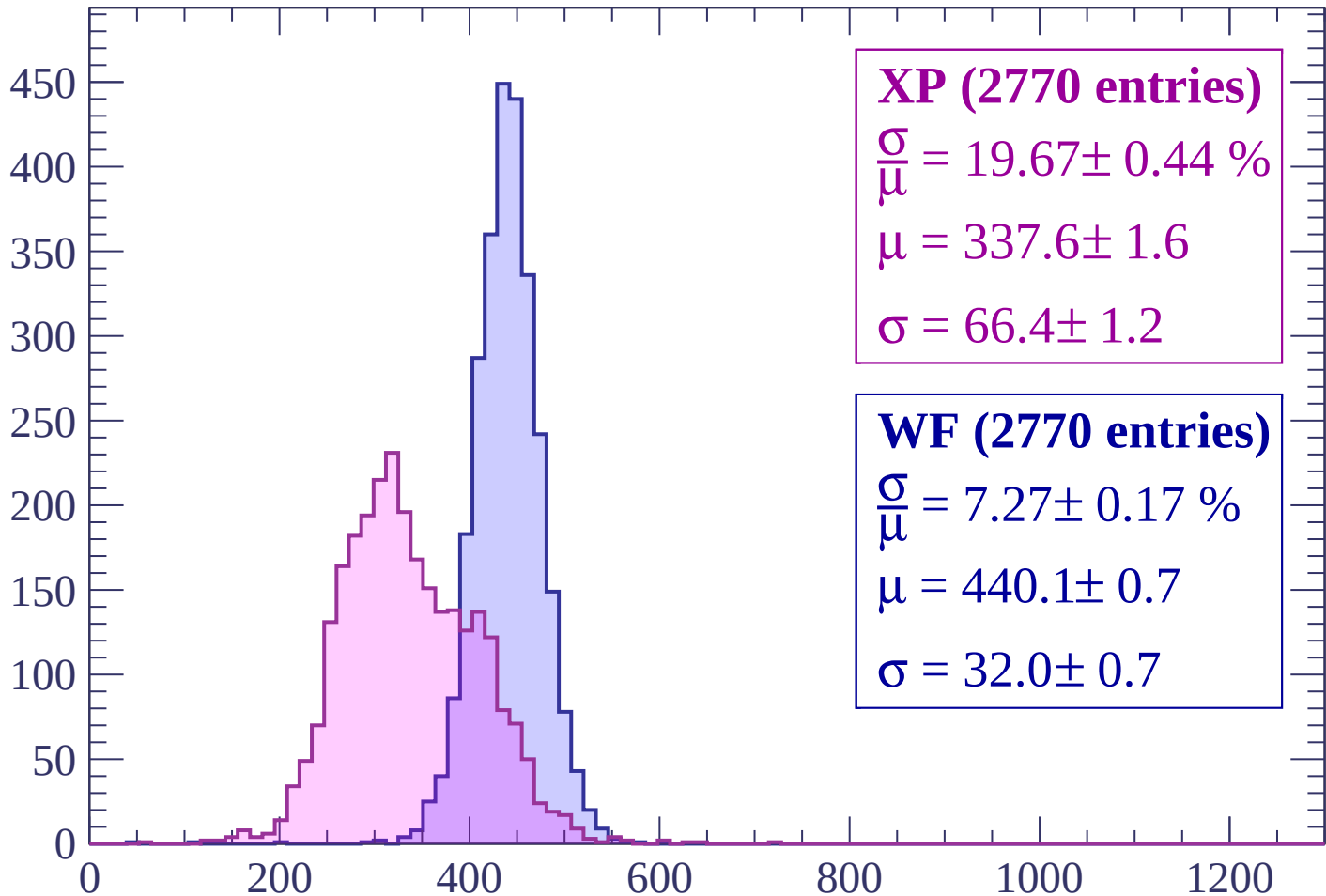
$\mu = 440.4 \pm 0.8$

$\sigma = 33.3 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | -2160 < p < -2100

Count



XP (2770 entries)

$$\frac{\sigma}{\mu} = 19.67 \pm 0.44 \%$$

$$\mu = 337.6 \pm 1.6$$

$$\sigma = 66.4 \pm 1.2$$

WF (2770 entries)

$$\frac{\sigma}{\mu} = 7.27 \pm 0.17 \%$$

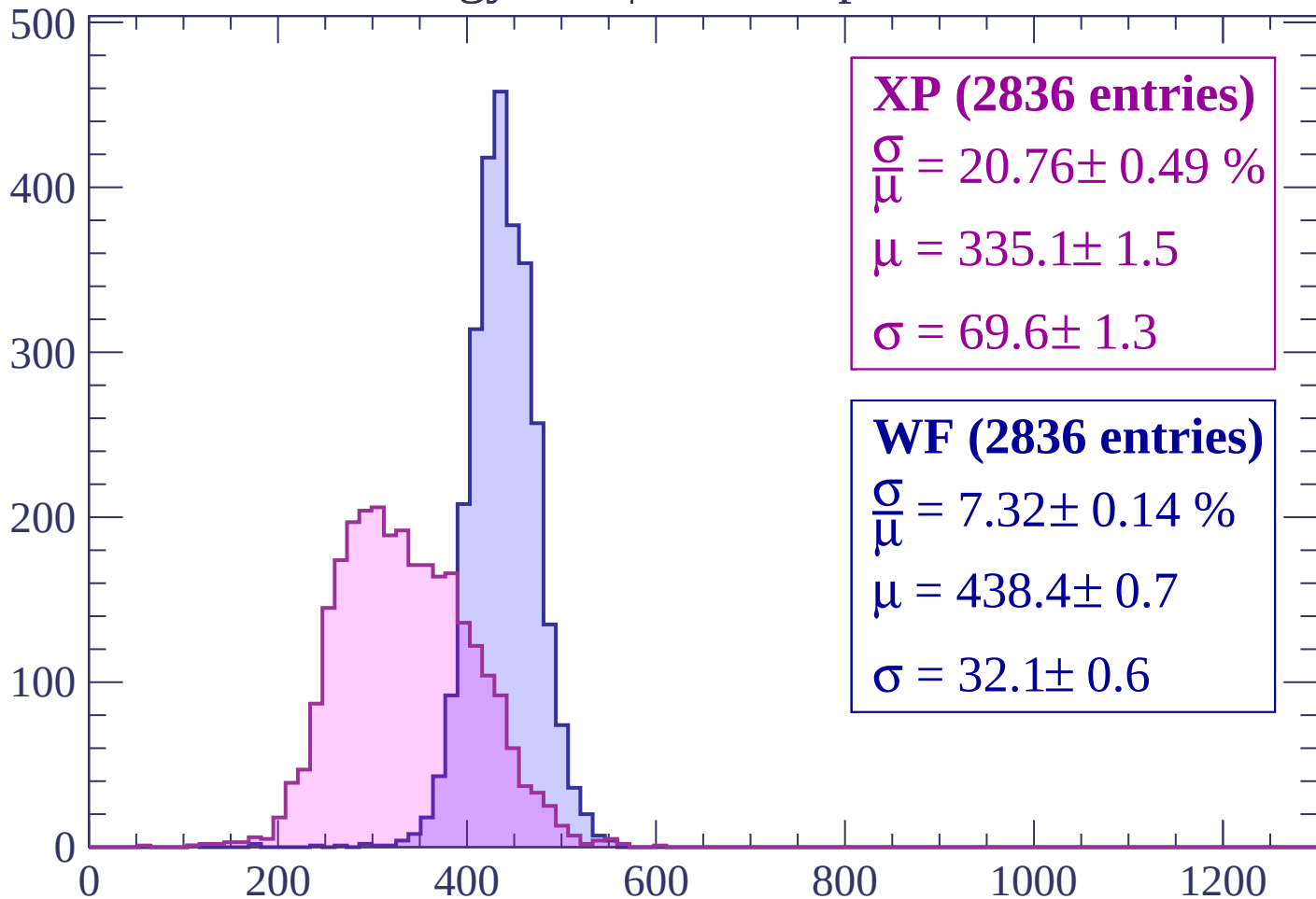
$$\mu = 440.1 \pm 0.7$$

$$\sigma = 32.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count



XP (2836 entries)

$$\frac{\sigma}{\mu} = 20.76 \pm 0.49 \%$$

$$\mu = 335.1 \pm 1.5$$

$$\sigma = 69.6 \pm 1.3$$

WF (2836 entries)

$$\frac{\sigma}{\mu} = 7.32 \pm 0.14 \%$$

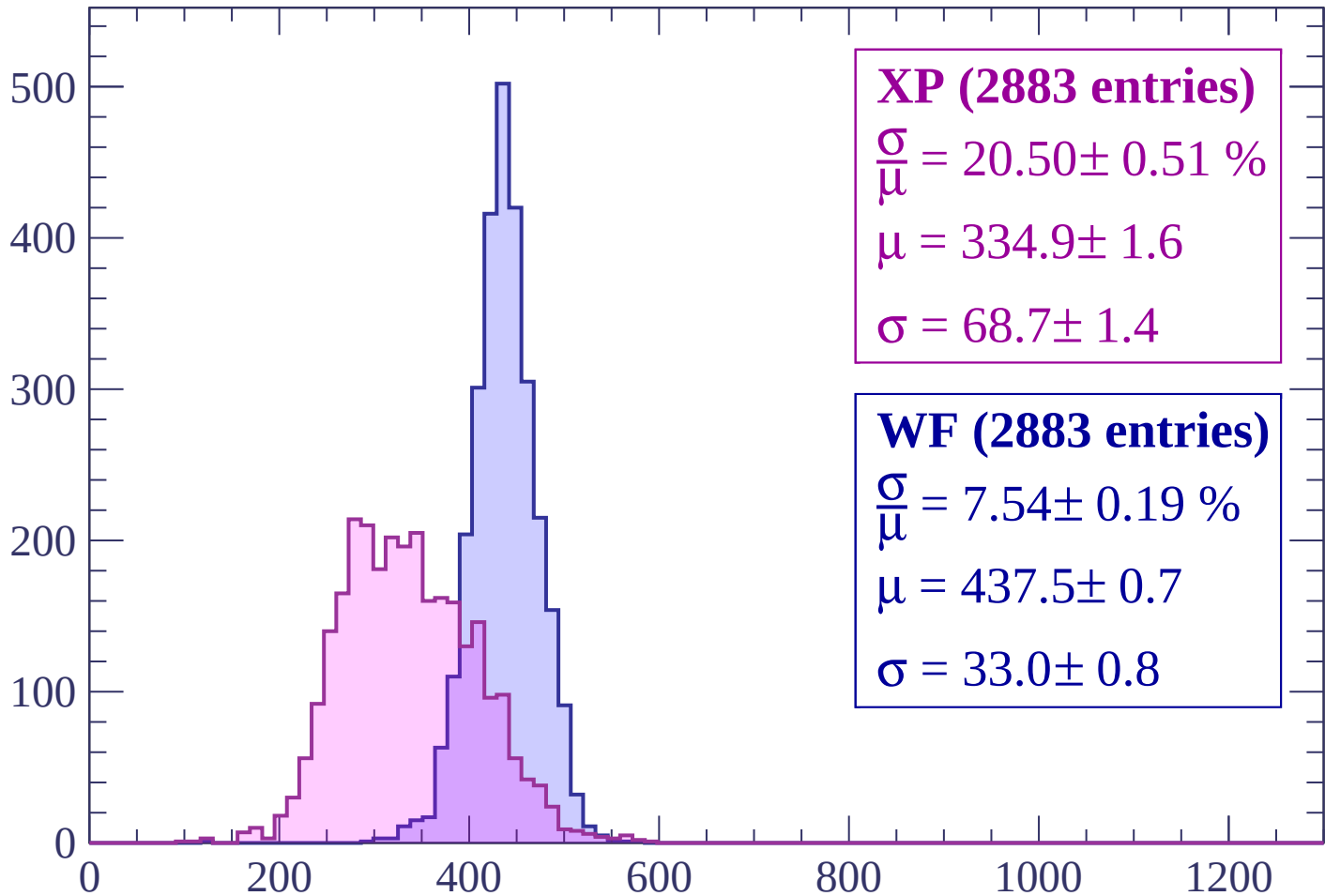
$$\mu = 438.4 \pm 0.7$$

$$\sigma = 32.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-2040 < p < -1980$

Count



XP (2883 entries)

$\frac{\sigma}{\mu} = 20.50 \pm 0.51 \%$

$\mu = 334.9 \pm 1.6$

$\sigma = 68.7 \pm 1.4$

WF (2883 entries)

$\frac{\sigma}{\mu} = 7.54 \pm 0.19 \%$

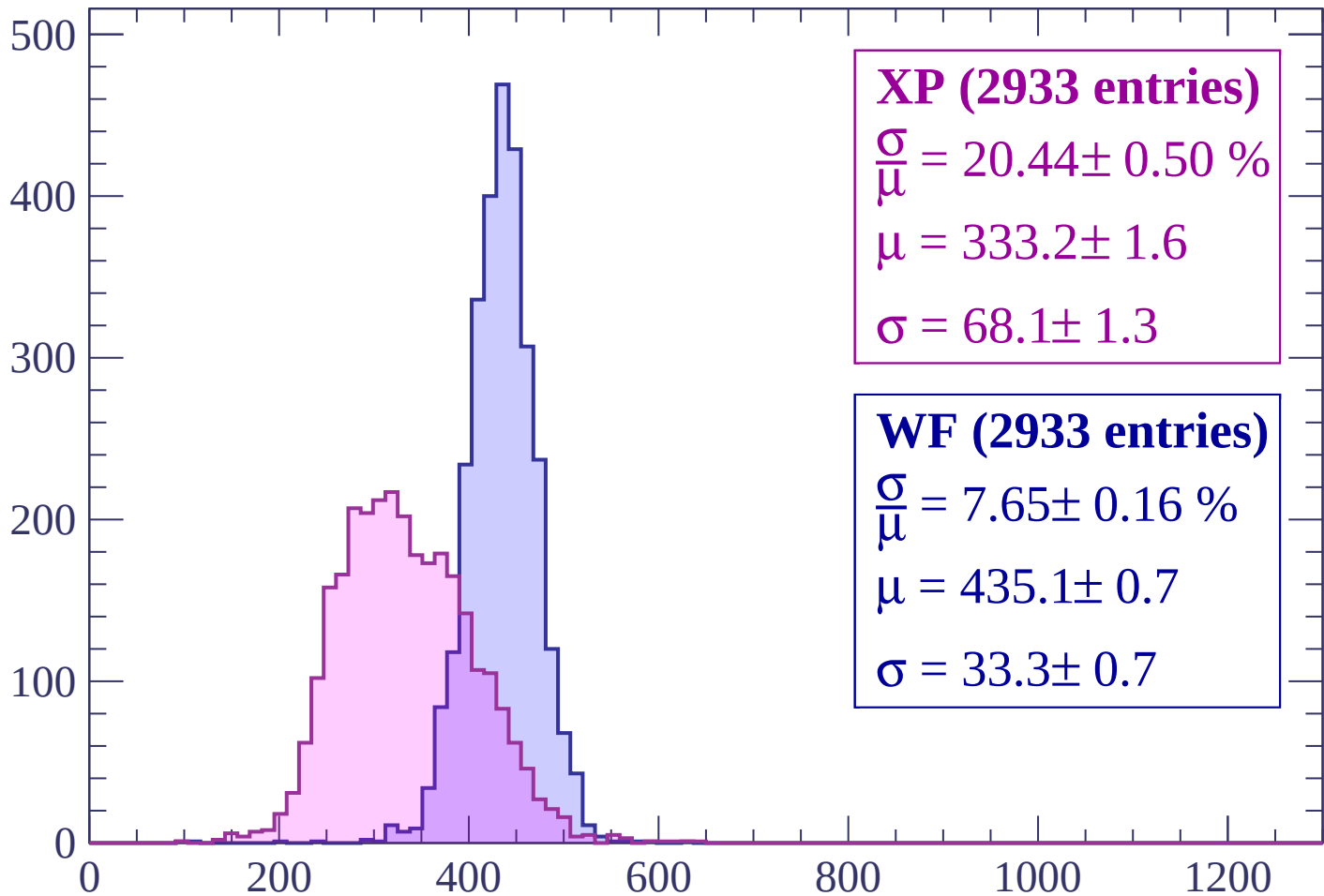
$\mu = 437.5 \pm 0.7$

$\sigma = 33.0 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | -1980 < p < -1920

Count



XP (2933 entries)

$$\frac{\sigma}{\mu} = 20.44 \pm 0.50 \%$$

$$\mu = 333.2 \pm 1.6$$

$$\sigma = 68.1 \pm 1.3$$

WF (2933 entries)

$$\frac{\sigma}{\mu} = 7.65 \pm 0.16 \%$$

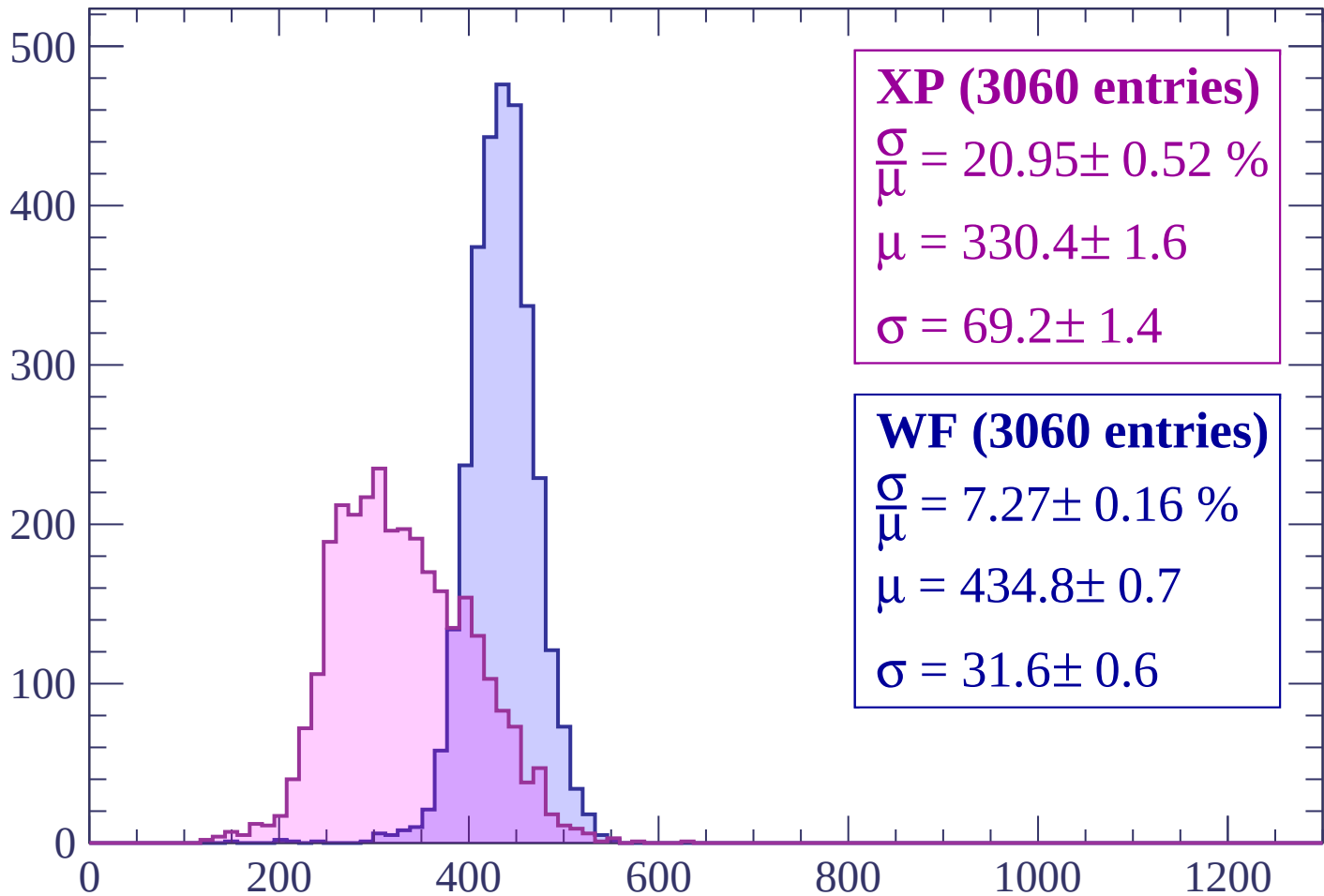
$$\mu = 435.1 \pm 0.7$$

$$\sigma = 33.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1860

Count



XP (3060 entries)

$$\frac{\sigma}{\mu} = 20.95 \pm 0.52 \%$$

$$\mu = 330.4 \pm 1.6$$

$$\sigma = 69.2 \pm 1.4$$

WF (3060 entries)

$$\frac{\sigma}{\mu} = 7.27 \pm 0.16 \%$$

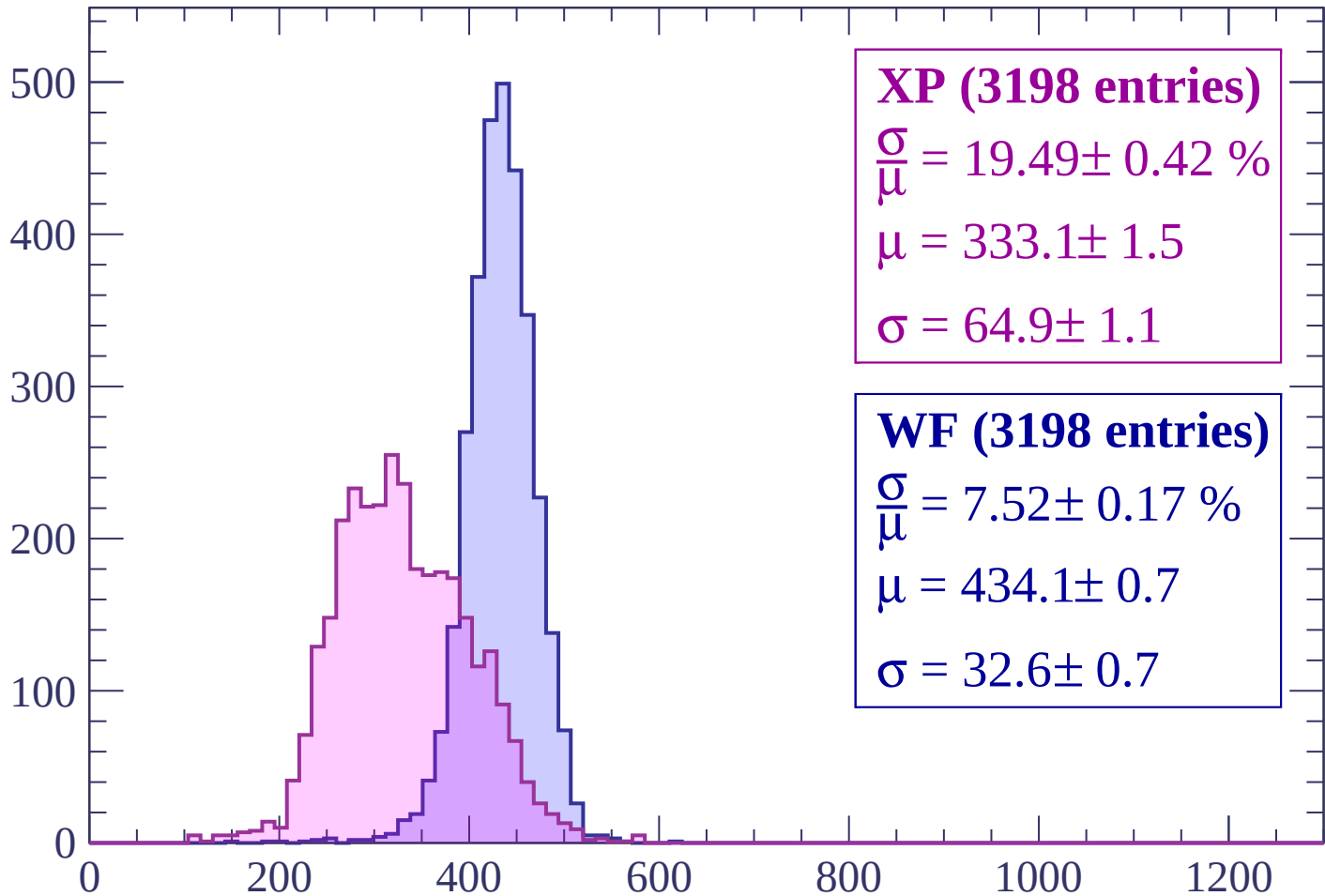
$$\mu = 434.8 \pm 0.7$$

$$\sigma = 31.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1860 < p < -1800$

Count



XP (3198 entries)

$$\frac{\sigma}{\mu} = 19.49 \pm 0.42 \%$$

$$\mu = 333.1 \pm 1.5$$

$$\sigma = 64.9 \pm 1.1$$

WF (3198 entries)

$$\frac{\sigma}{\mu} = 7.52 \pm 0.17 \%$$

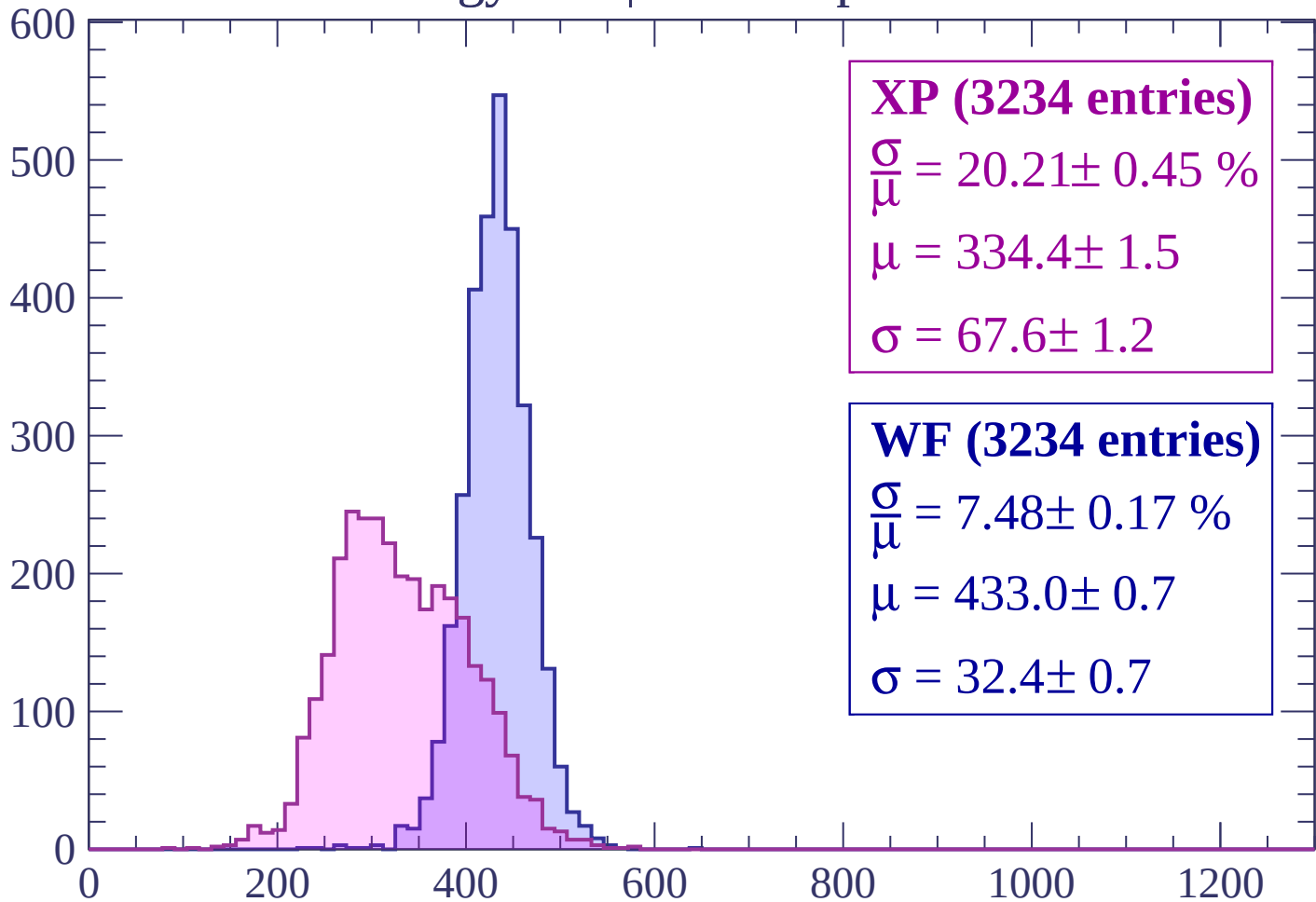
$$\mu = 434.1 \pm 0.7$$

$$\sigma = 32.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1740$

Count



XP (3234 entries)

$$\frac{\sigma}{\mu} = 20.21 \pm 0.45 \%$$

$$\mu = 334.4 \pm 1.5$$

$$\sigma = 67.6 \pm 1.2$$

WF (3234 entries)

$$\frac{\sigma}{\mu} = 7.48 \pm 0.17 \%$$

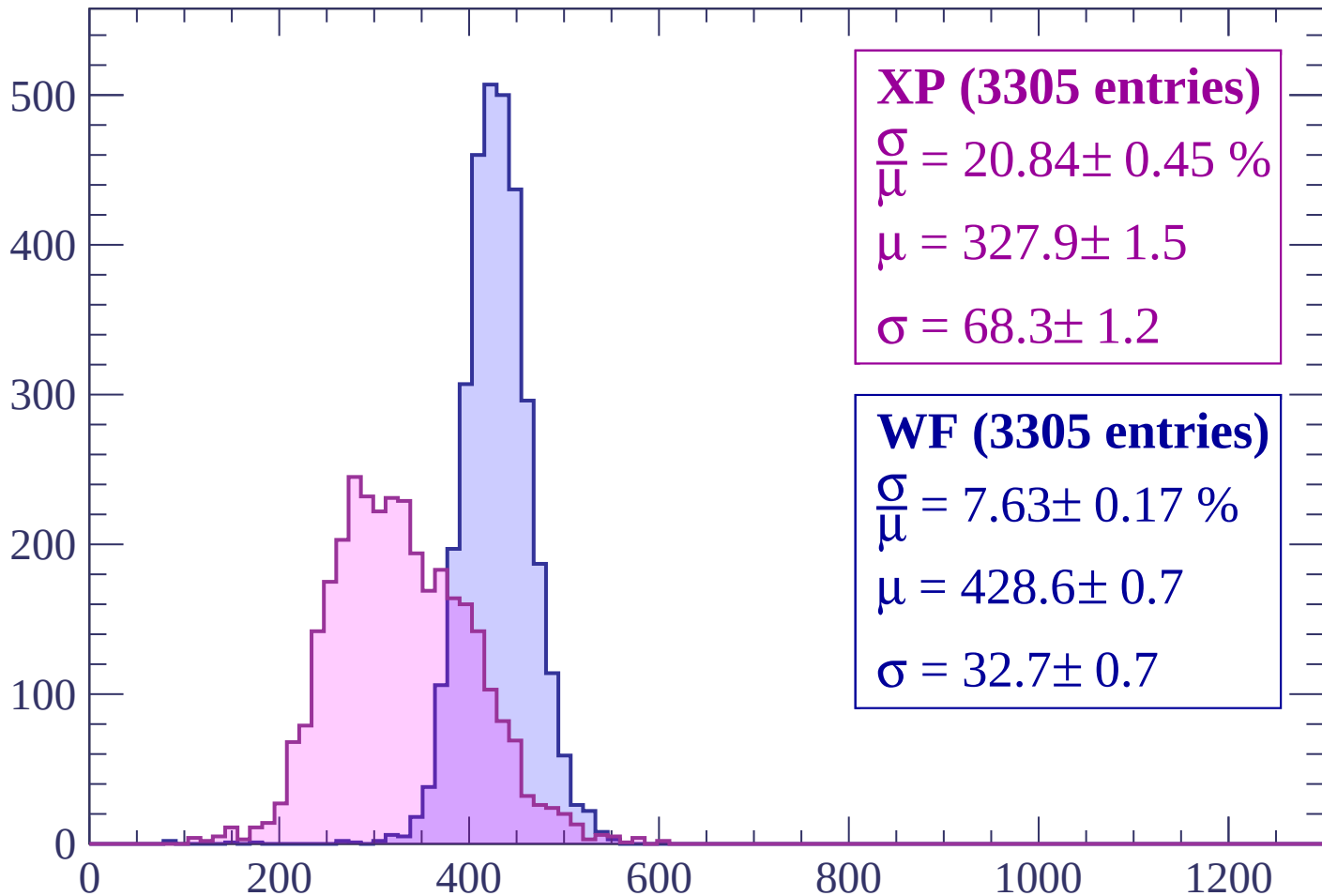
$$\mu = 433.0 \pm 0.7$$

$$\sigma = 32.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (3305 entries)

$$\frac{\sigma}{\mu} = 20.84 \pm 0.45 \%$$

$$\mu = 327.9 \pm 1.5$$

$$\sigma = 68.3 \pm 1.2$$

WF (3305 entries)

$$\frac{\sigma}{\mu} = 7.63 \pm 0.17 \%$$

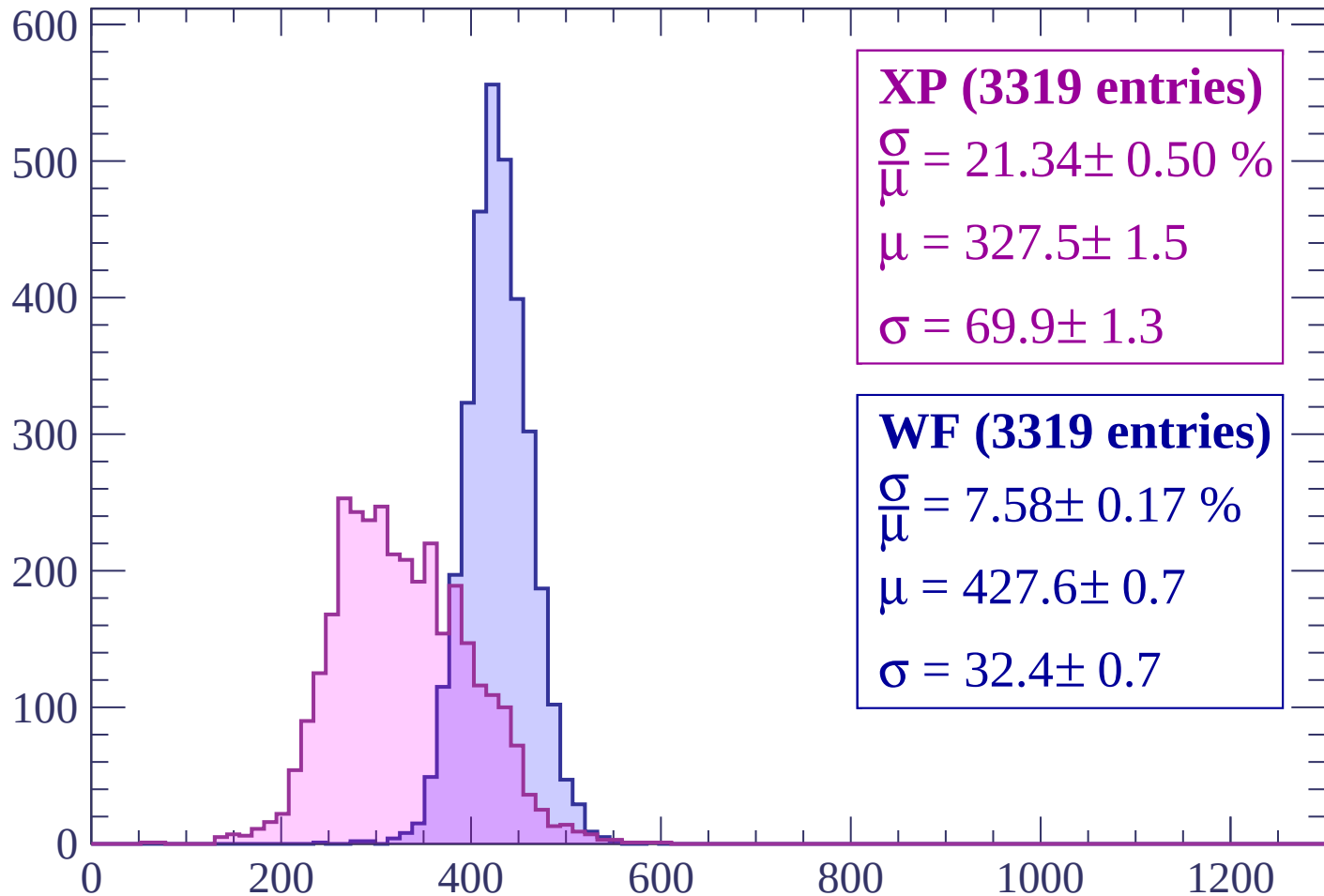
$$\mu = 428.6 \pm 0.7$$

$$\sigma = 32.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1620$

Count



XP (3319 entries)

$$\frac{\sigma}{\mu} = 21.34 \pm 0.50 \%$$

$$\mu = 327.5 \pm 1.5$$

$$\sigma = 69.9 \pm 1.3$$

WF (3319 entries)

$$\frac{\sigma}{\mu} = 7.58 \pm 0.17 \%$$

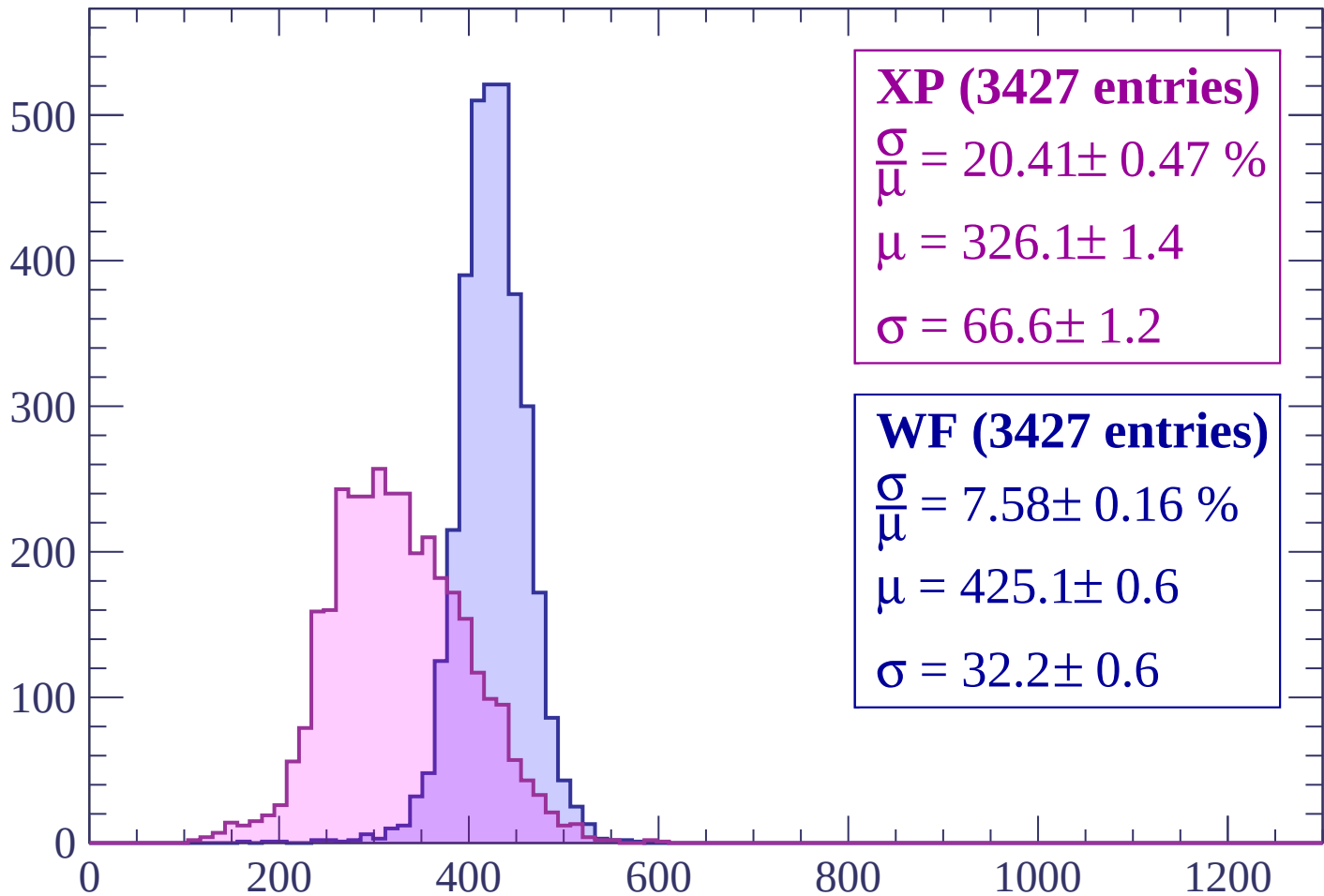
$$\mu = 427.6 \pm 0.7$$

$$\sigma = 32.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

Count



XP (3427 entries)

$$\frac{\sigma}{\mu} = 20.41 \pm 0.47 \%$$

$$\mu = 326.1 \pm 1.4$$

$$\sigma = 66.6 \pm 1.2$$

WF (3427 entries)

$$\frac{\sigma}{\mu} = 7.58 \pm 0.16 \%$$

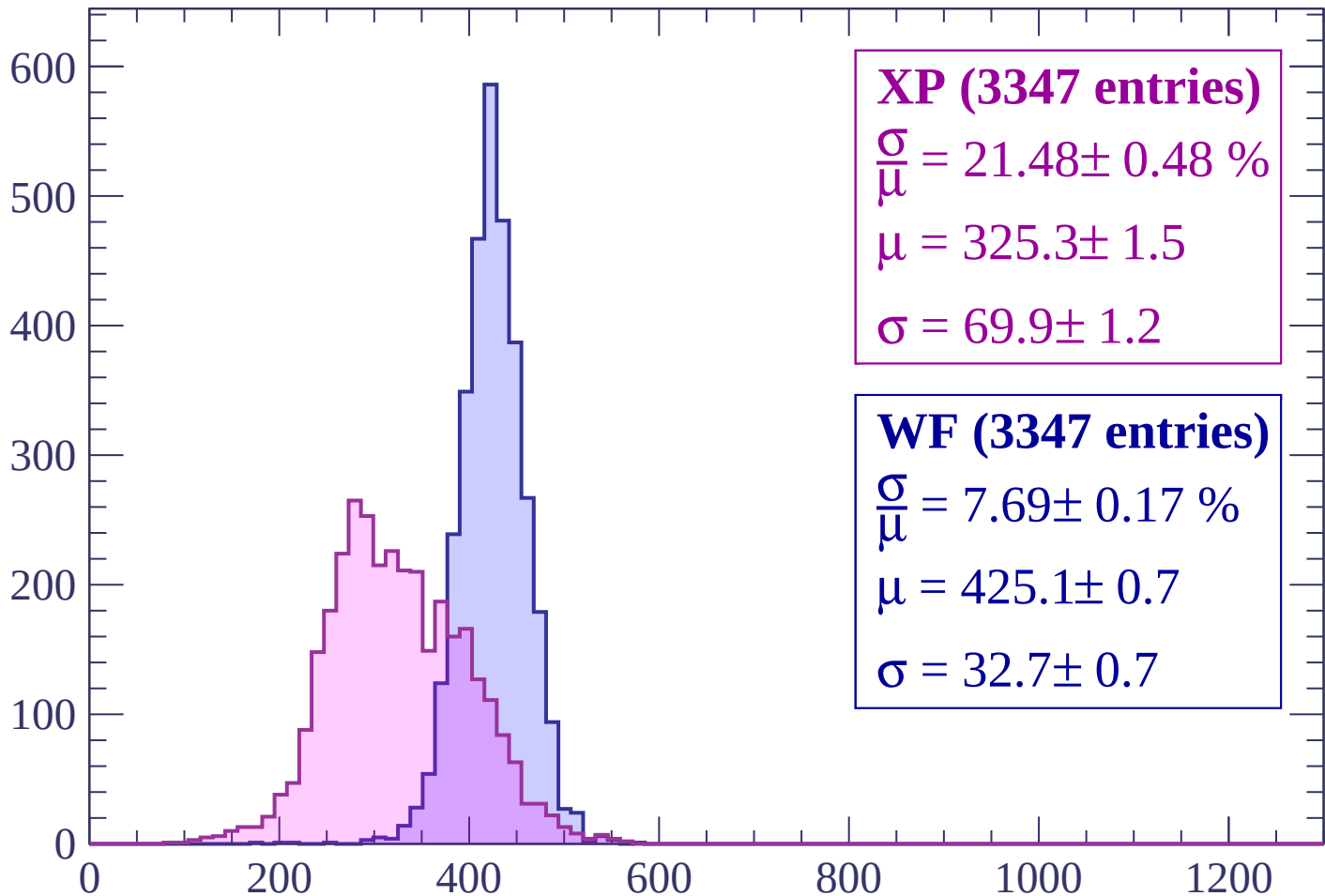
$$\mu = 425.1 \pm 0.6$$

$$\sigma = 32.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (3347 entries)

$$\frac{\sigma}{\mu} = 21.48 \pm 0.48 \%$$

$$\mu = 325.3 \pm 1.5$$

$$\sigma = 69.9 \pm 1.2$$

WF (3347 entries)

$$\frac{\sigma}{\mu} = 7.69 \pm 0.17 \%$$

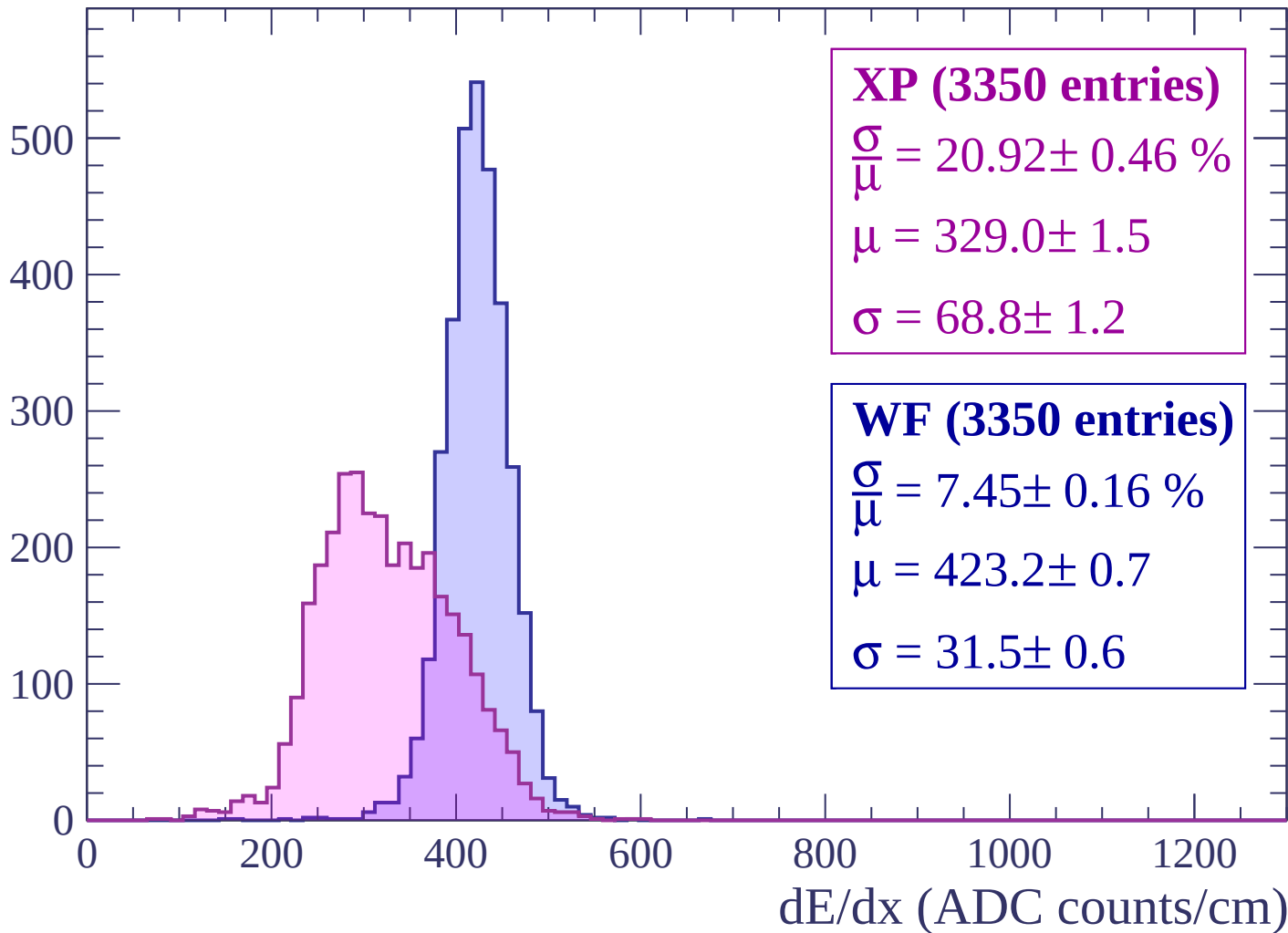
$$\mu = 425.1 \pm 0.7$$

$$\sigma = 32.7 \pm 0.7$$

dE/dx (ADC counts/cm)

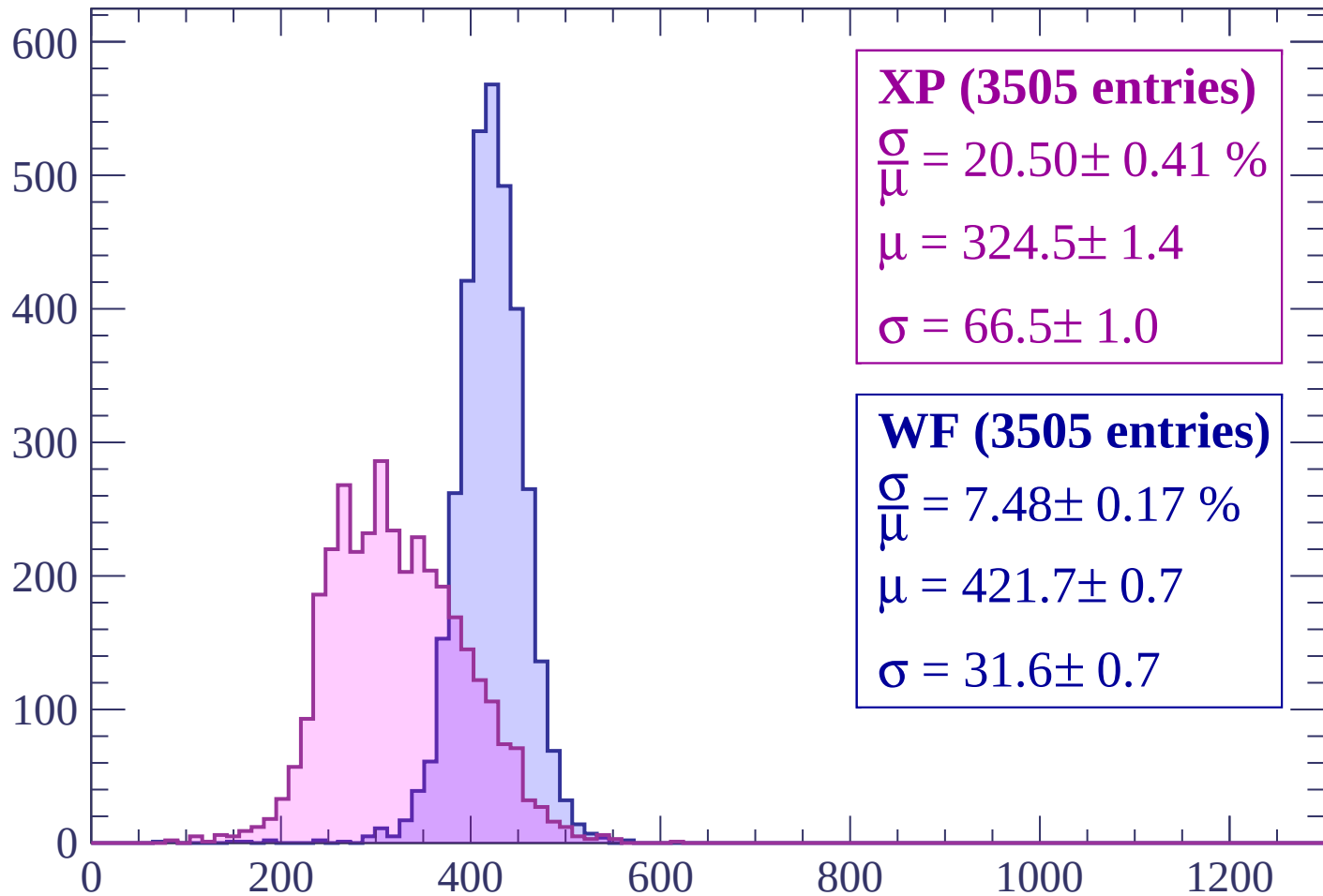
Energy loss | -1500 < p < -1440

Count



Energy loss | $-1440 < p < -1380$

Count



XP (3505 entries)

$$\frac{\sigma}{\mu} = 20.50 \pm 0.41 \%$$

$$\mu = 324.5 \pm 1.4$$

$$\sigma = 66.5 \pm 1.0$$

WF (3505 entries)

$$\frac{\sigma}{\mu} = 7.48 \pm 0.17 \%$$

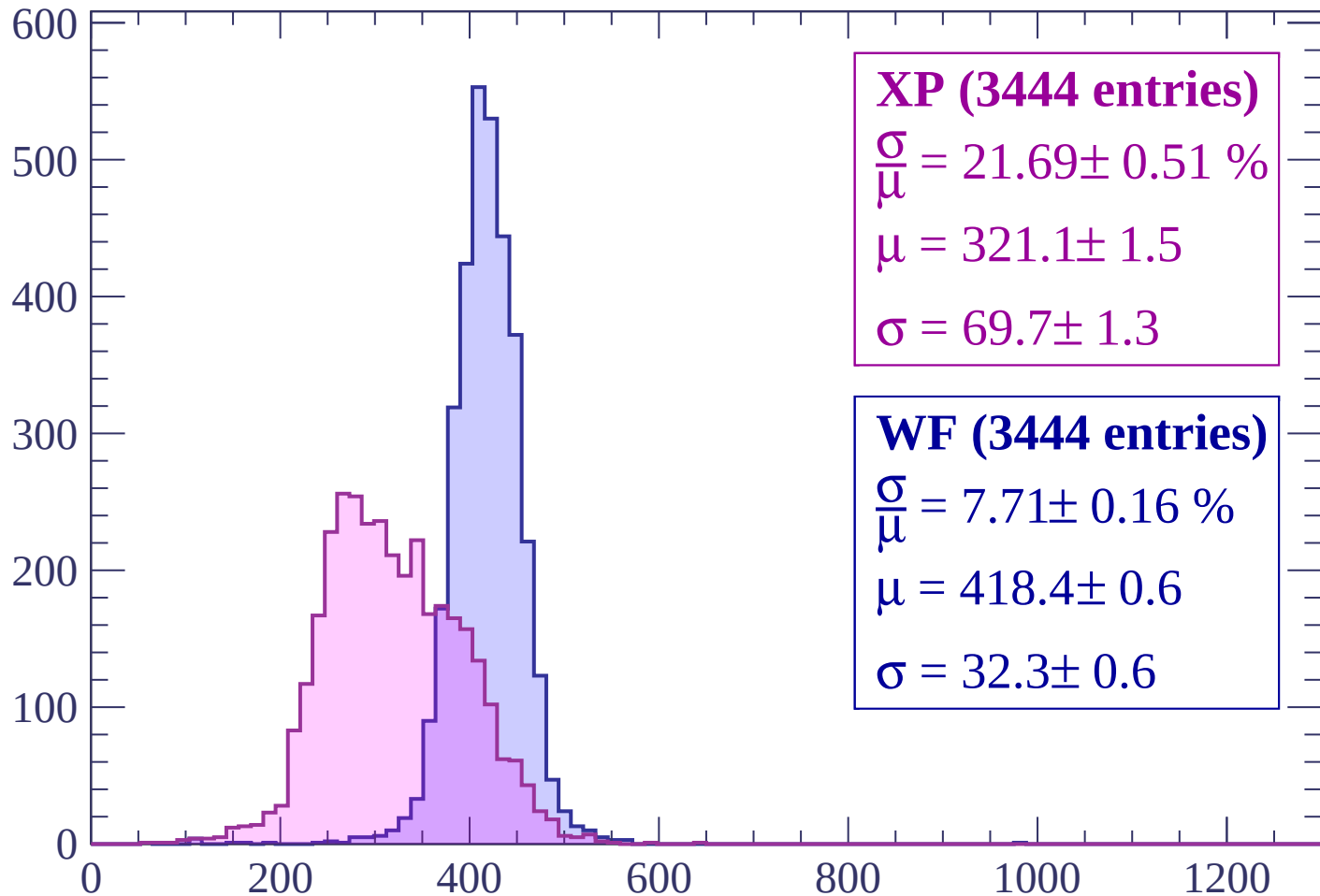
$$\mu = 421.7 \pm 0.7$$

$$\sigma = 31.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count



XP (3444 entries)

$$\frac{\sigma}{\mu} = 21.69 \pm 0.51 \%$$

$$\mu = 321.1 \pm 1.5$$

$$\sigma = 69.7 \pm 1.3$$

WF (3444 entries)

$$\frac{\sigma}{\mu} = 7.71 \pm 0.16 \%$$

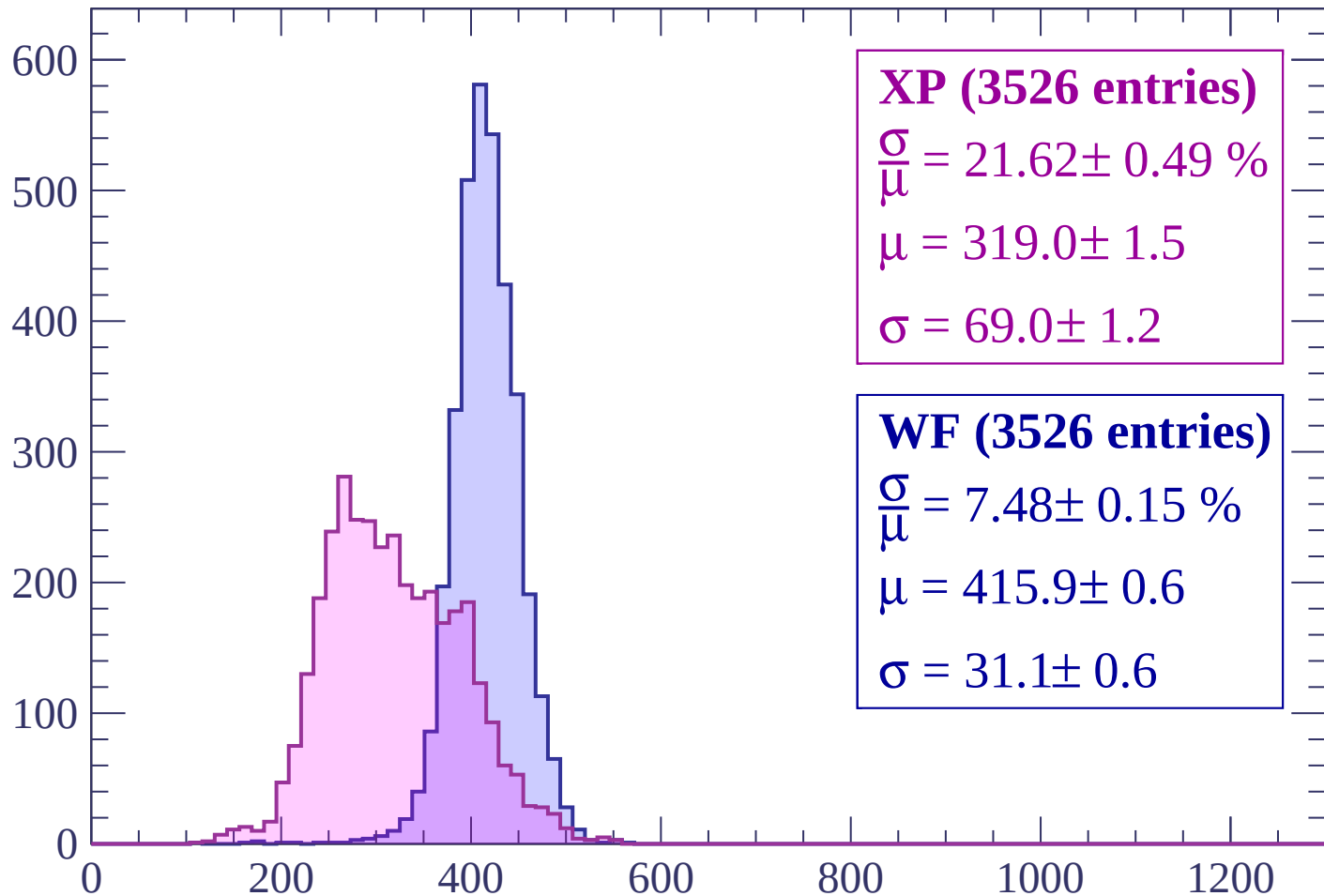
$$\mu = 418.4 \pm 0.6$$

$$\sigma = 32.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1260$

Count



XP (3526 entries)

$$\frac{\sigma}{\mu} = 21.62 \pm 0.49 \%$$

$$\mu = 319.0 \pm 1.5$$

$$\sigma = 69.0 \pm 1.2$$

WF (3526 entries)

$$\frac{\sigma}{\mu} = 7.48 \pm 0.15 \%$$

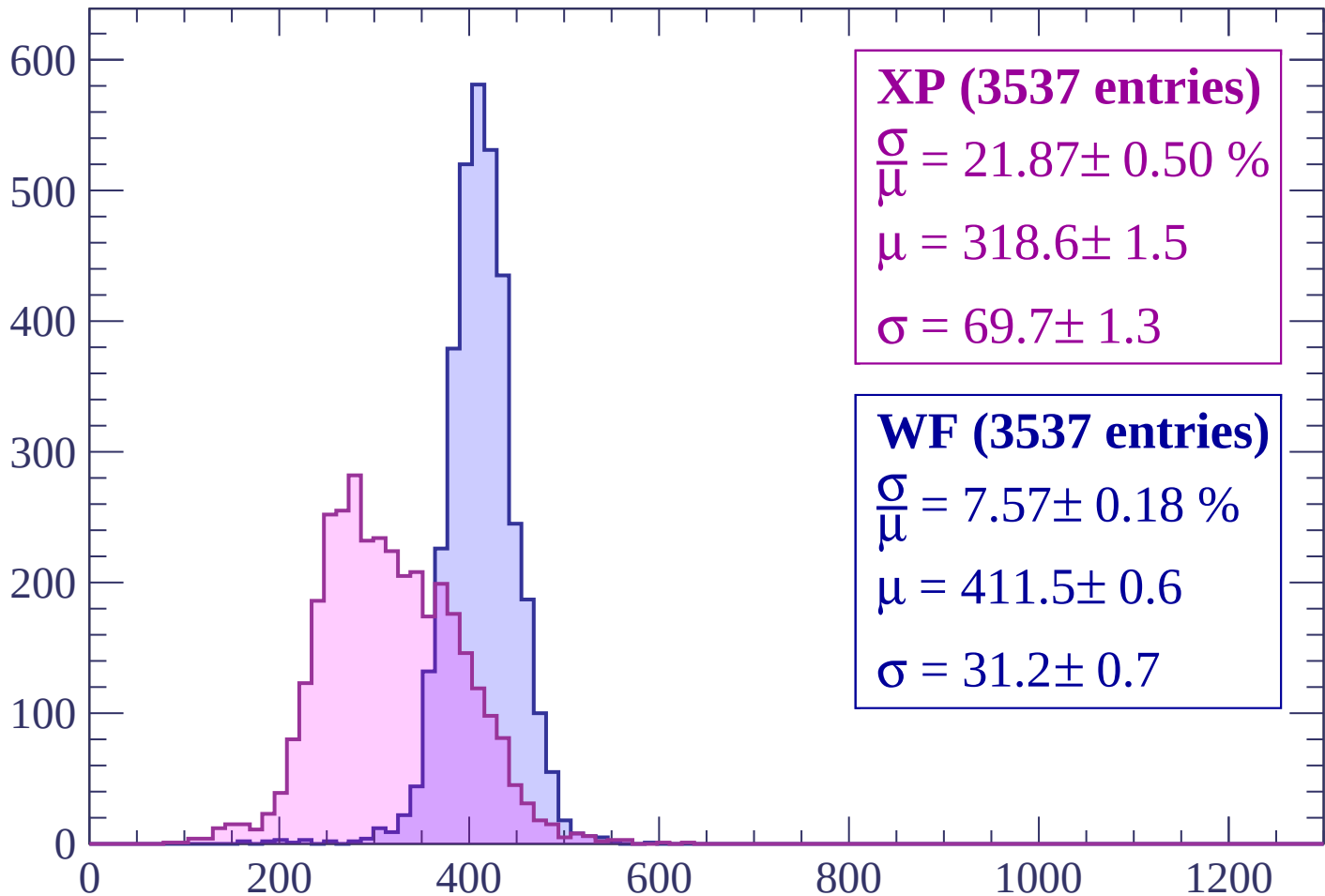
$$\mu = 415.9 \pm 0.6$$

$$\sigma = 31.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -1260 < p < -1200

Count



XP (3537 entries)

$$\frac{\sigma}{\mu} = 21.87 \pm 0.50 \%$$

$$\mu = 318.6 \pm 1.5$$

$$\sigma = 69.7 \pm 1.3$$

WF (3537 entries)

$$\frac{\sigma}{\mu} = 7.57 \pm 0.18 \%$$

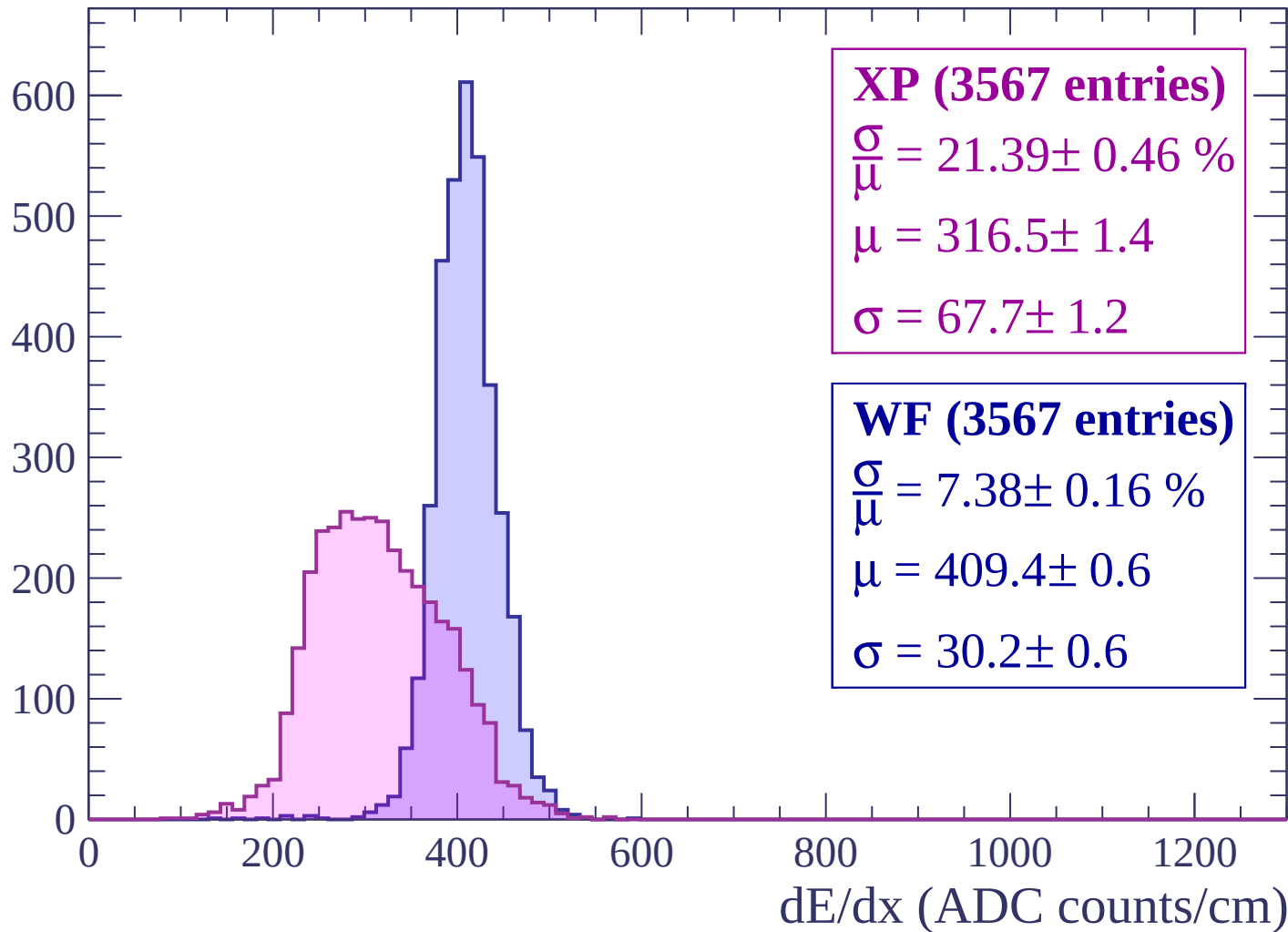
$$\mu = 411.5 \pm 0.6$$

$$\sigma = 31.2 \pm 0.7$$

dE/dx (ADC counts/cm)

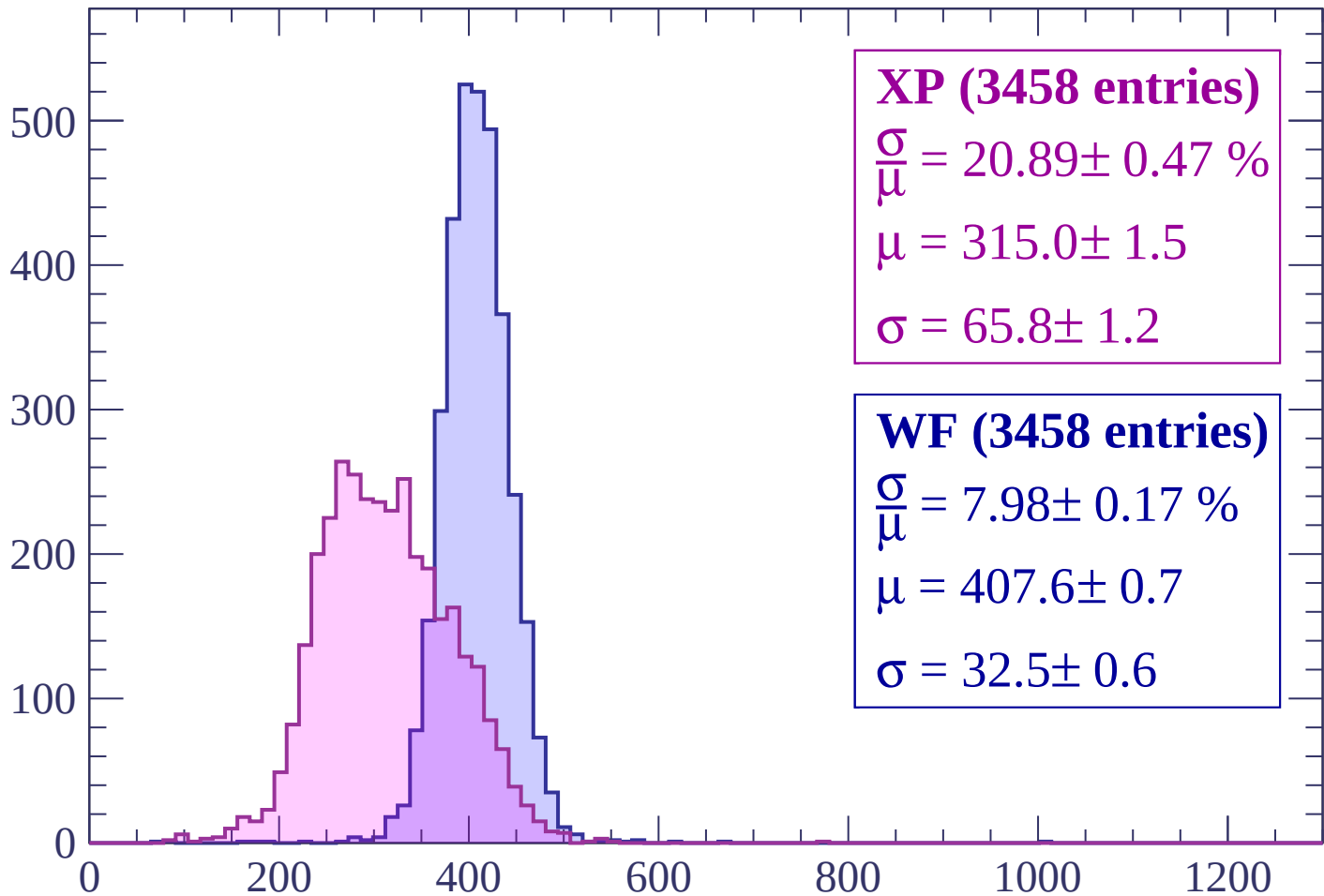
Energy loss | $-1200 < p < -1140$

Count



Energy loss | $-1140 < p < -1080$

Count



XP (3458 entries)

$$\frac{\sigma}{\mu} = 20.89 \pm 0.47 \%$$

$$\mu = 315.0 \pm 1.5$$

$$\sigma = 65.8 \pm 1.2$$

WF (3458 entries)

$$\frac{\sigma}{\mu} = 7.98 \pm 0.17 \%$$

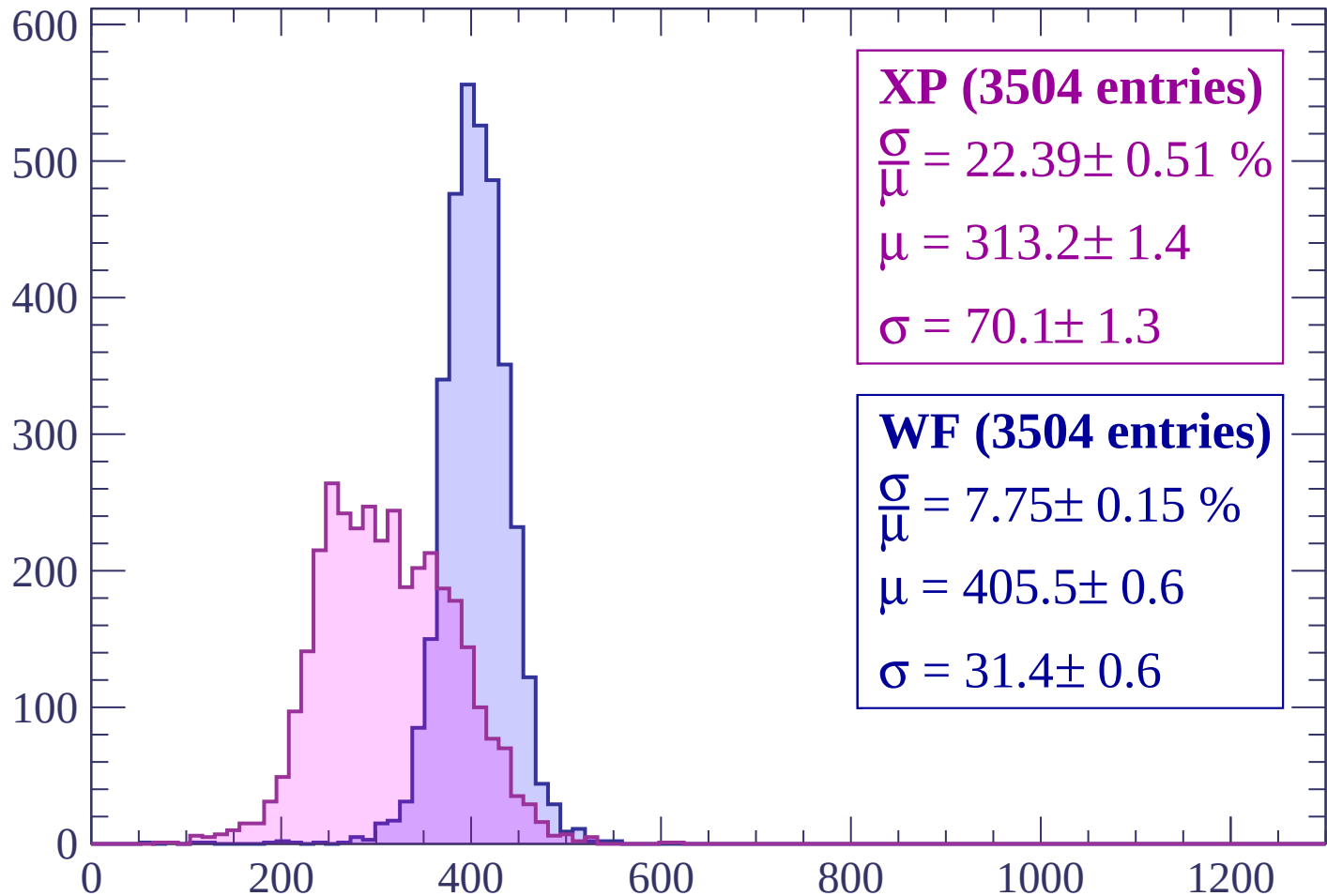
$$\mu = 407.6 \pm 0.7$$

$$\sigma = 32.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -1080 < p < -1020

Count



XP (3504 entries)

$$\frac{\sigma}{\mu} = 22.39 \pm 0.51 \%$$

$$\mu = 313.2 \pm 1.4$$

$$\sigma = 70.1 \pm 1.3$$

WF (3504 entries)

$$\frac{\sigma}{\mu} = 7.75 \pm 0.15 \%$$

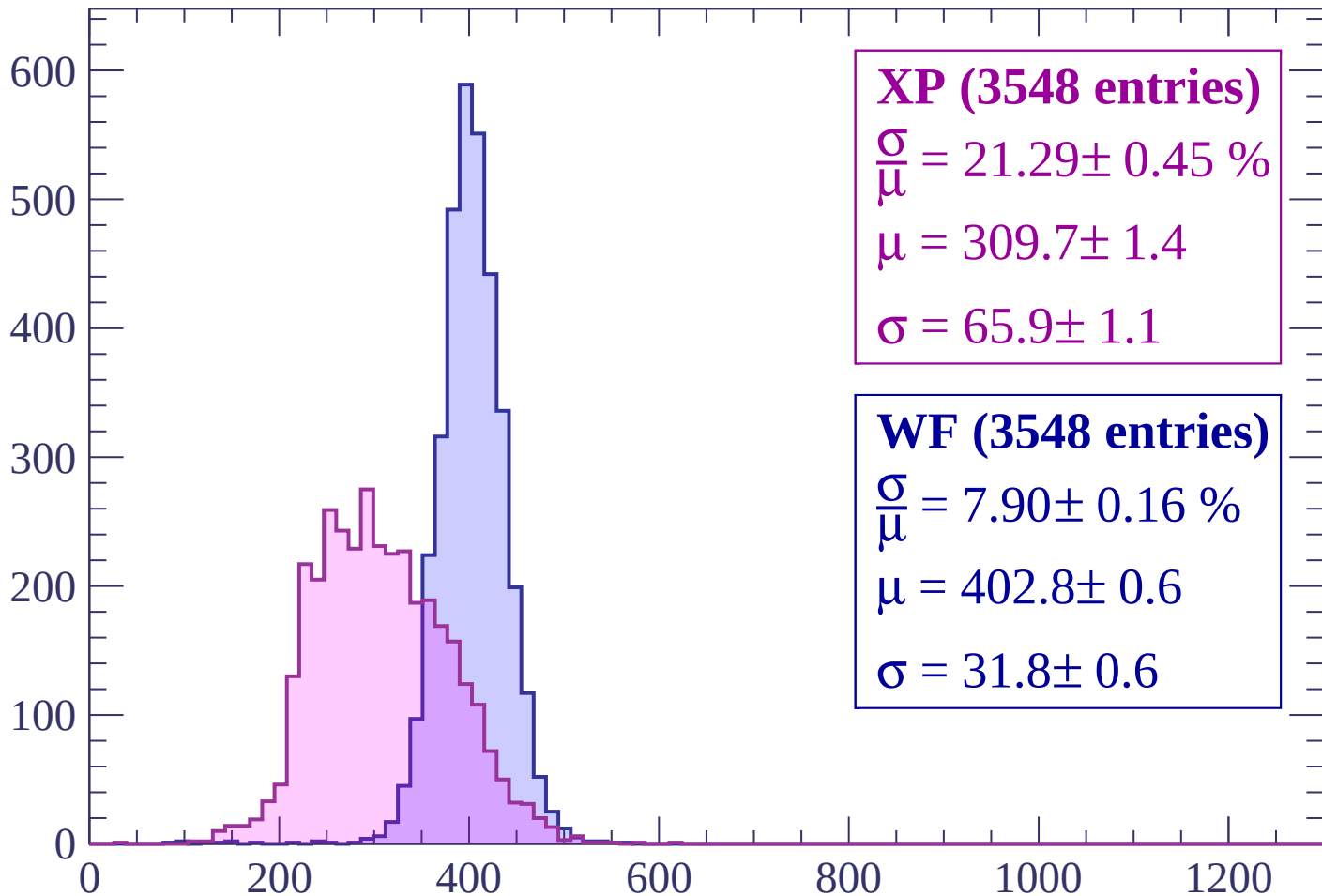
$$\mu = 405.5 \pm 0.6$$

$$\sigma = 31.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP (3548 entries)

$$\frac{\sigma}{\mu} = 21.29 \pm 0.45 \%$$

$$\mu = 309.7 \pm 1.4$$

$$\sigma = 65.9 \pm 1.1$$

WF (3548 entries)

$$\frac{\sigma}{\mu} = 7.90 \pm 0.16 \%$$

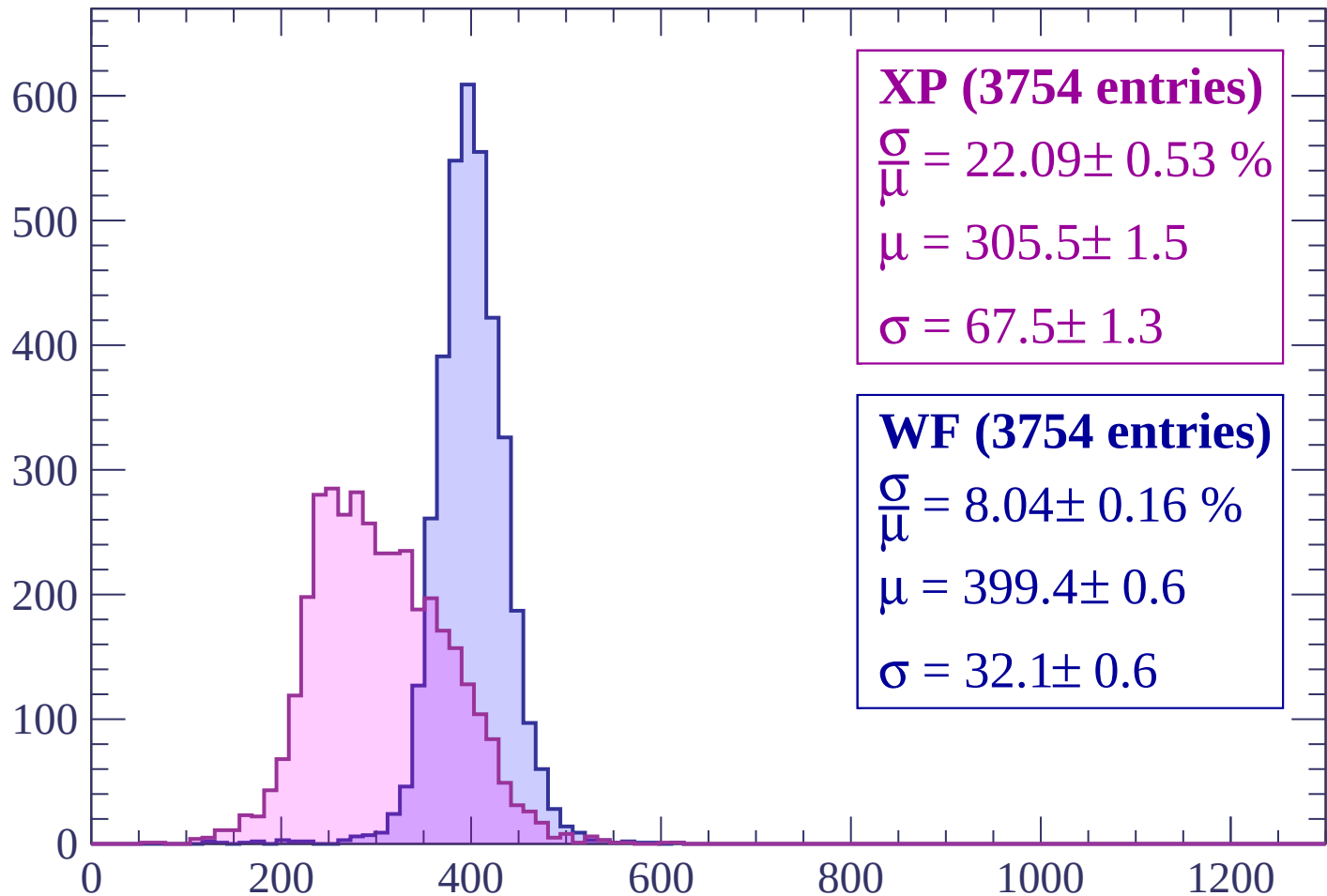
$$\mu = 402.8 \pm 0.6$$

$$\sigma = 31.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (3754 entries)

$$\frac{\sigma}{\mu} = 22.09 \pm 0.53 \%$$

$$\mu = 305.5 \pm 1.5$$

$$\sigma = 67.5 \pm 1.3$$

WF (3754 entries)

$$\frac{\sigma}{\mu} = 8.04 \pm 0.16 \%$$

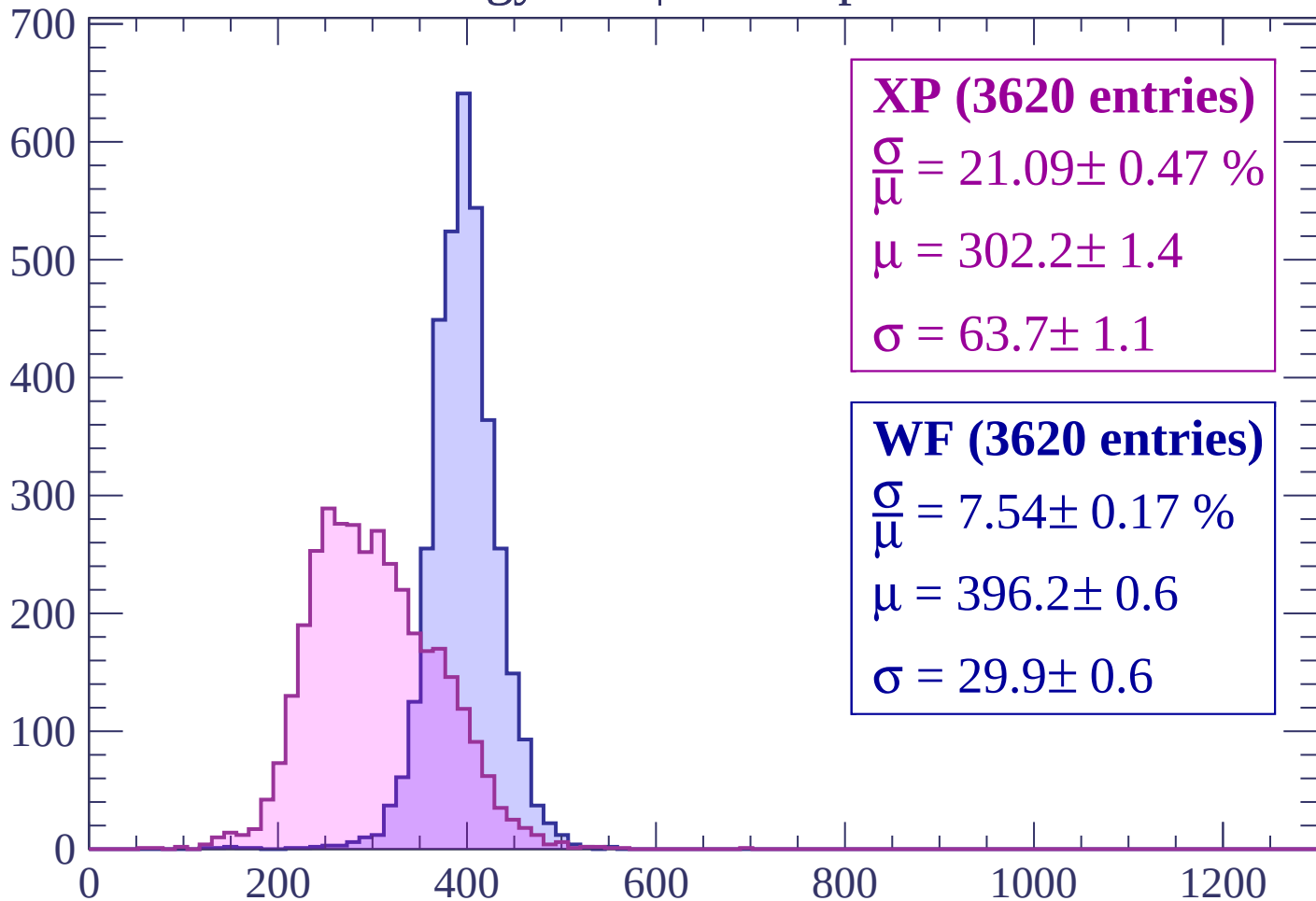
$$\mu = 399.4 \pm 0.6$$

$$\sigma = 32.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (3620 entries)

$$\frac{\sigma}{\mu} = 21.09 \pm 0.47 \%$$

$$\mu = 302.2 \pm 1.4$$

$$\sigma = 63.7 \pm 1.1$$

WF (3620 entries)

$$\frac{\sigma}{\mu} = 7.54 \pm 0.17 \%$$

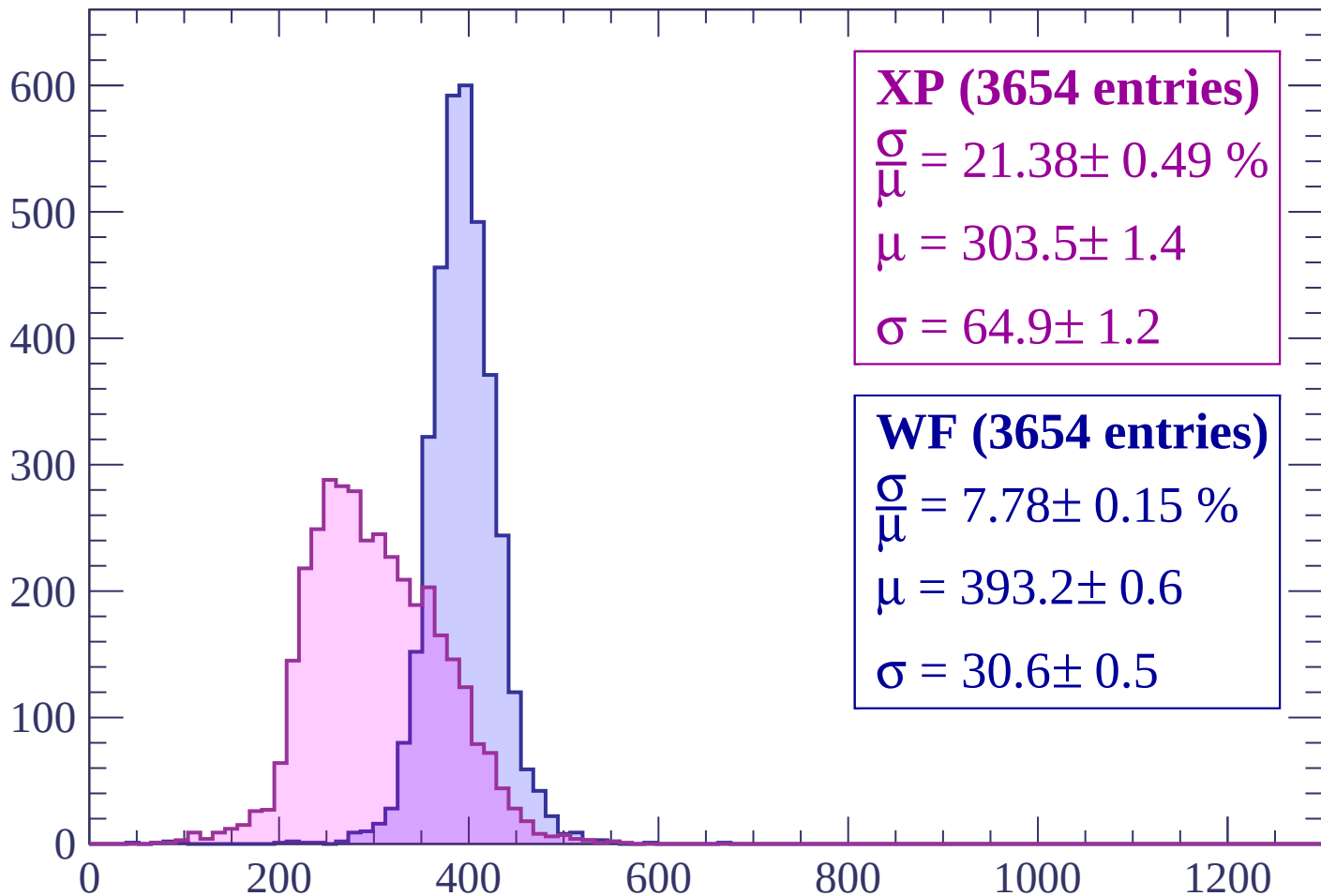
$$\mu = 396.2 \pm 0.6$$

$$\sigma = 29.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (3654 entries)

$$\frac{\sigma}{\mu} = 21.38 \pm 0.49 \%$$

$$\mu = 303.5 \pm 1.4$$

$$\sigma = 64.9 \pm 1.2$$

WF (3654 entries)

$$\frac{\sigma}{\mu} = 7.78 \pm 0.15 \%$$

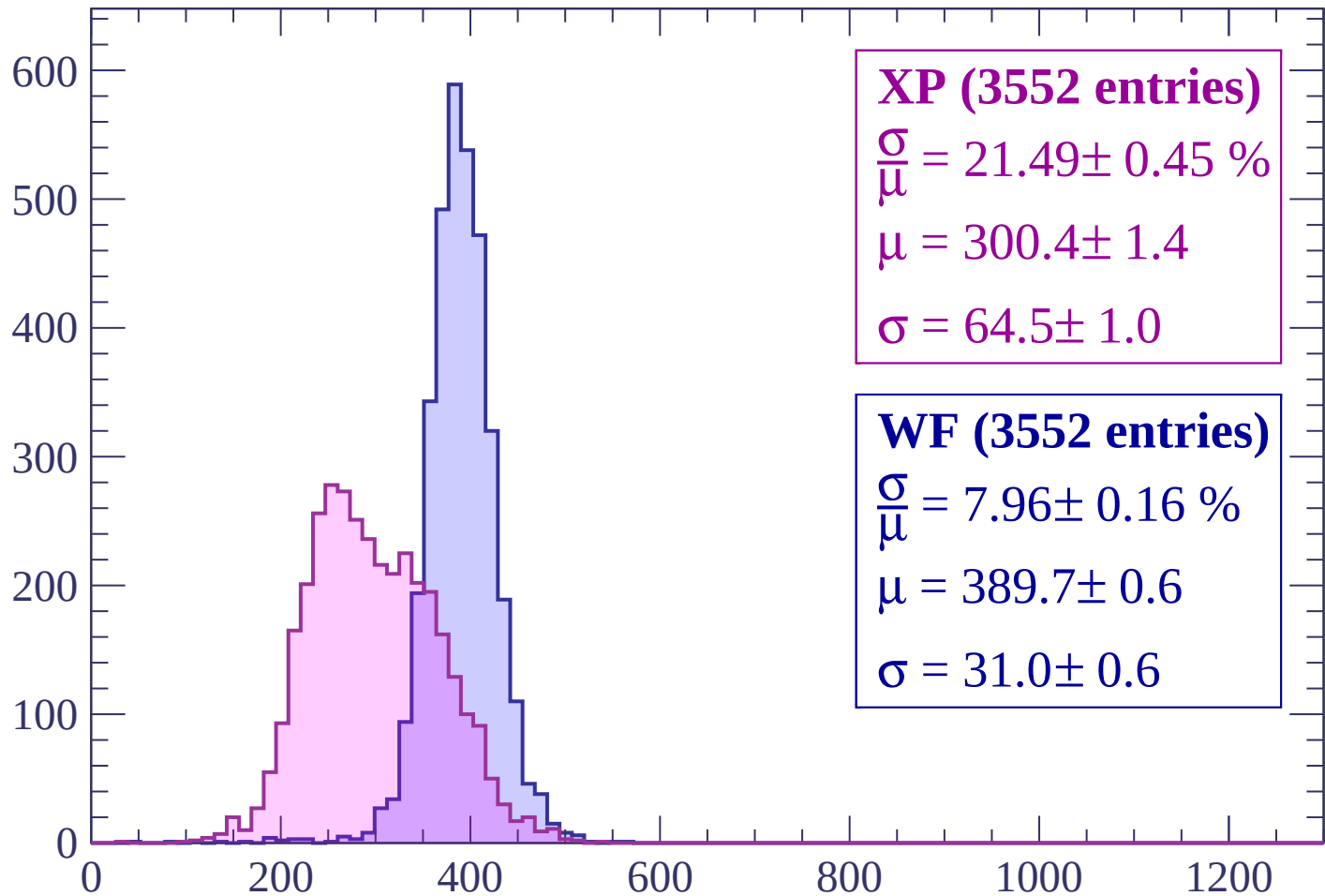
$$\mu = 393.2 \pm 0.6$$

$$\sigma = 30.6 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (3552 entries)

$$\frac{\sigma}{\mu} = 21.49 \pm 0.45 \%$$

$$\mu = 300.4 \pm 1.4$$

$$\sigma = 64.5 \pm 1.0$$

WF (3552 entries)

$$\frac{\sigma}{\mu} = 7.96 \pm 0.16 \%$$

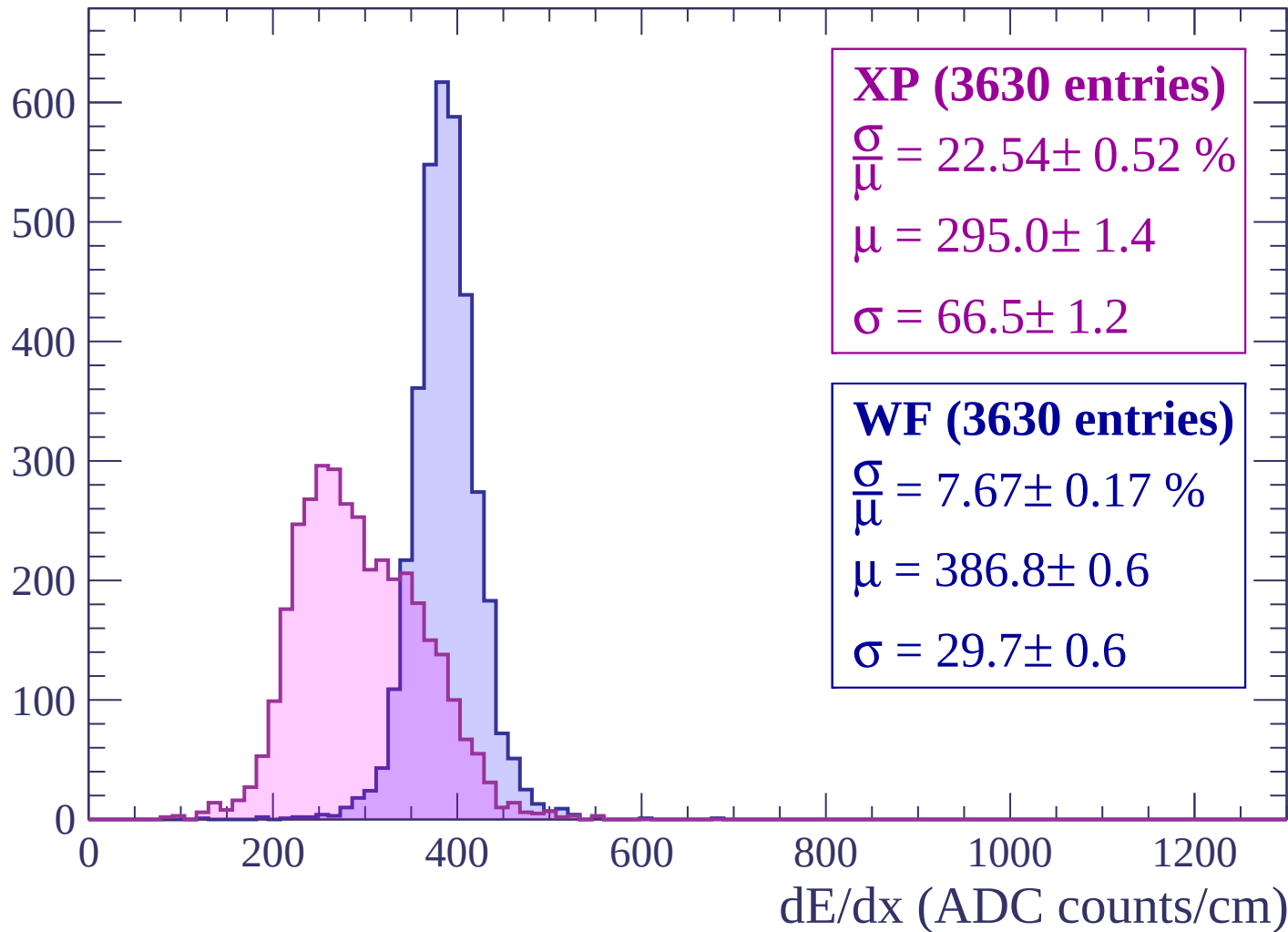
$$\mu = 389.7 \pm 0.6$$

$$\sigma = 31.0 \pm 0.6$$

dE/dx (ADC counts/cm)

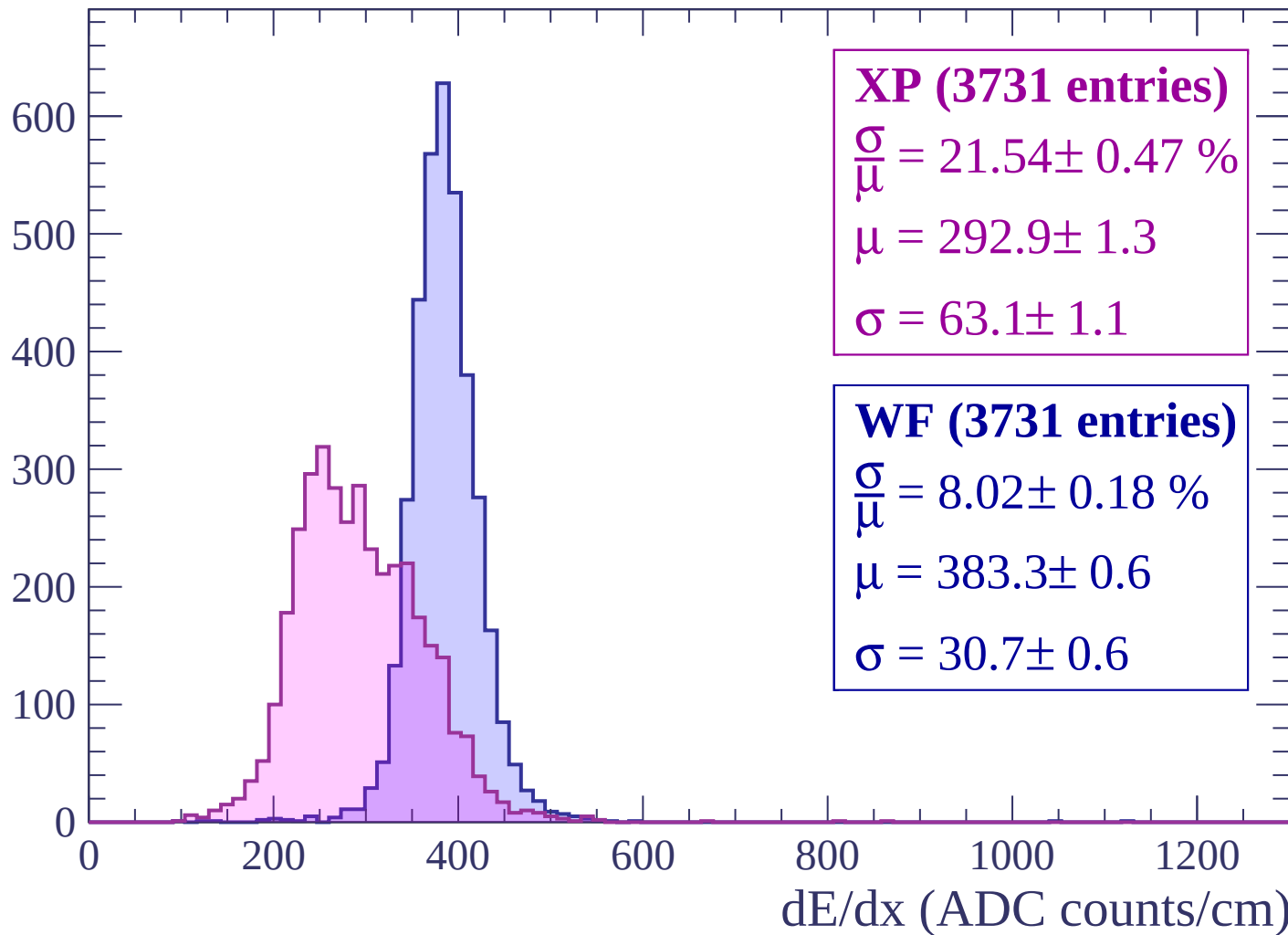
Energy loss | $-720 < p < -660$

Count



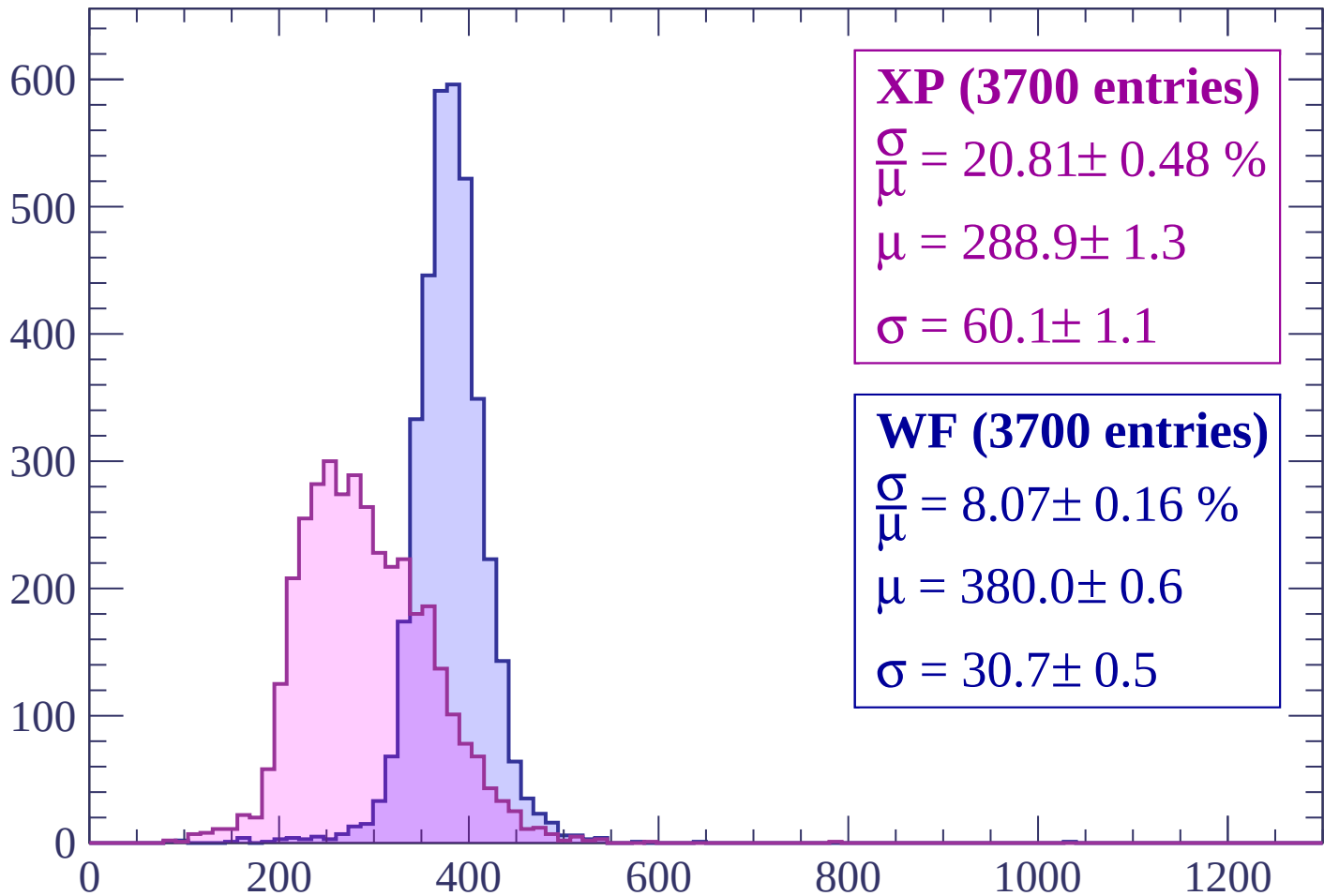
Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$

Count



XP (3700 entries)

$\frac{\sigma}{\mu} = 20.81 \pm 0.48 \%$

$\mu = 288.9 \pm 1.3$

$\sigma = 60.1 \pm 1.1$

WF (3700 entries)

$\frac{\sigma}{\mu} = 8.07 \pm 0.16 \%$

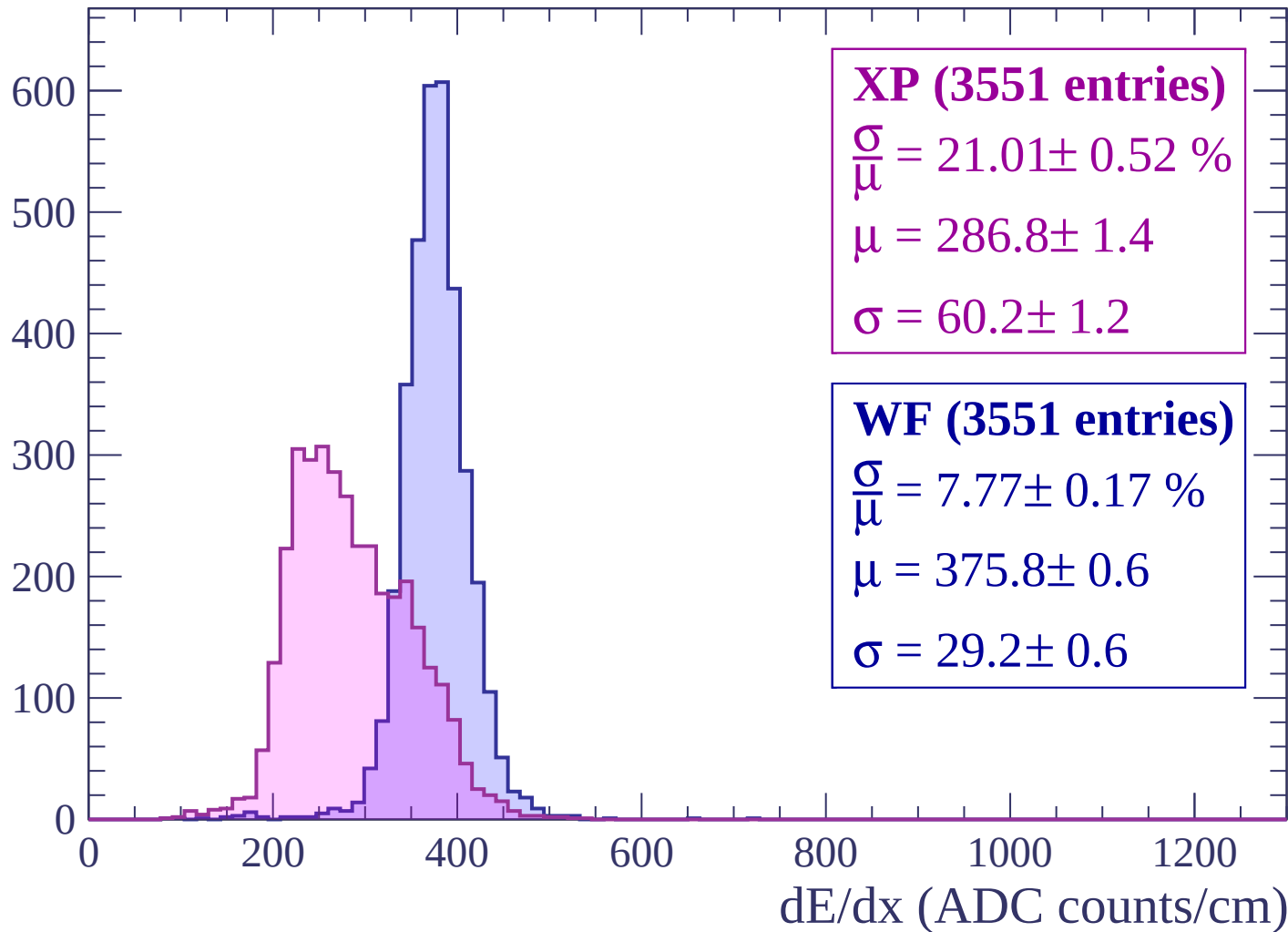
$\mu = 380.0 \pm 0.6$

$\sigma = 30.7 \pm 0.5$

dE/dx (ADC counts/cm)

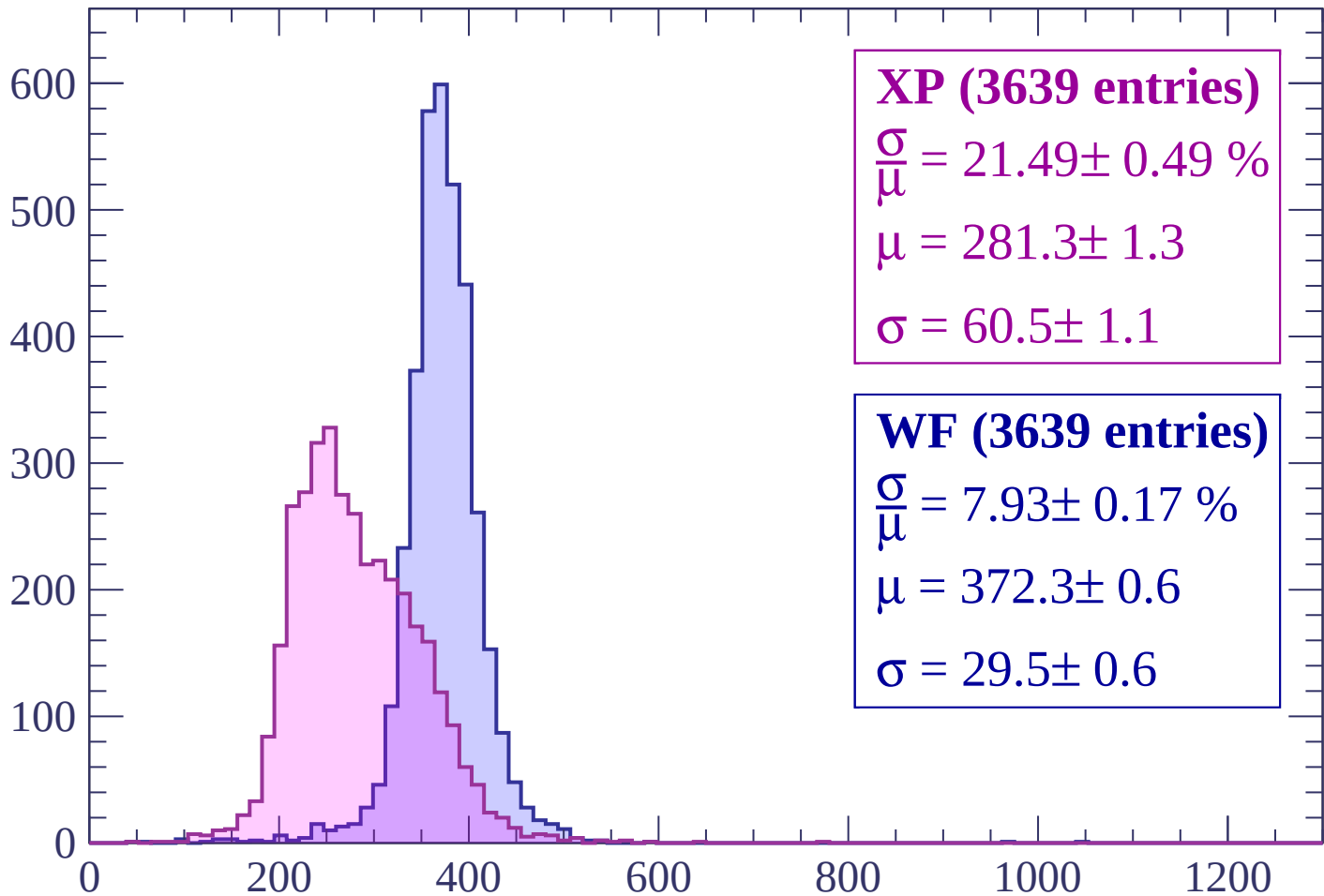
Energy loss | $-540 < p < -480$

Count



Energy loss | $-480 < p < -420$

Count



XP (3639 entries)

$$\frac{\sigma}{\mu} = 21.49 \pm 0.49 \%$$

$$\mu = 281.3 \pm 1.3$$

$$\sigma = 60.5 \pm 1.1$$

WF (3639 entries)

$$\frac{\sigma}{\mu} = 7.93 \pm 0.17 \%$$

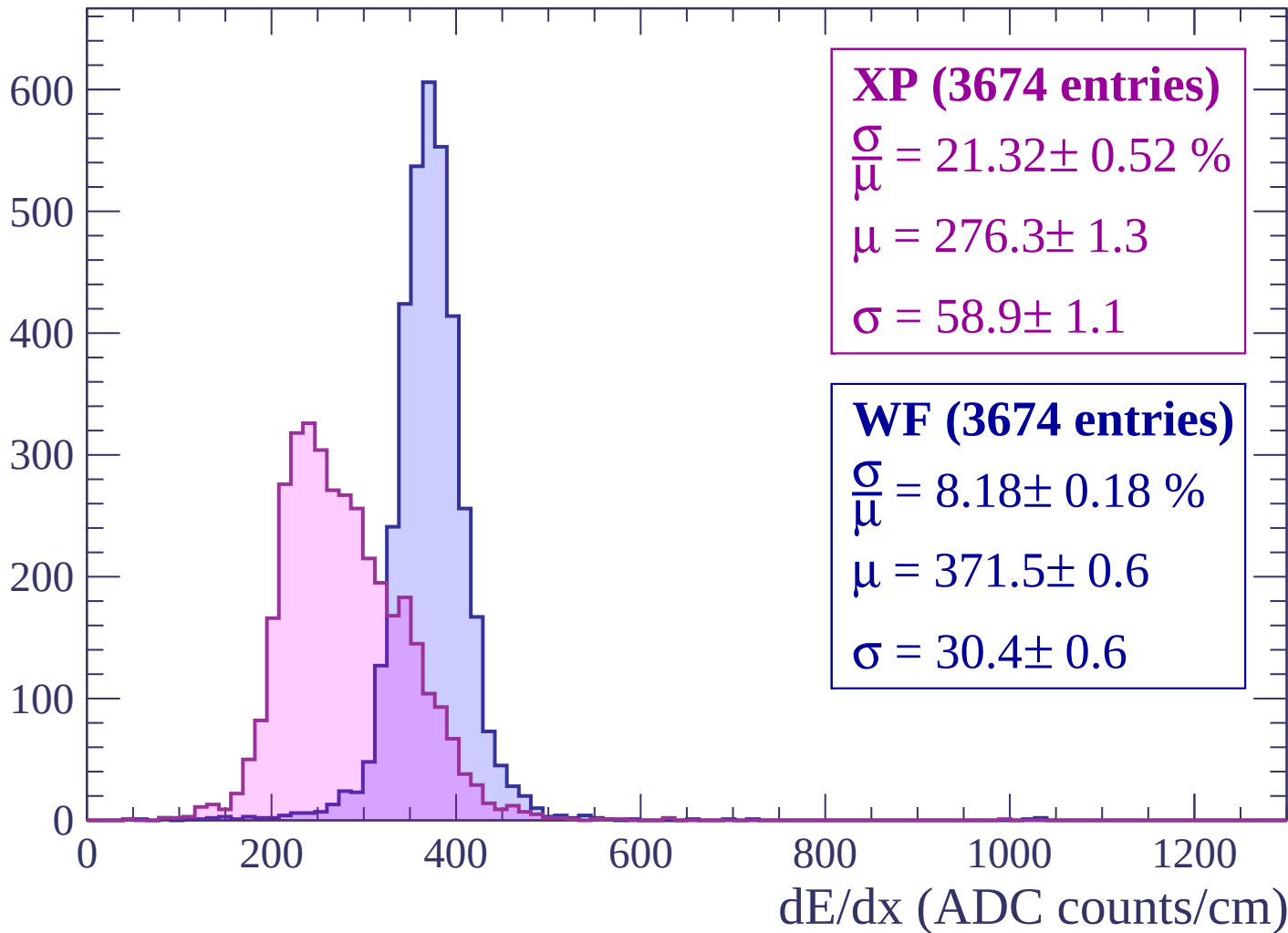
$$\mu = 372.3 \pm 0.6$$

$$\sigma = 29.5 \pm 0.6$$

dE/dx (ADC counts/cm)

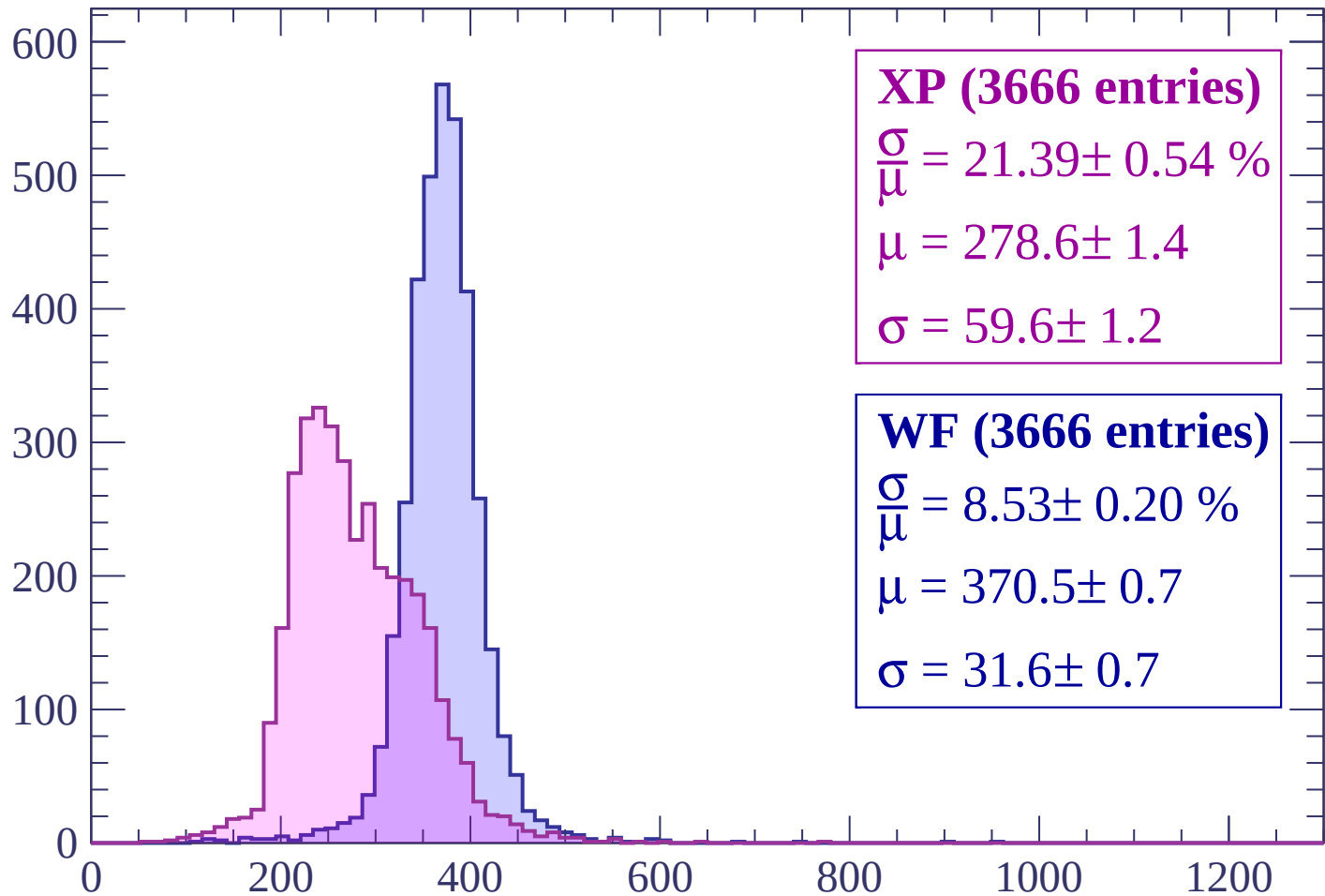
Energy loss | $-420 < p < -360$

Count



Energy loss | $-360 < p < -300$

Count



XP (3666 entries)

$$\frac{\sigma}{\mu} = 21.39 \pm 0.54 \%$$

$$\mu = 278.6 \pm 1.4$$

$$\sigma = 59.6 \pm 1.2$$

WF (3666 entries)

$$\frac{\sigma}{\mu} = 8.53 \pm 0.20 \%$$

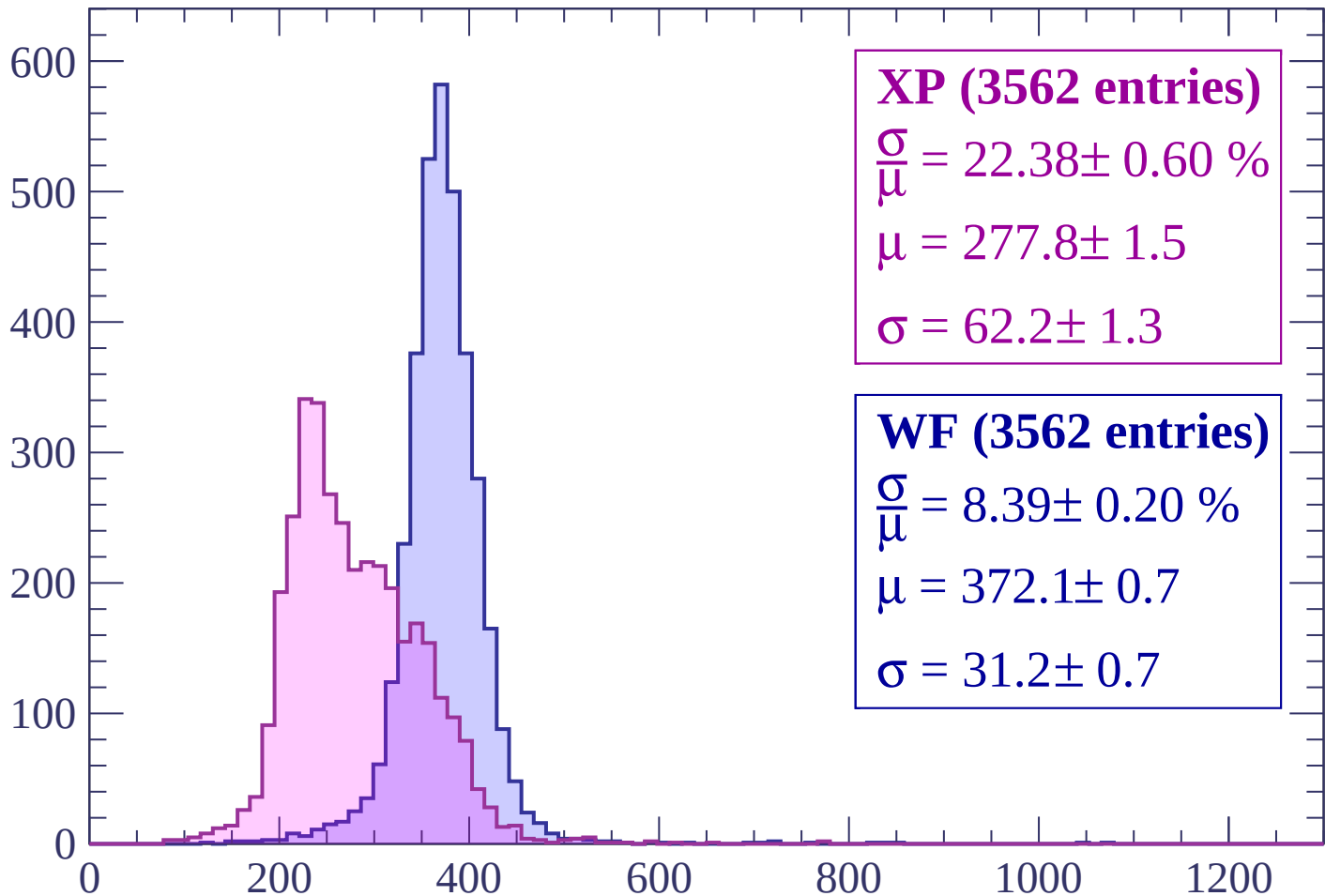
$$\mu = 370.5 \pm 0.7$$

$$\sigma = 31.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (3562 entries)

$$\frac{\sigma}{\mu} = 22.38 \pm 0.60 \%$$

$$\mu = 277.8 \pm 1.5$$

$$\sigma = 62.2 \pm 1.3$$

WF (3562 entries)

$$\frac{\sigma}{\mu} = 8.39 \pm 0.20 \%$$

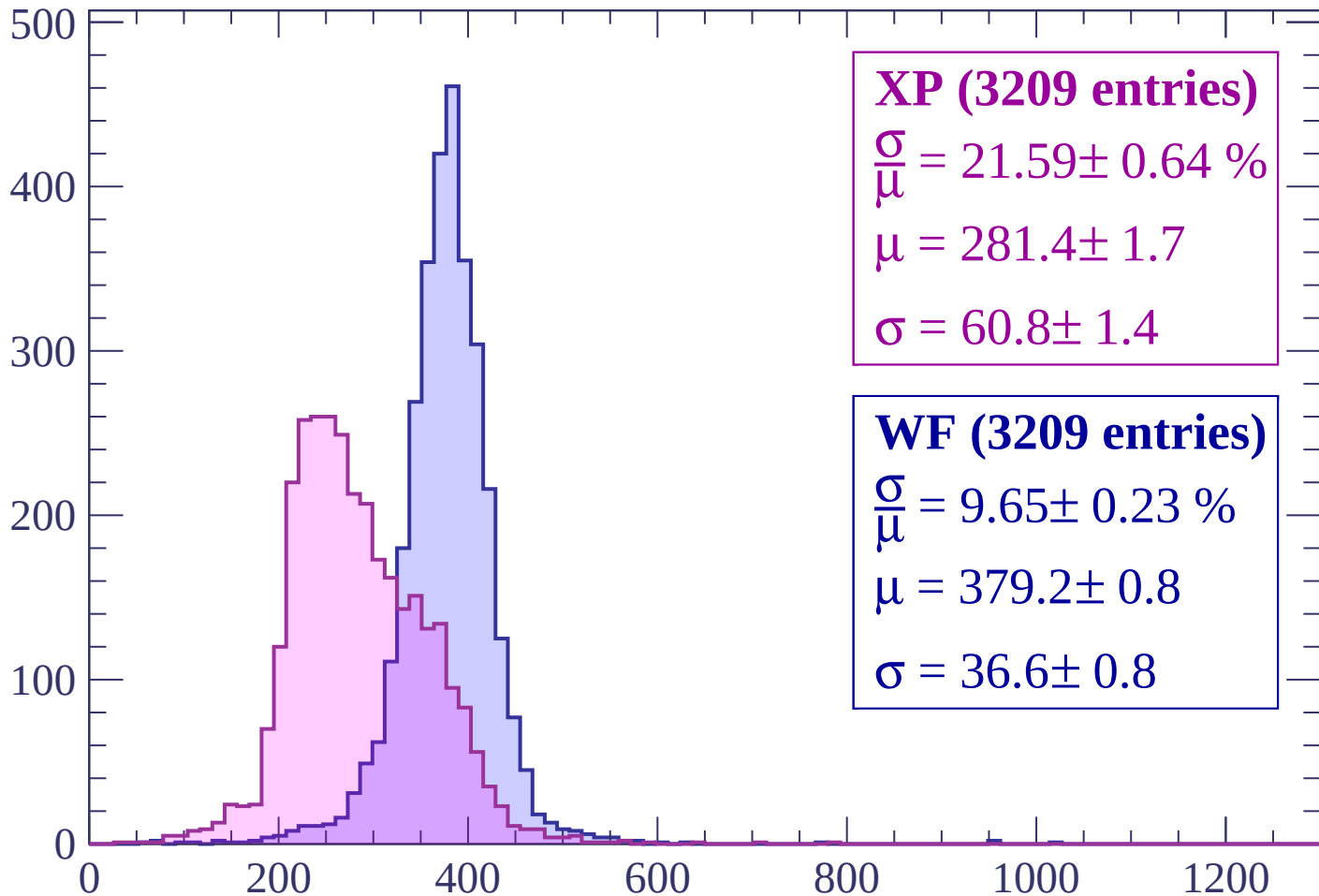
$$\mu = 372.1 \pm 0.7$$

$$\sigma = 31.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (3209 entries)

$$\frac{\sigma}{\mu} = 21.59 \pm 0.64 \%$$

$$\mu = 281.4 \pm 1.7$$

$$\sigma = 60.8 \pm 1.4$$

WF (3209 entries)

$$\frac{\sigma}{\mu} = 9.65 \pm 0.23 \%$$

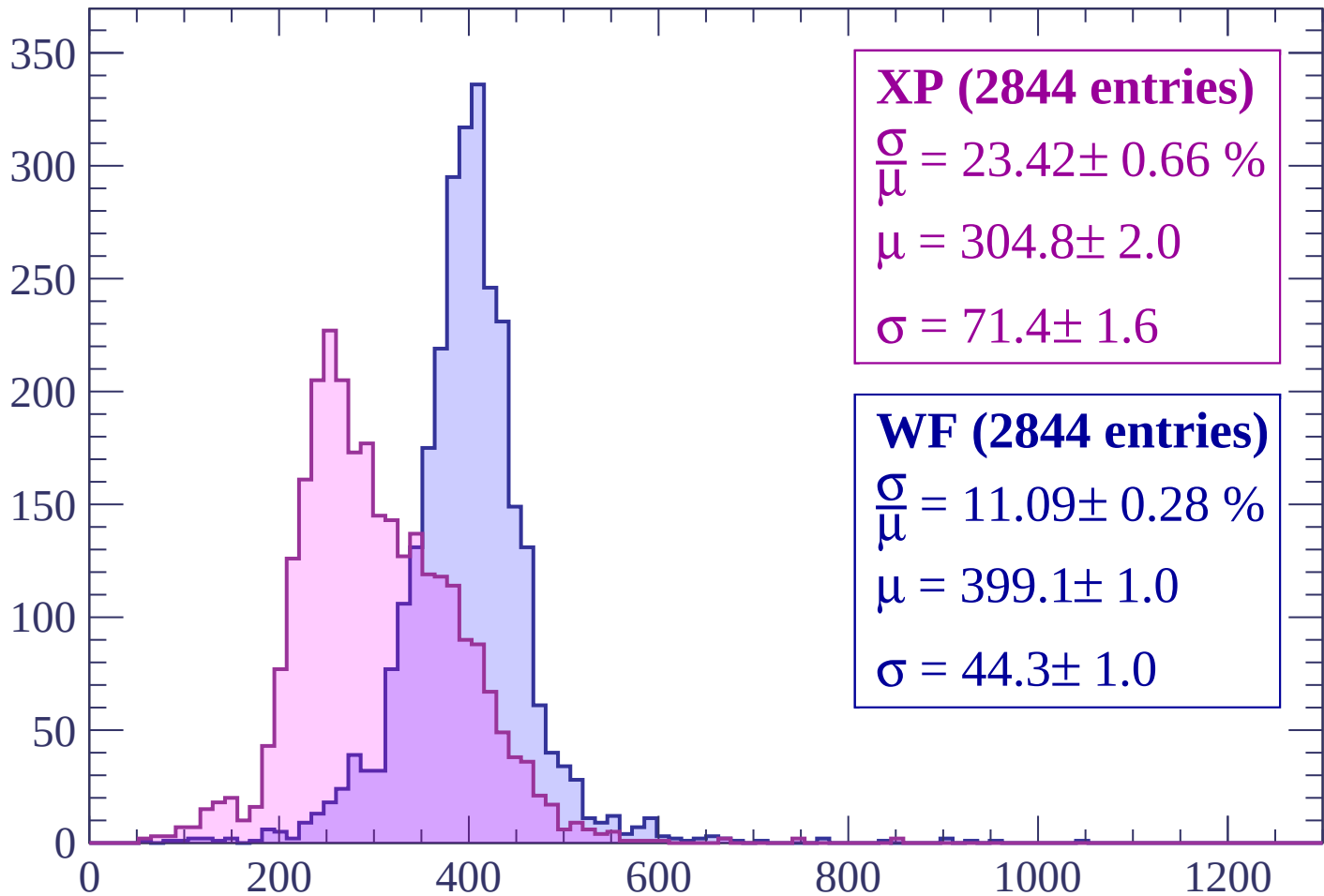
$$\mu = 379.2 \pm 0.8$$

$$\sigma = 36.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-180 < p < -120$

Count



XP (2844 entries)

$$\frac{\sigma}{\mu} = 23.42 \pm 0.66 \%$$

$$\mu = 304.8 \pm 2.0$$

$$\sigma = 71.4 \pm 1.6$$

WF (2844 entries)

$$\frac{\sigma}{\mu} = 11.09 \pm 0.28 \%$$

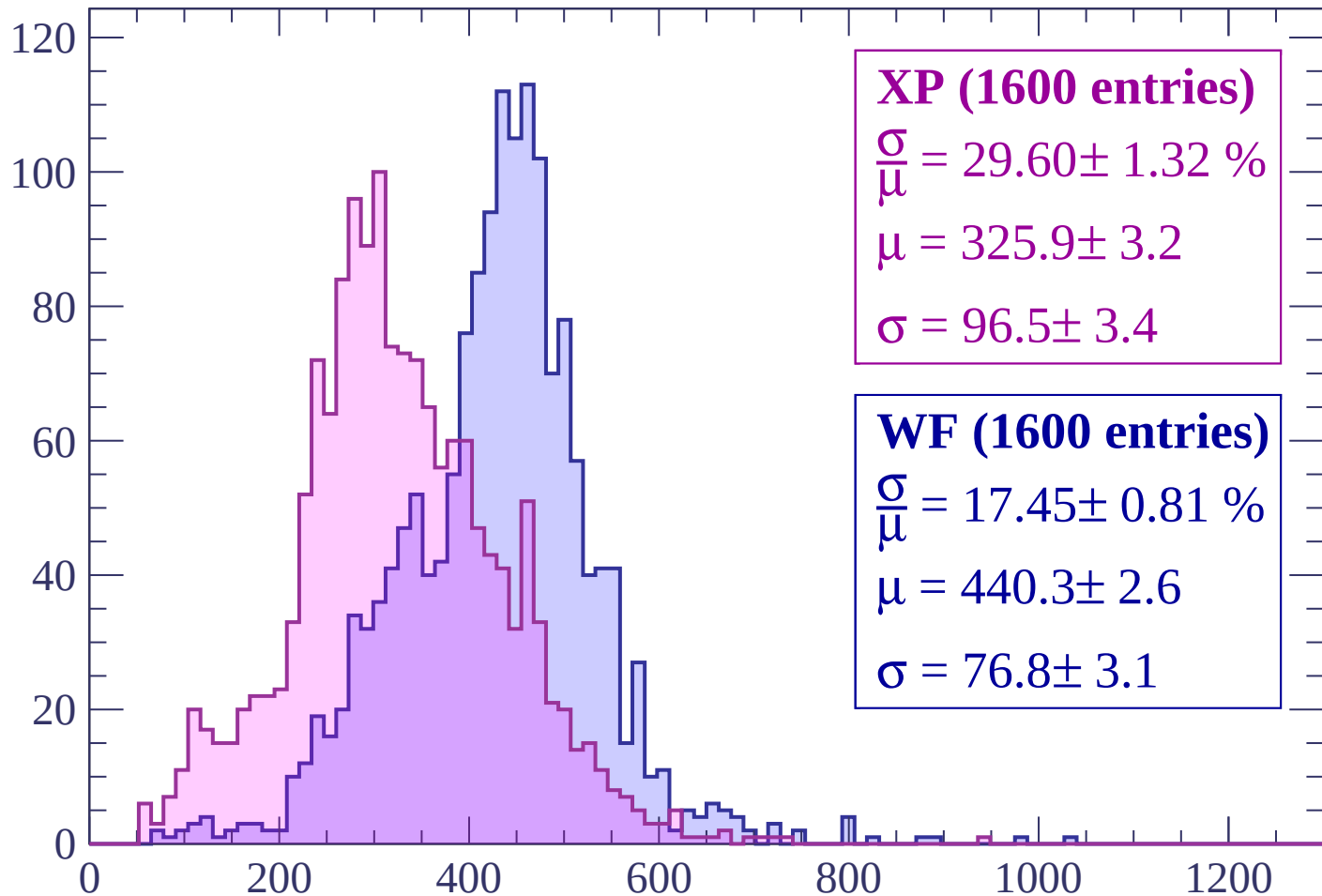
$$\mu = 399.1 \pm 1.0$$

$$\sigma = 44.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

Count



XP (1600 entries)

$$\frac{\sigma}{\mu} = 29.60 \pm 1.32 \%$$

$$\mu = 325.9 \pm 3.2$$

$$\sigma = 96.5 \pm 3.4$$

WF (1600 entries)

$$\frac{\sigma}{\mu} = 17.45 \pm 0.81 \%$$

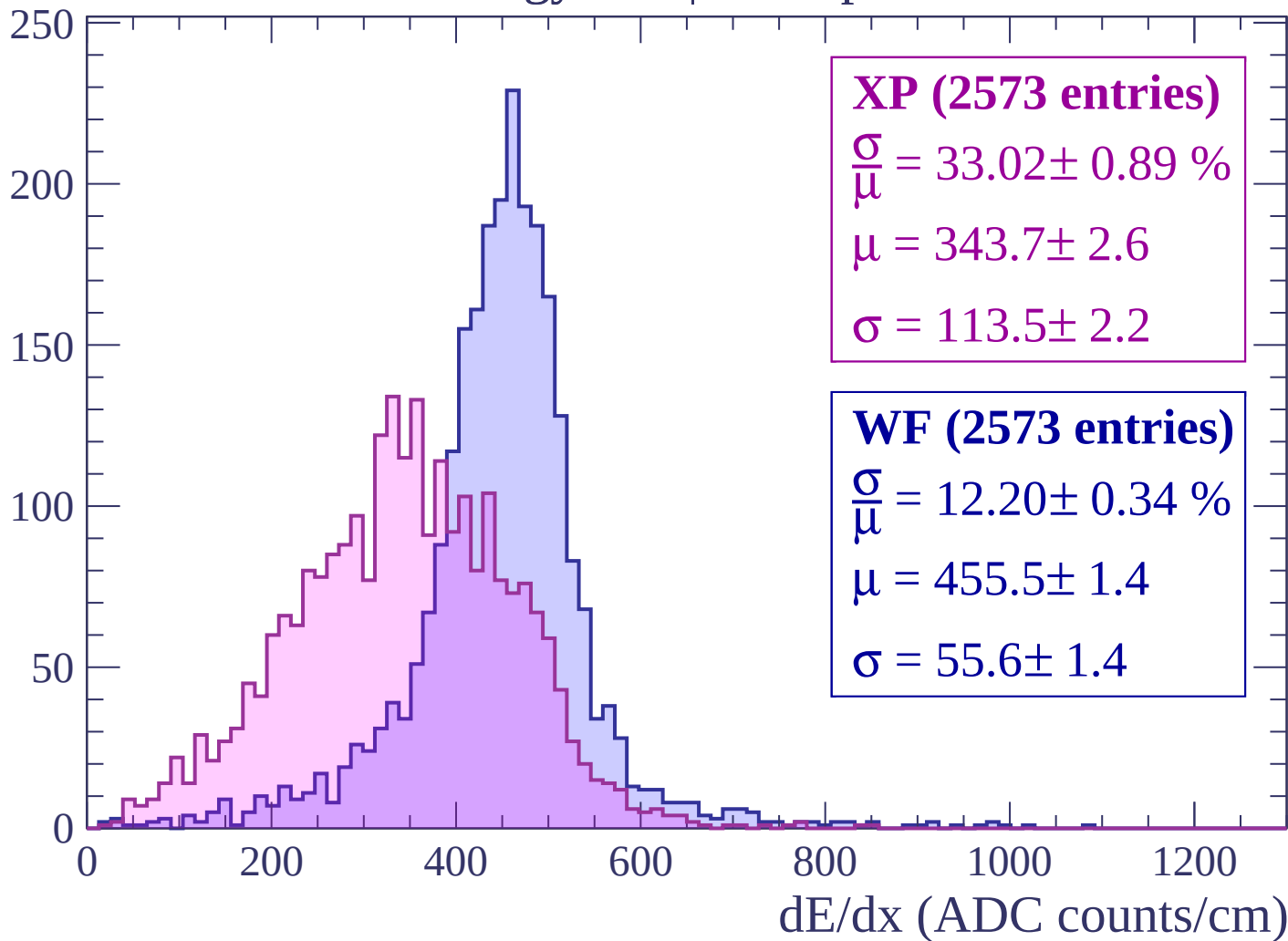
$$\mu = 440.3 \pm 2.6$$

$$\sigma = 76.8 \pm 3.1$$

dE/dx (ADC counts/cm)

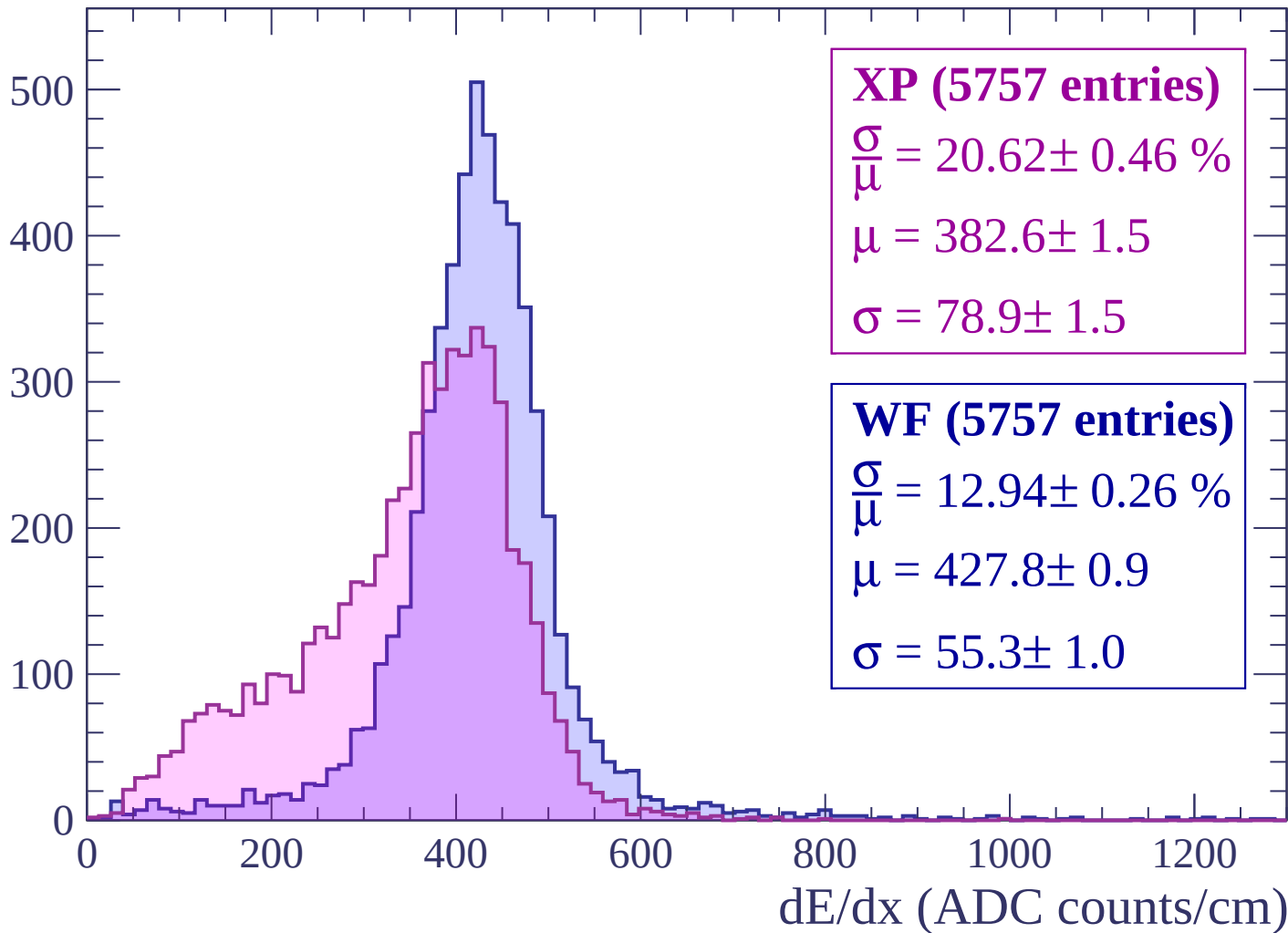
Energy loss | $-60 < p < 0$

Count



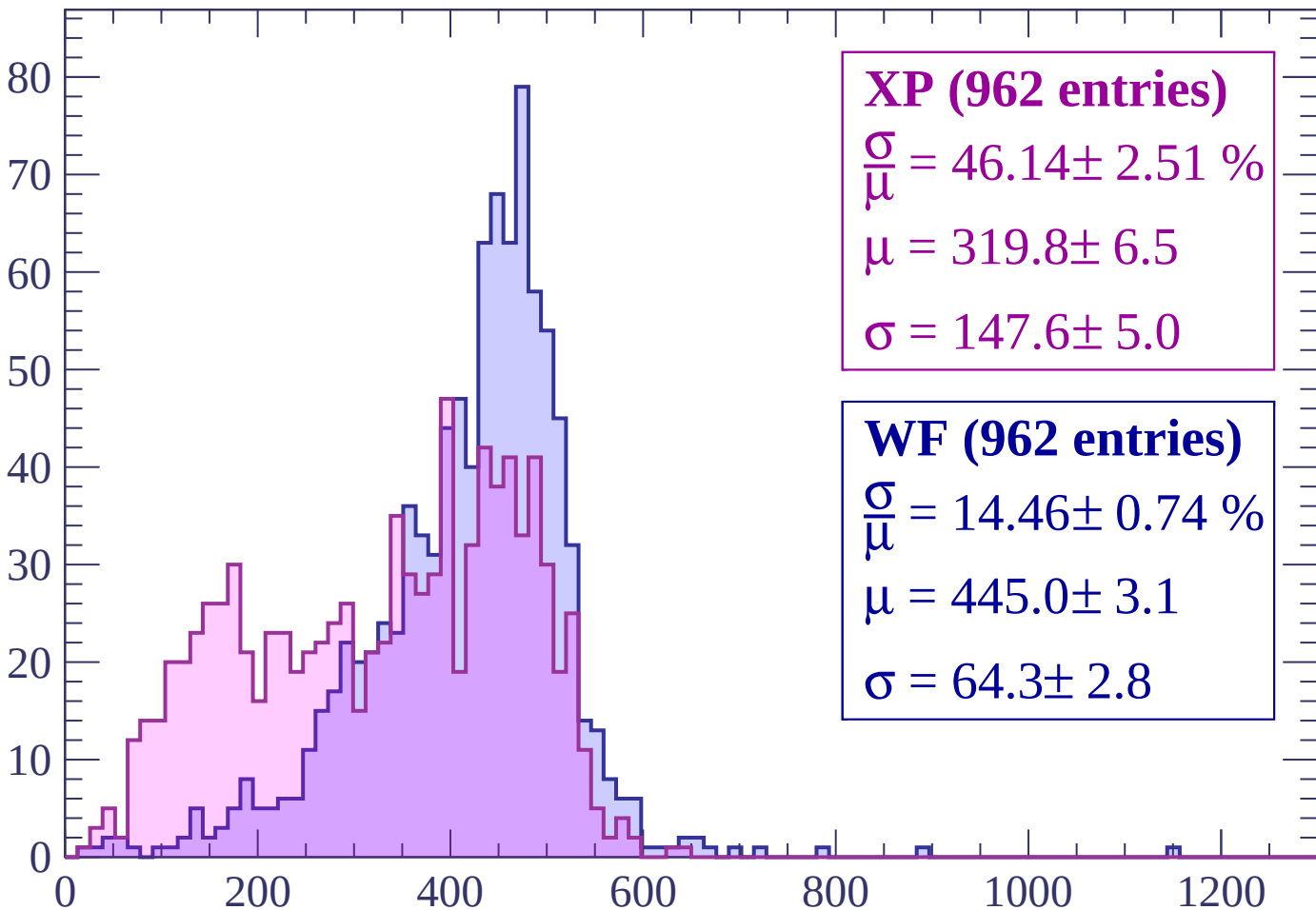
Energy loss | $0 < p < 60$

Count



Energy loss | $60 < p < 120$

Count



XP (962 entries)

$$\frac{\sigma}{\mu} = 46.14 \pm 2.51 \%$$

$$\mu = 319.8 \pm 6.5$$

$$\sigma = 147.6 \pm 5.0$$

WF (962 entries)

$$\frac{\sigma}{\mu} = 14.46 \pm 0.74 \%$$

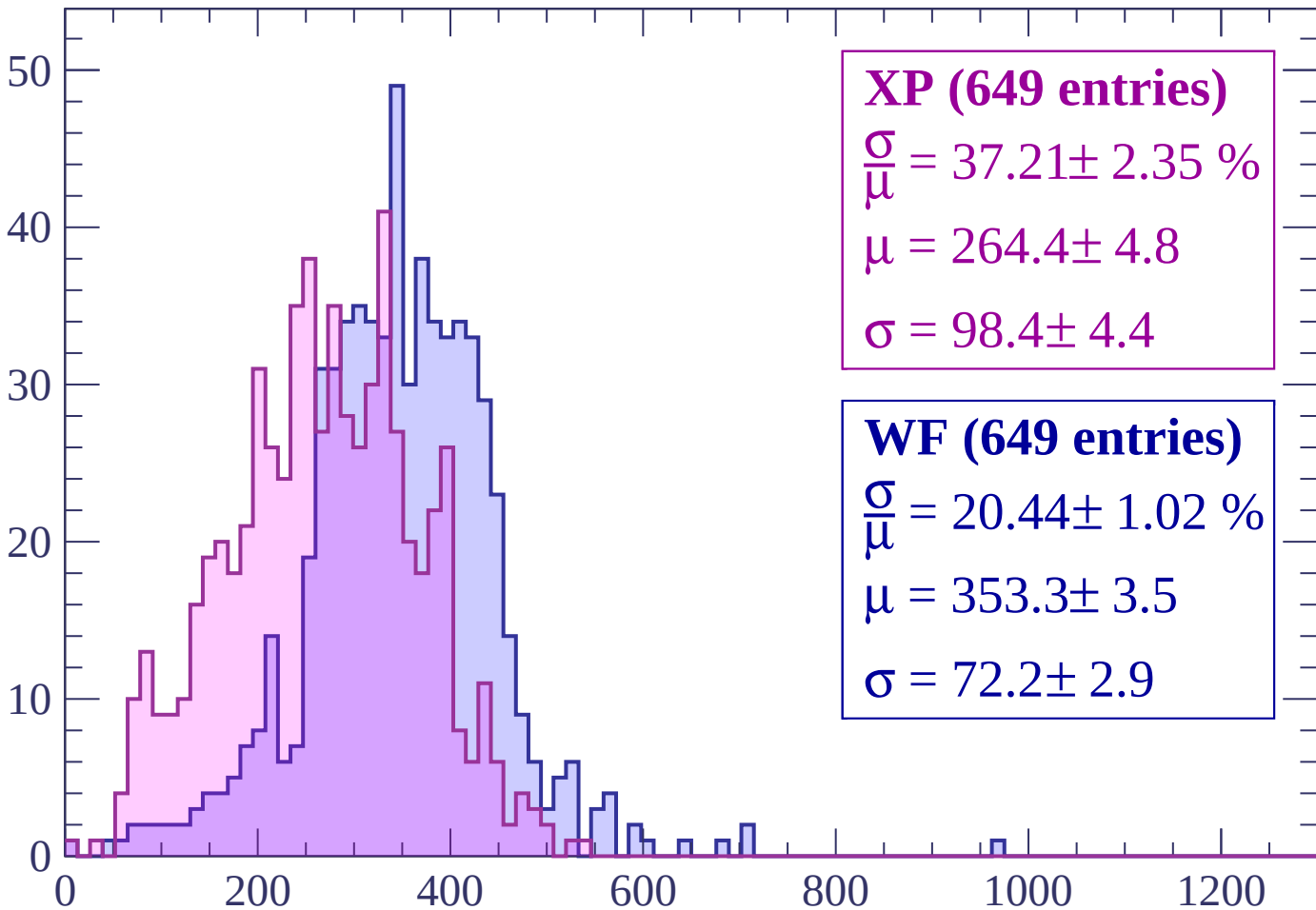
$$\mu = 445.0 \pm 3.1$$

$$\sigma = 64.3 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $120 < p < 180$

Count



XP (649 entries)

$$\frac{\sigma}{\mu} = 37.21 \pm 2.35 \%$$

$$\mu = 264.4 \pm 4.8$$

$$\sigma = 98.4 \pm 4.4$$

WF (649 entries)

$$\frac{\sigma}{\mu} = 20.44 \pm 1.02 \%$$

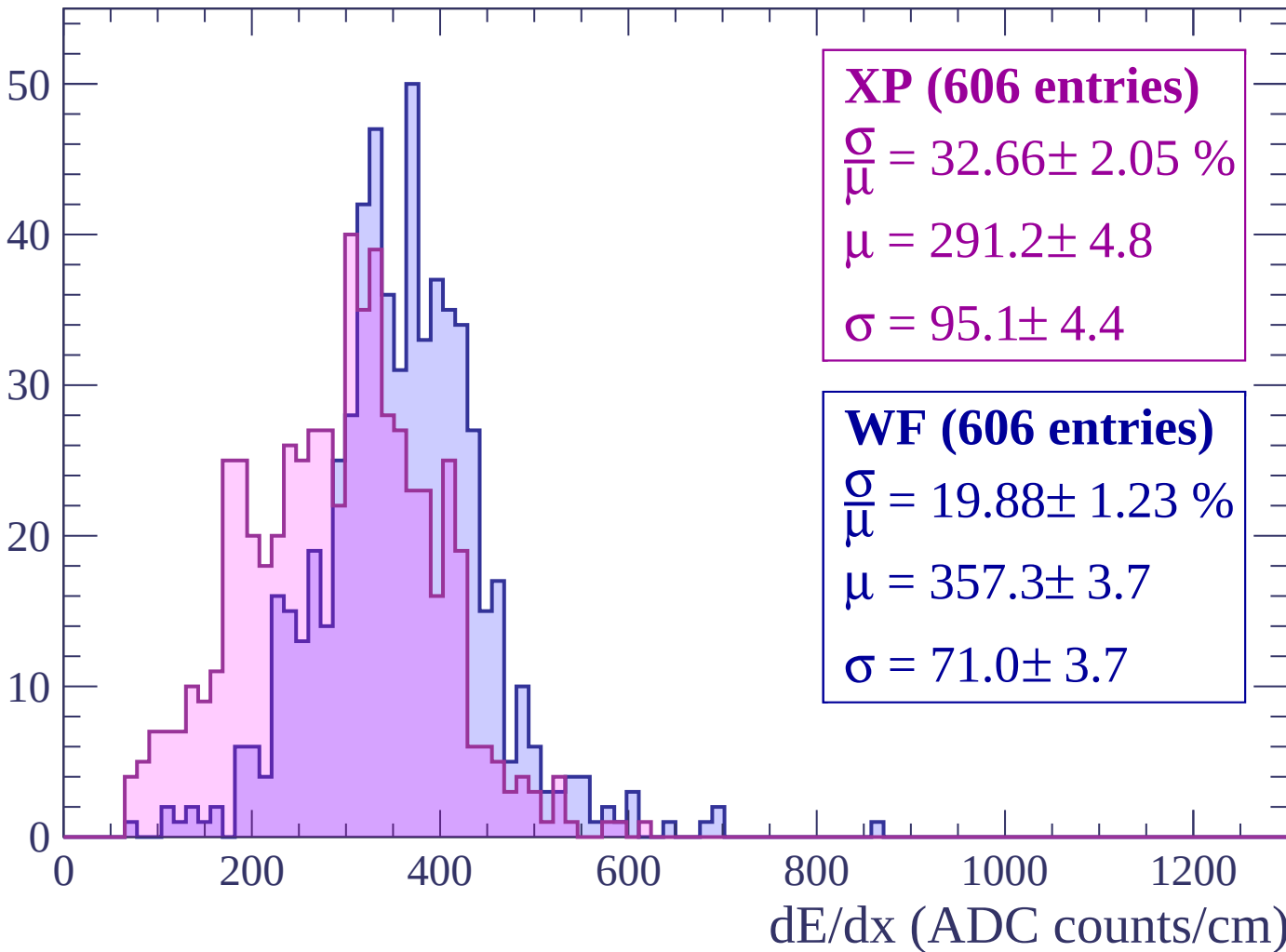
$$\mu = 353.3 \pm 3.5$$

$$\sigma = 72.2 \pm 2.9$$

dE/dx (ADC counts/cm)

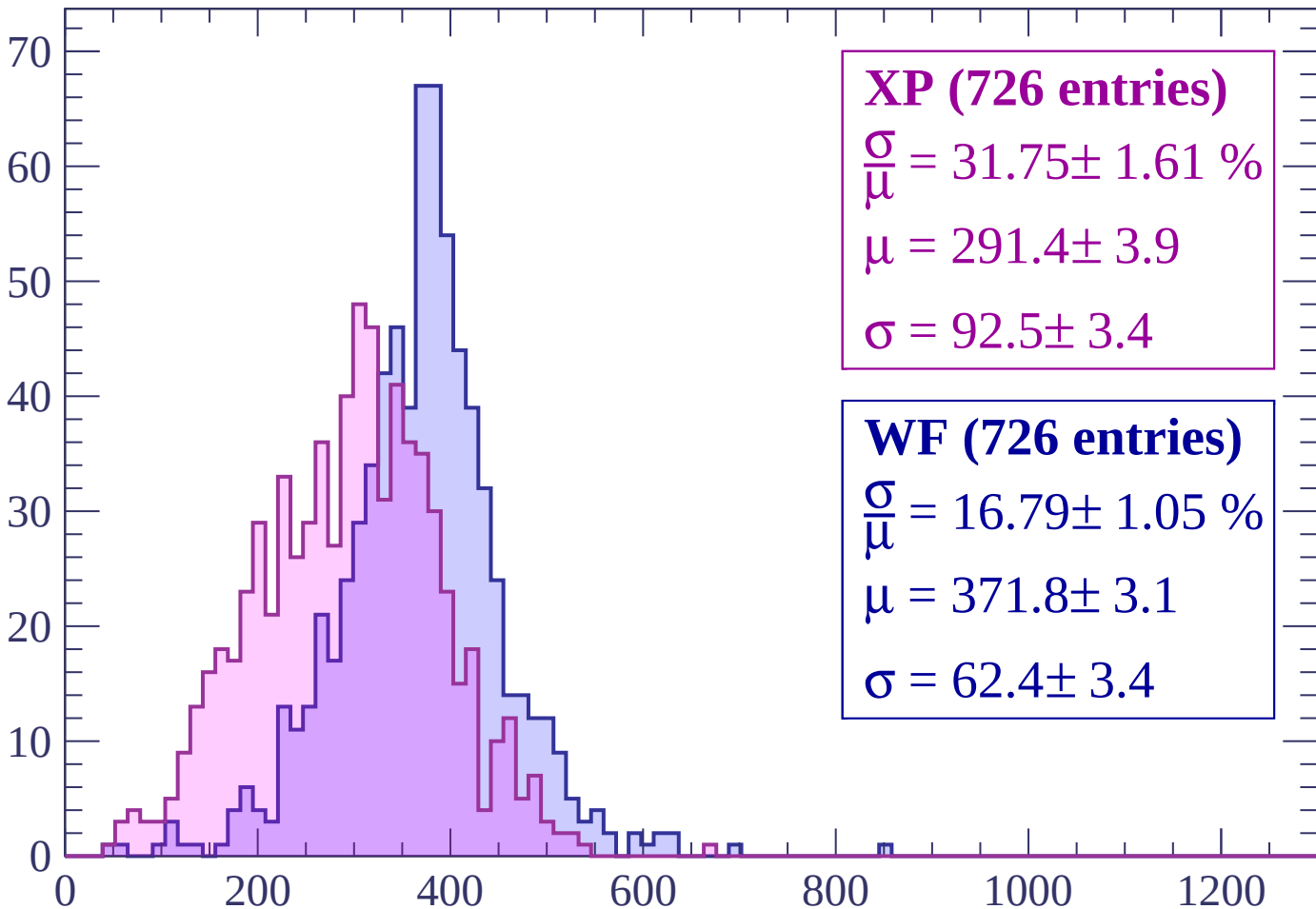
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



XP (726 entries)

$$\frac{\sigma}{\mu} = 31.75 \pm 1.61 \%$$

$$\mu = 291.4 \pm 3.9$$

$$\sigma = 92.5 \pm 3.4$$

WF (726 entries)

$$\frac{\sigma}{\mu} = 16.79 \pm 1.05 \%$$

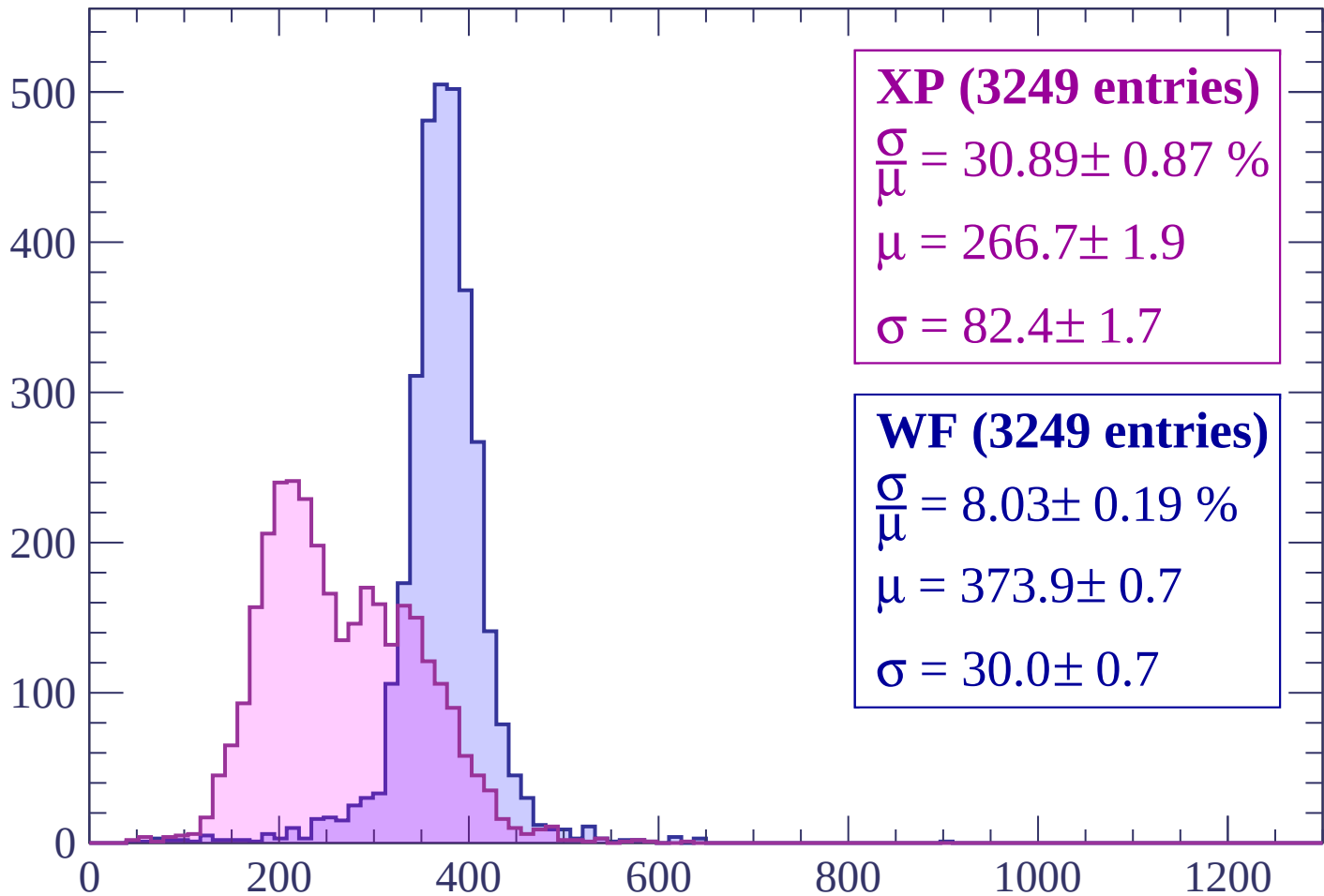
$$\mu = 371.8 \pm 3.1$$

$$\sigma = 62.4 \pm 3.4$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (3249 entries)

$$\frac{\sigma}{\mu} = 30.89 \pm 0.87 \%$$

$$\mu = 266.7 \pm 1.9$$

$$\sigma = 82.4 \pm 1.7$$

WF (3249 entries)

$$\frac{\sigma}{\mu} = 8.03 \pm 0.19 \%$$

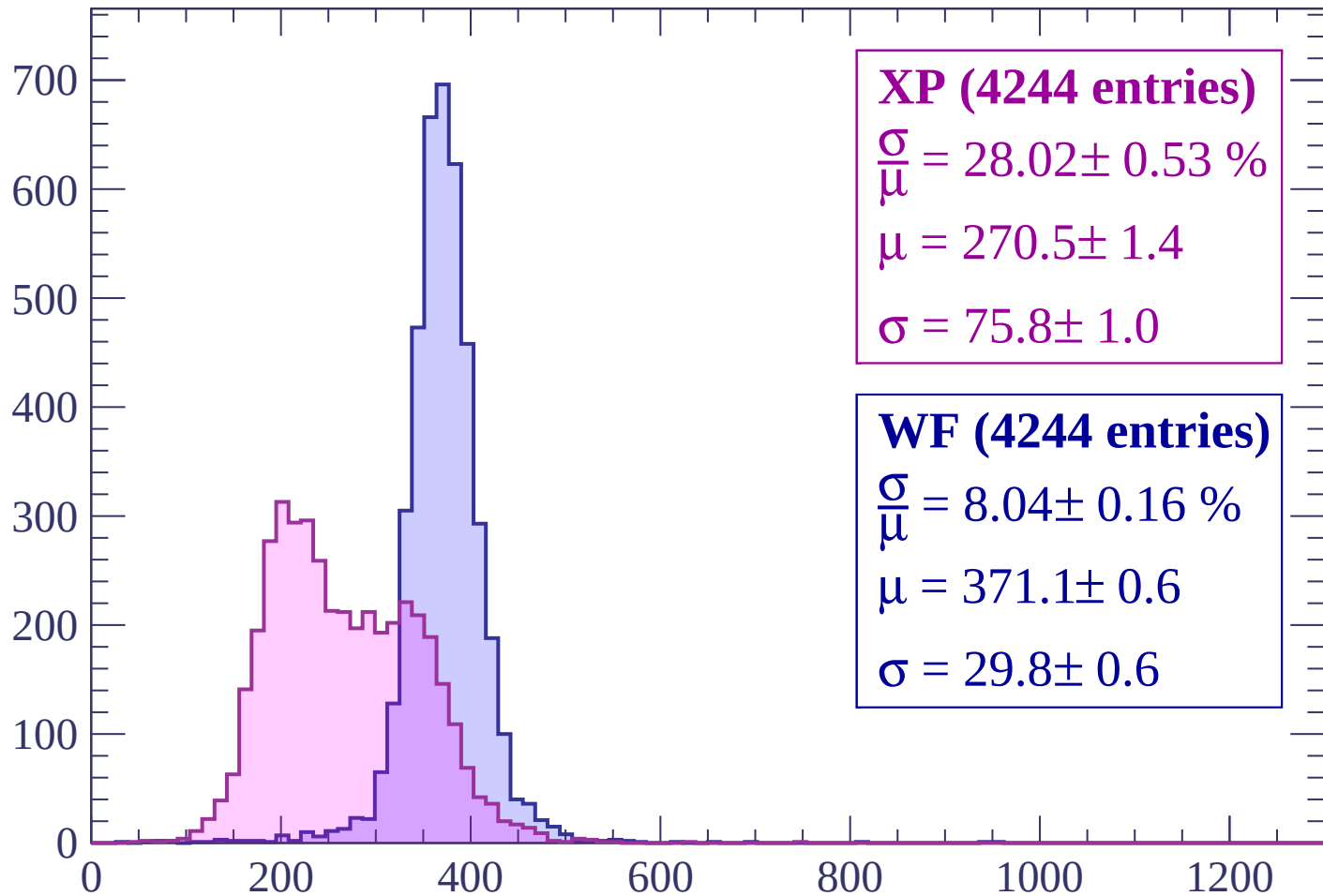
$$\mu = 373.9 \pm 0.7$$

$$\sigma = 30.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (4244 entries)

$$\frac{\sigma}{\mu} = 28.02 \pm 0.53 \%$$

$$\mu = 270.5 \pm 1.4$$

$$\sigma = 75.8 \pm 1.0$$

WF (4244 entries)

$$\frac{\sigma}{\mu} = 8.04 \pm 0.16 \%$$

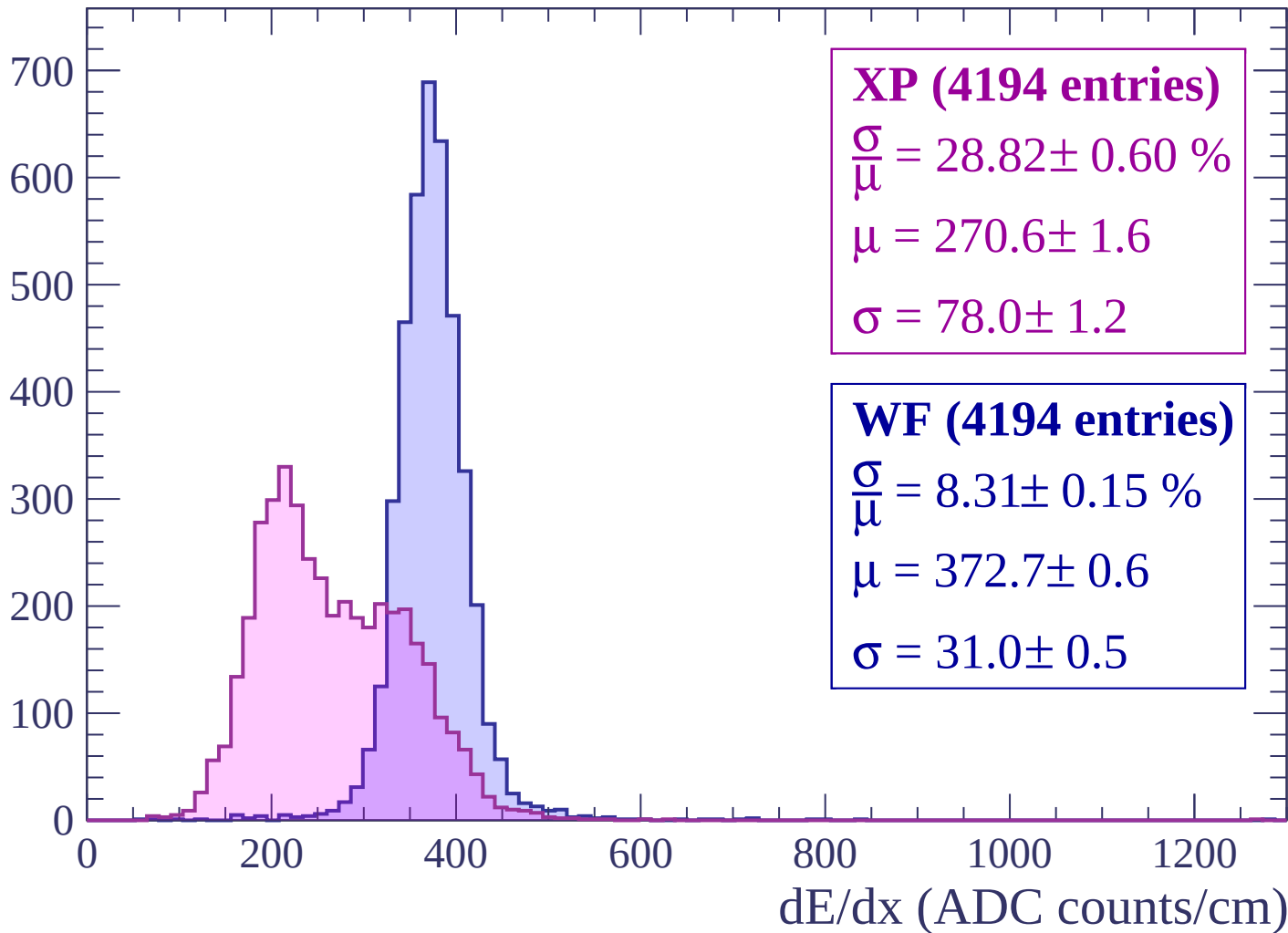
$$\mu = 371.1 \pm 0.6$$

$$\sigma = 29.8 \pm 0.6$$

dE/dx (ADC counts/cm)

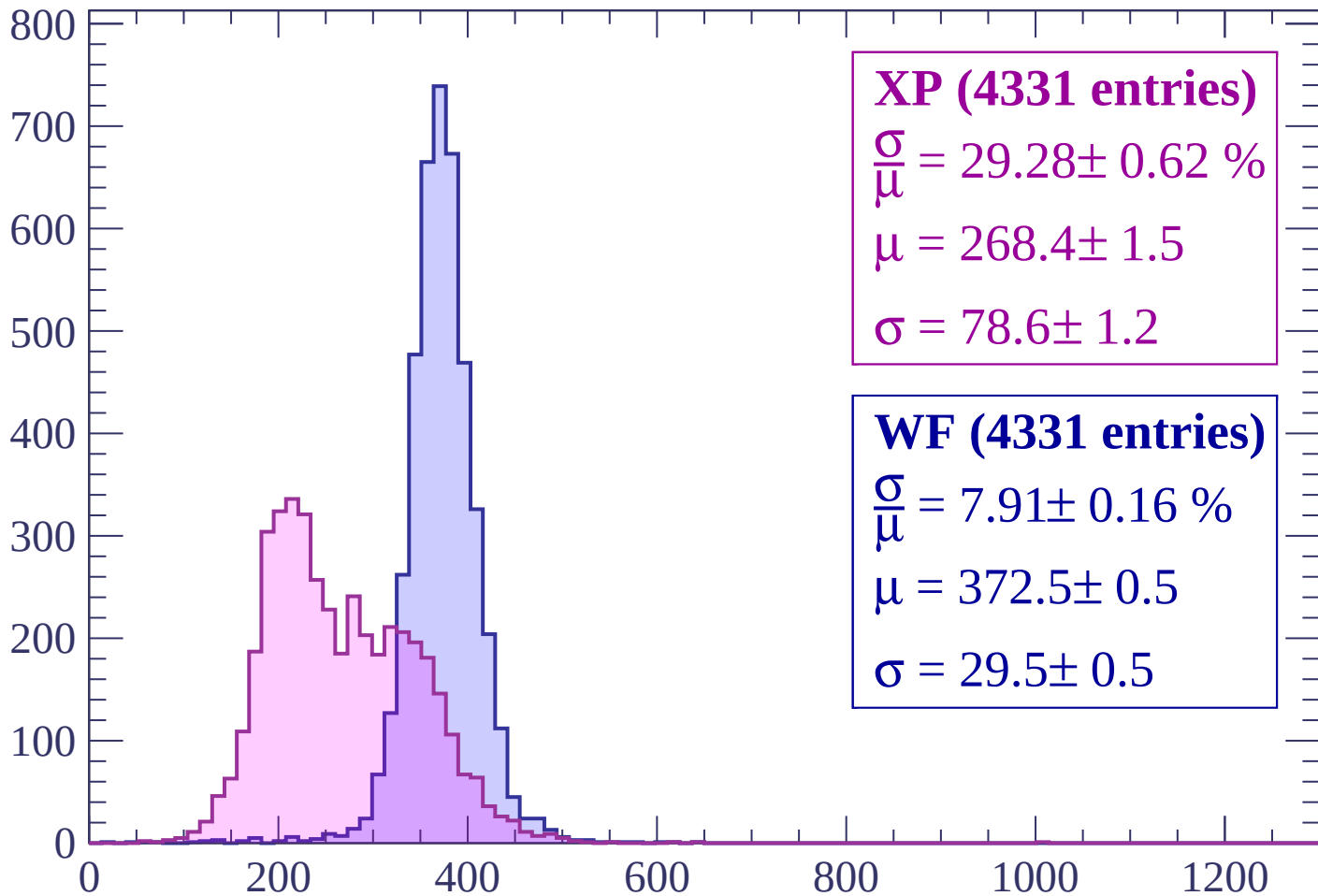
Energy loss | $420 < p < 480$

Count



Energy loss | $480 < p < 540$

Count



XP (4331 entries)

$$\frac{\sigma}{\mu} = 29.28 \pm 0.62 \%$$

$$\mu = 268.4 \pm 1.5$$

$$\sigma = 78.6 \pm 1.2$$

WF (4331 entries)

$$\frac{\sigma}{\mu} = 7.91 \pm 0.16 \%$$

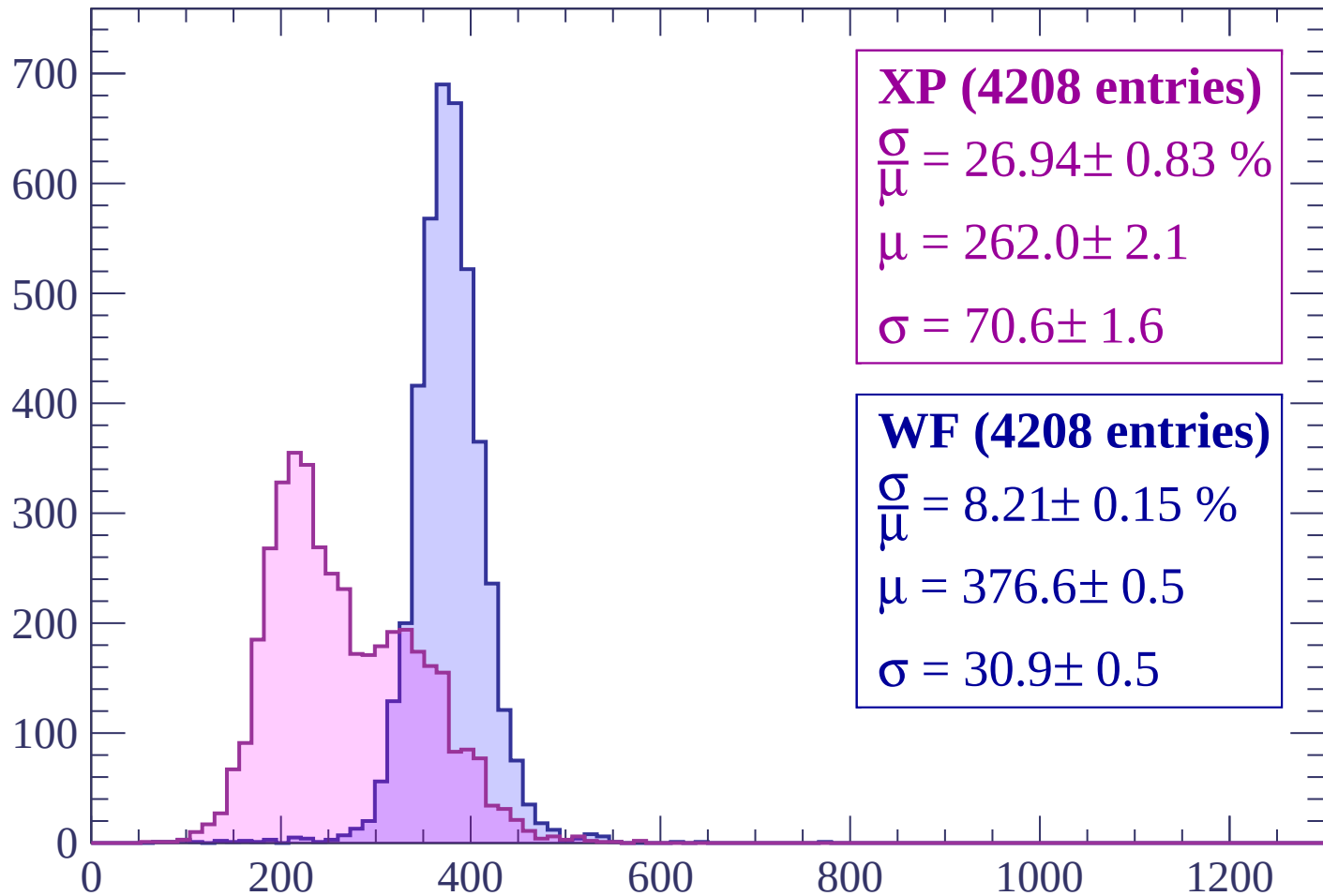
$$\mu = 372.5 \pm 0.5$$

$$\sigma = 29.5 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (4208 entries)

$$\frac{\sigma}{\mu} = 26.94 \pm 0.83 \%$$

$$\mu = 262.0 \pm 2.1$$

$$\sigma = 70.6 \pm 1.6$$

WF (4208 entries)

$$\frac{\sigma}{\mu} = 8.21 \pm 0.15 \%$$

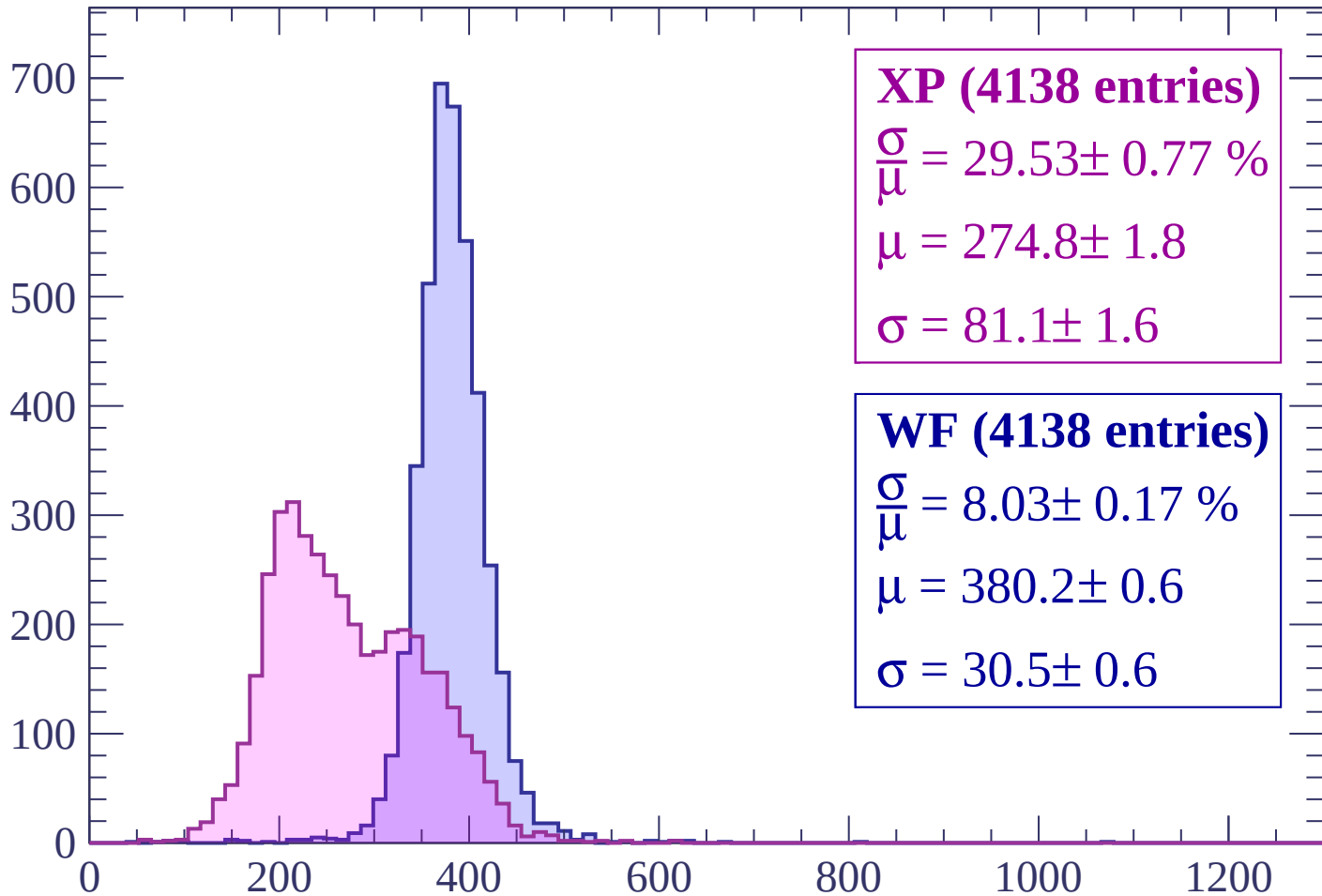
$$\mu = 376.6 \pm 0.5$$

$$\sigma = 30.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (4138 entries)

$$\frac{\sigma}{\mu} = 29.53 \pm 0.77 \%$$

$$\mu = 274.8 \pm 1.8$$

$$\sigma = 81.1 \pm 1.6$$

WF (4138 entries)

$$\frac{\sigma}{\mu} = 8.03 \pm 0.17 \%$$

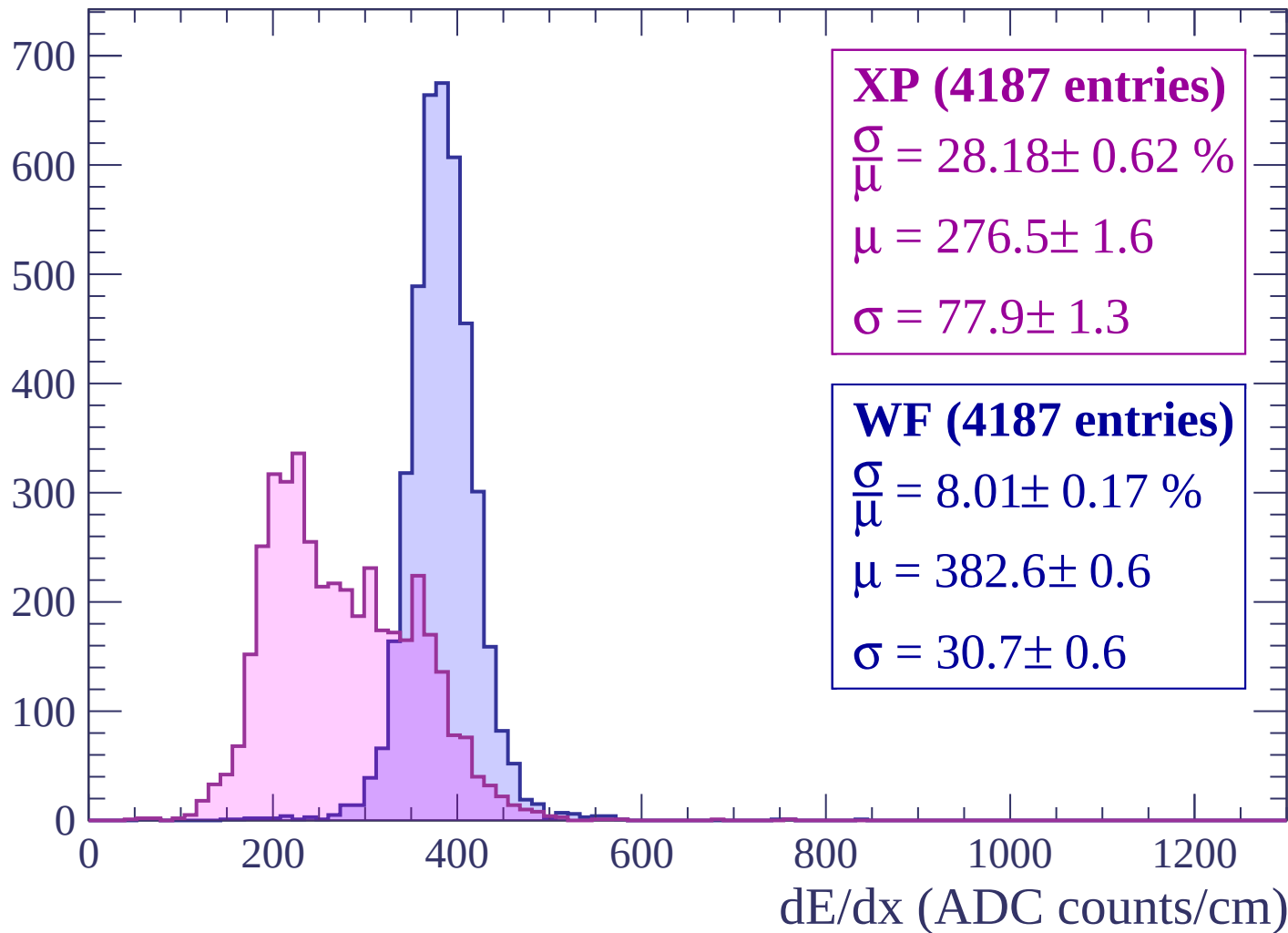
$$\mu = 380.2 \pm 0.6$$

$$\sigma = 30.5 \pm 0.6$$

dE/dx (ADC counts/cm)

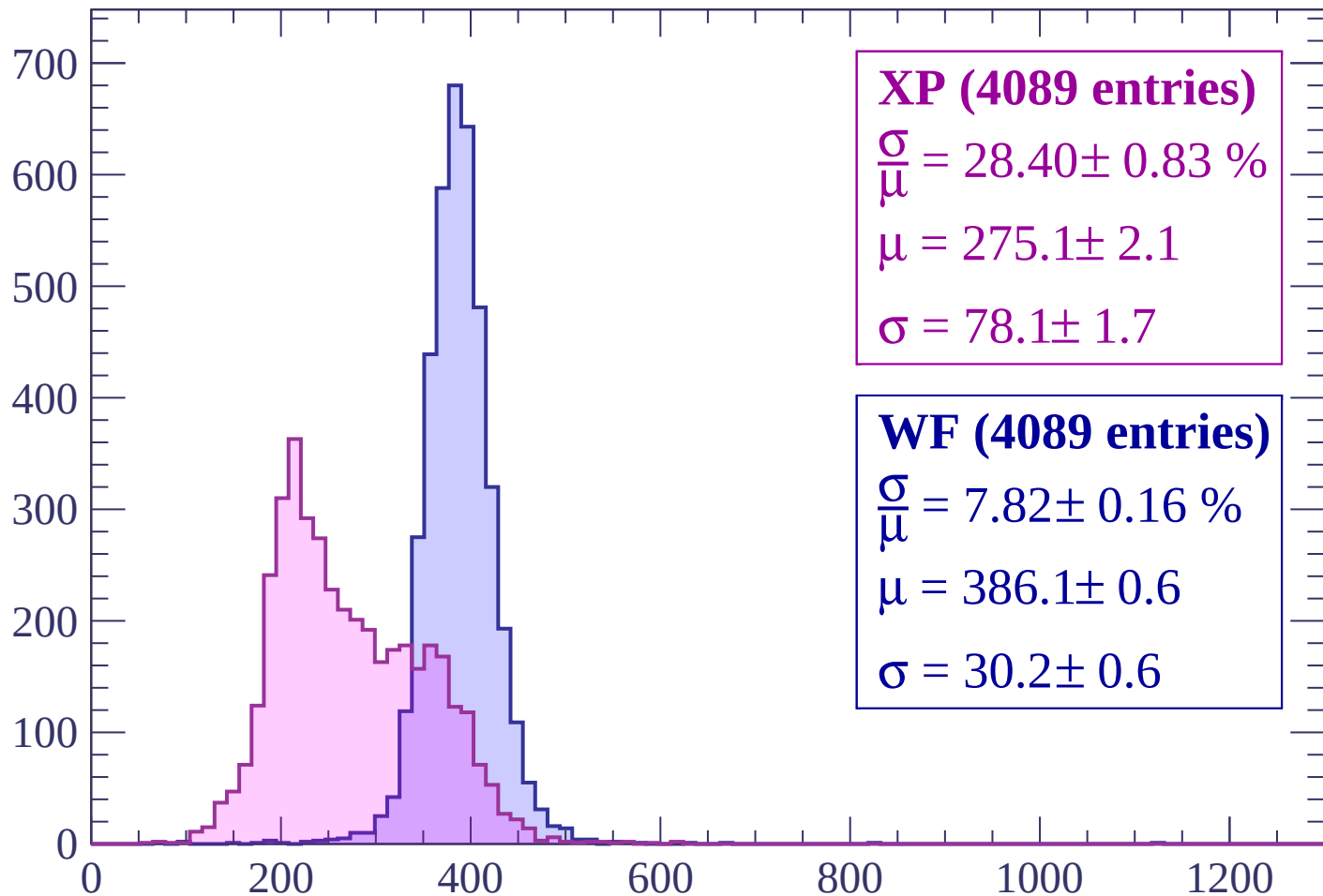
Energy loss | $660 < p < 720$

Count



Energy loss | $720 < p < 780$

Count



XP (4089 entries)

$$\frac{\sigma}{\mu} = 28.40 \pm 0.83 \%$$

$$\mu = 275.1 \pm 2.1$$

$$\sigma = 78.1 \pm 1.7$$

WF (4089 entries)

$$\frac{\sigma}{\mu} = 7.82 \pm 0.16 \%$$

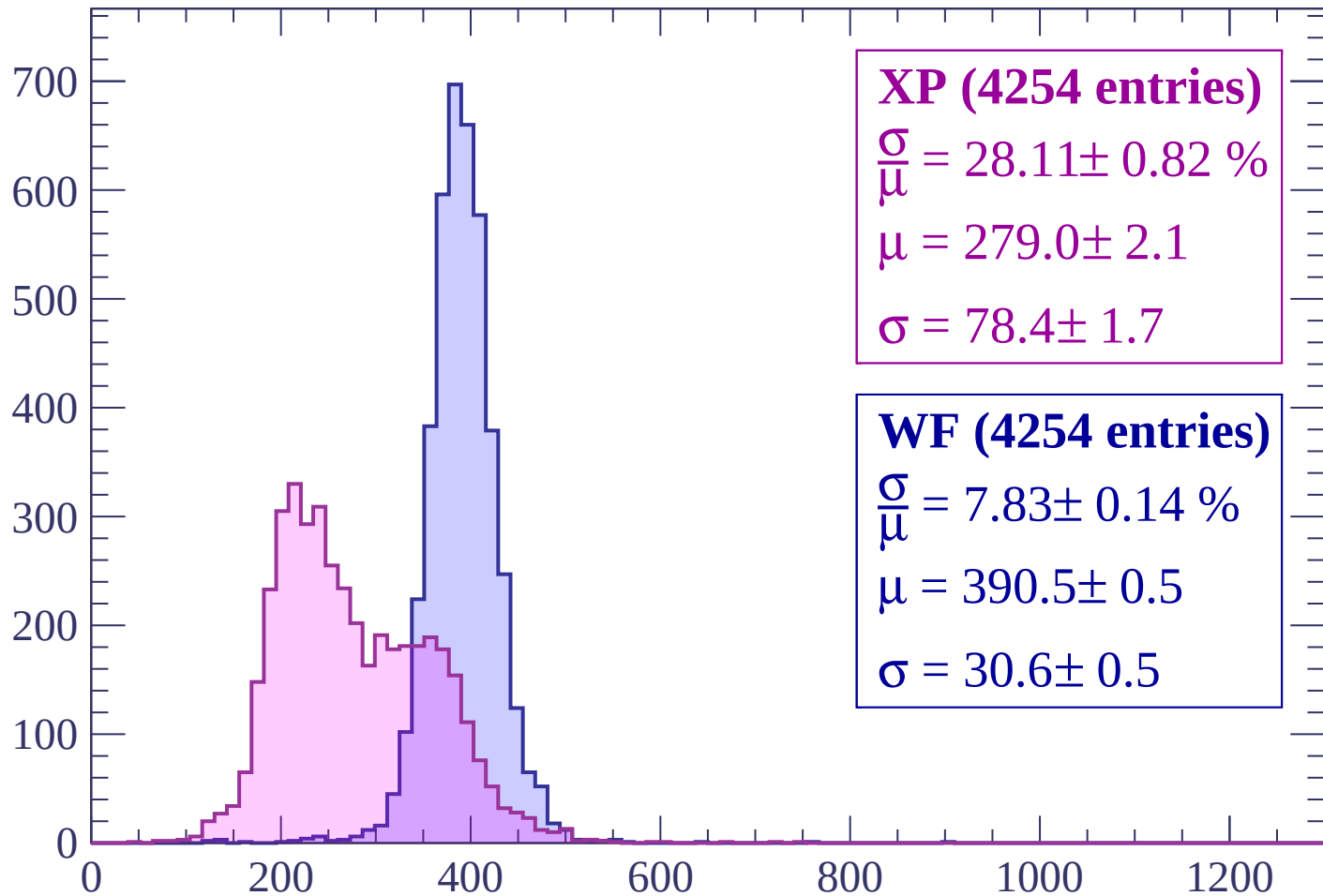
$$\mu = 386.1 \pm 0.6$$

$$\sigma = 30.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (4254 entries)

$\frac{\sigma}{\mu} = 28.11 \pm 0.82 \%$

$\mu = 279.0 \pm 2.1$

$\sigma = 78.4 \pm 1.7$

WF (4254 entries)

$\frac{\sigma}{\mu} = 7.83 \pm 0.14 \%$

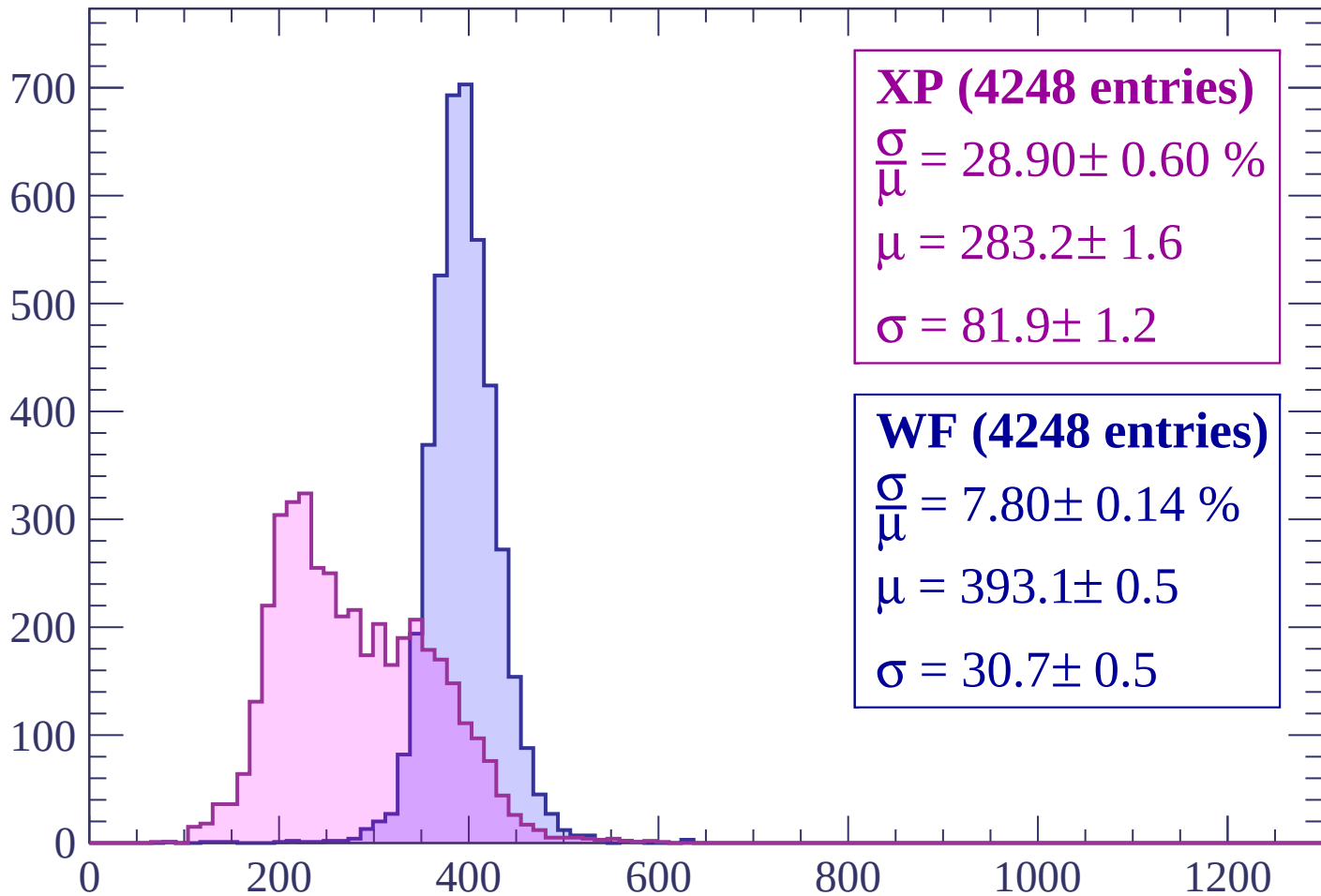
$\mu = 390.5 \pm 0.5$

$\sigma = 30.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (4248 entries)

$$\frac{\sigma}{\mu} = 28.90 \pm 0.60 \%$$

$$\mu = 283.2 \pm 1.6$$

$$\sigma = 81.9 \pm 1.2$$

WF (4248 entries)

$$\frac{\sigma}{\mu} = 7.80 \pm 0.14 \%$$

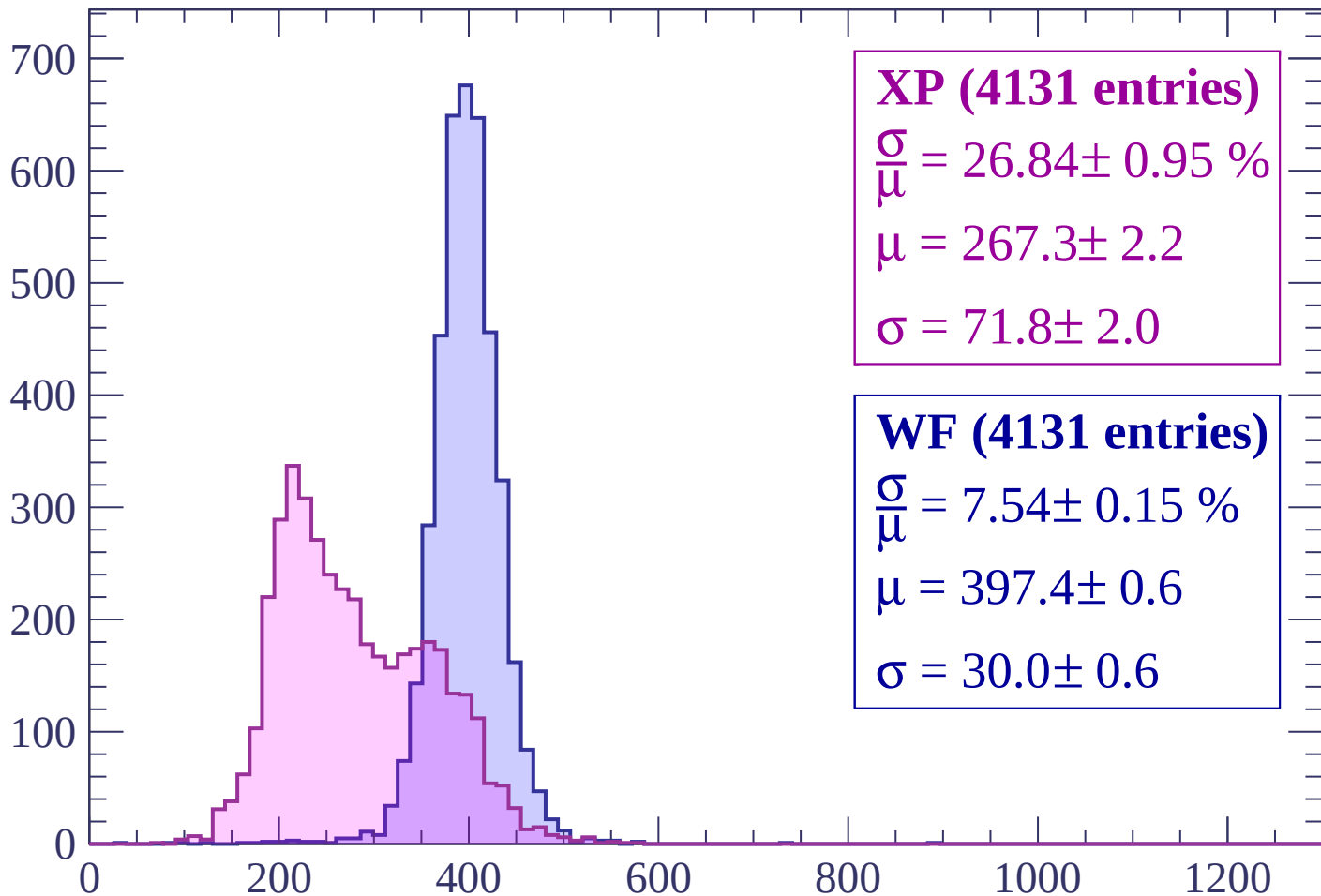
$$\mu = 393.1 \pm 0.5$$

$$\sigma = 30.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (4131 entries)

$$\frac{\sigma}{\mu} = 26.84 \pm 0.95 \%$$

$$\mu = 267.3 \pm 2.2$$

$$\sigma = 71.8 \pm 2.0$$

WF (4131 entries)

$$\frac{\sigma}{\mu} = 7.54 \pm 0.15 \%$$

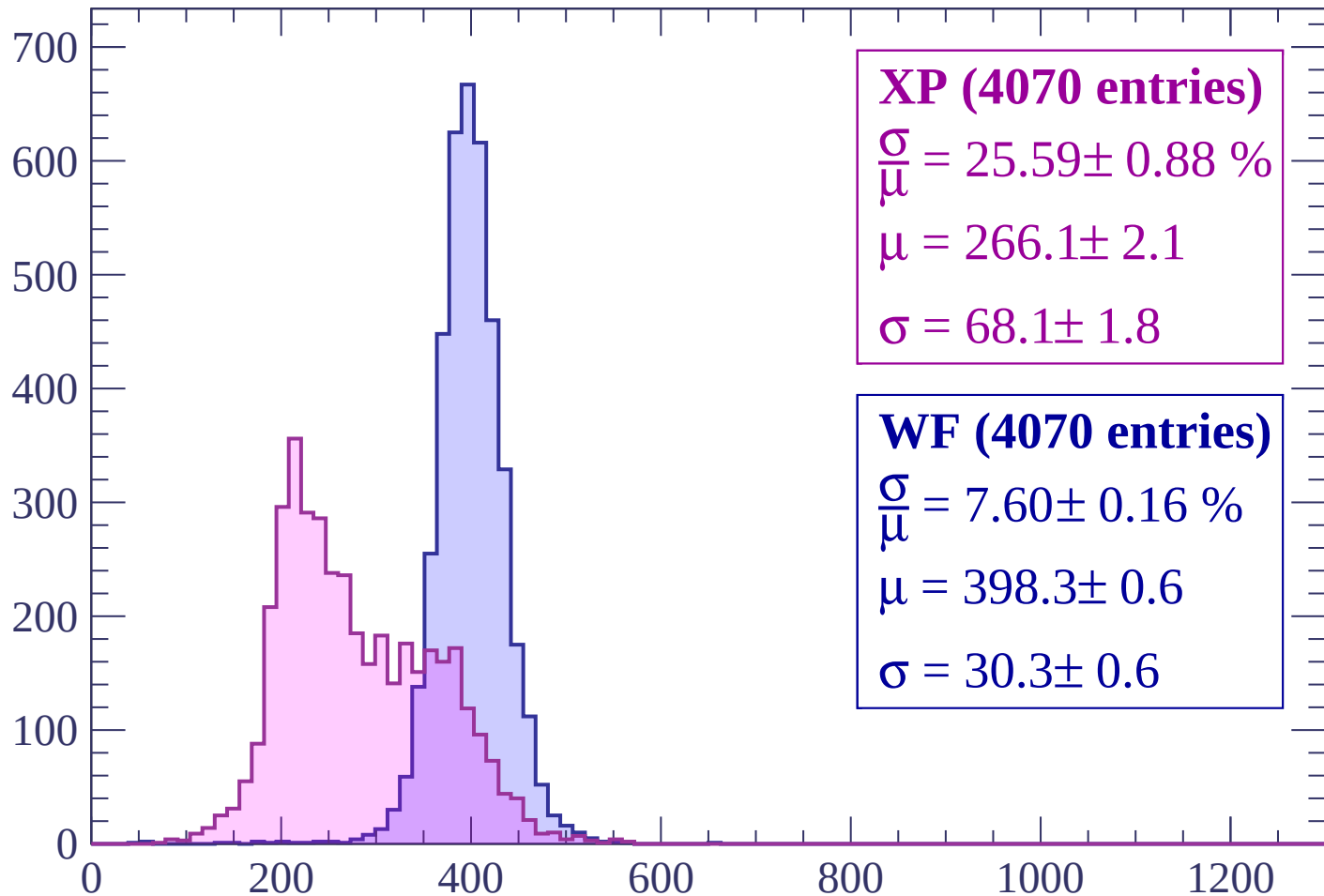
$$\mu = 397.4 \pm 0.6$$

$$\sigma = 30.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (4070 entries)

$$\frac{\sigma}{\mu} = 25.59 \pm 0.88 \%$$

$$\mu = 266.1 \pm 2.1$$

$$\sigma = 68.1 \pm 1.8$$

WF (4070 entries)

$$\frac{\sigma}{\mu} = 7.60 \pm 0.16 \%$$

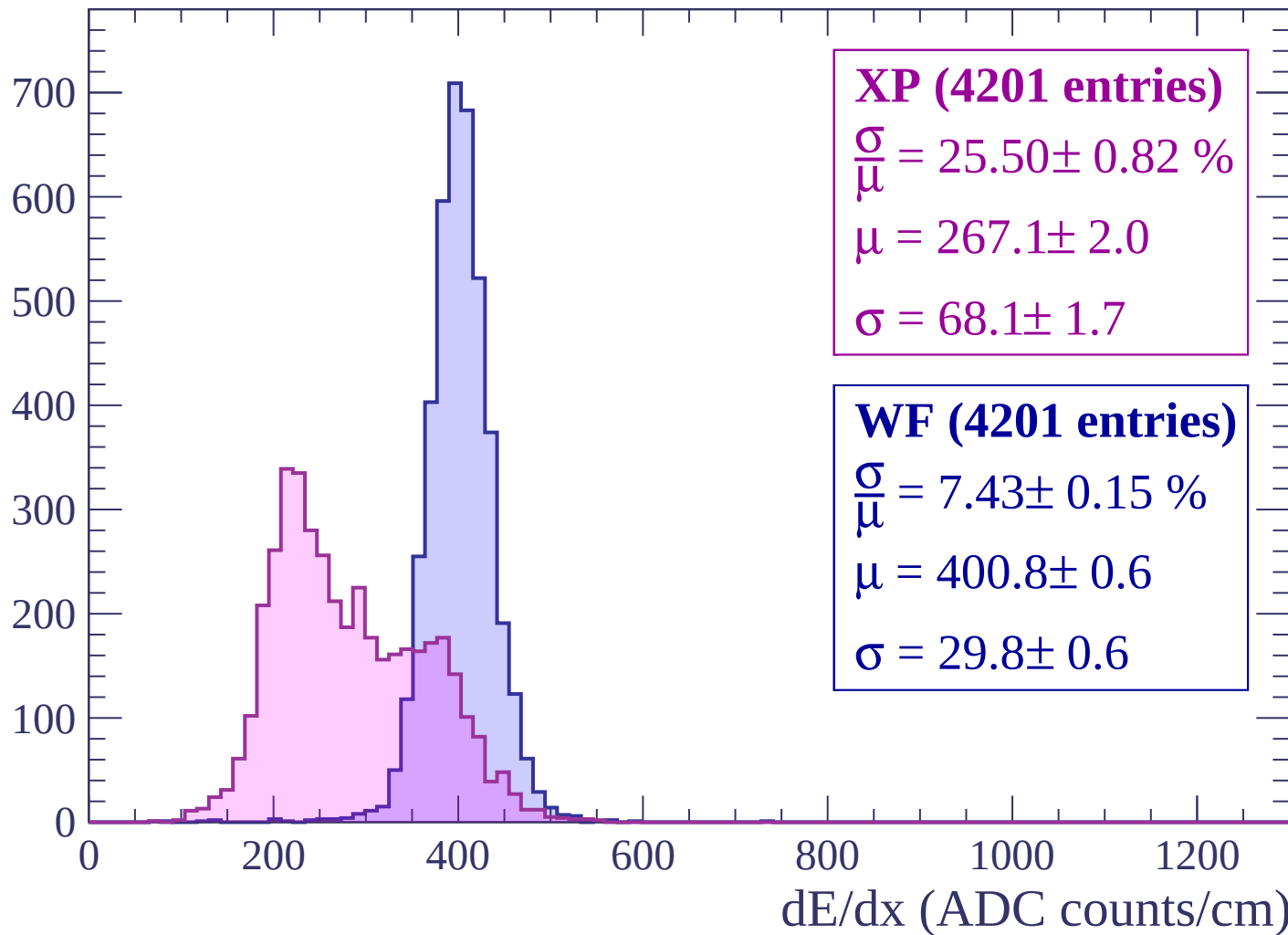
$$\mu = 398.3 \pm 0.6$$

$$\sigma = 30.3 \pm 0.6$$

dE/dx (ADC counts/cm)

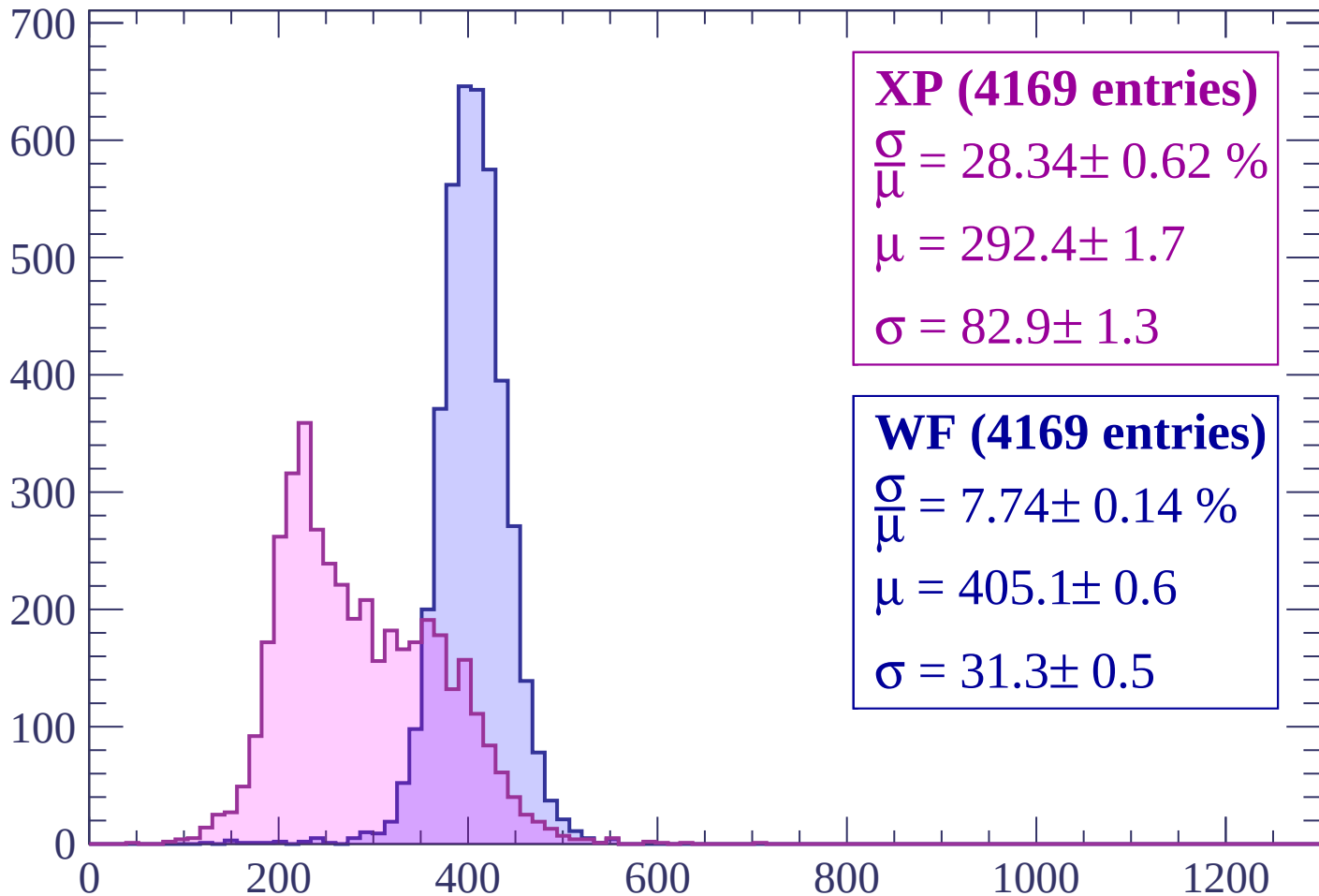
Energy loss | $1020 < p < 1080$

Count



Energy loss | $1080 < p < 1140$

Count



XP (4169 entries)

$$\frac{\sigma}{\mu} = 28.34 \pm 0.62 \%$$

$$\mu = 292.4 \pm 1.7$$

$$\sigma = 82.9 \pm 1.3$$

WF (4169 entries)

$$\frac{\sigma}{\mu} = 7.74 \pm 0.14 \%$$

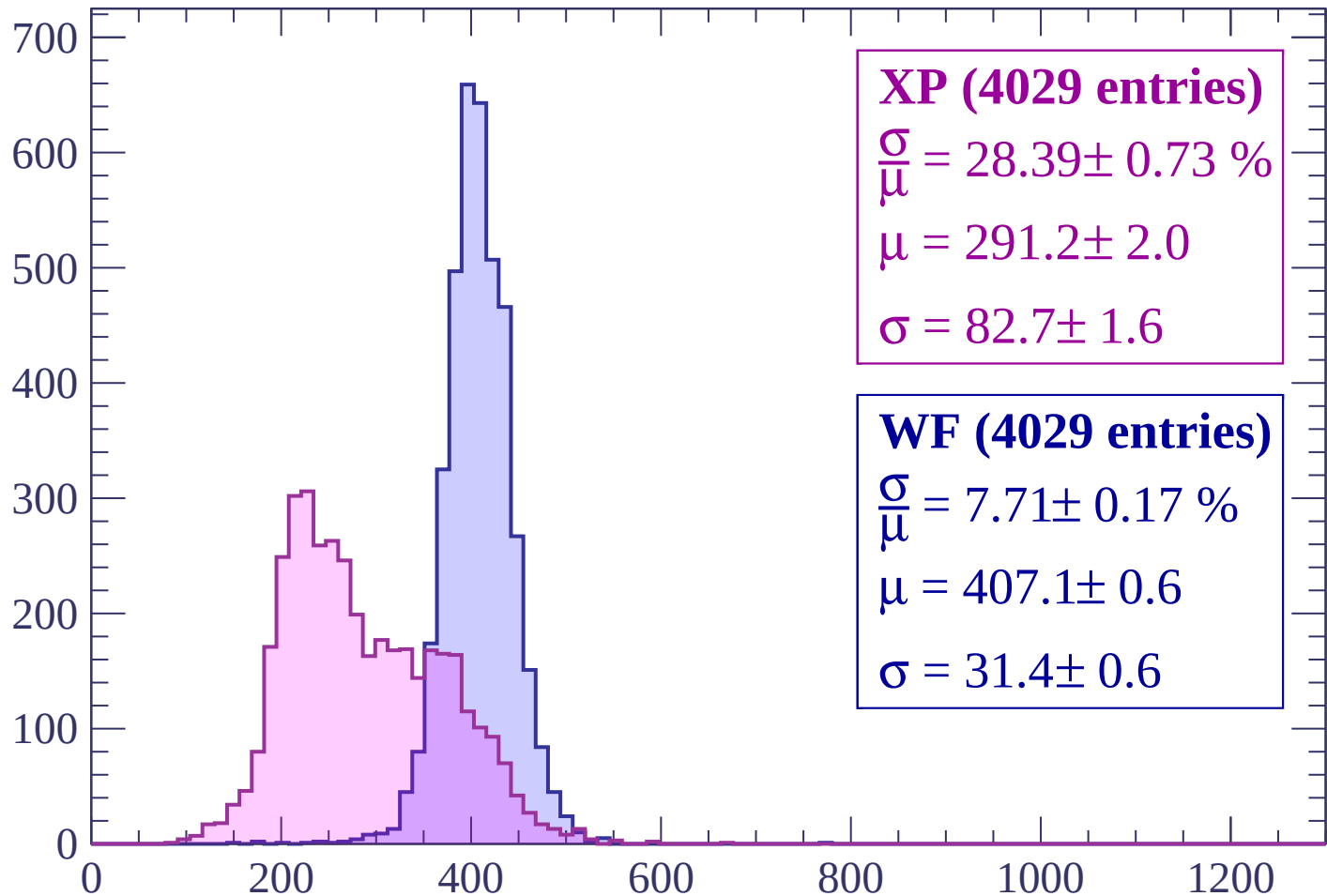
$$\mu = 405.1 \pm 0.6$$

$$\sigma = 31.3 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (4029 entries)

$$\frac{\sigma}{\mu} = 28.39 \pm 0.73 \%$$

$$\mu = 291.2 \pm 2.0$$

$$\sigma = 82.7 \pm 1.6$$

WF (4029 entries)

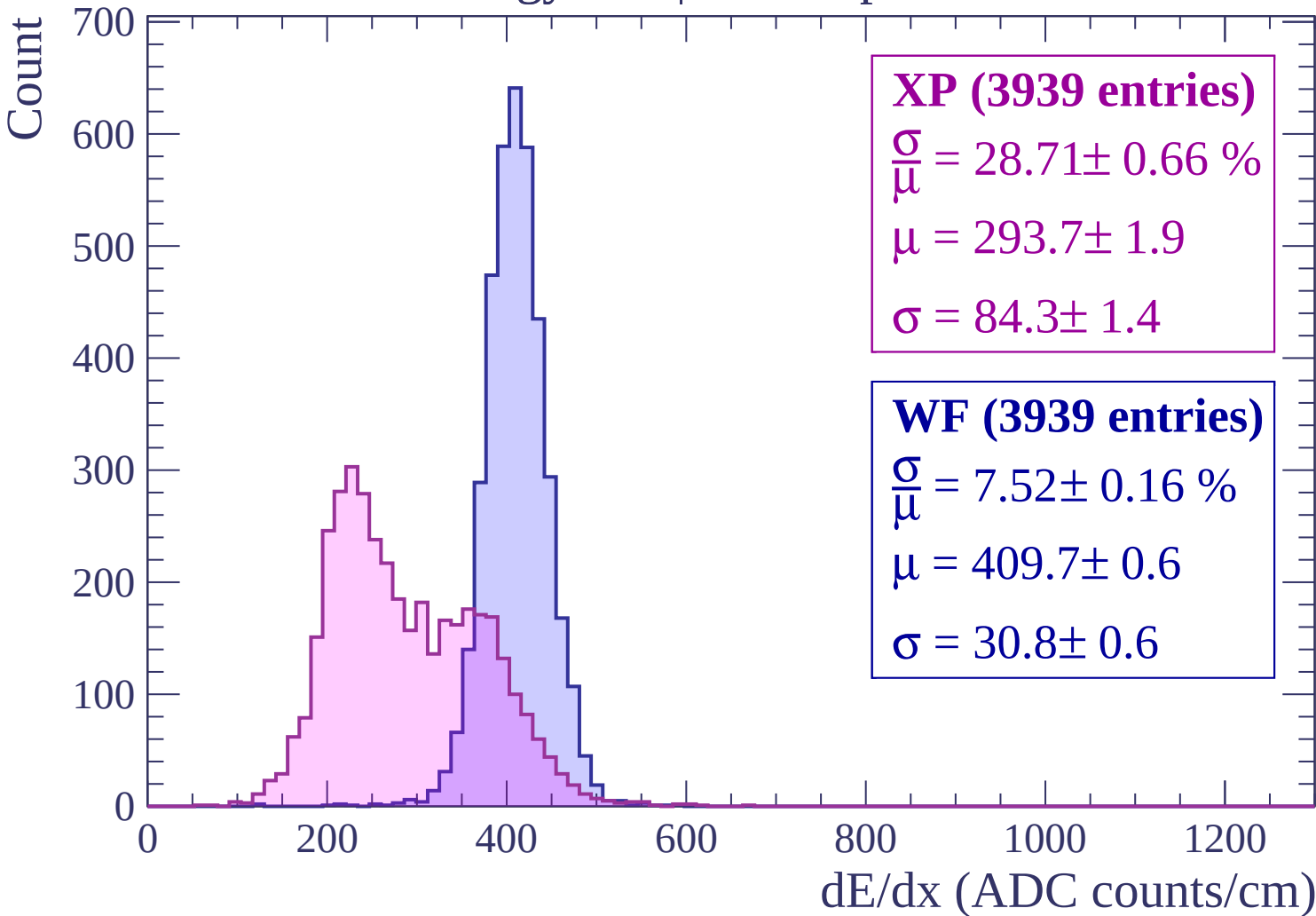
$$\frac{\sigma}{\mu} = 7.71 \pm 0.17 \%$$

$$\mu = 407.1 \pm 0.6$$

$$\sigma = 31.4 \pm 0.6$$

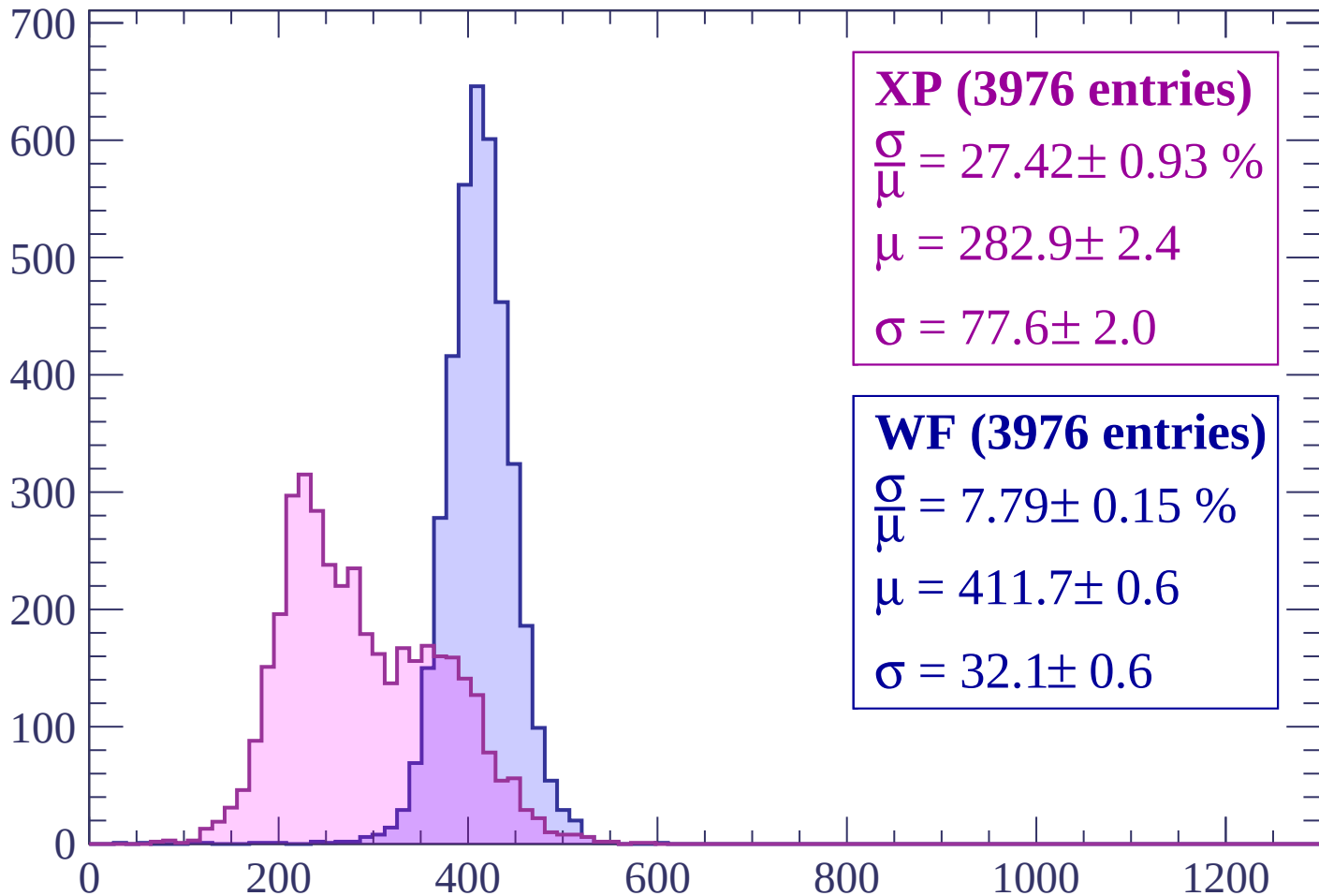
dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$



Energy loss | $1260 < p < 1320$

Count



XP (3976 entries)

$$\frac{\sigma}{\mu} = 27.42 \pm 0.93 \%$$

$$\mu = 282.9 \pm 2.4$$

$$\sigma = 77.6 \pm 2.0$$

WF (3976 entries)

$$\frac{\sigma}{\mu} = 7.79 \pm 0.15 \%$$

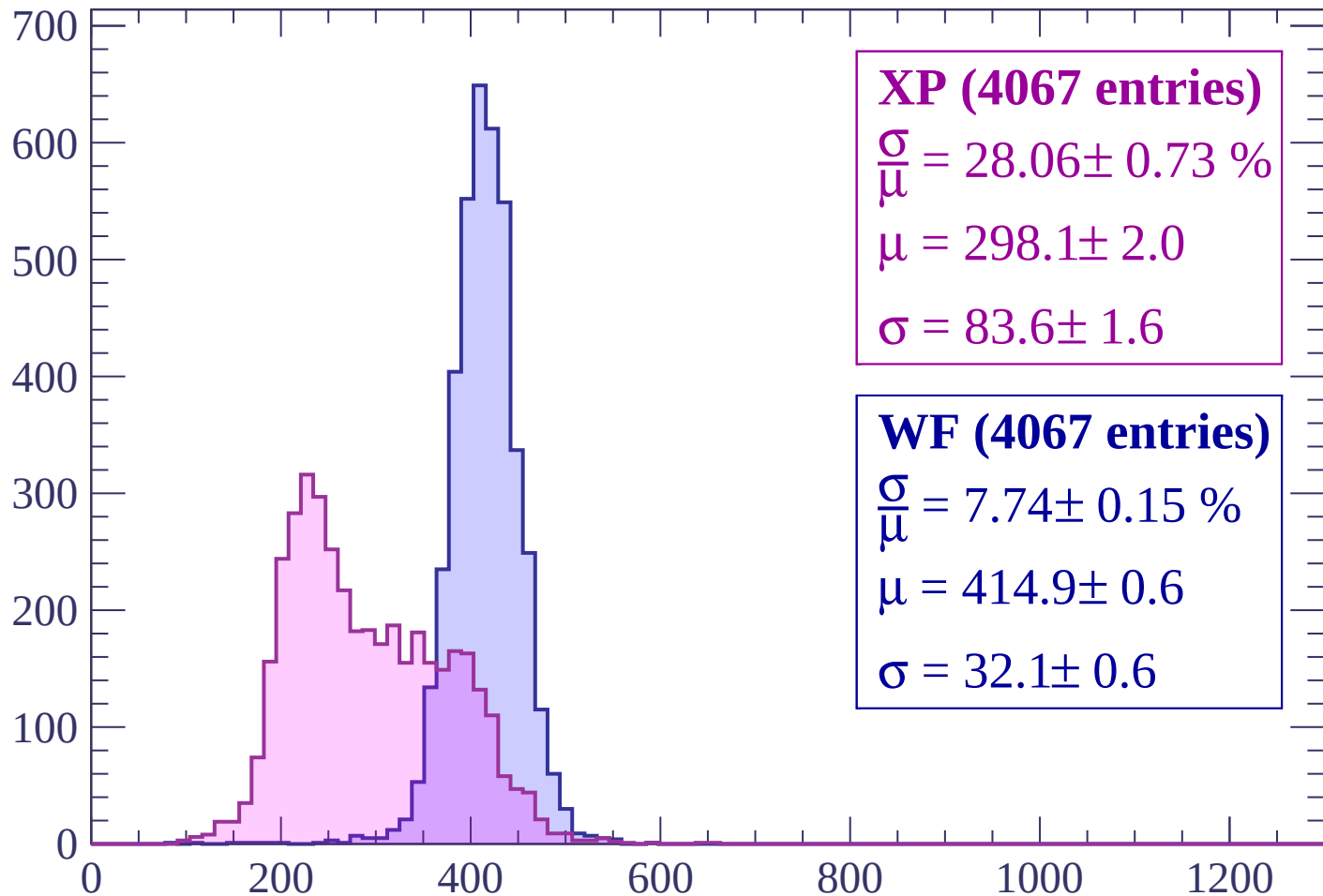
$$\mu = 411.7 \pm 0.6$$

$$\sigma = 32.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (4067 entries)

$$\frac{\sigma}{\mu} = 28.06 \pm 0.73 \%$$

$$\mu = 298.1 \pm 2.0$$

$$\sigma = 83.6 \pm 1.6$$

WF (4067 entries)

$$\frac{\sigma}{\mu} = 7.74 \pm 0.15 \%$$

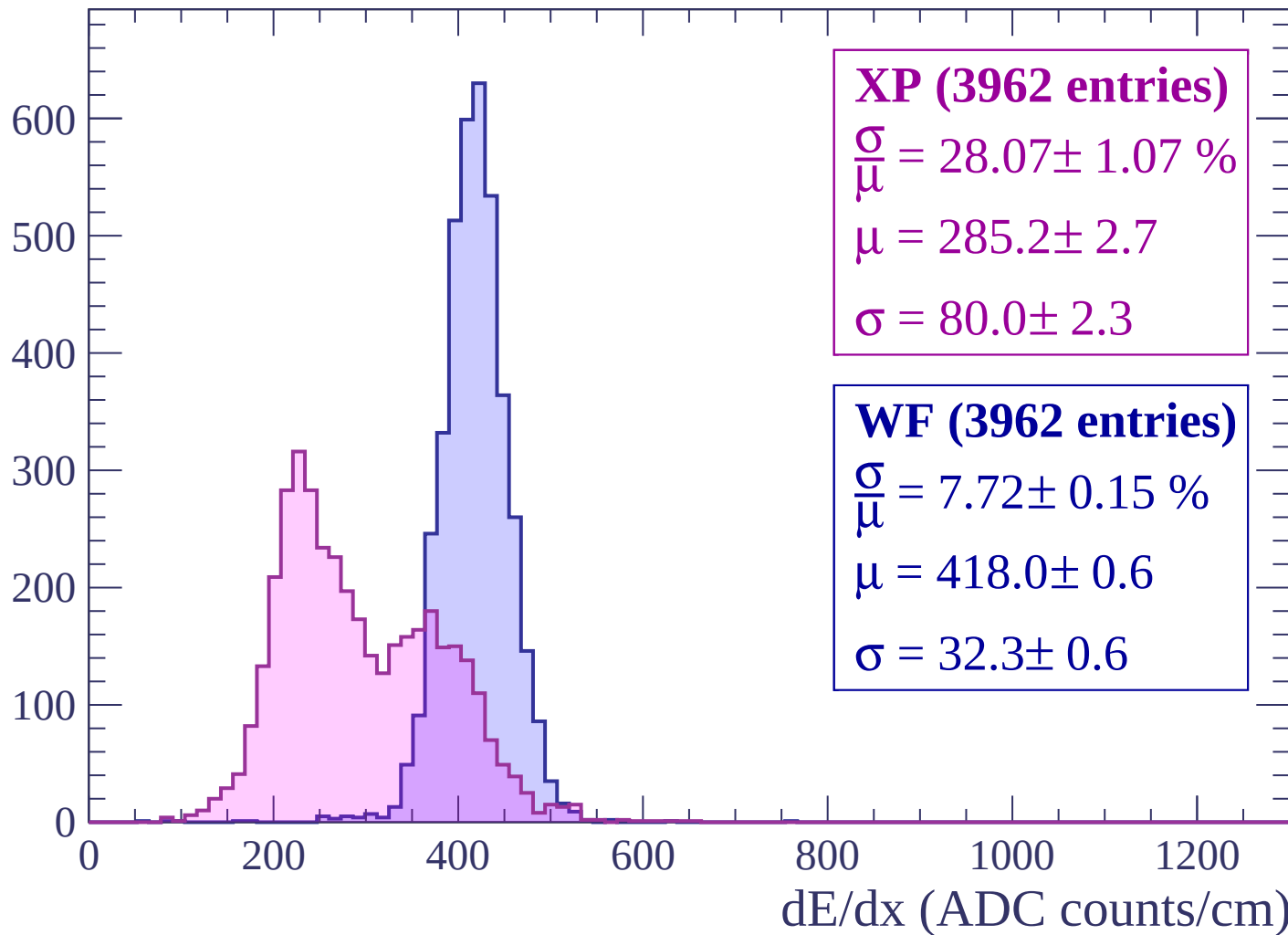
$$\mu = 414.9 \pm 0.6$$

$$\sigma = 32.1 \pm 0.6$$

dE/dx (ADC counts/cm)

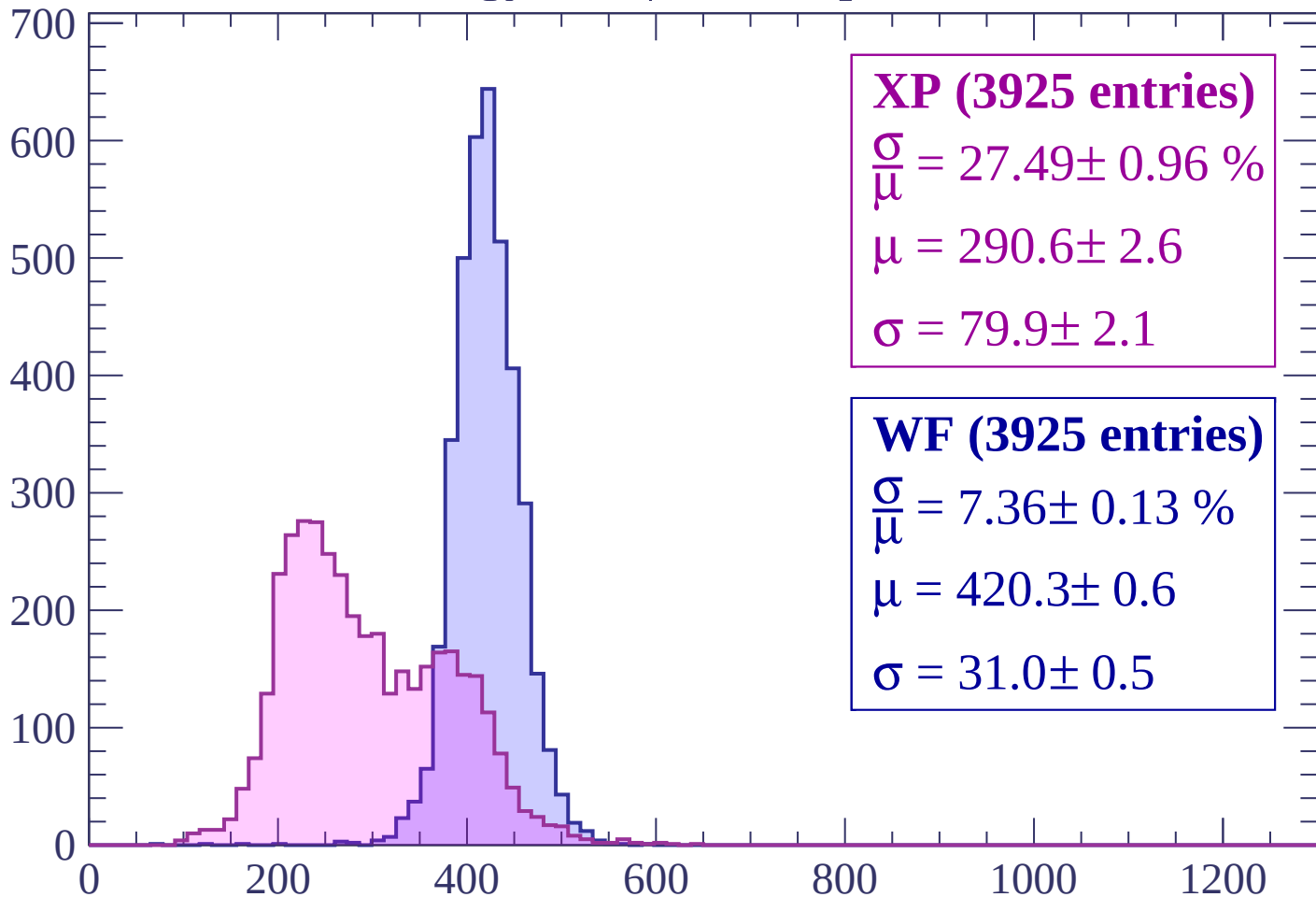
Energy loss | $1380 < p < 1440$

Count



Energy loss | $1440 < p < 1500$

Count



XP (3925 entries)

$$\frac{\sigma}{\mu} = 27.49 \pm 0.96 \%$$

$$\mu = 290.6 \pm 2.6$$

$$\sigma = 79.9 \pm 2.1$$

WF (3925 entries)

$$\frac{\sigma}{\mu} = 7.36 \pm 0.13 \%$$

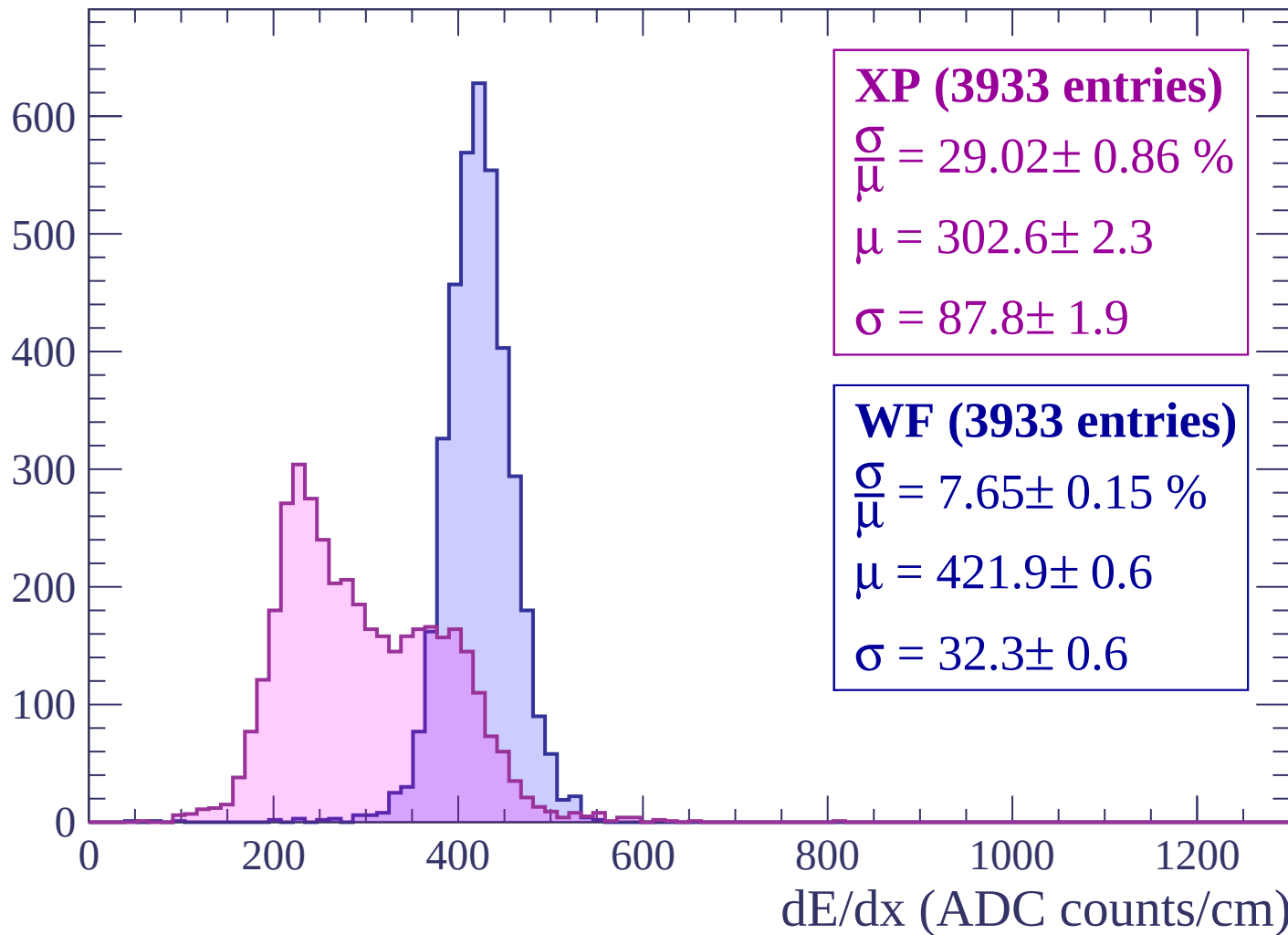
$$\mu = 420.3 \pm 0.6$$

$$\sigma = 31.0 \pm 0.5$$

dE/dx (ADC counts/cm)

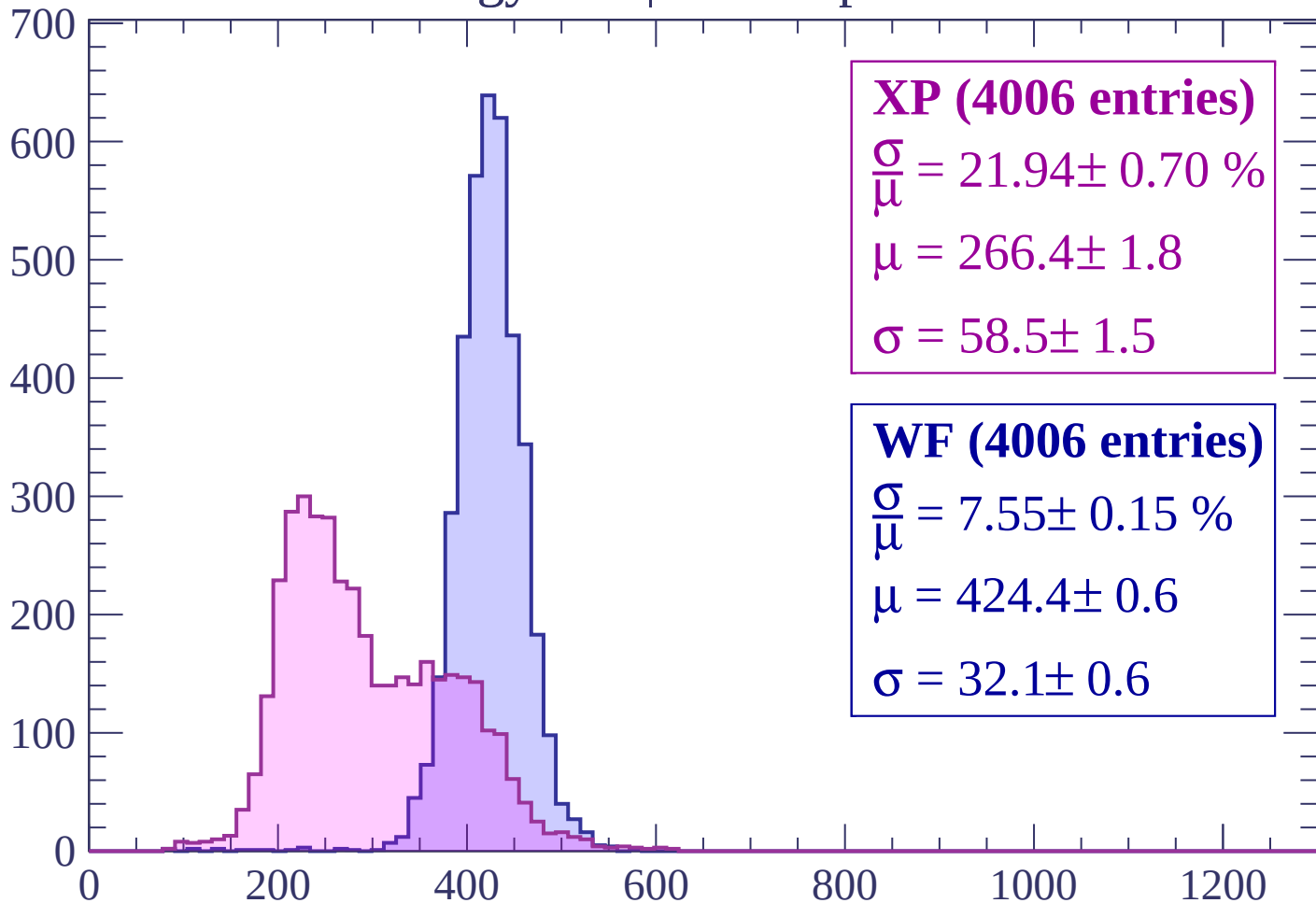
Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count



XP (4006 entries)

$$\frac{\sigma}{\mu} = 21.94 \pm 0.70 \%$$

$$\mu = 266.4 \pm 1.8$$

$$\sigma = 58.5 \pm 1.5$$

WF (4006 entries)

$$\frac{\sigma}{\mu} = 7.55 \pm 0.15 \%$$

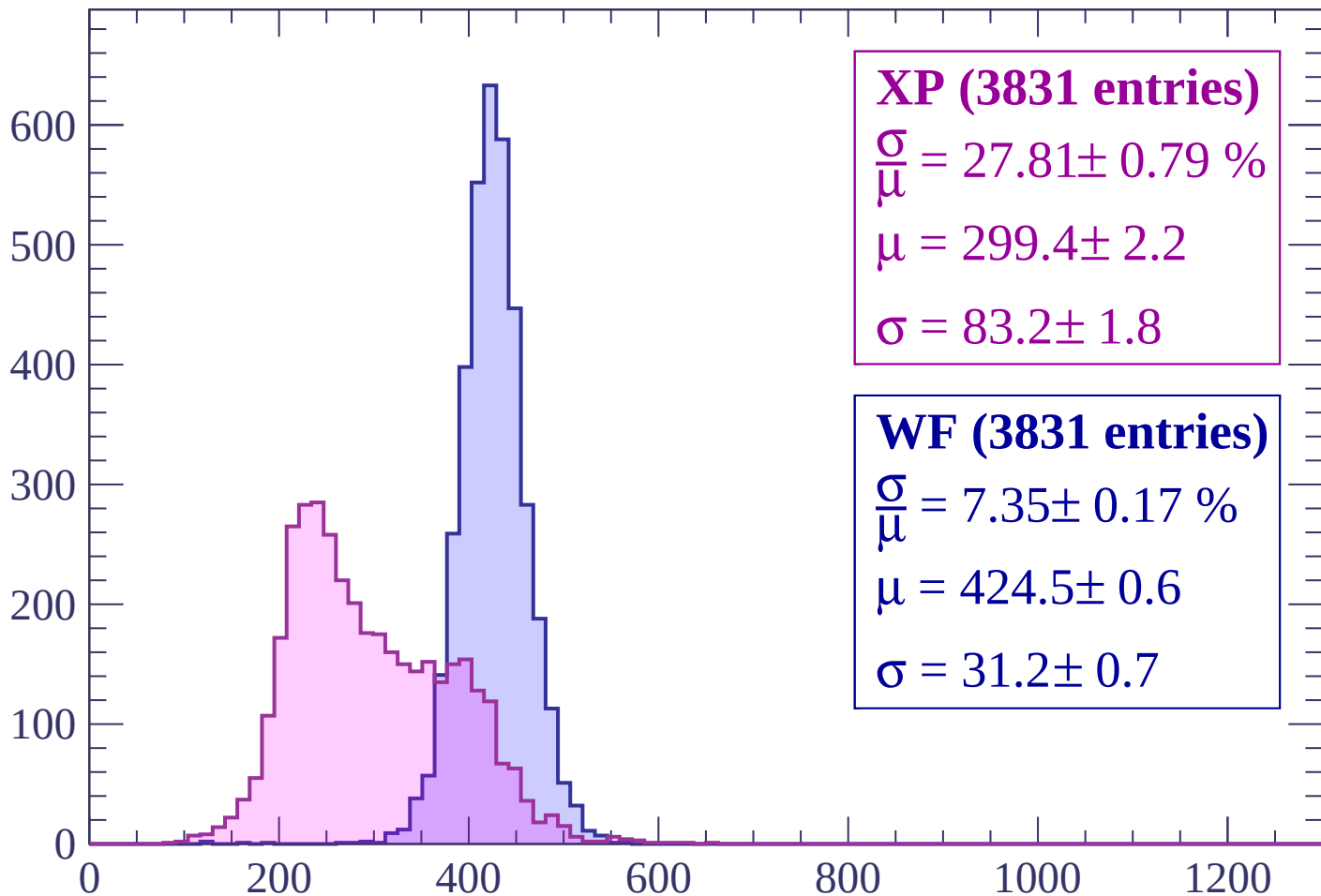
$$\mu = 424.4 \pm 0.6$$

$$\sigma = 32.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (3831 entries)

$\frac{\sigma}{\mu} = 27.81 \pm 0.79 \%$

$\mu = 299.4 \pm 2.2$

$\sigma = 83.2 \pm 1.8$

WF (3831 entries)

$\frac{\sigma}{\mu} = 7.35 \pm 0.17 \%$

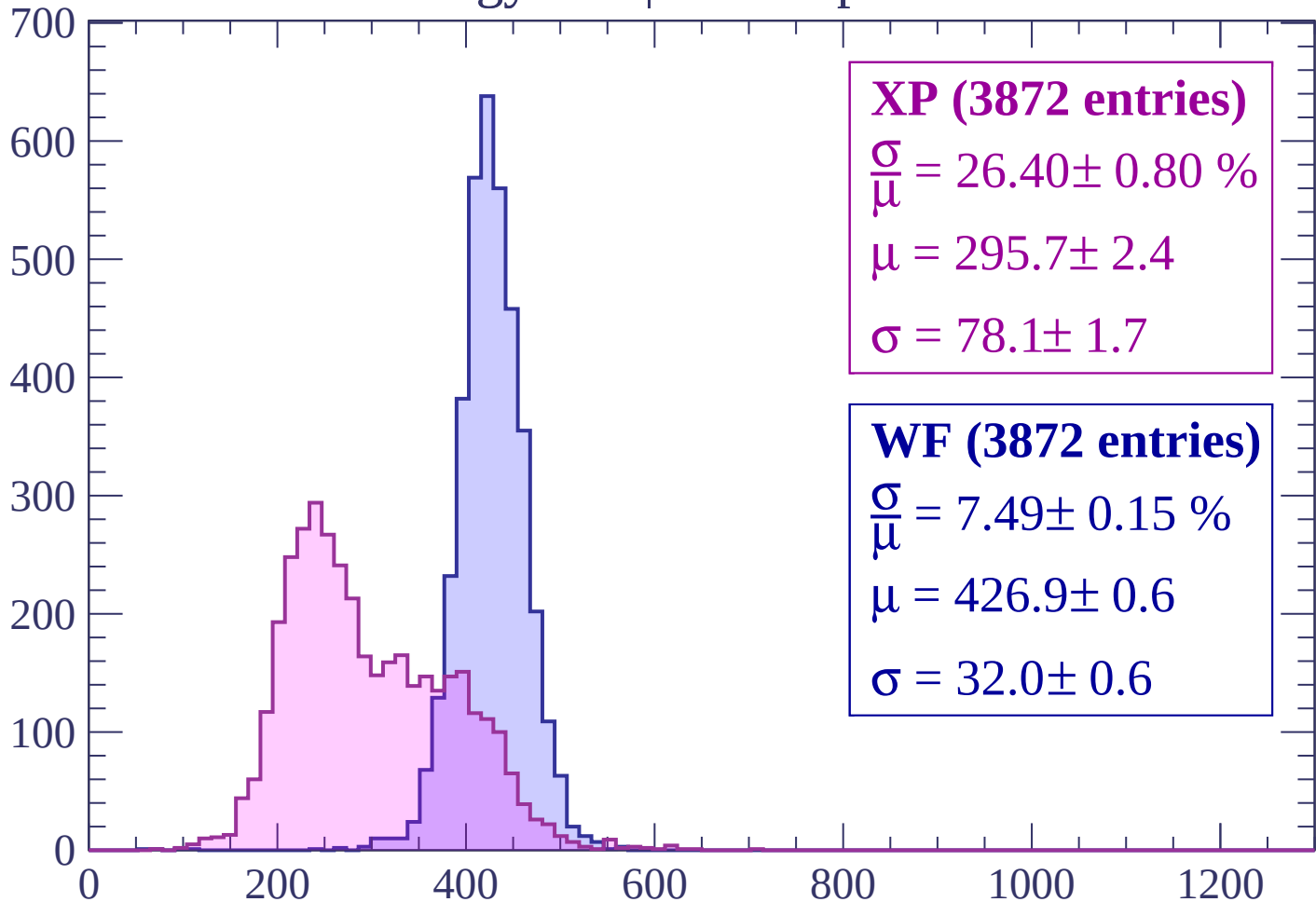
$\mu = 424.5 \pm 0.6$

$\sigma = 31.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (3872 entries)

$$\frac{\sigma}{\mu} = 26.40 \pm 0.80 \%$$

$$\mu = 295.7 \pm 2.4$$

$$\sigma = 78.1 \pm 1.7$$

WF (3872 entries)

$$\frac{\sigma}{\mu} = 7.49 \pm 0.15 \%$$

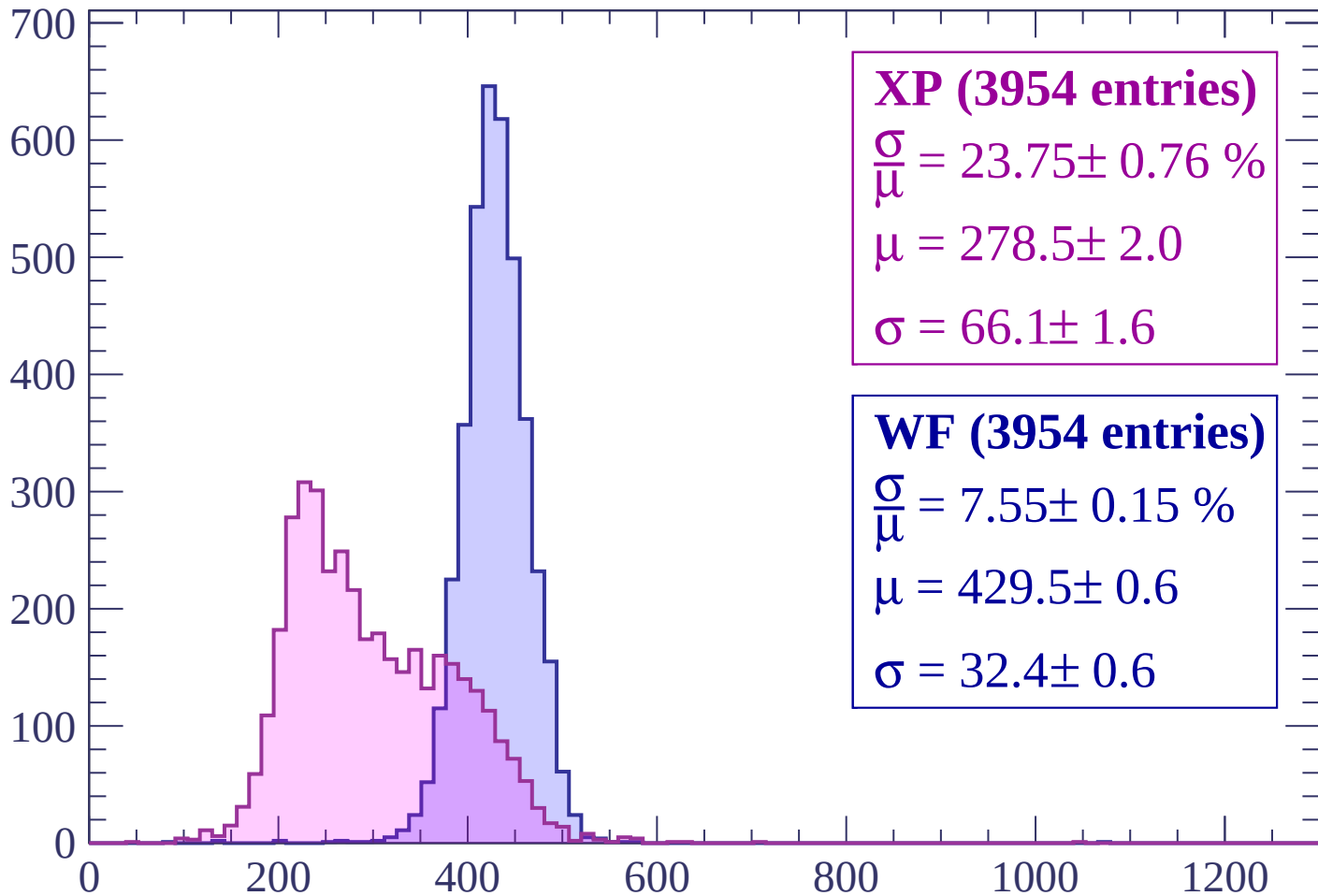
$$\mu = 426.9 \pm 0.6$$

$$\sigma = 32.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (3954 entries)

$$\frac{\sigma}{\mu} = 23.75 \pm 0.76 \%$$

$$\mu = 278.5 \pm 2.0$$

$$\sigma = 66.1 \pm 1.6$$

WF (3954 entries)

$$\frac{\sigma}{\mu} = 7.55 \pm 0.15 \%$$

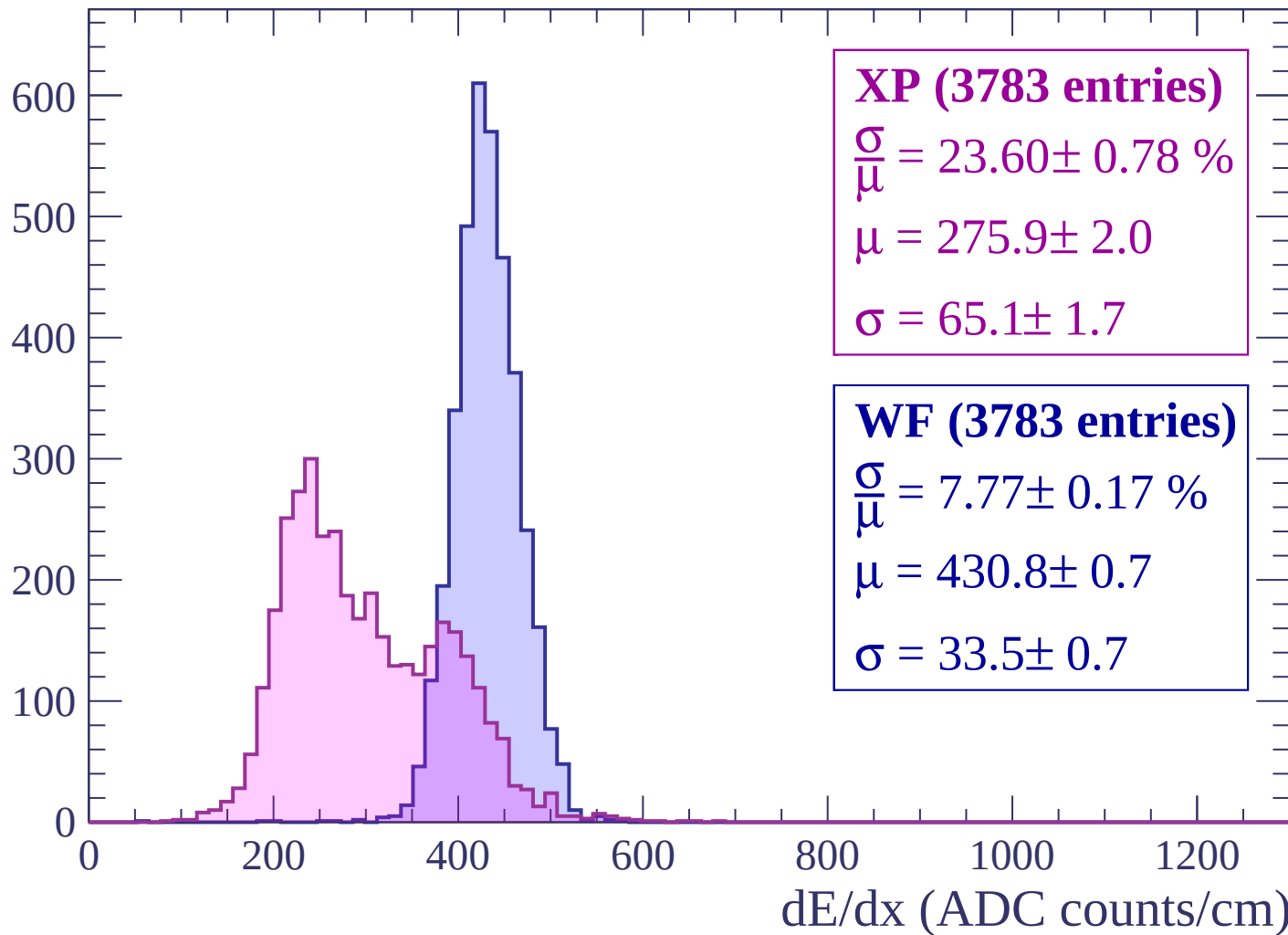
$$\mu = 429.5 \pm 0.6$$

$$\sigma = 32.4 \pm 0.6$$

dE/dx (ADC counts/cm)

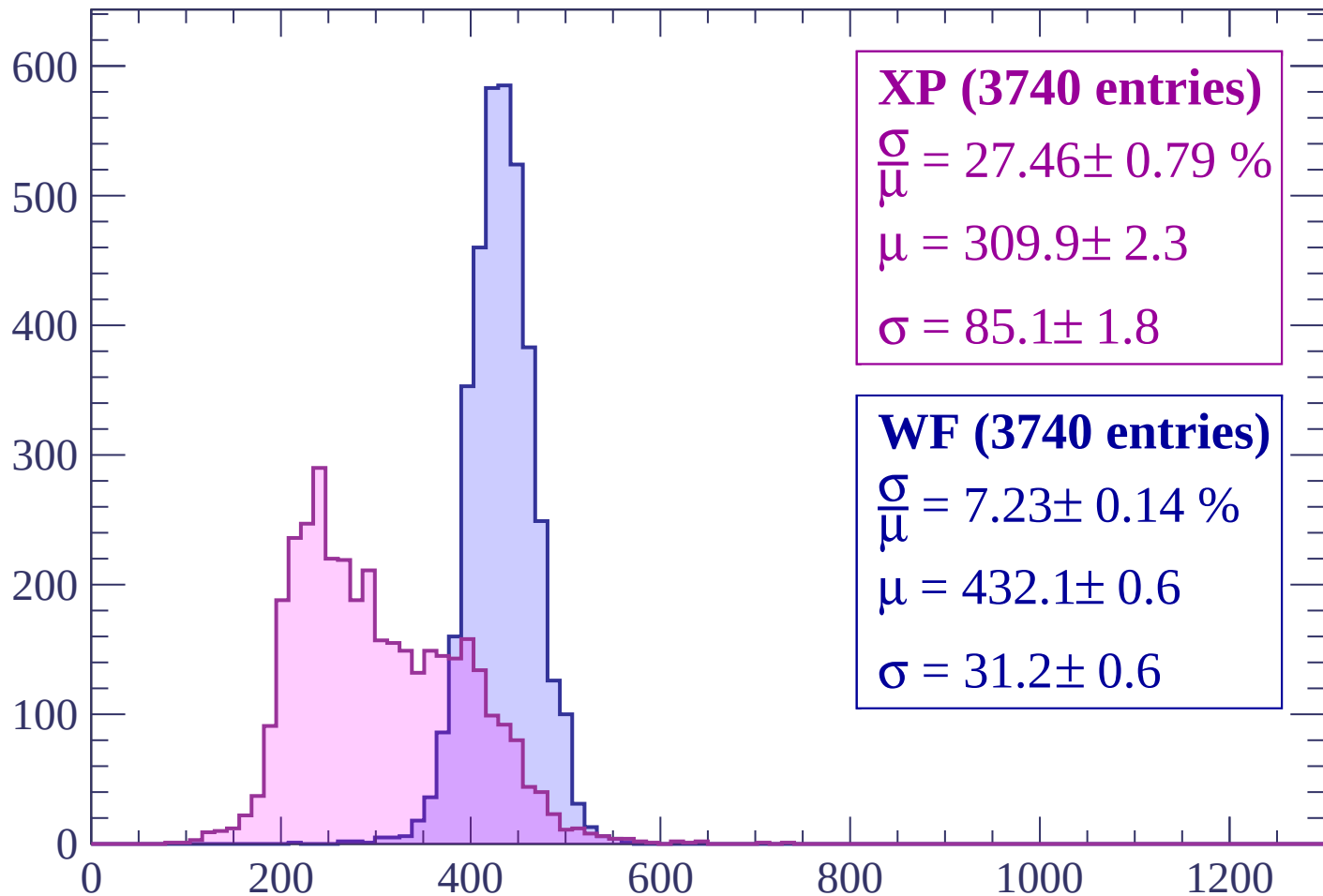
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (3740 entries)

$$\frac{\sigma}{\mu} = 27.46 \pm 0.79 \%$$

$$\mu = 309.9 \pm 2.3$$

$$\sigma = 85.1 \pm 1.8$$

WF (3740 entries)

$$\frac{\sigma}{\mu} = 7.23 \pm 0.14 \%$$

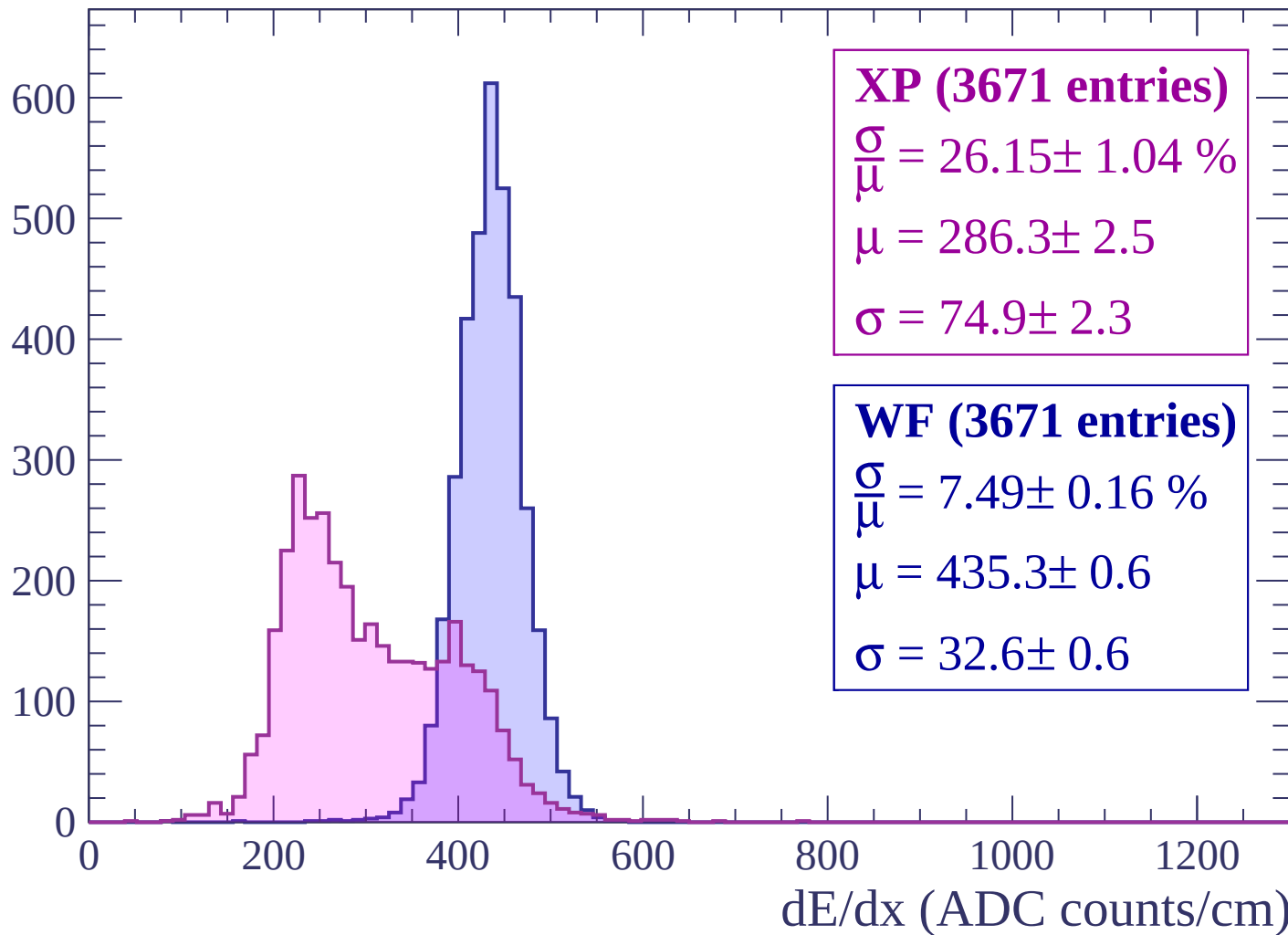
$$\mu = 432.1 \pm 0.6$$

$$\sigma = 31.2 \pm 0.6$$

dE/dx (ADC counts/cm)

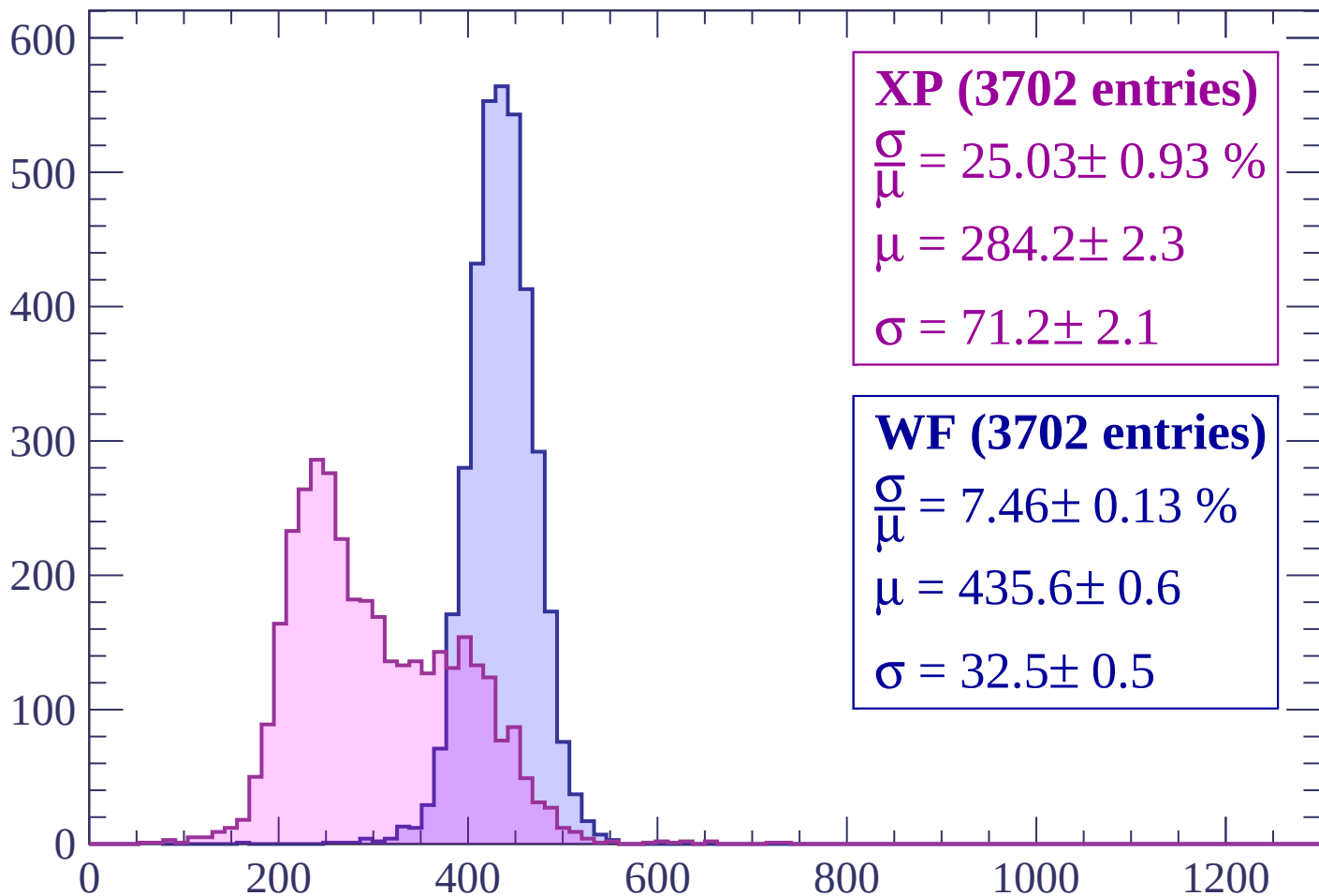
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count



XP (3702 entries)

$\frac{\sigma}{\mu} = 25.03 \pm 0.93 \%$

$\mu = 284.2 \pm 2.3$

$\sigma = 71.2 \pm 2.1$

WF (3702 entries)

$\frac{\sigma}{\mu} = 7.46 \pm 0.13 \%$

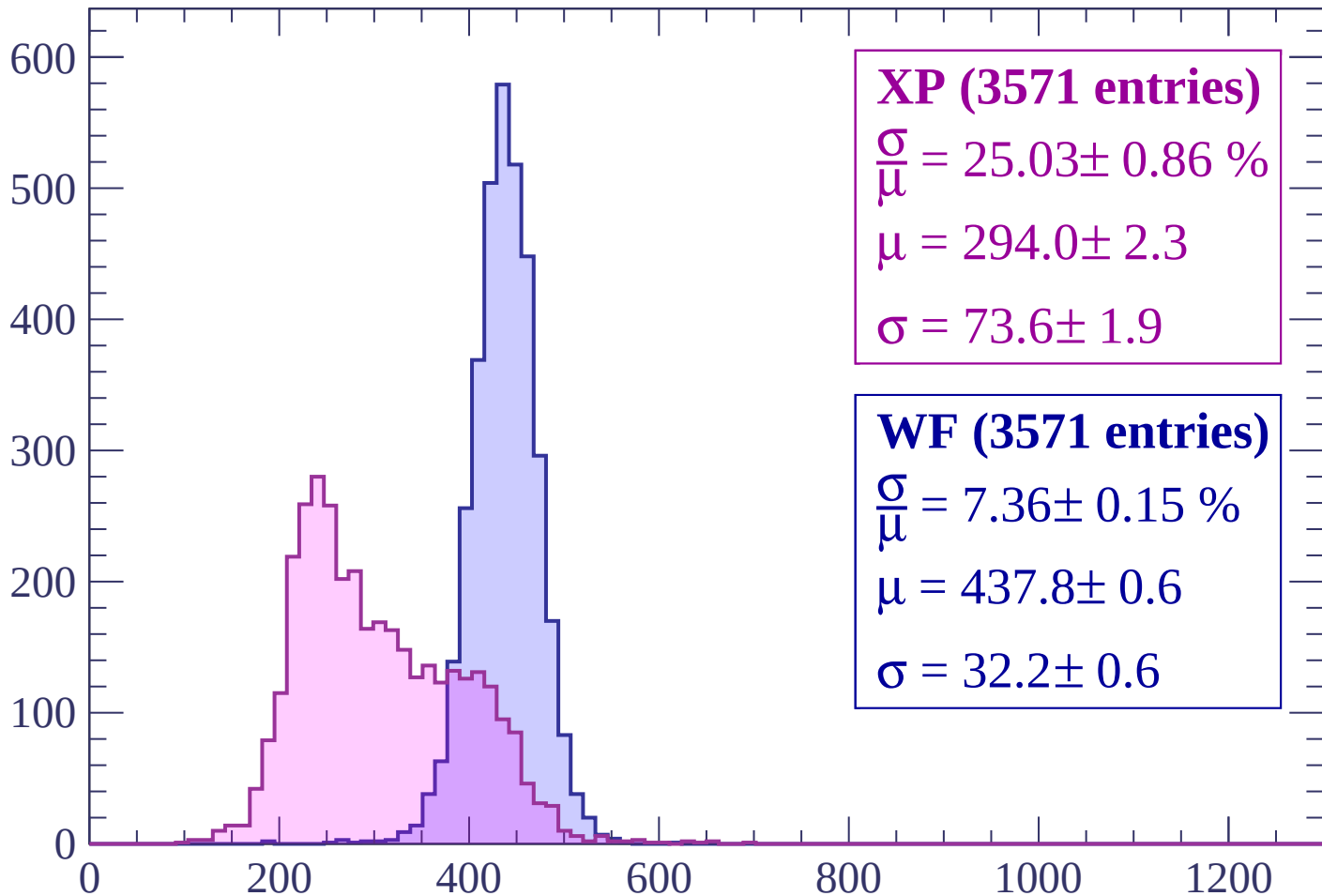
$\mu = 435.6 \pm 0.6$

$\sigma = 32.5 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (3571 entries)

$\frac{\sigma}{\mu} = 25.03 \pm 0.86 \%$

$\mu = 294.0 \pm 2.3$

$\sigma = 73.6 \pm 1.9$

WF (3571 entries)

$\frac{\sigma}{\mu} = 7.36 \pm 0.15 \%$

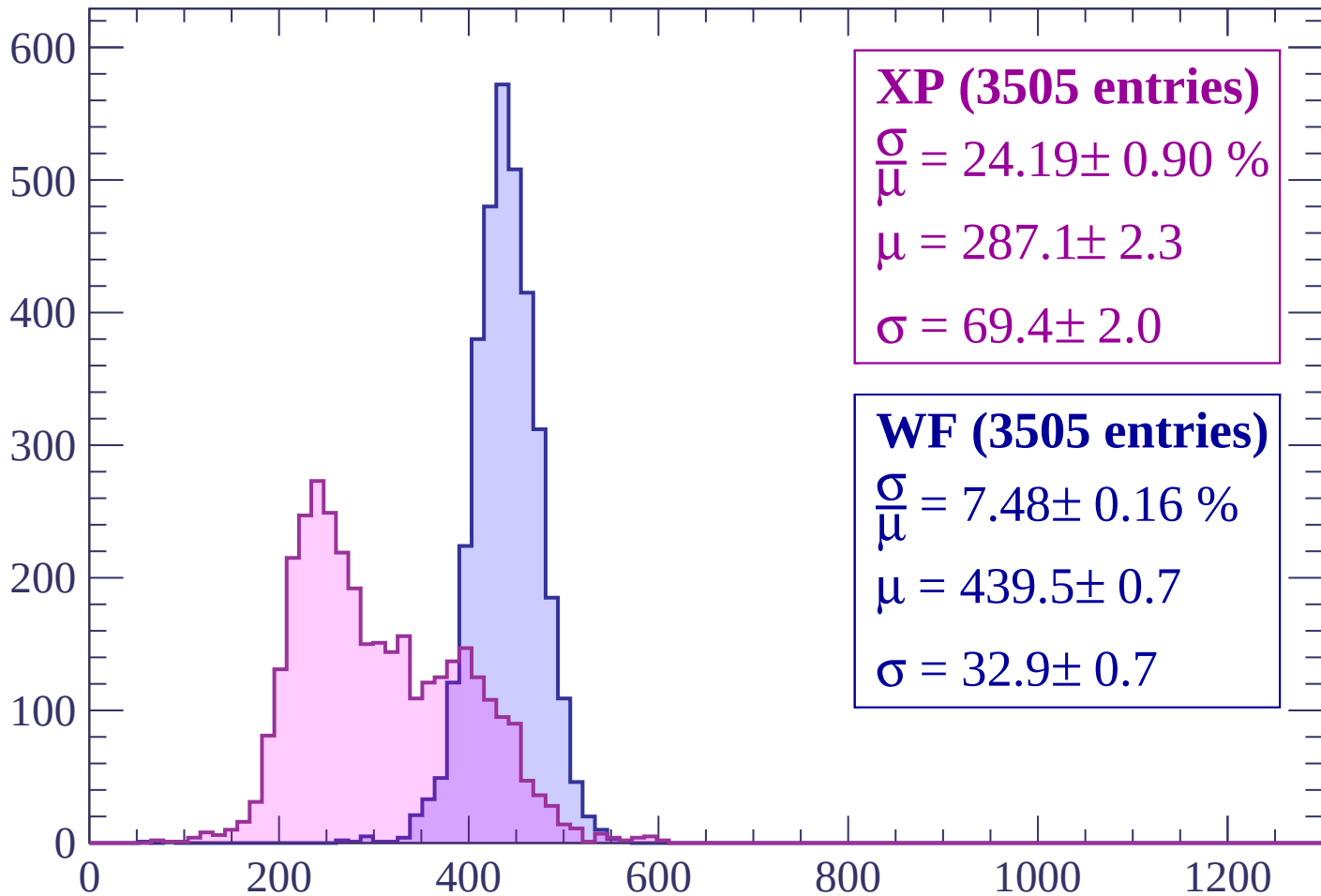
$\mu = 437.8 \pm 0.6$

$\sigma = 32.2 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (3505 entries)

$\frac{\sigma}{\mu} = 24.19 \pm 0.90 \%$

$\mu = 287.1 \pm 2.3$

$\sigma = 69.4 \pm 2.0$

WF (3505 entries)

$\frac{\sigma}{\mu} = 7.48 \pm 0.16 \%$

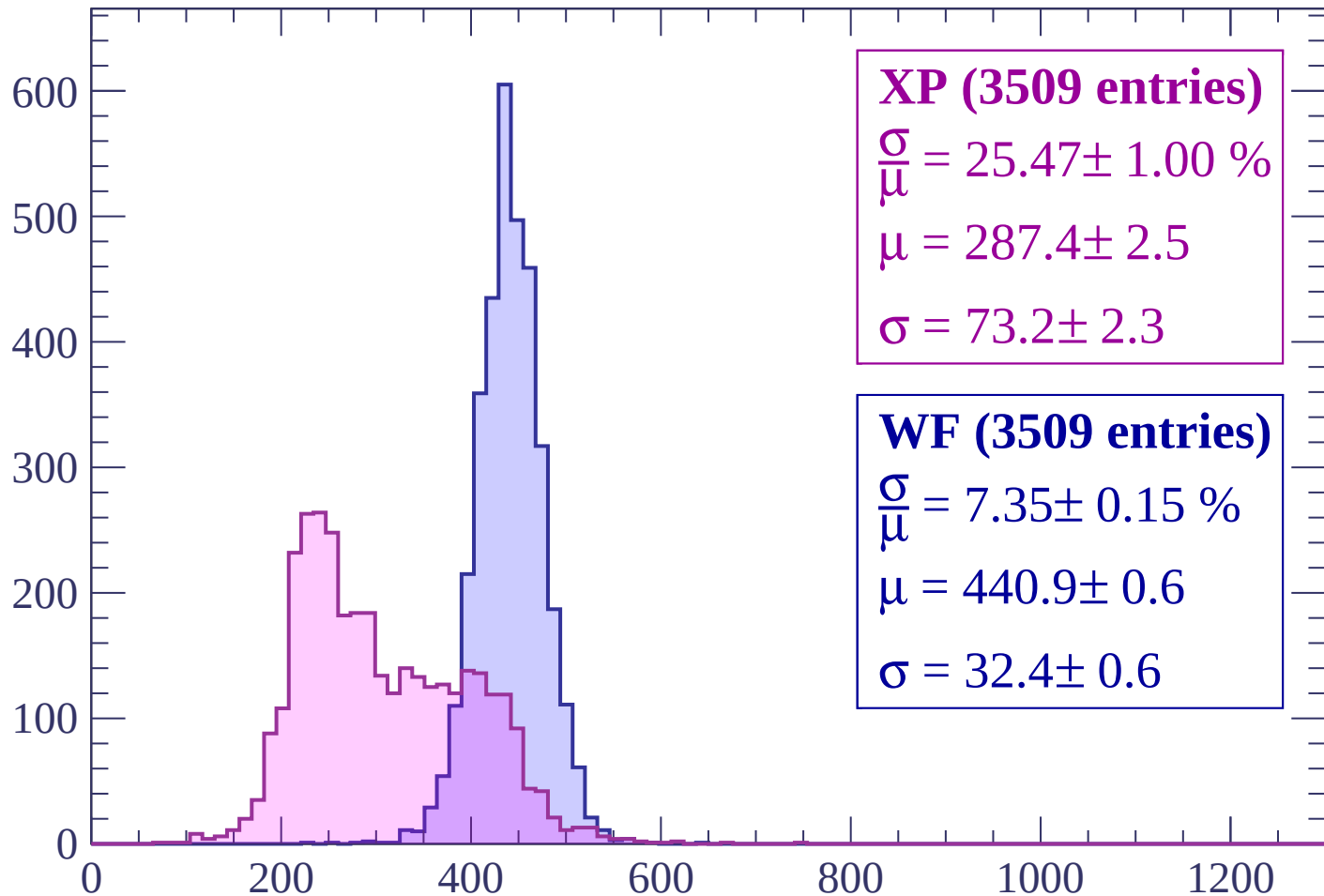
$\mu = 439.5 \pm 0.7$

$\sigma = 32.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (3509 entries)

$$\frac{\sigma}{\mu} = 25.47 \pm 1.00 \%$$

$$\mu = 287.4 \pm 2.5$$

$$\sigma = 73.2 \pm 2.3$$

WF (3509 entries)

$$\frac{\sigma}{\mu} = 7.35 \pm 0.15 \%$$

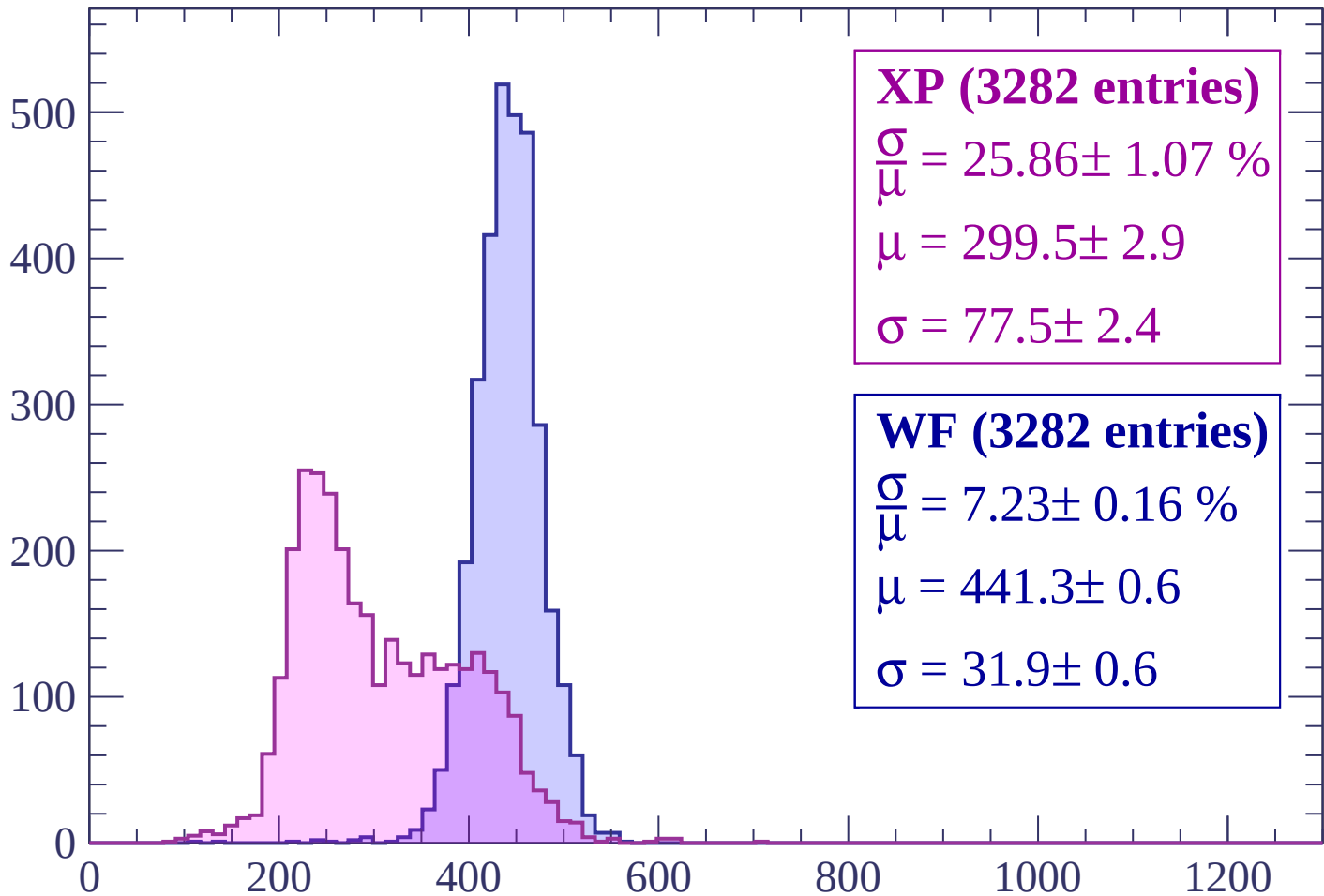
$$\mu = 440.9 \pm 0.6$$

$$\sigma = 32.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (3282 entries)

$$\frac{\sigma}{\mu} = 25.86 \pm 1.07 \%$$

$$\mu = 299.5 \pm 2.9$$

$$\sigma = 77.5 \pm 2.4$$

WF (3282 entries)

$$\frac{\sigma}{\mu} = 7.23 \pm 0.16 \%$$

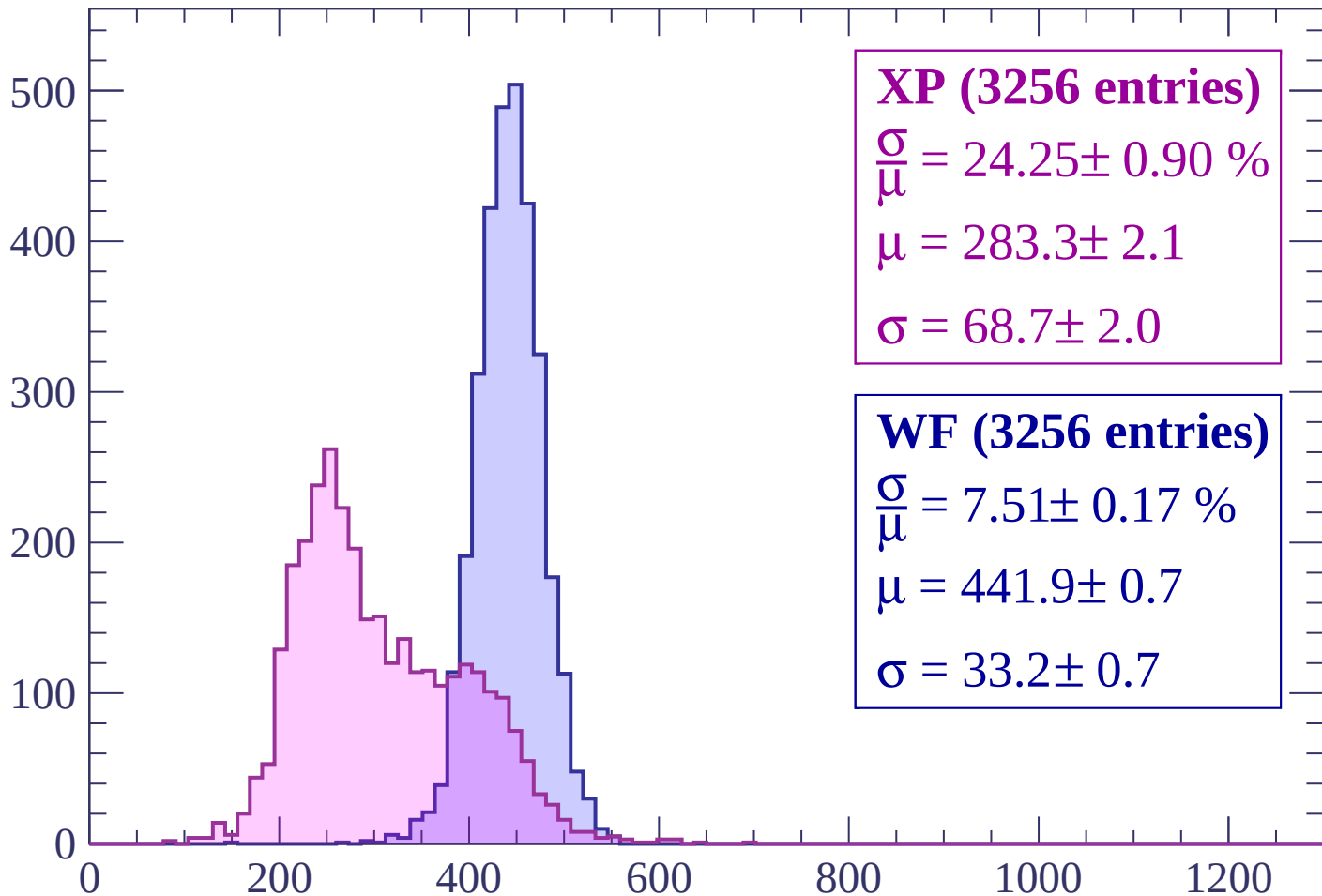
$$\mu = 441.3 \pm 0.6$$

$$\sigma = 31.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count



XP (3256 entries)

$\frac{\sigma}{\mu} = 24.25 \pm 0.90 \%$

$\mu = 283.3 \pm 2.1$

$\sigma = 68.7 \pm 2.0$

WF (3256 entries)

$\frac{\sigma}{\mu} = 7.51 \pm 0.17 \%$

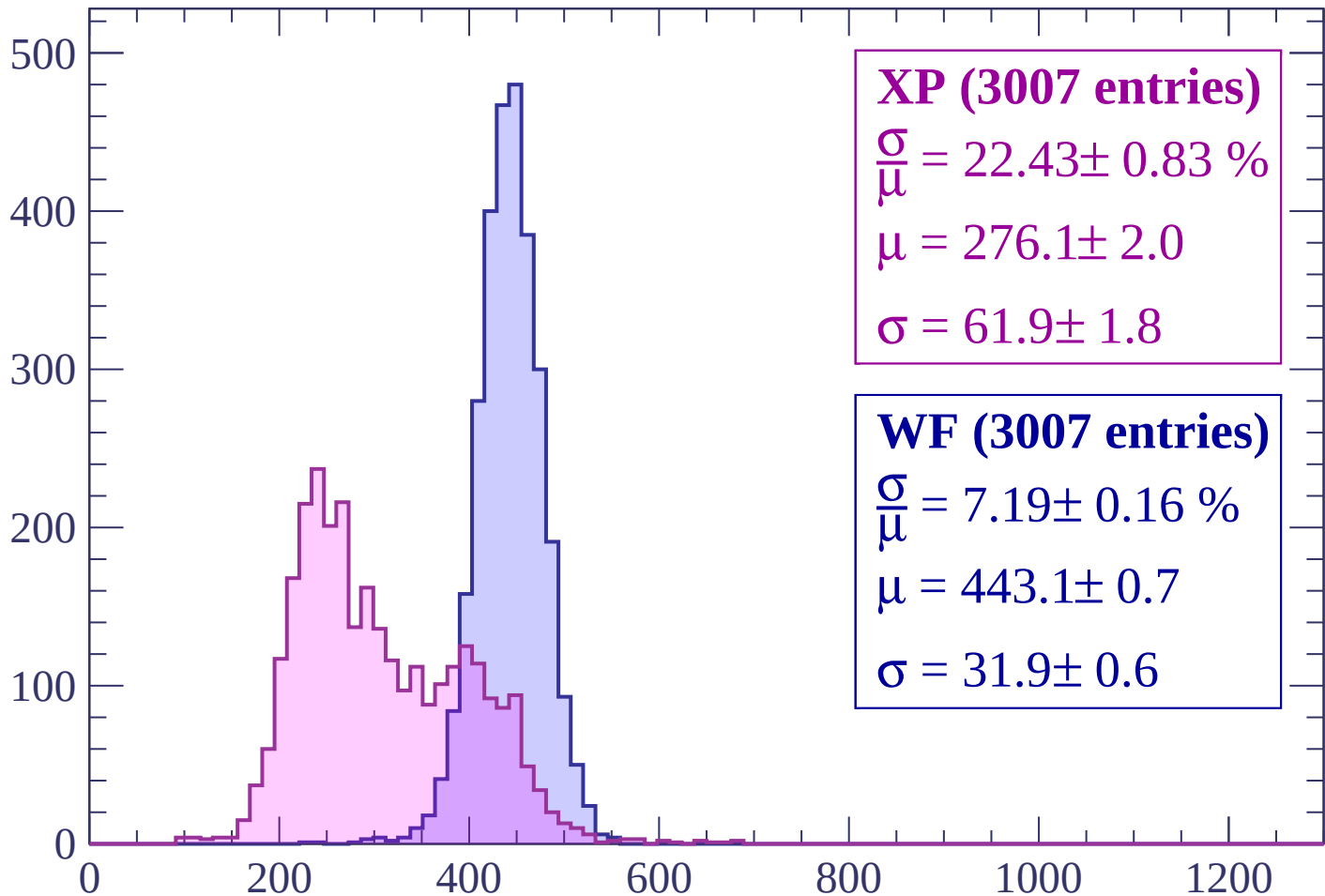
$\mu = 441.9 \pm 0.7$

$\sigma = 33.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $2340 < p < 2400$

Count



XP (3007 entries)

$$\frac{\sigma}{\mu} = 22.43 \pm 0.83 \%$$

$$\mu = 276.1 \pm 2.0$$

$$\sigma = 61.9 \pm 1.8$$

WF (3007 entries)

$$\frac{\sigma}{\mu} = 7.19 \pm 0.16 \%$$

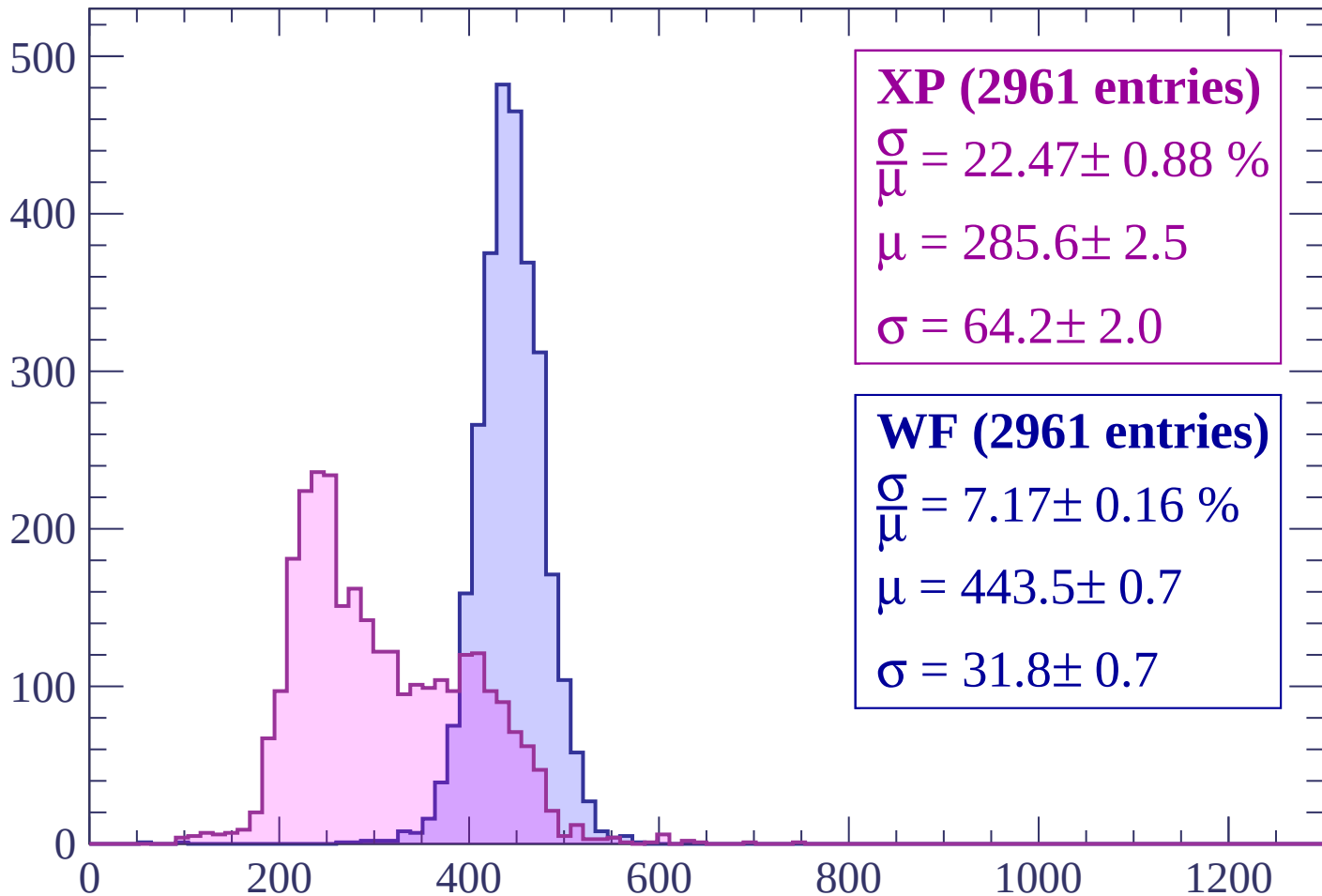
$$\mu = 443.1 \pm 0.7$$

$$\sigma = 31.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count



XP (2961 entries)

$$\frac{\sigma}{\mu} = 22.47 \pm 0.88 \%$$

$$\mu = 285.6 \pm 2.5$$

$$\sigma = 64.2 \pm 2.0$$

WF (2961 entries)

$$\frac{\sigma}{\mu} = 7.17 \pm 0.16 \%$$

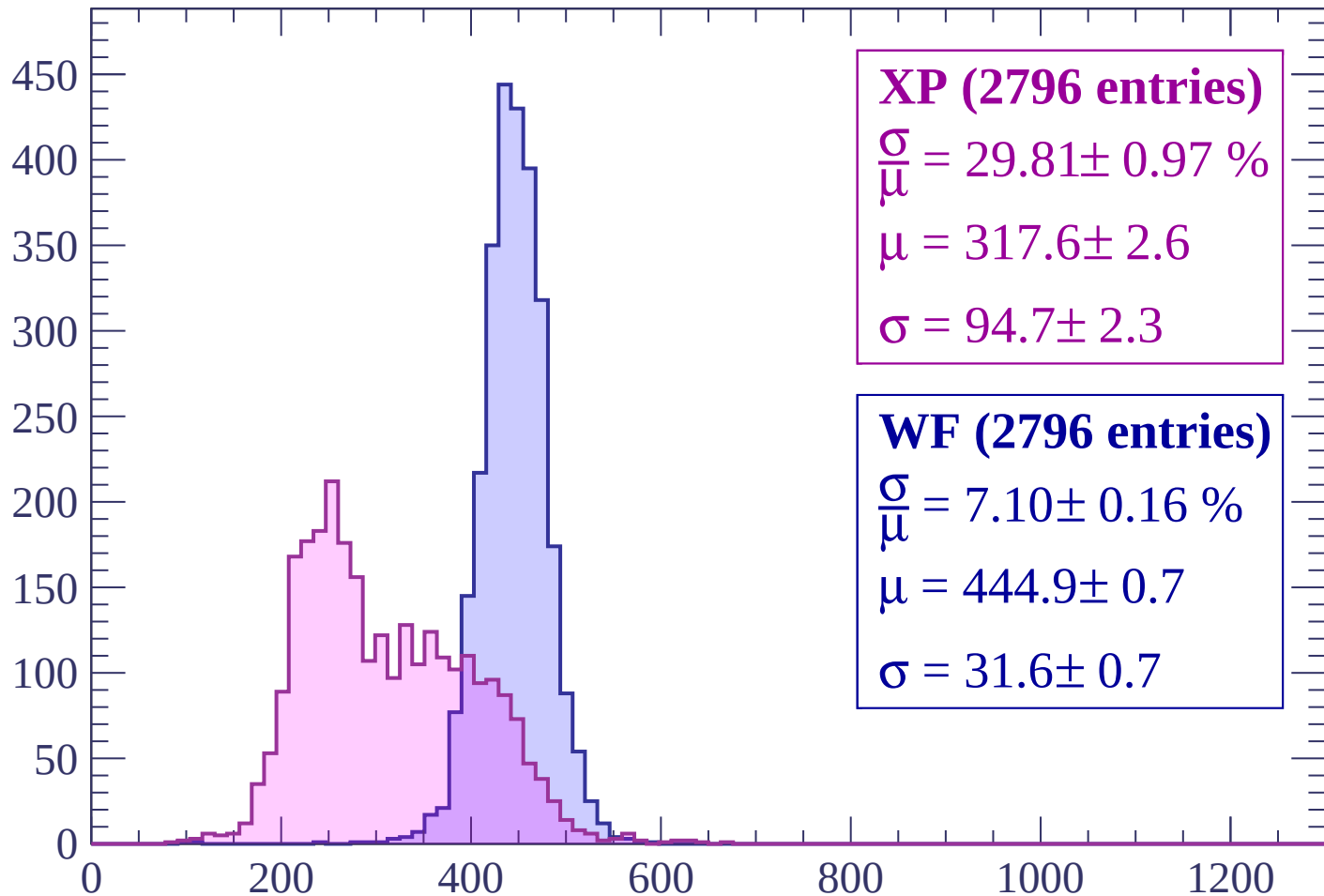
$$\mu = 443.5 \pm 0.7$$

$$\sigma = 31.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $2460 < p < 2520$

Count



XP (2796 entries)

$\frac{\sigma}{\mu} = 29.81 \pm 0.97 \%$

$\mu = 317.6 \pm 2.6$

$\sigma = 94.7 \pm 2.3$

WF (2796 entries)

$\frac{\sigma}{\mu} = 7.10 \pm 0.16 \%$

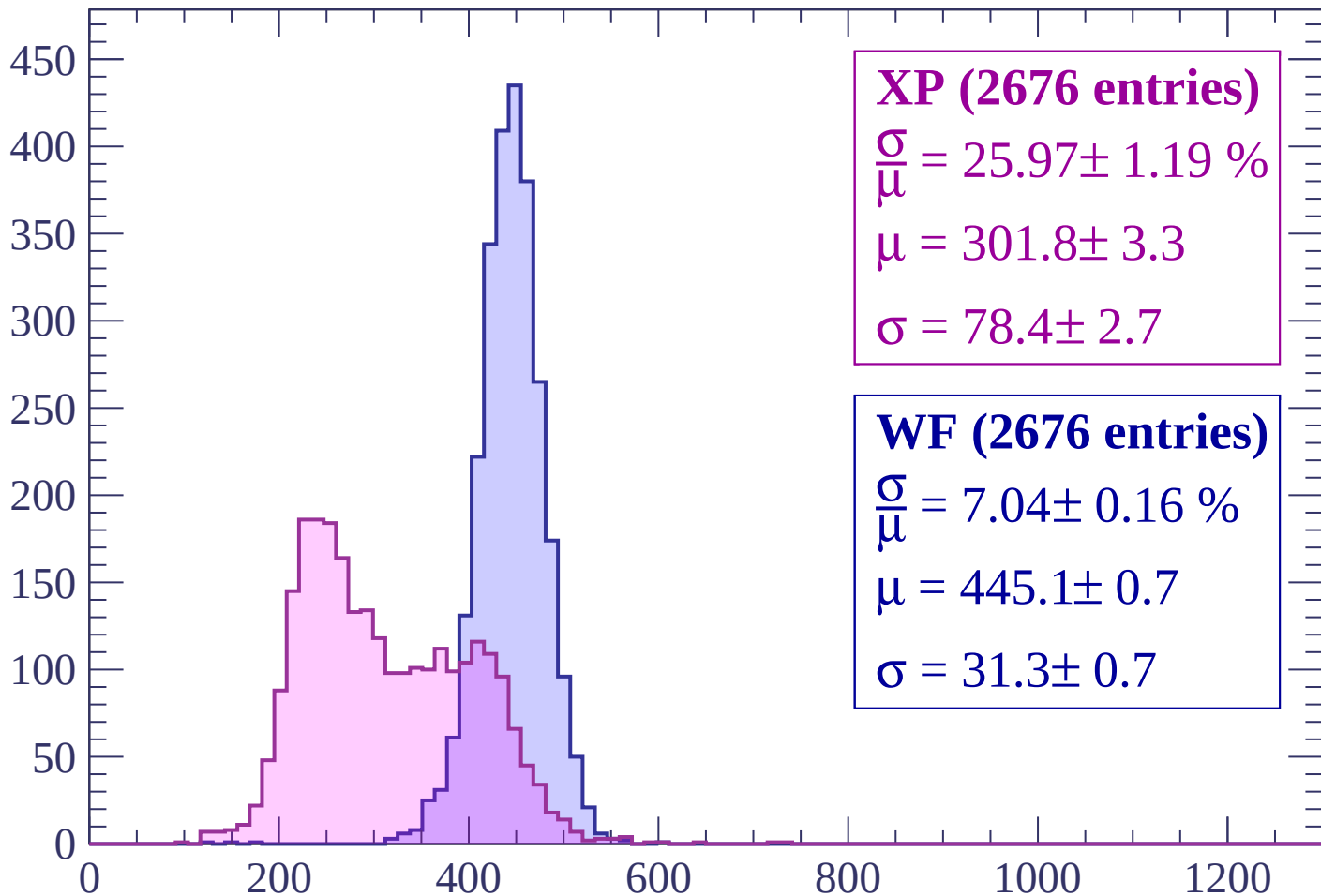
$\mu = 444.9 \pm 0.7$

$\sigma = 31.6 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $2520 < p < 2580$

Count



XP (2676 entries)

$\frac{\sigma}{\mu} = 25.97 \pm 1.19 \%$

$\mu = 301.8 \pm 3.3$

$\sigma = 78.4 \pm 2.7$

WF (2676 entries)

$\frac{\sigma}{\mu} = 7.04 \pm 0.16 \%$

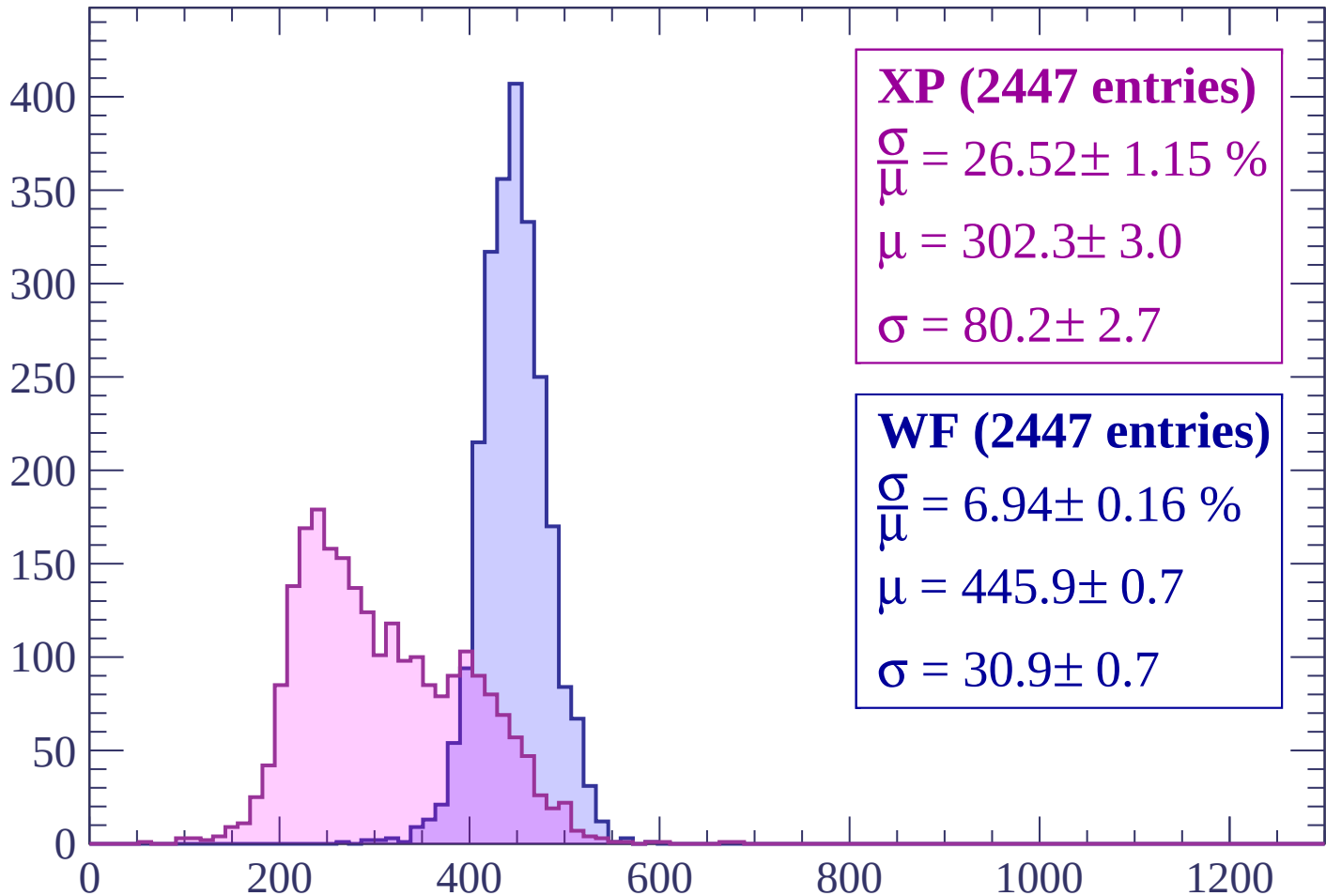
$\mu = 445.1 \pm 0.7$

$\sigma = 31.3 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (2447 entries)

$\frac{\sigma}{\mu} = 26.52 \pm 1.15 \%$

$\mu = 302.3 \pm 3.0$

$\sigma = 80.2 \pm 2.7$

WF (2447 entries)

$\frac{\sigma}{\mu} = 6.94 \pm 0.16 \%$

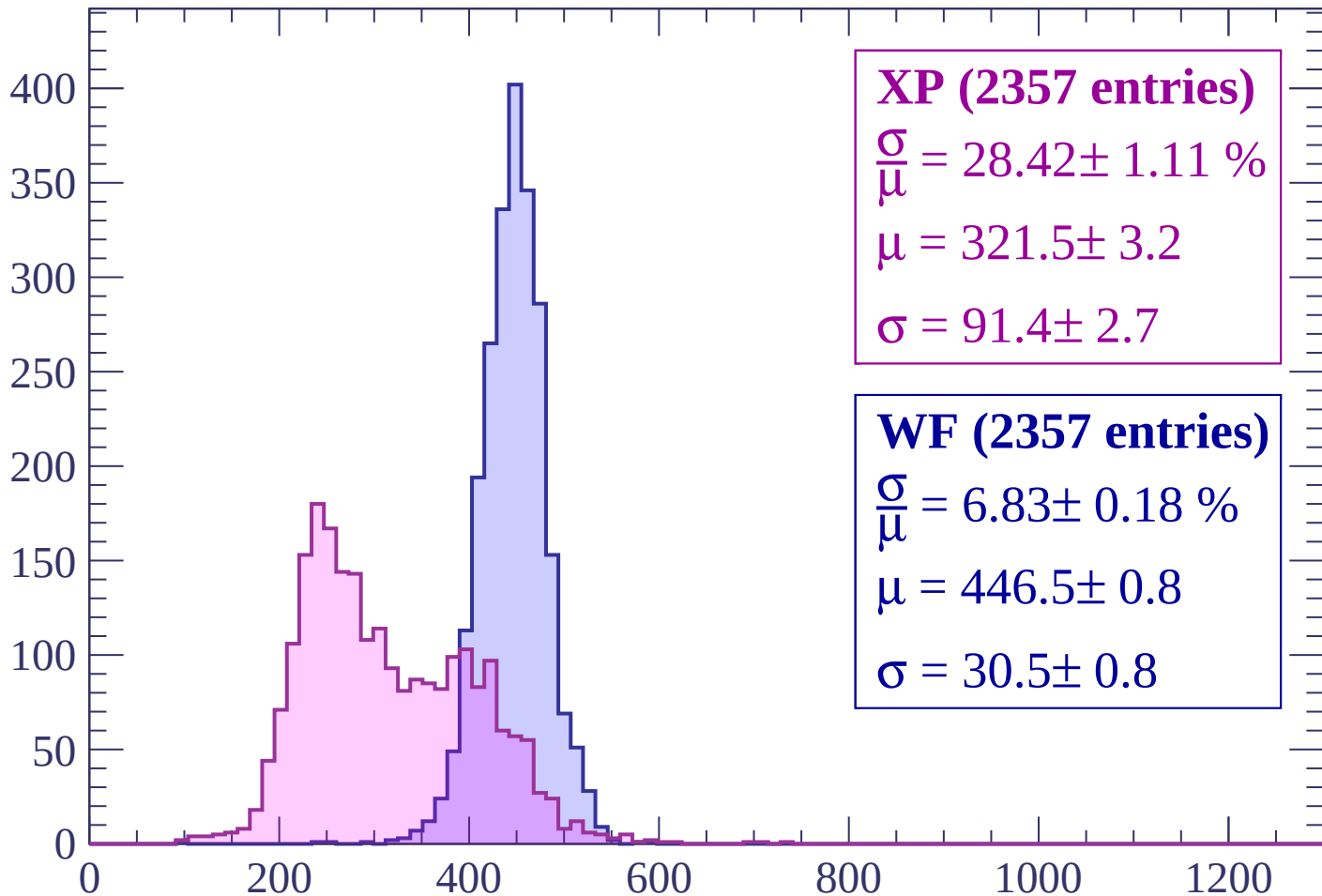
$\mu = 445.9 \pm 0.7$

$\sigma = 30.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $2640 < p < 2700$

Count



XP (2357 entries)

$$\frac{\sigma}{\mu} = 28.42 \pm 1.11 \%$$

$$\mu = 321.5 \pm 3.2$$

$$\sigma = 91.4 \pm 2.7$$

WF (2357 entries)

$$\frac{\sigma}{\mu} = 6.83 \pm 0.18 \%$$

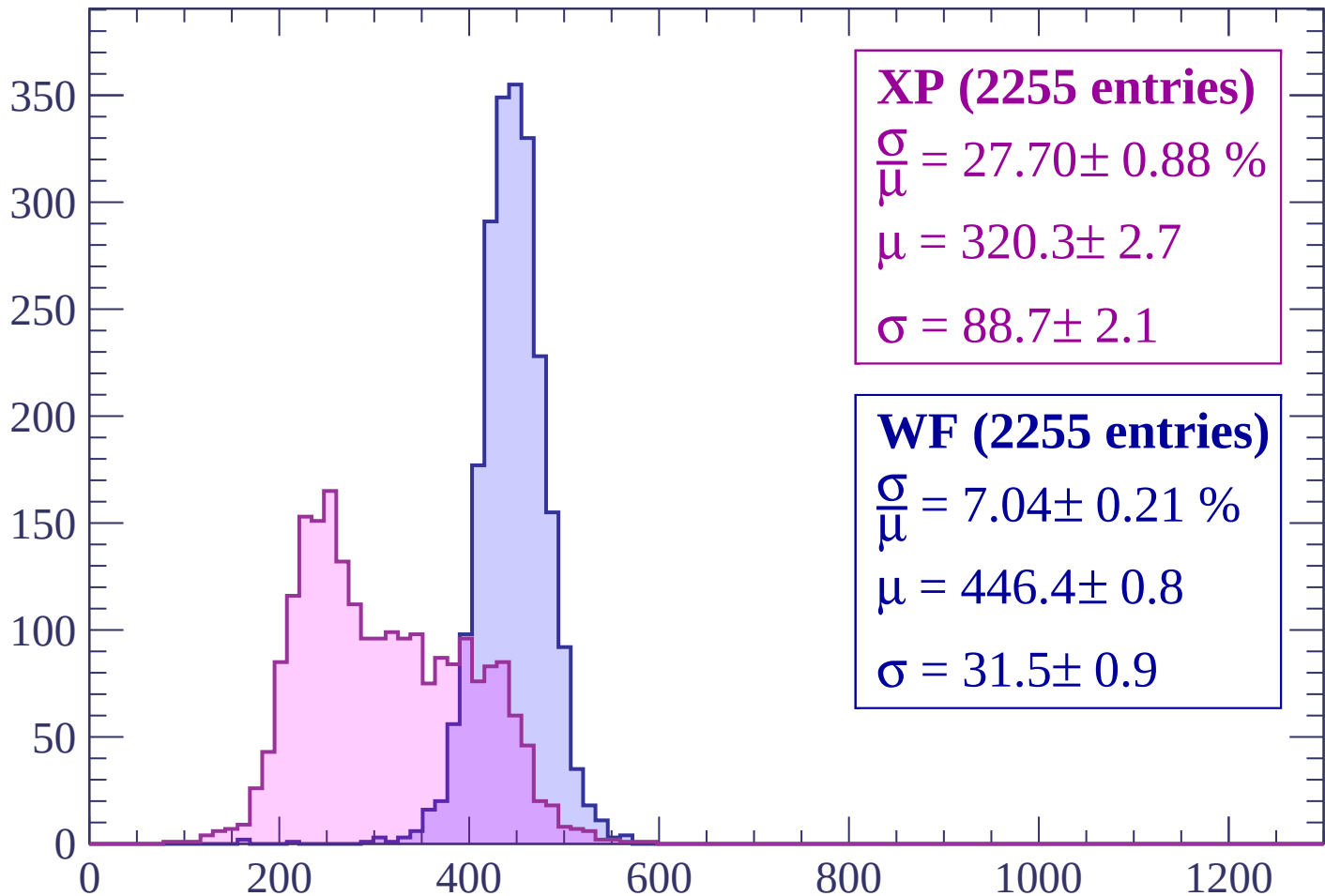
$$\mu = 446.5 \pm 0.8$$

$$\sigma = 30.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (2255 entries)

$$\frac{\sigma}{\mu} = 27.70 \pm 0.88 \%$$

$$\mu = 320.3 \pm 2.7$$

$$\sigma = 88.7 \pm 2.1$$

WF (2255 entries)

$$\frac{\sigma}{\mu} = 7.04 \pm 0.21 \%$$

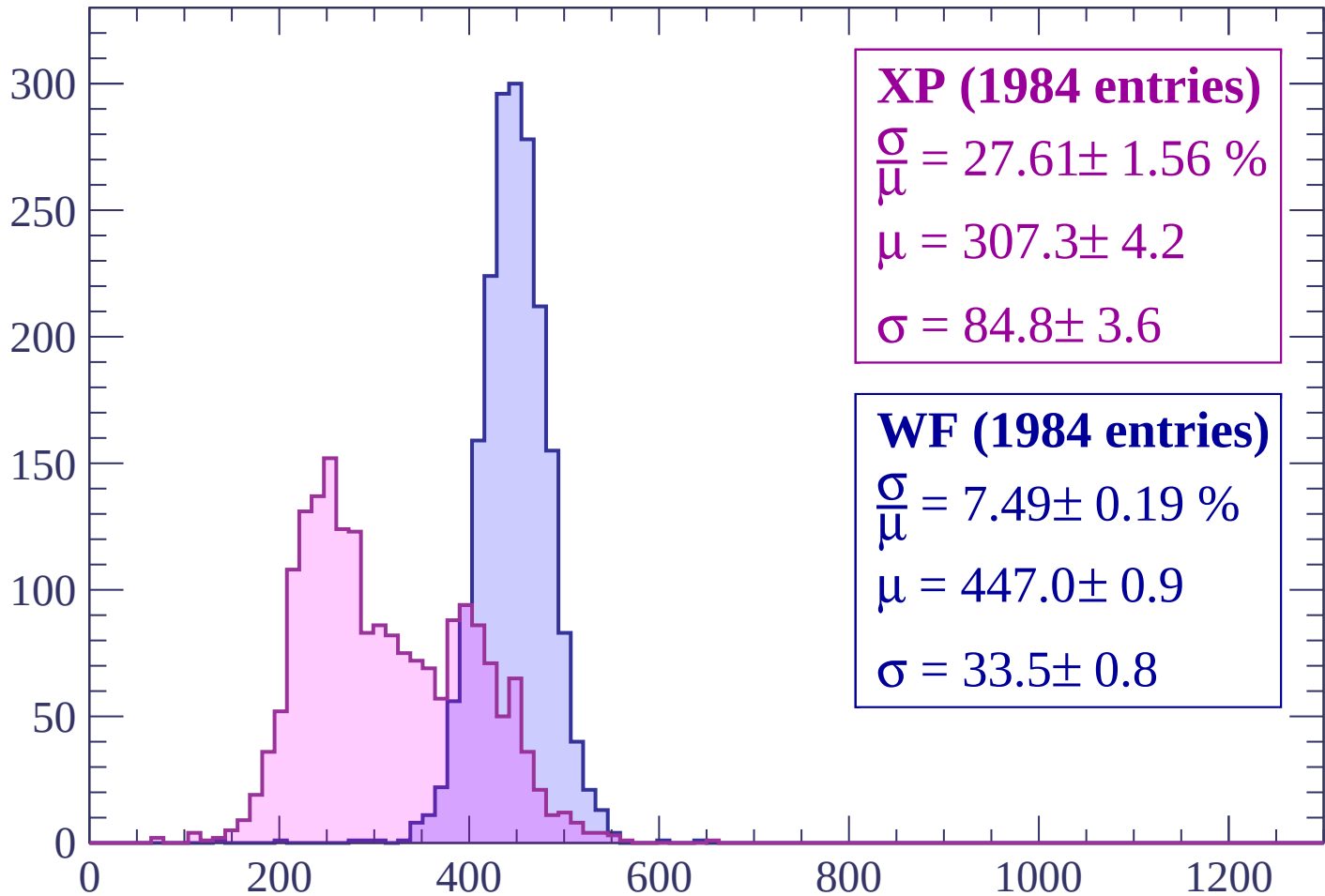
$$\mu = 446.4 \pm 0.8$$

$$\sigma = 31.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2760 < p < 2820$

Count



XP (1984 entries)

$$\frac{\sigma}{\mu} = 27.61 \pm 1.56 \%$$

$$\mu = 307.3 \pm 4.2$$

$$\sigma = 84.8 \pm 3.6$$

WF (1984 entries)

$$\frac{\sigma}{\mu} = 7.49 \pm 0.19 \%$$

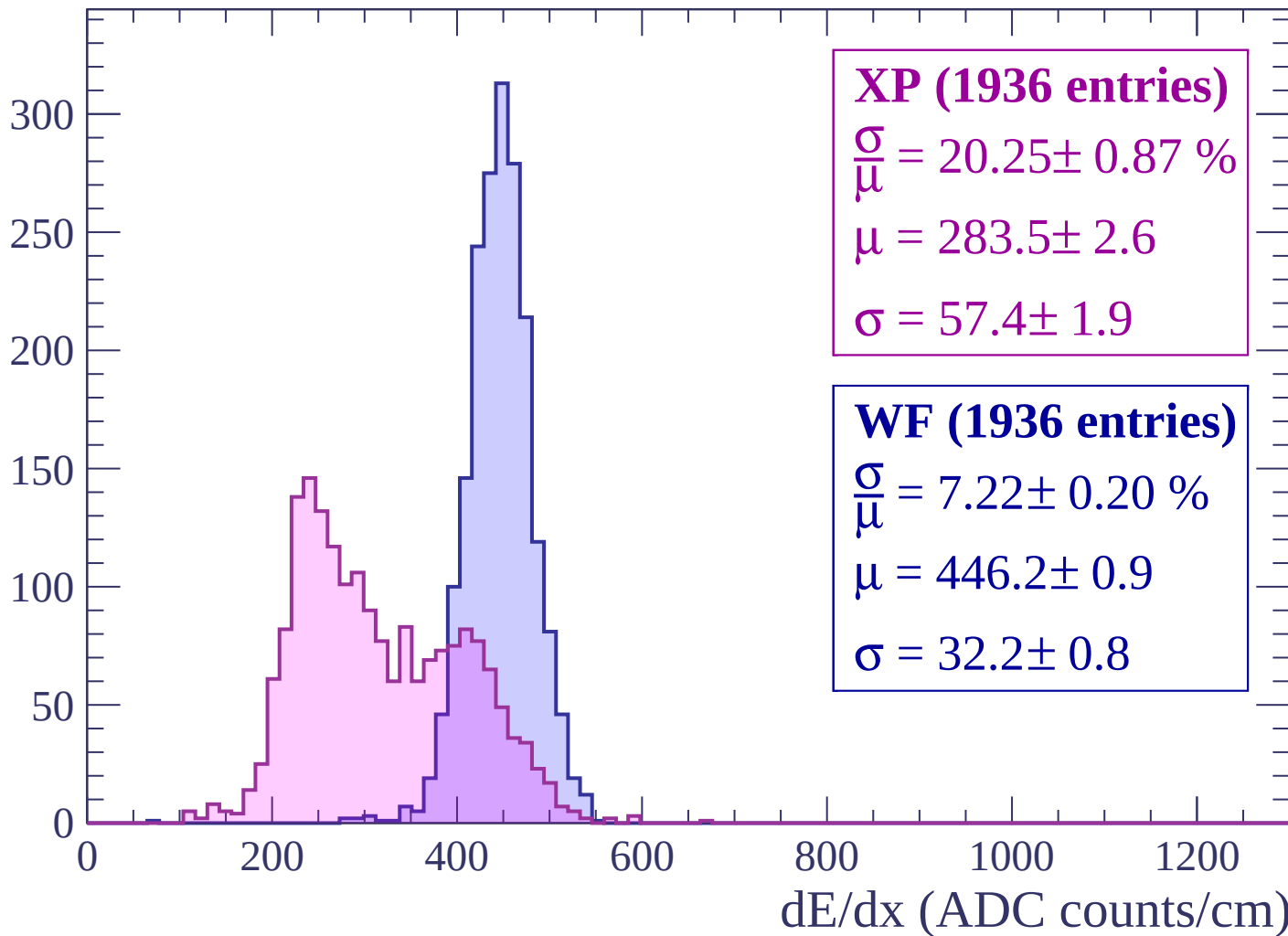
$$\mu = 447.0 \pm 0.9$$

$$\sigma = 33.5 \pm 0.8$$

dE/dx (ADC counts/cm)

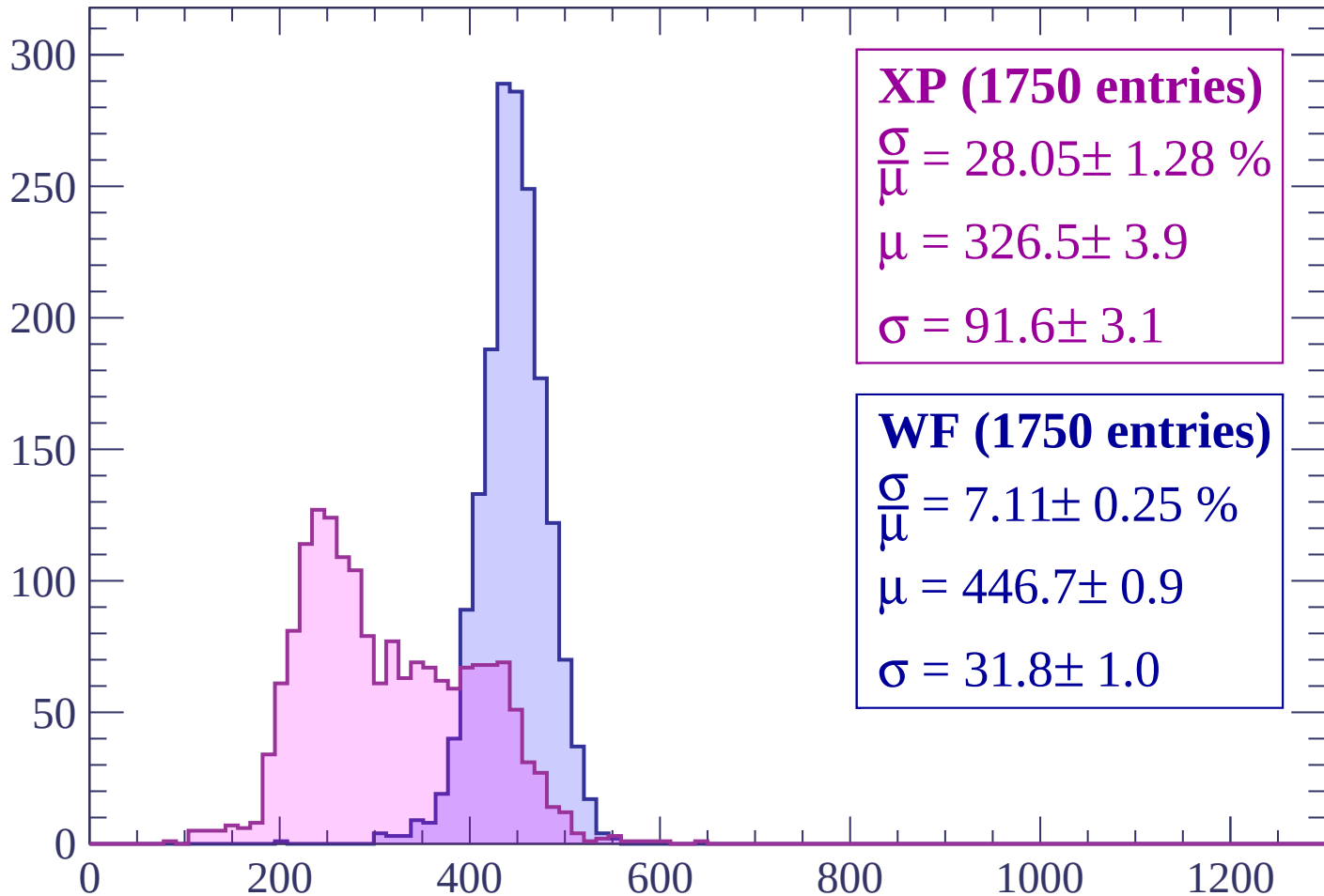
Energy loss | $2820 < p < 2880$

Count



Energy loss | $2880 < p < 2940$

Count



XP (1750 entries)

$$\frac{\sigma}{\mu} = 28.05 \pm 1.28 \%$$

$$\mu = 326.5 \pm 3.9$$

$$\sigma = 91.6 \pm 3.1$$

WF (1750 entries)

$$\frac{\sigma}{\mu} = 7.11 \pm 0.25 \%$$

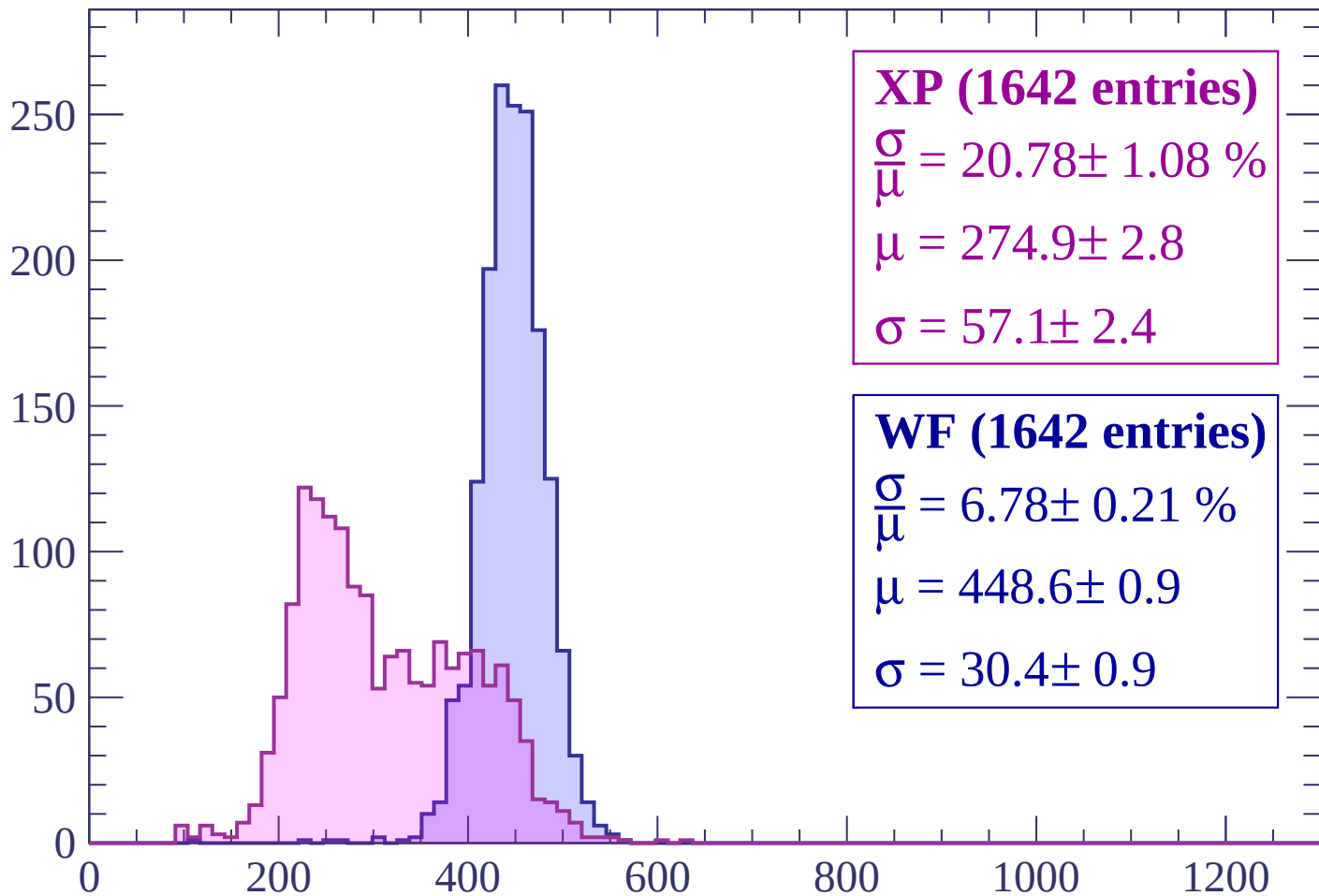
$$\mu = 446.7 \pm 0.9$$

$$\sigma = 31.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count



XP (1642 entries)

$$\frac{\sigma}{\mu} = 20.78 \pm 1.08 \%$$

$$\mu = 274.9 \pm 2.8$$

$$\sigma = 57.1 \pm 2.4$$

WF (1642 entries)

$$\frac{\sigma}{\mu} = 6.78 \pm 0.21 \%$$

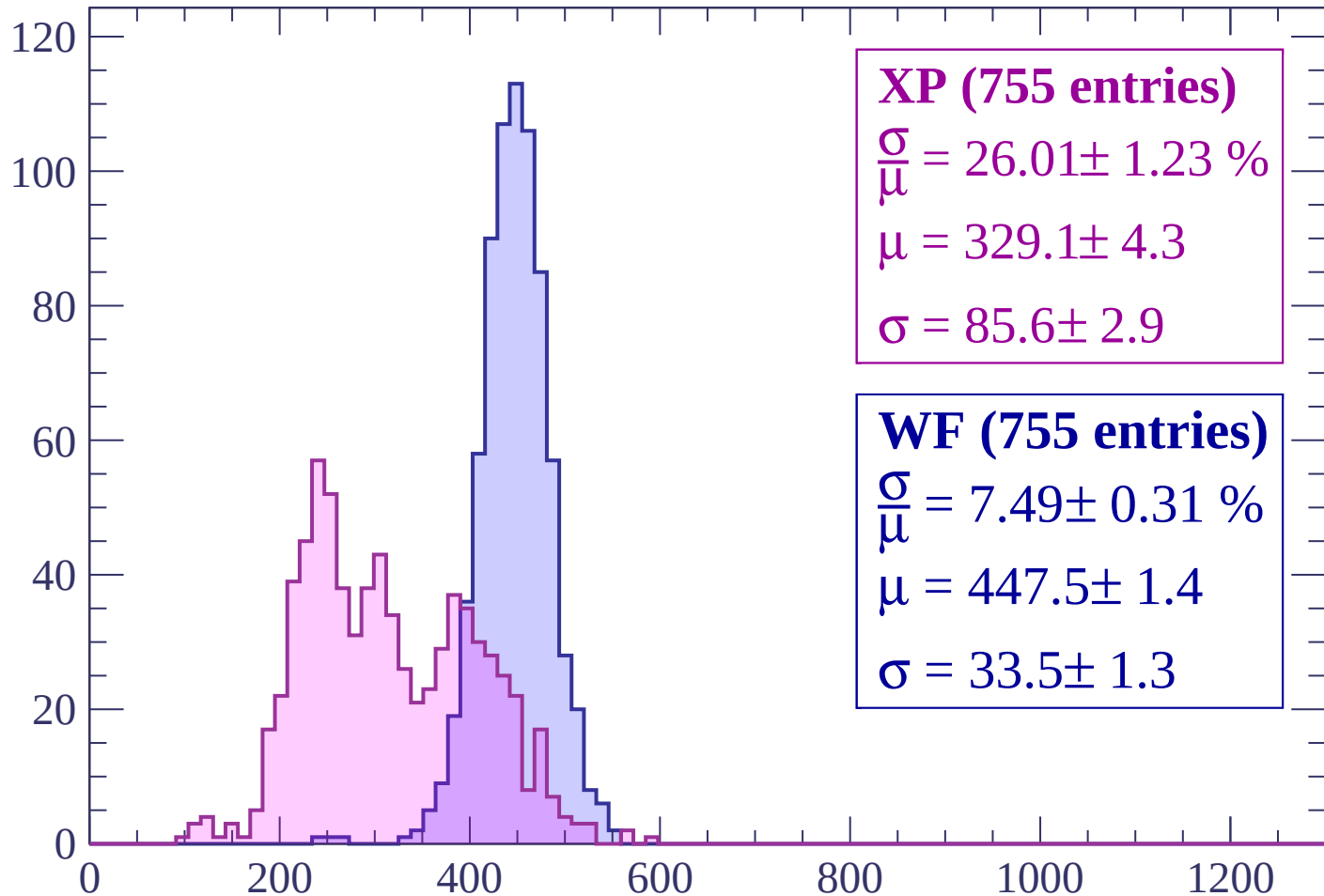
$$\mu = 448.6 \pm 0.9$$

$$\sigma = 30.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



XP (755 entries)

$$\frac{\sigma}{\mu} = 26.01 \pm 1.23 \%$$

$$\mu = 329.1 \pm 4.3$$

$$\sigma = 85.6 \pm 2.9$$

WF (755 entries)

$$\frac{\sigma}{\mu} = 7.49 \pm 0.31 \%$$

$$\mu = 447.5 \pm 1.4$$

$$\sigma = 33.5 \pm 1.3$$

dE/dx (ADC counts/cm)

